

Federated LPV Identification and Control

Development Report

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Abstract

This living report develops and tests a federated LPV approach that separates measurable operating-condition variation from persistent client heterogeneity. The first stage is an oracle feasibility study. Only after the structural benefit has been established will limited local data, learned clustering, uncertainty, and communication constraints be introduced. The report will progressively become the technical basis for a consolidated paper manuscript.

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1 Core idea

The project separates two sources of variation that a frozen-model workflow can confound:

$$\text{measurable operating variation} \neq \text{persistent client heterogeneity}. \quad (1)$$

For vehicle dynamics, longitudinal speed $\rho = v_x$ is measurable and varies within a drive. Structural parameters such as mass, yaw inertia, and tire cornering stiffness remain fixed for a client in the initial study. Client i is represented as

$$x_{i,k+1} = A_i(\rho_k)x_{i,k} + B_i(\rho_k)u_{i,k}, \quad (2)$$

and is assumed to belong approximately to one persistent dynamics family $c(i)$:

$$A_i(\rho) \approx A_{c(i)}(\rho), \quad B_i(\rho) \approx B_{c(i)}(\rho). \quad (3)$$

The central hypothesis is that LPV scheduling should explain operating-point variation, whereas clustered federated learning should handle unknown persistent structural variation. In geometric terms, the goal is to learn families of model manifolds rather than clusters of unrelated frozen-model points.

1.1 Research question and hypothesis

Research question. Can federated system identification disentangle measurable operating-condition variation from persistent fleet heterogeneity, enabling a heterogeneous fleet to be represented and controlled by a small number of reusable LPV model and controller families?

Hypothesis. Explicit LPV scheduling combined with structural specialization yields better generalization and lower model/controller proliferation than either one global LPV model or purely clustered LTI models.

2 Motivation and practical value

A frozen LTI model changes when the same vehicle is observed at different speeds. Clustering those models without accounting for speed can therefore confuse an operating point with a vehicle identity. An LPV description encodes that these observations belong to one structural system evaluated at different values of a known scheduling variable.

The distinction may reduce model and controller proliferation. A purely LTI architecture can require approximately

$$N_{\text{families}} N_{\text{speed regions}} \quad (4)$$

calibrated controllers, whereas the proposed representation requires roughly one scheduled controller family per relevant structural class. This matters for calibration, deployment, validation, maintenance, and eventual certification.

Federation becomes particularly useful when individual clients cover only parts of the scheduling range. Several compatible clients can jointly identify an LPV family over an envelope that no single client has explored, without transferring raw trajectories. Structural specialization simultaneously limits negative transfer from physically incompatible clients. The full architecture separates the three questions summarized in table 1.

Table 1: Distinct information roles in the proposed architecture.

Quantity	Meaning	Mechanism
$\rho(t)$	current operating condition	LPV scheduling
c_i	persistent dynamics family	clustering/personalization
Σ_i	reliability of local evidence	uncertainty-aware federation

This work continues the uncertainty-aware clustered identification and LQI design pipeline developed for the IFAC study. Its new contribution is to give scheduling variation and persistent fleet heterogeneity distinct mathematical roles. The relation to LPV control, heterogeneous federated learning, and federated system identification is developed in section 3.

3 Related work and positioning

3.1 LPV modeling and vehicle control

LPV models represent dynamics whose coefficients depend on measurable scheduling variables, providing a systematic alternative to independently designed frozen controllers. Established LPV references cover identification, embedding choices, gain scheduling, and stability analysis [14, 1, 6]. Vehicle-control studies have used LPV formulations to expose velocity dependence and tire nonlinearities to scheduled controllers, including path-tracking designs validated in high-fidelity simulation [3, 5]. The bicycle and tire models used here follow the standard vehicle-dynamics hierarchy from linear cornering stiffness to nonlinear saturation [11].

This project does not claim novelty for representing vehicle dynamics as LPV or for interpolating gains over speed. Its LPV contribution is representational: known within-client operating variation is assigned to scheduling, so that persistent between-client differences can be learned without proliferating unrelated frozen-speed models.

3.2 Federated learning under heterogeneity

Federated averaging established the shared-model paradigm in which clients retain their raw data and communicate locally computed updates [10]. When client objectives differ, federated multi-task and clustered methods replace a single compromise model with structured sharing [13, 4, 12]. These methods motivate family-specific aggregation, but generally treat heterogeneity as a statistical or task-level property rather than separating a measured scheduling coordinate from a persistent physical phenotype.

3.3 Federated system identification and the research gap

FedSysID studies collaborative identification when clients observe similar but non-identical linear dynamical systems and quantifies the tradeoff between added samples and system heterogeneity [15]. This directly supports the premise that a fleet can improve system identification through

collaboration, while also showing why unstructured global aggregation can become harmful as physical differences increase.

The gap addressed here lies at the intersection of these bodies of work. Existing LPV vehicle-control studies primarily assume a prescribed plant or centralized identification; clustered FL methods learn multiple client models but do not encode continuous operating-point variation; and current federated system identification centers on shared or nearby fixed-system models. The proposed architecture instead seeks a small number of *LPV model manifolds*: each family shares a speed-dependent model, while family membership captures persistent vehicle heterogeneity. The oracle experiments in sections 7 to 10 first test whether this decomposition is useful before any claim is made about recovering it from decentralized data.

4 Problem formulation

Let the structural parameters of client i be ϑ_i and the measured scheduling variable be ρ_k . During the oracle study,

$$\vartheta_i(t) = \vartheta_i, \quad \rho_k = v_{x,k} \in [10, 30] \text{ m s}^{-1}. \quad (5)$$

The eventual hierarchical model is

$$\Theta_i(\rho) = \Theta_{\text{shared}}(\rho) + \Delta\Theta_{c(i)}(\rho) + \epsilon_i(\rho), \quad \Theta = [A \ B]. \quad (6)$$

The first finite-dimensional LPV basis to test is

$$\Theta(\rho) = \Theta_0 + \rho^{-1}\Theta_1 + \rho^{-2}\Theta_2. \quad (7)$$

Phase 1 must validate this approximation before it is used for learning.

The controlled output is yaw rate r , with sideslip β constrained through the cost. A two-degree-of-freedom scheduled LQI architecture is considered:

$$u = -K_x(\rho)x - K_I(\rho)\chi + k_r(\rho)r^*. \quad (8)$$

The same Q and R weights are used for all methods so that improvements cannot be attributed to method-specific tuning.

5 Vehicle benchmark

The plant is the linear dynamic bicycle model with $x = [\beta \ r]^\top$ and steering input $u = \delta$. Its parameters are $\vartheta = [m, I_z, l_f, l_r, C_f, C_r]$. The initial centers are a nominal balanced vehicle, a heavy/high-inertia vehicle, and a vehicle with a changed front–rear handling balance. Ten clients per family receive modest fixed parameter scatter. These values are initial hypotheses and will be checked for physical plausibility and nontrivial but non-extreme separability during plant validation. The relative centers used in the oracle study are listed in table 2; their absolute values and validated handling properties appear later in table 5.

Table 2: Initial family centers relative to the nominal vehicle.

Family	m/m_0	$I_z/I_{z,0}$	$C_f/C_{f,0}$	$C_r/C_{r,0}$
Nominal	1.00	1.00	1.00	1.00
Heavy	1.15	1.20	1.05	1.00
Handling balance	0.95	0.90	0.90	1.10

The training set combines independent lateral maneuvers and smooth speed profiles. It also includes frozen-speed identification runs at 12, 16, 20, 24, 28 m s^{−1}. Test data contain unseen

intermediate speeds, unseen speed sequences, and a lateral maneuver with different timing and amplitude. This separation is required to test dynamics generalization rather than memorization of a speed–maneuver pairing.

6 Experimental methodology

The study follows a gated progression:

$$\begin{aligned} &\text{plant validation} \rightarrow \text{LPV representation validation} \rightarrow \text{oracle model comparison} \\ &\rightarrow \text{oracle controller comparison} \rightarrow \text{nonlinear-plant escalation} \rightarrow \text{federated identification.} \end{aligned} \quad (9)$$

The early stages deliberately remove learning imperfections. This allows a negative result to reject the structural premise before a federated algorithm is developed. Results are first obtained for one deterministic development seed and then repeated for ten independently sampled fleets and test-trajectory variants.

6.1 Plant hierarchy

The first plant is the speed-dependent linear bicycle model from section 5. It is LTI at a fixed speed but LPV across a drive:

$$\dot{x} = A(v_x, \vartheta)x + B(v_x, \vartheta)\delta. \quad (10)$$

It supports a clean test of the scheduling–specialization decomposition. After this oracle test, the plant is replaced by a nonlinear bicycle model with saturating hyperbolic-tangent tire forces. The identified models remain linear LTI or LPV models. Thus, the second plant tests whether the compact abstraction survives deliberate model mismatch.

6.2 Compared model architectures

The five oracle architectures are defined in table 3. M1–M4 form a two-factor comparison between scheduling and structural specialization, while M5 tests whether a larger bank of frozen LTI prototypes can substitute for continuous scheduling. All methods receive equivalent data

Table 3: Core oracle model architectures.

Method	Scheduling	Specialization	Interpretation	Families
M1: Global LTI	none	none	ignores both factors	1
M2: Clustered LTI	none	oracle family	structure only	3
M3: Global LPV	continuous	none	scheduling only	1
M4: Clustered LPV	continuous	oracle family	both factors	3
M5: Gridded LTI	discrete	family and speed	complex oracle	15

support. Controller comparisons use common state and input weights. Method-specific tuning is prohibited unless it is explicitly reported as a separate sensitivity analysis.

6.3 Training and test separation

Frozen-speed identification uses $v_x \in \{12, 16, 20, 24, 28\} \text{ m s}^{-1}$. Interpolation tests include $v_x \in \{14, 18, 22, 26\} \text{ m s}^{-1}$. Varying-speed training trajectories combine smooth speed profiles and lateral maneuvers independently so that speed and maneuver identity are not confounded. Test data include unseen speeds, unseen speed sequences, unseen lateral timing and amplitude, and unseen speed–maneuver combinations.

6.4 Common metrics

Model metrics include one-step and free-run errors in sideslip and yaw rate, and matrix errors

$$E_A(v) = \|A_i(v) - \hat{A}(v)\|_F, \quad E_B(v) = \|B_i(v) - \hat{B}(v)\|_F. \quad (11)$$

Closed-loop metrics include yaw-rate tracking RMSE, sideslip RMS, steering RMS, peak steering, peak steering rate, and the frozen-grid closed-loop spectral radius. Every main result reports fleet mean, dispersion, worst-family result, and family gap

$$\Delta J_{\text{family}} = \max_c J_c - \min_c J_c. \quad (12)$$

6.5 Experiment documentation contract

Every experiment below is documented before execution using five fields: objective, predeclared expectation, setup and evidence, decision gate, and results/interpretation. The latter remains marked as pending until the experiment has been run. Configuration, random seeds, output paths, and the producing Git commit will be added with each result. Failed experiments and changed decisions remain in this development report even if they are omitted from the final paper.

7 Phase 0: plant and fleet validation

Phase 0 verifies the equations, discretization, operating envelope, and physical family design before identification or control is introduced.

7.1 Experiment 0A: nominal-vehicle validation

Objective. Validate the nominal continuous and zero-order-hold-discretized bicycle model on $v_x \in [10, 30] \text{ m s}^{-1}$ with initial sample time $T_s = 0.01 \text{ s}$.

Predeclared expectation. Matrix entries, poles, time constants, and steady-state gains vary smoothly with speed. The nominal model is physically plausible and numerically well conditioned over the complete envelope.

Setup and planned evidence. On a fine speed grid, compute continuous and discrete eigenvalues, controllability, steady-state steering-to-yaw-rate and steering-to-sideslip gains, and an understeer or equivalent handling measure. Plot selected A and B entries, pole trajectories, and steady-state gains versus speed. Check equations, units, and sign conventions independently.

Decision gate. No unintended instability, singularity, loss of controllability, discontinuity, or implausible gain may occur. Any failure blocks all following experiments.

Results, interpretation, and decision. The audit was executed on 201 uniformly spaced frozen speeds. The implementation uses an augmented matrix exponential for exact zero-order-hold discretization. All continuous-time models are asymptotically stable. The least-negative pole real part is -3.911 s^{-1} at the high-speed end of the envelope. The corresponding maximum discrete-time spectral radius is 0.96164, which remains below one with a clear numerical margin. The pole pair remains complex throughout the envelope. Its real part moves from -11.733 to -3.911 s^{-1} , while its imaginary magnitude changes from 2.989 to 3.517 s^{-1} . The model therefore becomes less strongly damped in absolute time as speed increases, without approaching open-loop instability. The complete speed-dependent audit is shown in figure 1, with representative numerical values collected in table 4.

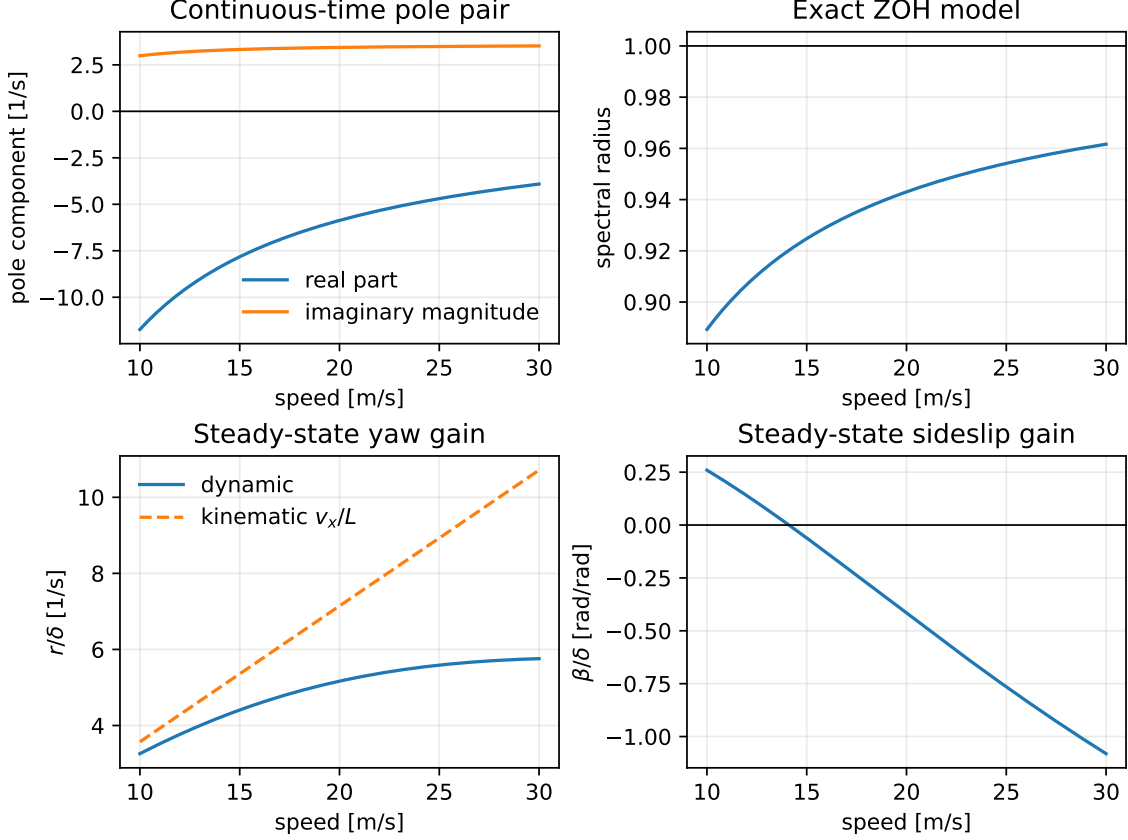


Figure 1: Experiment 0A nominal-vehicle audit. The continuous pole pair remains in the open left half-plane and its exact-ZOH image remains within the unit circle. The yaw-rate gain increasingly departs from the kinematic gain as speed rises, consistently with positive understeer. The sideslip gain changes sign because the body sideslip at the center of gravity changes direction under steady-state cornering; this is not an instability.

The controllability matrix has full rank at every grid point. Its smallest singular value is 2.506. The largest reported condition number is 173.1 at 10 m s^{-1} . Because the controllability columns mix quantities with physical units, this condition number is retained only as a numerical diagnostic, not as an invariant measure of control authority. Direct matrix conditioning is benign: $\max_v \kappa_2(A) = 6.93$ and $\max_v \kappa_2(A_d) = 1.13$.

The steady-state yaw-rate gain rises from 3.260 to 5.757 s^{-1} , but its ratio to the kinematic value v_x/L falls from 0.913 to 0.537 . This agrees exactly with the positive understeer coefficient

$$K_u = \frac{m(C_r l_r - C_f l_f)}{C_f C_r (l_f + l_r)} = 2.679 \times 10^{-3} \text{ s}^2 \text{ m}^{-1}, \quad (13)$$

for which the steady-state relation is $r/\delta = v_x/(L + K_u v_x^2)$. The sideslip gain changes from 0.260 to -1.081 rad/rad and crosses zero at approximately 14.11 m s^{-1} . Independent steady-state solution of the force and yaw-moment balances gives the same gains and signs.

Experiment 0A therefore **passes**. The equations, sign convention, exact discretization, sample time, and speed envelope can be retained for Experiment 0B. This pass establishes internal physical and numerical consistency of the benchmark. It does not claim calibration to a particular production vehicle; absolute parameter realism remains a limitation to revisit before the eventual paper-level nonlinear study. Machine-readable speed-grid results and the summary statistics are stored with the experiment outputs.

Table 4: Selected Experiment 0A values at three frozen speeds.

v_x [m s ⁻¹]	$\Re\{\lambda\}$ [s ⁻¹]	$\rho(A_d)$	r/δ [s ⁻¹]	β/δ
10	-11.733	0.8893	3.260	0.260
20	-5.867	0.9430	5.166	-0.417
30	-3.911	0.9616	5.757	-1.081

7.2 Experiment 0B: structural-family validation

Objective. Verify that the nominal, heavy/high-inertia, and changed-handling-balance centers create distinct but physically interpretable dynamics.

Predeclared expectation. The heavy family differs mainly in response speed and gain. The handling family changes yaw/sideslip balance rather than merely rescaling the nominal response.

Setup and planned evidence. At $v_x \in \{12, 20, 28\}$ m s⁻¹, apply a small steering step, a smooth sine sweep, and a lane-change-like input. Compare yaw rate, sideslip, poles, steady-state gains, and family-to-family dynamic distance. Report a compact table of absolute physical parameters and relative changes.

Decision gate. All centers must remain controllable and physically plausible. Differences must be large enough to affect prediction and control, but not so extreme that the benchmark makes family recovery trivial or unrealistic.

Results, interpretation, and decision. All three centers remain asymptotically stable and controllable at all 201 tested speeds. Their worst discrete spectral radii are 0.9616 (nominal), 0.9666 (heavy), and 0.9578 (handling). Thus, the heavy center has the slowest absolute decay, while the changed-handling center has the fastest. The minimum controllability singular values are 2.506, 2.854, and 1.942, respectively. As in Experiment 0A, these dimensioned singular values are used only to detect loss of rank or gross numerical degeneration.

Table 5: Absolute parameters and steady-state handling measure of the three vehicle-family centers. Axle cornering stiffness values use positive magnitudes.

Family	m [kg]	I_z [kg m ²]	C_f [kN/rad]	C_r [kN/rad]	K_u [10 ⁻³ s ² /m]
Nominal	1500	2500	80	80	2.679
Heavy	1725	3000	84	80	2.494
Handling	1425	2250	72	88	4.370

All three centers are understeering, but their balance is distinct. Relative to the nominal center, the heavy model has a slightly smaller understeer coefficient and therefore a slightly higher steady-state yaw gain. Its least-negative pole is closer to zero and its high-speed sideslip magnitude is larger. The handling center has a 63% larger understeer coefficient. Its yaw gain consequently flattens and then decreases slightly at high speed, while its sideslip magnitude is substantially reduced. These effects agree with the intended physical roles; the heavy family is not simply a uniform scaling, and the handling family changes the yaw-sideslip balance.

The normalized frozen-model distance from nominal is nearly independent of speed: 0.086–0.089 for the heavy center and approximately 0.056 for the handling center. This is desirable for the experimental decomposition. Speed moves each vehicle along its own model curve, whereas structural offsets persist throughout the scheduling envelope, as shown in figure 2.

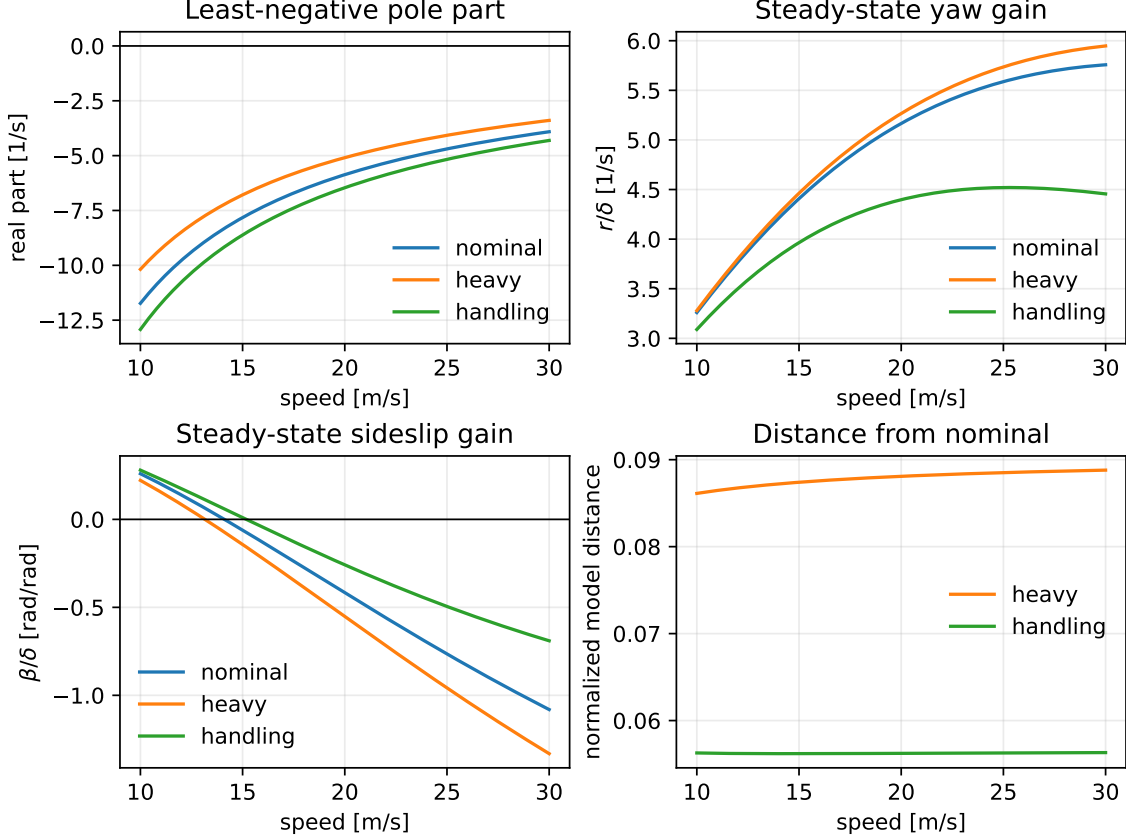


Figure 2: Frozen-speed characteristics of the three family centers. Structural differences persist across speed. The handling center is more strongly understeering, while the heavy center has slower pole decay and larger high-speed sideslip magnitude than the nominal center. The normalized model distance combines separately normalized A_d and B_d differences.

The step, sine-sweep, and lane-change-like inputs produce the same qualitative ordering at 12, 20, and 28 m s^{-1} . For the representative lane-change-like maneuver at 28 m s^{-1} , the heavy center increases peak yaw rate by 2.7% and peak sideslip magnitude by 23.6% relative to nominal. The handling center reduces the corresponding peaks by 15.2% and 33.4%. At 20 m s^{-1} , the respective changes are +2.0% and +27.6% for the heavy center, and -12.6% and -33.5% for the handling center. The complete response metrics for all 27 family-speed-maneuver cases are stored in machine-readable form. The representative time responses in figure 3 make this family ordering visible for the lane-change-like input.

Experiment 0B therefore **passes provisionally**. The centers are stable, controllable, interpretable, and dynamically distinct without being separated by orders of magnitude. The qualifier is important: the handling center is only about 5.6% from nominal in the normalized frozen-model metric. Experiment 0C must therefore determine whether the proposed within-family scatter obscures this separation. The family centers are retained unchanged until that population-level test is complete.

7.3 Experiment 0C: within-family client population

Objective. Construct a 30-client fleet with ten clients per family and determine whether the desired hierarchical structure is actually present.

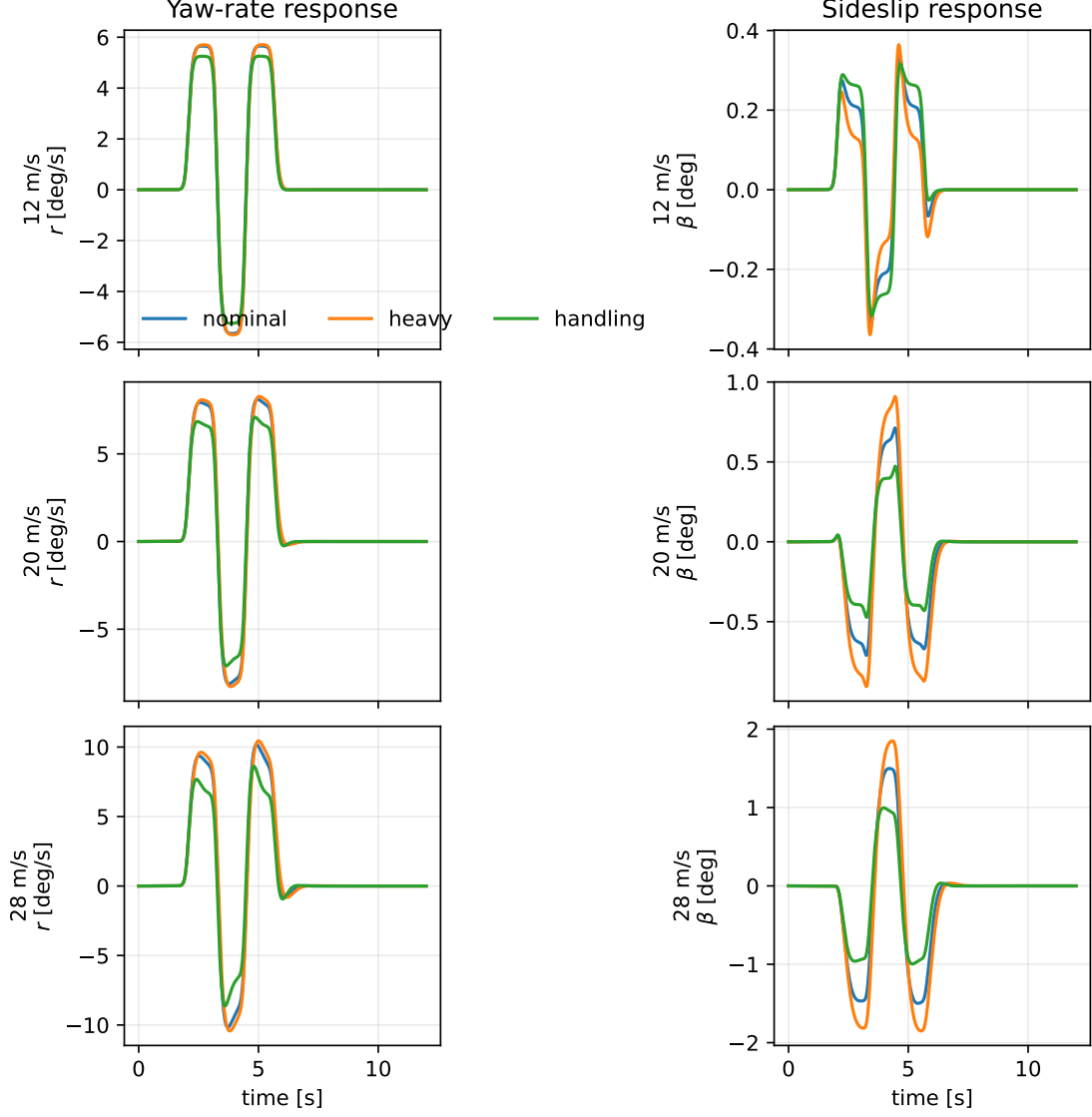


Figure 3: Yaw-rate and sideslip responses to the same lane-change-like steering input at the three predeclared test speeds. Family differences are modest at low speed and become practically visible at medium and high speed.

Predeclared expectation. Client envelopes overlap modestly within each family. Persistent differences between families remain visible over relevant parts of the speed range.

Setup and planned evidence. Sample bounded, fixed client parameters with initial ranges $m, I_z : \pm 2.5\%$ and $C_f, C_r : \pm 5\%$. Inspect physical parameters, frozen-model distances, response envelopes, and controllability over speed. The primary figure shows family centers and within-family bands for selected dynamic characteristics.

Decision gate. Within-family variation should be smaller than between-family variation without perfect separation on every scalar feature. Invalid samples are not silently discarded; the sampling rule must be revised if they occur systematically.

Results, interpretation, and decision. The bounded sampling rule was evaluated for the development fleet (seed 1) and nine additional fleet seeds, giving 300 clients in the robustness

audit. All clients are stable and controllable at every one of the 101 evaluated speeds. No sample was rejected or repaired. The exact fleet-generation settings are now stored in the Phase-0 configuration file.

For structural comparison, each client is represented by its exact-ZOH A_d and B_d matrices at 10, 15, 20, 25, and 30 m s^{-1} . The distance separately normalizes the A_d and B_d differences at each speed before averaging over matrices and speeds. Across the ten fleets, the mean between-family distance is 2.88–3.57 times the mean within-family distance, with mean ratio 3.28. The true-label silhouette ranges from 0.548 to 0.623 and has mean 0.590. Hence the hierarchy is consistent across seeds rather than being an artifact of the development realization.

Nearest-center classification using the three exact, known family centers is correct for all 300 clients. The smallest correct-to-incorrect center-distance margin is 0.00585. This perfect oracle-center result does not imply that cluster learning from short noisy trajectories will be trivial. It uses full-envelope, noise-free matrices and known centers. Moreover, in every tested fleet the largest within-family pairwise distance exceeds the smallest between-family distance. The population is therefore not separated by a universal scalar-distance threshold.

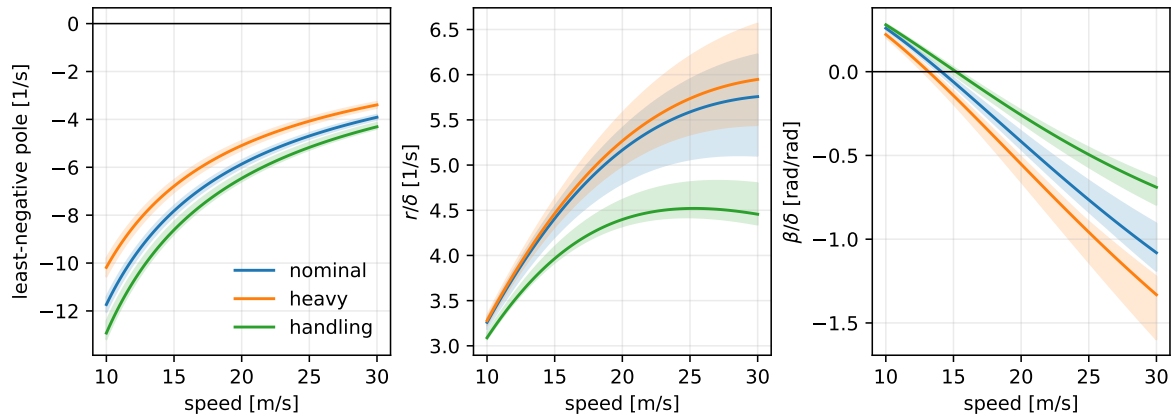


Figure 4: Seed-1 family centers and client envelopes under the predeclared bounded scatter. Some scalar response bands overlap, especially for nominal and heavy clients, while persistent multivariate family structure remains.

The response envelopes in figure 4 show the intended balance. A single frozen scalar characteristic is not sufficient everywhere: nominal and heavy yaw-gain bands overlap substantially, and their high-speed sideslip bands overlap. The handling family becomes distinct mainly through the joint speed-dependent yaw and sideslip behavior. The pole envelopes remain stable and relatively compact for all families.

For visualization only, the seed-1 model vectors were coefficient-wise standardized and projected with PCA. The first two components explain 98.2% of the variance (78.6% and 19.6% individually) and show three coherent groups in figure 5. PCA is not used to define the oracle labels or to claim performance of the future clustering algorithm.

Experiment 0C therefore **passes**. The proposed scatter preserves a stable, reproducible hierarchy in multivariate dynamics while retaining overlap in individual characteristics and pairwise distances. Together, Experiments 0A–0C support freezing the linear benchmark for Phase 1. This conclusion remains scoped to exact full-envelope models. Limited coverage, estimation noise, and learned clustering are deliberately deferred and may substantially reduce recoverability.

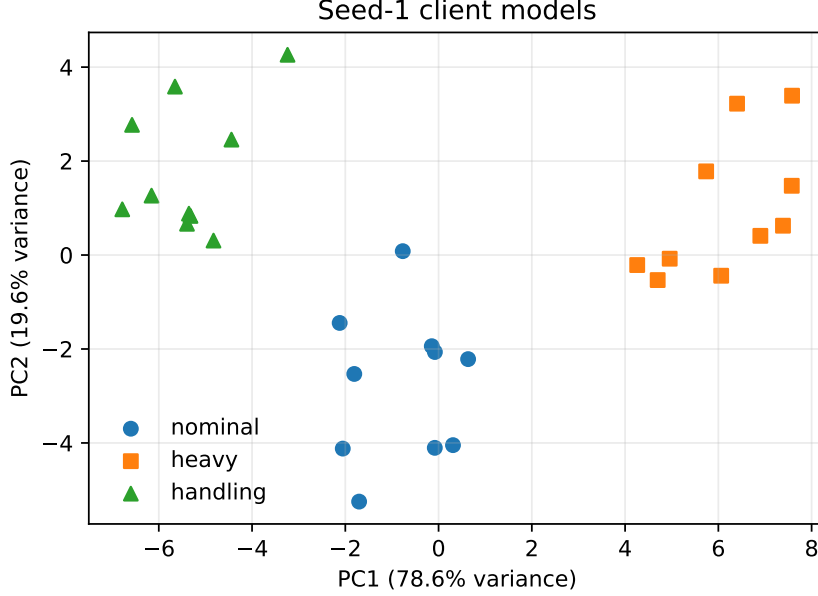


Figure 5: Diagnostic PCA projection of the 30 exact full-envelope client models in the development fleet. True family labels are shown only to inspect whether the designed structural information is recoverable in principle.

7.4 Phase-0 benchmark freeze

After Experiments 0A–0C pass, the equations, sample time, speed envelope, family centers, scatter model, and development seed are frozen. Later changes require an explicit report entry explaining the failure or new evidence that motivated them.

All three experiments passed. The frozen Phase-0 benchmark uses $T_s = 0.01\text{ s}$, $v_x \in [10, 30]\text{ ms}^{-1}$, the three centers in table 5, ten clients per family, bounded independent $\pm 2.5\%$ scatter in mass and yaw inertia, bounded independent $\pm 5\%$ scatter in front and rear cornering stiffness, and development seed 1. The next task is Experiment 1A: frozen-grid LPV basis validation.

8 Phase 1: LPV representation validation

8.1 Experiment 1A: frozen-grid matrix approximation

Objective. Select the smallest LPV basis that accurately represents the discretized bicycle models between measured speed anchors.

Predeclared expectation. The physically motivated reciprocal-speed basis is likely effective, but exact discretization may make a richer mixed basis necessary.

Setup and planned evidence. For every client, calculate exact frozen discrete matrices and fit

$$\Phi_1(v) = [1, v], \quad (14)$$

$$\Phi_2(v) = [1, v^{-1}, v^{-2}], \quad (15)$$

$$\Phi_3(v) = [1, v, v^{-1}, v^{-2}]. \quad (16)$$

Fit on the five training anchors and evaluate at unseen intermediate speeds and a dense grid. Report absolute and normalized $E_A(v)$ and $E_B(v)$, worst-client error, coefficient magnitude, and regression conditioning.

Decision gate. Choose the smallest basis with consistently low interpolation error and acceptable conditioning. The originally proposed basis is rejected if the evidence favors a different representation.

Results, interpretation, and decision. The comparison was run on all 30 clients for each of the ten robustness seeds, giving 300 independently perturbed vehicles. Each model was fit to exact ZOH matrices at $v = \{12, 16, 20, 24, 28\} \text{ m s}^{-1}$. The unseen points $v = \{14, 18, 22, 26\} \text{ m s}^{-1}$ measure interpolation, while a 201-point grid over $[10, 30] \text{ m s}^{-1}$ also tests the two edge intervals outside the fitting anchors. Nonconstant feature columns were standardized before least squares. Table 6 reports the combined relative error

$$e_{AB} = \sqrt{\frac{1}{2} (e_A^2 + e_B^2)}, \quad e_A = \frac{\|\hat{A}_d - A_d\|_F}{\|A_d\|_F}, \quad (17)$$

with e_B defined analogously.

Table 6: Experiment 1A basis comparison over 300 clients. Error entries are percentages; “full worst” includes edge extrapolation to 10 and 30 m s^{-1} .

Basis	Terms	κ_2	Interp. mean	Interp. worst	Full worst
$[1, v]$	2	1.00	3.222×10^{-1}	6.607×10^{-1}	2.255
$[1, v^{-1}, v^{-2}]$	3	17.51	6.814×10^{-5}	2.711×10^{-4}	2.266×10^{-3}
Mixed	4	140.72	1.820×10^{-5}	7.137×10^{-5}	1.065×10^{-3}

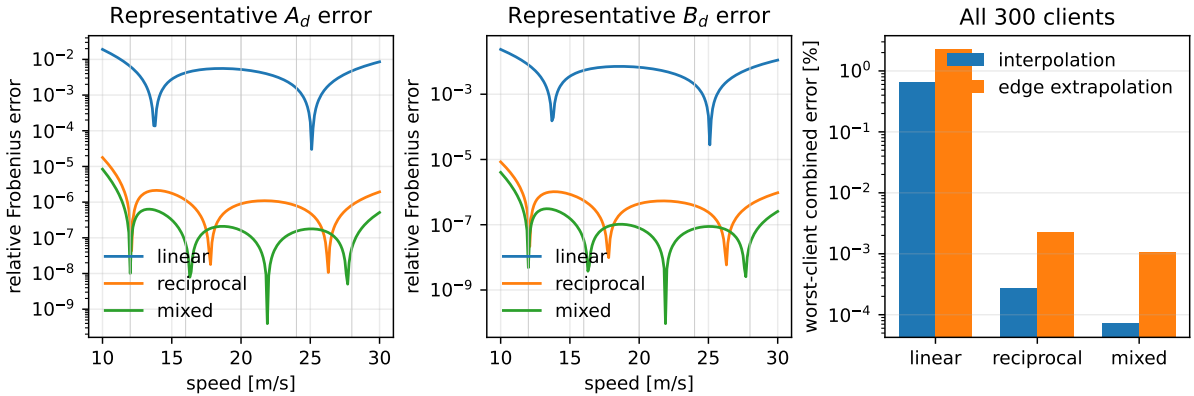


Figure 6: Experiment 1A relative matrix errors. The first two panels show a representative nominal client; gray lines mark fitting anchors. The final panel gives population worst cases on a logarithmic scale.

The error curves and population worst cases in figure 6 confirm the numerical ranking in table 6 over the full speed envelope.

The linear basis reaches a 2.26% worst-case full-envelope error, which is too large compared with the method differences that later experiments seek to resolve. The reciprocal basis reduces this error by roughly three orders of magnitude to 0.00227%. Adding the v term halves an already negligible error, but increases the standardized design-matrix condition number from

17.5 to 140.7 and adds another coefficient matrix. The largest reciprocal-basis error occurs in edge extrapolation; its worst unseen-interpolation error is only $2.71 \times 10^{-4}\%$.

Experiment 1A therefore passes. We select and provisionally freeze

$$\boxed{\Phi(v) = [1, v^{-1}, v^{-2}]} \quad (18)$$

as the smallest sufficiently accurate representation. Experiment 1B must now verify that this pointwise result remains accurate under time-varying simulation.

8.2 Experiment 1B: varying-speed trajectory prediction

Objective. Confirm that pointwise matrix accuracy translates into accurate speed-varying simulation.

Predeclared expectation. The chosen LPV basis gives nearly exact one-step prediction and small, bounded free-run error on the linear plant.

Setup and planned evidence. Use moderate persistently exciting steering with independent smooth profiles such as $12 \rightarrow 20 \rightarrow 15$, $20 \rightarrow 28 \rightarrow 18$, and $15 \rightarrow 25 \rightarrow 12 \text{ ms}^{-1}$. Compare the true speed-varying plant, exact frozen-matrix lookup, and fitted LPV model. Plot representative states and error versus time, speed, and client family.

Decision gate. One-step errors must be negligible relative to the later between-method effects, and free-run errors must not grow pathologically. Otherwise the basis or the time-varying discretization treatment is revised before Phase 2.

Results, interpretation, and decision. The experiment used three smooth speed profiles, $12 \rightarrow 20 \rightarrow 15$, $20 \rightarrow 28 \rightarrow 18$, and $15 \rightarrow 25 \rightarrow 12 \text{ ms}^{-1}$, each lasting 12 seconds. All profiles were paired with the same independent three-frequency multisine steering signal, whose peak magnitude is below 2.2° . This common input prevents speed and maneuver differences from being confounded. The evaluation again covered all 300 vehicles from the ten robustness seeds, for 900 trajectories in total.

The reference trajectory was integrated by fourth-order Runge–Kutta while evaluating $A(v(t))$ and $B(v(t))$ continuously inside each sample. Two scheduled discrete models were propagated recursively: exact ZOH matrices frozen at the left sample speed and the fitted reciprocal LPV matrices. A third comparison evaluates the LPV trajectory directly against the exact scheduled discrete trajectory, thereby isolating basis approximation from the shared sample-wise freezing error. Relative errors are normalized by the RMS norm of the corresponding reference state trajectory.

Table 7: Experiment 1B one-step and free-run relative RMSE over 900 trajectories. All error entries are percentages.

Comparison	1-step avg.	1-step max.	Free avg.	Free max.
Exact frozen vs. continuous	2.102×10^{-3}	3.275×10^{-3}	2.139×10^{-2}	3.339×10^{-2}
Reciprocal LPV vs. continuous	2.116×10^{-3}	3.297×10^{-3}	2.184×10^{-2}	3.358×10^{-2}
Reciprocal LPV vs. exact frozen	8.532×10^{-5}	1.725×10^{-4}	2.461×10^{-3}	6.112×10^{-3}

The representative trajectories and separated error components are shown in figure 7; the aggregate errors are reported in table 7.

The fitted LPV model is visually indistinguishable from both references. Its worst free-run error against the continuous-speed plant is only 0.0336%, with worst physical RMSEs of

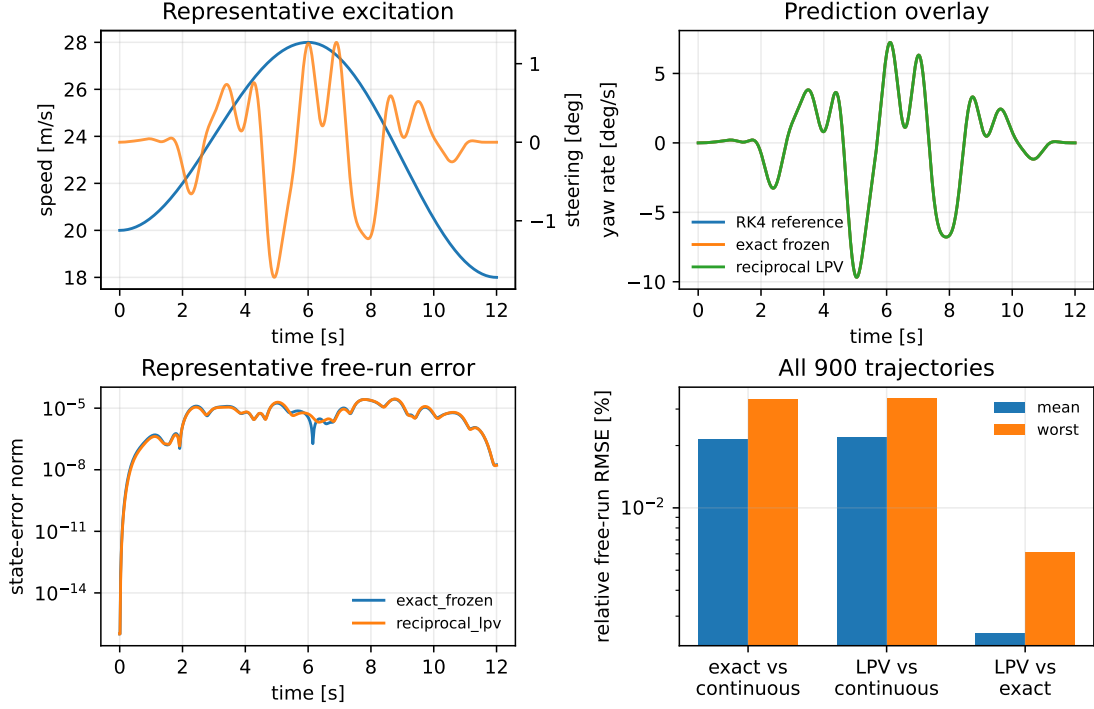


Figure 7: Experiment 1B varying-speed validation. The scheduled predictions overlap visually with the continuous-time reference. The lower panels expose the small free-run errors and distinguish the LPV basis error from the common within-sample speed-freezing error.

4.78×10^{-4} degrees in sideslip and $1.15 \times 10^{-3} \text{ deg s}^{-1}$ in yaw rate. Most of this discrepancy is already present for exact frozen matrices. The isolated reciprocal-basis contribution has 0.00246% mean and 0.00611% worst free-run RMSE. Its worst one-step error is $1.73 \times 10^{-4}\%$. Errors remain bounded throughout each maneuver and decay with the faded input, rather than accumulating pathologically.

Experiment 1B therefore passes. The reciprocal representation is sufficiently accurate that its approximation error is negligible relative to the effects targeted in the oracle architecture comparison.

8.3 Phase-1 representation freeze

Experiments 1A and 1B jointly validate and freeze the Phase-1 representation:

- scheduling basis $\Phi(v) = [1, v^{-1}, v^{-2}]$;
- standardized nonconstant regression features;
- exact ZOH target matrices at $T_s = 0.01 \text{ s}$;
- fitting anchors $v = \{12, 16, 20, 24, 28\} \text{ m s}^{-1}$.

These choices remain fixed in the Phase-2 oracle comparison. The validated model is still the linear-tire bicycle plant; the nonlinear tanh-tire benchmark remains a later robustness layer after the core architectural claim is tested.

9 Phase 2: oracle model comparison

This is the first direct test of the central hypothesis. Family labels are known, data are long and low-noise, and learning imperfections are intentionally absent.

9.1 Experiment 2A: construct M1–M5

Objective. Construct the architectures in table 3 under matched data support.

Setup and fairness rules. Global models pool all clients. Clustered models use true family labels. LTI models are fitted across the same operating data as LPV models rather than at one favorable speed. Aggregation is uniform initially. Uncertainty weighting and learned clusters remain outside the oracle comparison.

Decision gate. All methods must pass numerical checks and use identical train/test splits. Model errors are decomposed by method, speed, family, client, and trajectory.

Results, interpretation, and decision. M1–M5 were constructed independently for each of the ten fleet seeds. Each seed contains 30 clients, ten from each oracle family. Every architecture uses the same exact ZOH matrices at the five Phase-1 fitting anchors. M1 averages over clients and speed, M2 averages over speed within each family, M3 fits one pooled reciprocal LPV model, and M4 fits one reciprocal LPV model per family. M5 averages clients within each family and anchor, then selects the nearest of the resulting 15 frozen prototypes at evaluation time. All required matrix dimensions, family assignments, model counts, and prediction interfaces pass automated tests. Experiment 2A therefore passes without changing the predeclared architectures or support.

9.2 Experiment 2B: isolate scheduling and specialization

Objective. Determine whether scheduling and structural specialization solve distinct and complementary errors.

Predeclared expectation. For speed variation within one family, LPV models outperform LTI models. At fixed speed across families, specialized models outperform global models. With both effects present, M4 outperforms M2 and M3.

Setup and planned evidence. Three diagnostic conditions are used:

1. D1: one family over the full speed envelope;
2. D2: all families at $v_x = 20 \text{ m s}^{-1}$;
3. D3: all families under varying speed.

Report one-step and free-run errors, error-versus-speed curves, worst-family error, and family gap. After development, repeat with ten independently sampled fleets.

Decision gate. M4 must improve on both M2 and M3 in D3. A useful working gate is more than 10% for LPV relevance under broad speed variation and approximately 5–10% for specialization, with practical scale and seed consistency taking precedence over the numerical thresholds. If M3 and M4 are equivalent, clustering is unnecessary. If M2 and M4 are equivalent, LPV scheduling is unnecessary.

Table 8: Phase-2 oracle model comparison over ten fleet seeds.

Method	Models	Matrix avg.	Fixed-20 avg.	Free avg.	Free max.	Family gap
M1	1	6.070	5.591	16.230	34.070	7.388
M2	3	2.987	2.070	13.090	28.564	3.271
M3	1	5.537	5.546	10.859	26.441	6.846
M4	3	1.949	1.952	3.532	12.821	1.056
M5	15	2.058	1.952	4.422	12.988	1.059

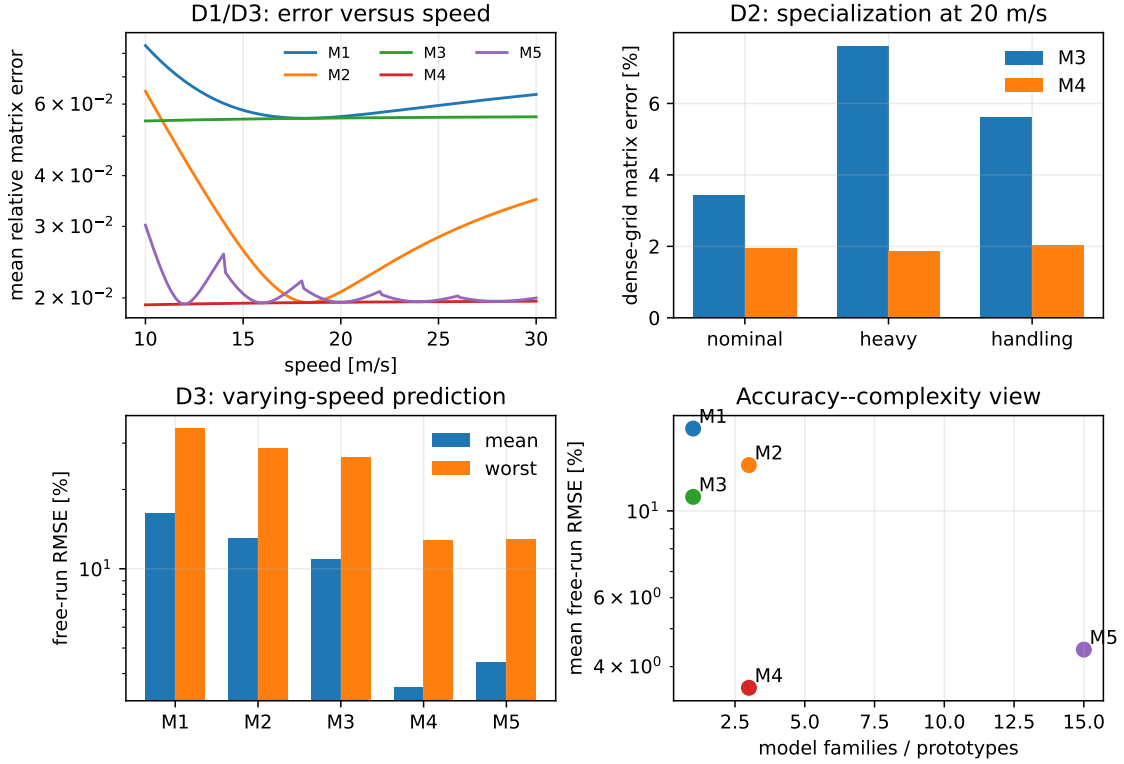


Figure 8: Phase-2 oracle comparison. Scheduling removes speed-dependent error, family specialization removes structural bias, and M4 provides the best accuracy–complexity combination.

Results and decision. Evaluation uses a 201-point speed grid over $[10, 30] \text{ m s}^{-1}$ and the three independent varying-speed profiles from Experiment 1B. The trajectory plant is the exact client-specific speed-scheduled ZOH model. Table 8 aggregates 300 clients, 60,300 client-speed evaluations per method, and 900 trajectories per method. All errors are relative percentages. The family gap is the difference between the largest and smallest family-mean free-run errors in percentage points.

The complementary effects are summarized visually in figure 8 and numerically in table 8.

The three diagnostics support the intended two-factor interpretation. Under varying speed, M3 reduces mean free-run error by 33.1% relative to M1. When family specialization is already present, M4 improves on M2 by 73.0%. At the fixed 20 m s^{-1} diagnostic, M4 reduces matrix error relative to M3 by 64.8%, so the specialization effect is not an artifact of changing speed. Finally, M4 improves varying-speed free-run error over M3 by 67.5%. Thus, neither scheduling nor clustering subsumes the other.

These effects are seed-consistent. M1→M3 improves all ten fleets by 28.1–36.9%, M2→M4 by 69.3–76.3%, and M3→M4 by 62.0–71.3%. M4 also reduces the family gap from 6.85 to 1.06 percentage points. Its remaining error is primarily the deliberately retained within-family client scatter, since one family model is not expected to reproduce every individual vehicle exactly.

M5 is a strong discrete approximation, but its 15 prototypes still have 20.1% higher mean free-run error than M4’s three continuous LPV families. M4 is better than M5 for every fleet seed, with improvements of 16.4–24.8%. Their worst trajectory errors are similar, indicating that rare within-family deviations, rather than speed discretization alone, determine the remaining tail.

Experiment 2B therefore passes all gates. The result establishes the central oracle claim on the linear plant:

LPV handles operating-point variation, while specialization handles persistent fleet variation.

(19)

Controller synthesis may now proceed without changing the validated model architecture. The later nonlinear-plant layer remains necessary to determine whether this conclusion survives tire saturation and model mismatch.

10 Phase 3: linear-plant controller oracle

10.1 Experiment 3A: common controller architecture

Objective. Translate model differences into a fair closed-loop comparison.

Setup. Augment the state with yaw-rate tracking error,

$$\chi_{k+1} = \chi_k + T_s(r_k - r_k^*), \quad (20)$$

and use

$$\delta_k = -K_x(v_k)x_k - K_I(v_k)\chi_k + k_r(v_k)r_k^*. \quad (21)$$

For LPV methods, calculate LQI gains at $v \in \{10, 15, 20, 25, 30\} \text{ m s}^{-1}$ and interpolate them. Select common Q and R once on the nominal vehicle and freeze them.

Planned evidence and decision gate. Audit controllability and closed-loop spectral radius for every client on a dense speed grid. Gain interpolation must be continuous and all evaluated closed loops must satisfy the predefined stability margin before simulation.

Results, interpretation, and decision. The common design uses

$$Q = \text{diag}(100, 10, 500), \quad R = 10, \quad (22)$$

selected once and held fixed for M1–M5. For a unit constant yaw-rate target, the design model is solved for (x^*, δ^*) , and the prefilter is

$$k_r = \delta^* + K_x x^*. \quad (23)$$

This makes $(x^*, \chi^* = 0)$ consistent with the full feedback law. All augmented design pairs are controllable. M3 and M4 gains are linearly interpolated over the five controller speeds, while M5 deliberately retains nearest-prototype switching.

An implementation audit caught and rejected an initial prefilter $k_r = 1/G_r(1)$ that supplied only equilibrium steering and omitted $K_x x^*$. That version forced the integrator to reconstruct the missing feedback command and reversed the expected tracking ranking. The correction above is the standard equilibrium-consistent 2-DOF construction and was applied identically to every method before accepting any Phase-3 result. It is not method-specific retuning.

The frozen-grid audit covers all 300 clients, five methods, and 201 speeds. All closed loops are stable. The worst spectral radius is 0.969 for M4, leaving a minimum frozen-model margin of 0.031. Interpolated gains are continuous by construction and tested at grid boundaries. Experiment 3A therefore passes.

10.2 Experiment 3B: closed-loop scenarios C1–C4

Objective. Test scheduling, specialization, their combination, and out-of-distribution generalization in closed loop.

Predeclared expectation. LPV methods dominate for a fixed family under changing speed (C1), specialized methods dominate across families at fixed speed (C2), M4 performs best when both effects vary (C3), and its advantage survives unseen speed–maneuver combinations (C4).

Setup and planned evidence. Use a smooth yaw-rate sine, single lane change, and double lane change/S-curve. Measure yaw-rate RMSE, sideslip RMS, steering RMS, peak steering, peak steering rate, worst-family result, and family gap. Plot one representative varying-speed trajectory and aggregate distributions across clients and seeds.

Decision gate. M4 must improve tracking or family consistency without materially greater steering effort. Benefits must survive C4. A method that wins only through aggressive input or one favorable maneuver does not pass.

Results, interpretation, and decision. The four scenarios use independent yaw-rate references and speed profiles. C1 uses the nominal family, a smooth sine reference, and $12 \rightarrow 28 \rightarrow 16 \text{ ms}^{-1}$. C2 uses all families at 20 ms^{-1} with a lane-change reference. C3 combines all families with the three Phase-1 speed profiles. C4 pairs the unseen $28 \rightarrow 16 \rightarrow 24 \text{ ms}^{-1}$ sequence with a different-amplitude and different-timing S-curve. Table 9 reports aggregate tracking and the C3 effort/stability diagnostics.

The aggregate comparison in figure 9 complements the scenario and effort metrics in table 9.

The intended factor isolation is visible in closed loop. In C1, adding scheduling reduces M1’s tracking error by 45.2% for M3. In C2, family specialization reduces M3’s error by 17.2% for M4. In the combined C3 test, M4 improves on M2 by 40.7% and on M3 by 21.7%. All improvements occur for 10/10 fleet seeds. M4 also reduces the C3 family gap from 0.00682 to 0.000849 deg s^{-1} relative to M3, an 87.6% reduction.

Table 9: Phase-3 oracle controller results over ten fleet seeds. Tracking columns are mean yaw-rate RMSE in deg s^{-1} ; u is C3 steering RMS in degrees and $\dot{u}_{\max}$ is the C3 worst peak rate in deg s^{-1} .

Method	C1	C2	C3	C4	u	$\dot{u}_{\max}$	$\rho_{\max}$
M1	0.0496	0.0222	0.0299	0.0362	0.5667	1.25	0.965
M2	0.0408	0.0187	0.0270	0.0313	0.5666	1.25	0.967
M3	0.0272	0.0204	0.0204	0.0374	0.5659	1.25	0.967
M4	0.0220	0.0169	0.0160	0.0324	0.5658	1.25	0.969
M5	0.0233	0.0169	0.0182	0.0388	0.5656	28.02	0.969

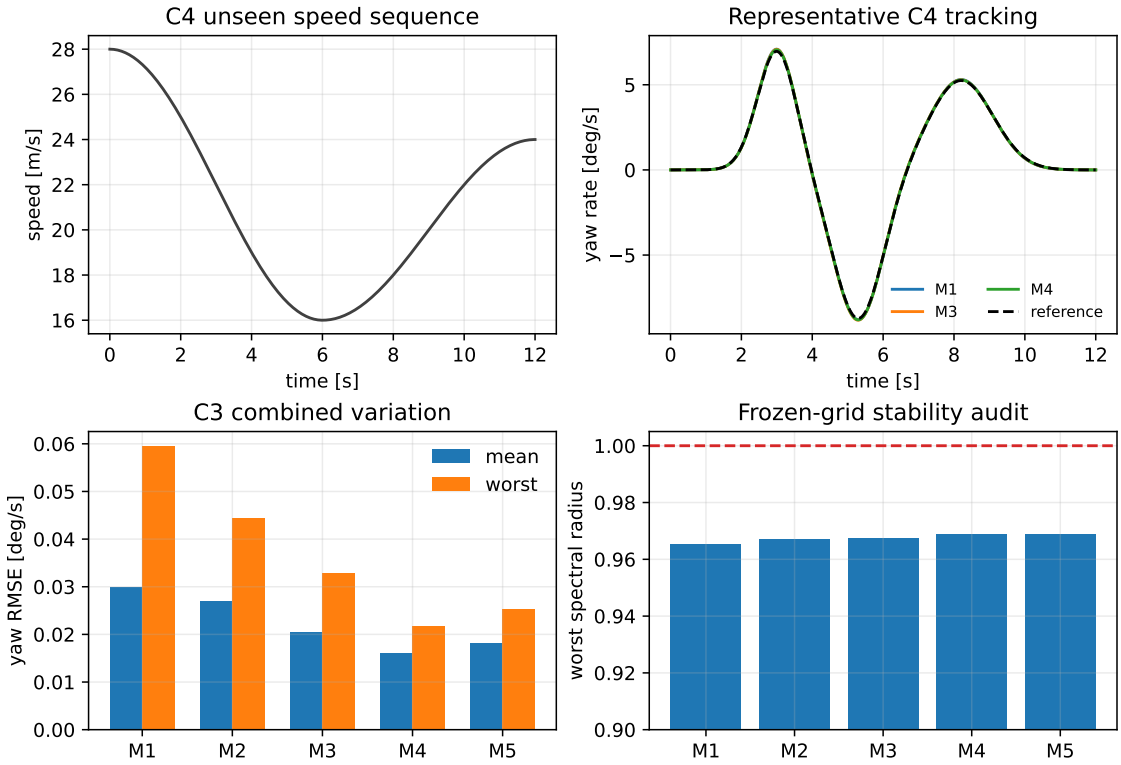


Figure 9: Phase-3 oracle controller comparison. M4 gives the lowest aggregate C3 tracking error with smooth control, while M5’s discrete gain changes create steering-rate spikes. All frozen closed loops remain stable.

These tracking gains do not come from greater control effort. M3 and M4 have nearly identical C3 steering RMS, peak steering, and peak steering rate. M4 also improves on M5's C3 tracking by 11.9%. Although M5 has similar steering RMS, nearest-grid gain switching raises its worst peak steering rate to 28.0 deg s^{-1} , compared with 1.25 deg s^{-1} for M4. This is a concrete practical advantage of a continuous controller family over discrete prototypes.

The C4 result supports a narrower generalization claim. M4 improves on global LPV M3 by 13.4% for every seed and reduces its family gap by approximately 77%, so specialization remains useful on the unseen combination. However, clustered LTI M2 is 3.76% better than M4 on mean C4 tracking for all ten seeds. Thus M4 is not universally optimal on every trajectory, and this negative comparison is retained rather than hidden. The particular C4 sequence does not reward scheduling enough to overcome that small difference.

Experiment 3B is therefore a qualified pass. The central complementarity claim holds strongly in C1–C3 and survives C4 relative to the appropriate global LPV baseline, with unchanged control effort and improved family consistency. The data do not support the stronger claim that clustered LPV must beat clustered LTI on every individual speed–maneuver combination. Phase 4 should now test the LTI-complexity tradeoff systematically rather than relying on a single five-point grid.

11 Phase 4: nonlinear-plant escalation

Phase 4 replaces the linear plant, not the compact identified model class. It tests whether the structural conclusion survives realistic plant–model mismatch. Experiments 4A–4B validate the plant and excitation, 4C tests prediction, and 4D separates transfer from redesign. The unsuccessful 4D-R correction motivates the diagnosis and structured redesign in 4E–4F, followed by independent-fleet validation in 4G. Earlier negative results remain part of the record. Throughout this phase, identification is centralized batch fitting with oracle family labels. Streaming RLS, federation, and learned clustering have not yet been introduced.

11.1 Experiment 4A: nonlinear tire-model validation

Objective. Implement and validate saturating tire forces

$$F_{y,j} = C_j \alpha_{\text{sat},j} \tanh\left(\frac{\alpha_j}{\alpha_{\text{sat},j}}\right), \quad j \in \{f, r\}. \quad (24)$$

Predeclared expectation. The nonlinear plant agrees with the linear model at small slip and departs smoothly as tire saturation becomes relevant.

Setup and planned evidence. Check the force–slip curves, small-signal slope, force limits, integration accuracy, sideslip, yaw rate, and lateral acceleration. Preserve the Phase-0 structural families initially so that the plant change is isolated.

Decision gate. The model must have correct small-slip behavior, smooth saturation, plausible force limits, and numerically converged simulations. Otherwise Phase 4 is blocked.

Results, interpretation, and decision. The nonlinear plant retains the Phase-0 state, input, structural family parameters, and sign conventions. A common dry-road friction coefficient $\mu = 0.9$ is used. Static axle loads are

$$F_{z,f} = mg \frac{l_r}{l_f + l_r}, \quad F_{z,r} = mg \frac{l_f}{l_f + l_r}, \quad (25)$$

and the saturation angles are selected as

$$\alpha_{\text{sat},j} = \frac{\mu F_{z,j}}{C_j}. \quad (26)$$

Consequently, each tanh law has the original cornering stiffness at zero slip and approaches the axle friction limit $\mu F_{z,j}$ smoothly. The resulting family-center scales and force limits are listed in table 10.

Table 10: Experiment 4A tanh-tire parameters for the three Phase-0 family centers at $\mu = 0.9$.

Family	$\alpha_{\text{sat},f}$ [deg]	$\alpha_{\text{sat},r}$ [deg]	$F_{y,f}^{\text{max}}$ [kN]	$F_{y,r}^{\text{max}}$ [kN]
Nominal	5.420	4.065	7.568	5.676
Heavy	5.936	4.675	8.703	6.527
Handling	5.721	3.511	7.189	5.392

Central finite differences were used to linearize the nonlinear right-hand side at zero sideslip, yaw rate, and steering for all family centers at $v_x \in \{12, 20, 28\} \text{ m s}^{-1}$. The maximum relative Jacobian error against the analytical Phase-0 matrices is 9.74×10^{-13} . Numerical force curves are odd, monotone, have the prescribed zero-slip slope, and remain within their friction limits.

Open-loop simulations use a smooth steering doublet at the same three speeds. Three diagnostic amplitudes expose the transition from the linear limit to tire saturation. Their worst results over all family-speed combinations are reported in table 11. These amplitudes validate the plant only; Experiment 4B will independently select the final near-linear, moderate, and strong-but-feasible maneuver levels.

Table 11: Worst Experiment 4A nonlinear response diagnostics over the three family centers and speeds 12, 20, and 28 m s^{-1} . The model difference is trajectory RMSE relative to the corresponding linear response.

Input	Amplitude [deg]	Model difference [%]	Force utilization	$ a_y _{\text{max}}$ [m s^{-2}]
Small	0.1	0.0084	0.032	0.274
Moderate	2.0	3.705	0.595	5.121
Strong	6.0	173.7	1.000	8.826

The small input produces virtually identical linear and nonlinear trajectories. The moderate input introduces a visible but controlled departure. The strongest case drives at least one axle close to its friction limit and therefore separates sharply from the unbounded linear-tire prediction, especially at high speed. All 27 nonlinear family-speed-amplitude simulations remain finite. These behaviors, together with the tire curves, are summarized in figure 10.

Integration was audited for the nominal 20 m s^{-1} , 6° case using RK4 steps from 40 to 5 ms against a 1.25 ms reference. The relative trajectory error decreases by approximately a factor of 16 whenever the step is halved. At the retained 10 ms step it is $7.10 \times 10^{-7}\%$, well below the predeclared 0.1% gate.

Experiment 4A therefore **passes**. The nonlinear equations recover the validated linear plant locally, provide bounded and physically interpretable tire forces, and are numerically converged at the existing sample time. Experiment 4B may proceed without changing the Phase-0 family definitions.

11.2 Experiment 4B: maneuver-severity calibration

Objective. Define reproducible near-linear, moderately nonlinear, and strongly nonlinear but feasible operating levels.

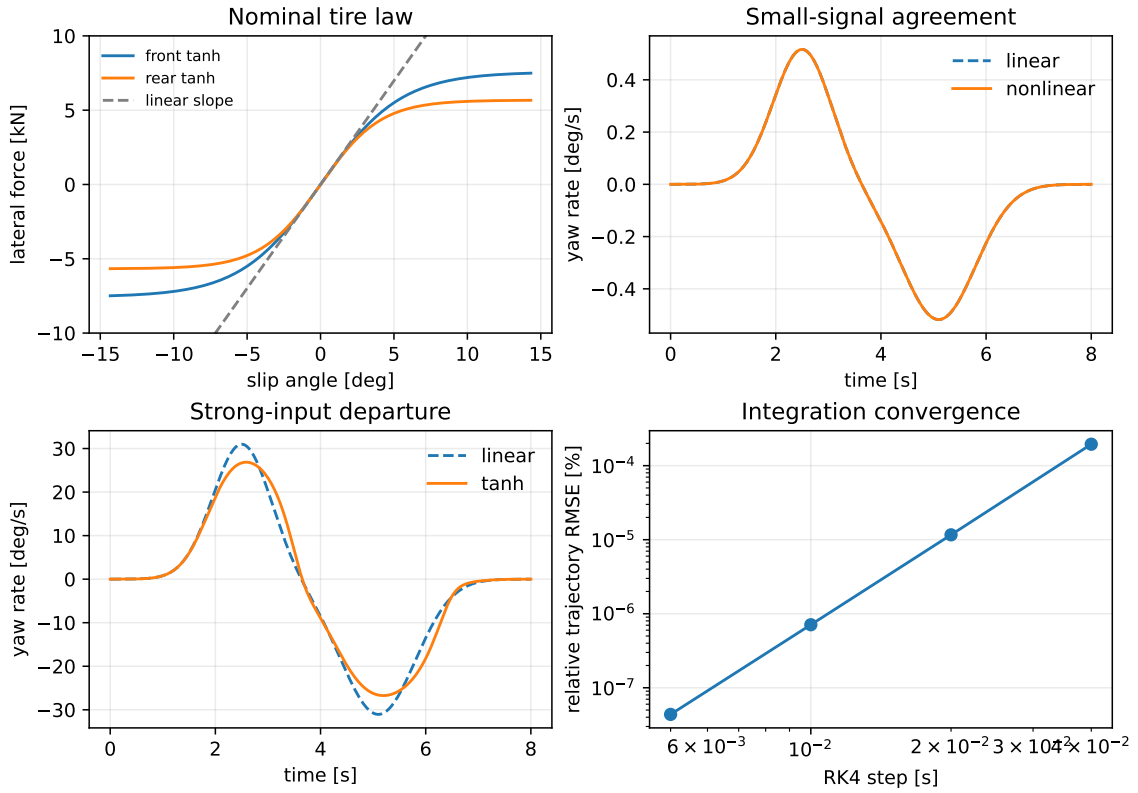


Figure 10: Experiment 4A nonlinear-plant validation. The tanh tire law recovers the prescribed small-slip slope and saturates smoothly. The nonlinear vehicle matches the linear model under weak excitation and departs under strong excitation. The RK4 trajectory error decreases at the expected fourth-order rate.

Setup and planned evidence. Scale the lateral reference by $\gamma_m \in \{0.5, 1.0, 1.5\}$. Report peak slip, lateral acceleration, tire-force utilization, steering limits, and the fraction of time outside a predeclared near-linear tire region. Adjust the absolute base maneuver once, globally, if the three intended regimes are not obtained.

The test uses the unseen smooth S-curve from the Phase-3 generalization case and the M4 family-specific scheduled controller without retuning. Two continuously varying speed profiles are applied to every family center: $12 \rightarrow 28 \rightarrow 16 \text{ m s}^{-1}$ and $28 \rightarrow 16 \rightarrow 24 \text{ m s}^{-1}$. The tire is declared near-linear while its local slope retains at least 95% of the zero-slip cornering stiffness. Because

$$\frac{\partial F_{y,j}}{\partial \alpha_j} / C_j = \text{sech}^2(\alpha_j / \alpha_{\text{sat},j}), \quad (27)$$

this definition is independent of axle stiffness and load scaling.

Decision gate. Severity levels must represent distinct regimes without making the strongest case physically meaningless or infeasible for every linear controller.

Results, interpretation, and decision. An initial coefficient of 8.0 deg s^{-1} did not quite reach the strong-regime utilization gate. In accordance with the protocol, the coefficient was adjusted once, globally, to 8.5 deg s^{-1} and then frozen. Due to the asymmetric S-curve shape, the resulting unscaled reference peak is 9.735 deg s^{-1} . The calibrated worst-case results across both speed profiles and all family centers are given in table 12.

Table 12: Experiment 4B maneuver-severity calibration. “Outside” is the largest trajectory fraction for which at least one tire has lost more than 5% of its zero-slip slope.

Regime	γ_m	Peak $ r^* $ [deg/s]	Utilization	Outside [%]	Peak $ a_y $ [m s^{-2}]
Near-linear	0.5	4.867	0.262	7.75	2.276
Moderate	1.0	9.735	0.517	36.75	4.489
Strong	1.5	14.602	0.753	50.92	6.497

The maximum steering magnitudes are 1.205° , 2.428° , and 3.692° , respectively. Thus even the strongest tier remains below the predeclared 12° steering limit and the $0.9g$ lateral-acceleration limit. All 18 family–profile–severity simulations remain finite. Maximum utilization increases monotonically through the declared bands: at most 0.35 for the near-linear tier, between 0.35 and 0.75 for the moderate tier, and between 0.75 and 0.98 for the strong tier. The largest near-linear outside fraction is 7.75%, below its 10% gate. The trajectories and regime separation are shown in figure 11.

Experiment 4B therefore **passes**. These three reference scales are now frozen for Experiments 4C and 4D. They are calibrated excitation regimes, not claims that the entire strong trajectory remains inside the linear tire region.

11.3 Experiment 4C: oracle models under nonlinear mismatch

Objective. Repeat M1–M5 using nonlinear-plant trajectories while retaining linear LTI/LPV identified models.

Predeclared expectation. Absolute errors grow with maneuver severity, but the complementary benefits of scheduling and specialization remain under moderate nonlinear excitation.

Setup and decision gate. Evaluate one-step and free-run error versus speed, family, and severity. The M4 advantage must survive at least the moderate regime and unseen speed–maneuver combinations. If all linear LPV models fail, revise the representation before FL.

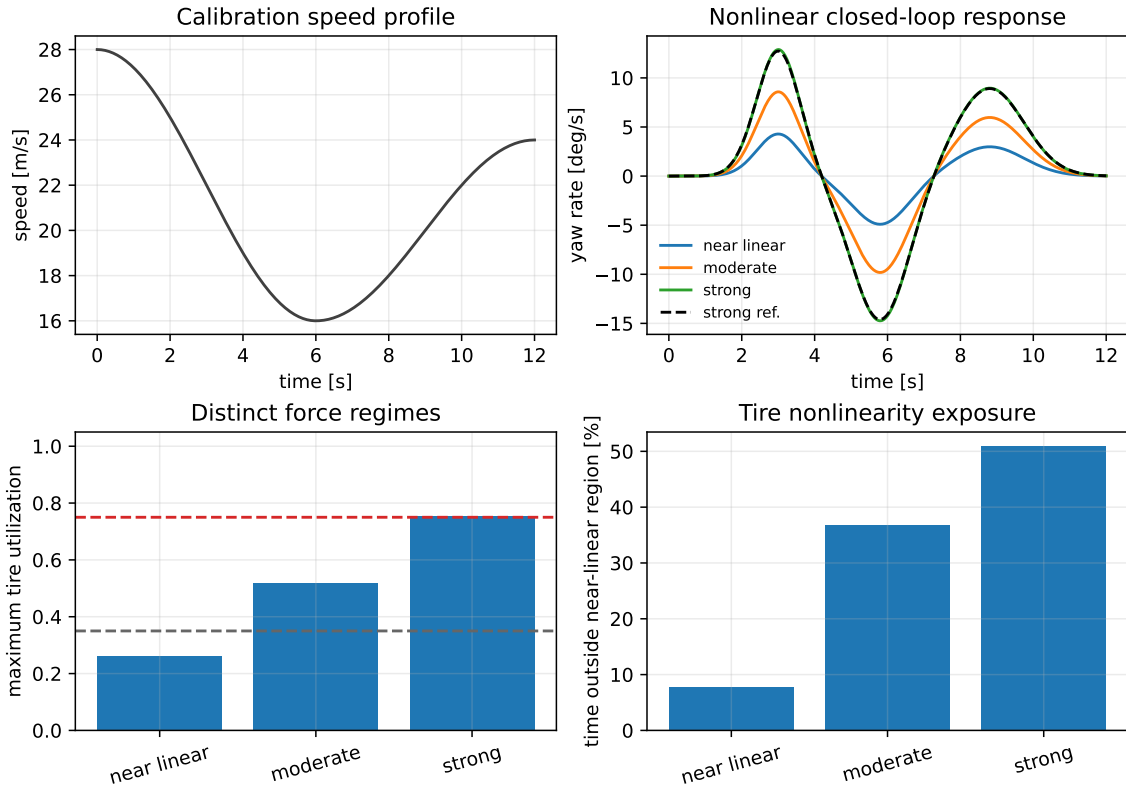


Figure 11: Experiment 4B severity calibration under the M4 scheduled controller. The same S-curve is scaled globally. Tire-force utilization and exposure beyond the declared near-linear region increase monotonically, while the strong case remains within the feasibility limits.

Results, interpretation, and decision. The same 30-client fleet from seed 1 is simulated with the validated nonlinear plant. Training data combine a lane change under $12 \rightarrow 20 \rightarrow 15 \text{ m s}^{-1}$ and a sinusoidal maneuver under $20 \rightarrow 28 \rightarrow 18 \text{ m s}^{-1}$. Each maneuver is run at all three frozen 4B severities. The M4 controller generates bounded closed-loop excitation, but only the resulting (x_k, u_k, v_k) samples and oracle family labels are used for identification.

M1–M5 are refitted directly by least squares on these nonlinear trajectories. M1 and M2 use constant matrices, M3 and M4 use the reciprocal-speed basis, and M5 uses five family-specific nearest-speed models at $\{12, 16, 20, 24, 28\} \text{ m s}^{-1}$. The held-out test combines the unseen $28 \rightarrow 16 \rightarrow 24 \text{ m s}^{-1}$ profile with the unseen S-curve. One-step predictions are reset to the measured state at every sample. Free-run predictions start at the true initial state and replay only the recorded steering and speed sequences. Thus the latter expose accumulated model mismatch without conflating it with controller redesign.

For comparability between β and r , normalized RMSE divides each state error by that trajectory’s state standard deviation, with small fixed floors for nearly unexcited signals. The moderate-severity results are summarized in table 13.

Table 13: Experiment 4C held-out moderate-severity model errors, averaged over 30 clients. The final column is the largest family-mean free-run NRMSE.

Method	One-step NRMSE	Free-run NRMSE	Worst family
M1 global LTI	0.01418	1.0093	1.5173
M2 clustered LTI	0.01289	0.7305	0.7806
M3 global LPV	0.00699	0.5540	0.7549
M4 clustered LPV	0.00281	0.1122	0.1441
M5 gridded LTI	0.00840	0.2965	0.3782

Against M3, M4 reduces moderate one-step NRMSE by 59.9% and free-run NRMSE by 79.8%. Against M2, the reductions are 78.2% and 84.6%. M4 also outperforms the 15-model M5 representation. Its worst family mean is less than one fifth of M3’s in free run. This supports the intended complementarity: continuous scheduling removes operating-point error, while family specialization removes persistent structural bias.

The normalized M4 errors remain low at all three severities. Nevertheless, the absolute yaw-rate mismatch grows monotonically, as physically expected. M4’s one-step yaw RMSE is $0.0054, 0.0108, \text{ and } 0.0184 \text{ deg s}^{-1}$ from near-linear to strong excitation. Its free-run yaw RMSE similarly rises from 0.136 through 0.266 to 0.388 deg s^{-1} . The nonlinear mismatch is therefore visible, but does not erase the architecture ranking. The full severity trends, representative free-run trace, and worst-family comparison appear in figure 12.

Experiment 4C therefore **passes**. The benefit of jointly representing speed scheduling and structural families survives moderate and strong tanh-tire nonlinearity on an unseen speed–maneuver combination. Experiment 4D may now test controller transfer and nonlinear-data redesign without revising the compact model representation.

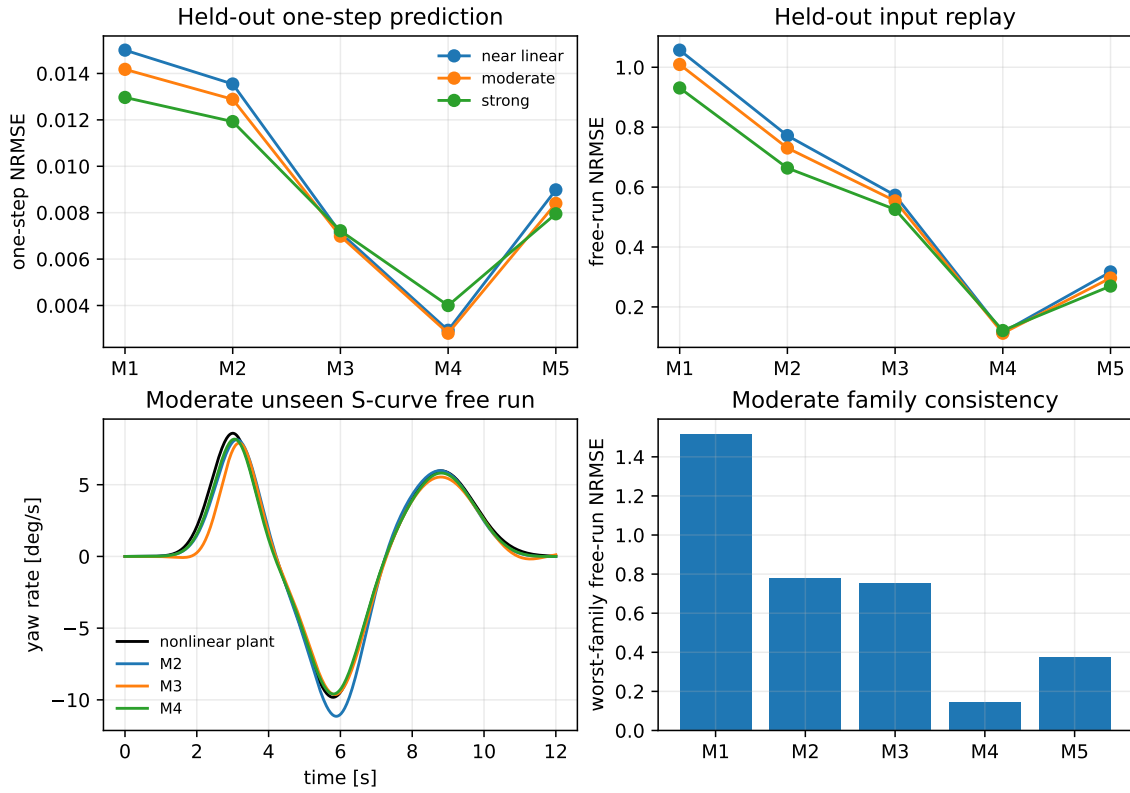


Figure 12: Experiment 4C oracle linear-model comparison on nonlinear-plant data. M4 retains the lowest held-out one-step and free-run errors across all calibrated severities and sharply reduces the worst-family error.

11.4 Experiment 4D: controller transfer and redesign

Objective. Separate robustness of the Phase-3 controller from the achievable performance of controllers redesigned using LPV approximations identified on nonlinear data.

Setup. The transfer test applies Phase-3 controllers unchanged. The redesign test refits the compact models on nonlinear trajectories and synthesizes controllers with the same common weights. The redesign test is primary; transfer is diagnostic.

Decision gate. The qualitative M4 benefit should survive moderate excitation. If it survives only near the linear region, the claim and operating envelope are restricted explicitly. Frozen-grid checks are necessary but are not presented as a formal LPV stability certificate.

Results, interpretation, and decision. The transfer and redesign controllers are evaluated on all 30 clients using the same held-out $28 \rightarrow 16 \rightarrow 24 \text{ m s}^{-1}$ S-curve and the three frozen severity scales. No controller receives method-specific tuning. The original Phase-3 controllers retain their $10\text{--}30 \text{ m s}^{-1}$ design grid. Redesigned controllers are synthesized only on the $12\text{--}28 \text{ m s}^{-1}$ identification support. An initial redesign attempt that extrapolated the identified LPV models to 10 and 30 m s^{-1} was rejected because the global LPV controller became unstable. Restricting synthesis to observed support is a common methodological correction, applied identically to M3 and M4, rather than controller tuning.

The moderate-severity tracking results are shown in table 14. Feasibility requires finite trajectories, peak steering below 12° , and peak lateral acceleration below $0.9g$.

Table 14: Experiment 4D moderate-severity yaw-rate tracking RMSE averaged over 30 clients. “Transfer” uses the unchanged Phase-3 controllers. “Redesign” uses controllers synthesized from the 4C nonlinear-data models.

Method	Transfer	Redesign	Worst family	Feasible
M1 global LTI	0.04205	0.07879	0.08853	1.00
M2 clustered LTI	0.03763	0.03758	0.05320	1.00
M3 global LPV	0.03942	0.44160	0.46787	0.00
M4 clustered LPV	0.03464	0.05737	0.07781	1.00
M5 gridded LTI	0.04262	0.11602	0.21879	1.00

The transfer diagnostic is positive. Unchanged M4 reduces moderate tracking error by 12.1% relative to M3 and 8.0% relative to M2. All transferred controllers remain feasible through strong excitation. This shows that the Phase-3 clustered scheduled controller is robust to the tested tire nonlinearity.

The primary redesign gate, however, **fails**. Redesigned M4 remains feasible and improves on redesigned M3 by 87.0%, but its moderate tracking error is 52.7% higher than redesigned M2. Redesigned M3 is infeasible for all moderate test clients. M5 remains amplitude-feasible, but nearest-grid switching produces steering-rate peaks above 800 deg s^{-1} , which further weakens it as a practical alternative.

A frozen small-signal audit over every client and 161 speeds from 12 to 28 m s^{-1} explains why prediction quality from Experiment 4C is not sufficient for controller synthesis. All transferred designs have worst spectral radius below 0.967. For the redesigned controllers, the worst radii are 0.967, 0.960, 5.888, 1.032, and 19.869 for M1–M5. Thus the unconstrained trajectory regressions can fit one-step and replay behavior well while perturbing control-relevant matrix directions enough to destroy robust closed-loop stability. M4 happens to remain bounded on the tested maneuver, but its frozen-grid radius above one prevents treating the redesign as acceptable. The tracking, severity, and family effects are summarized in figure 13.

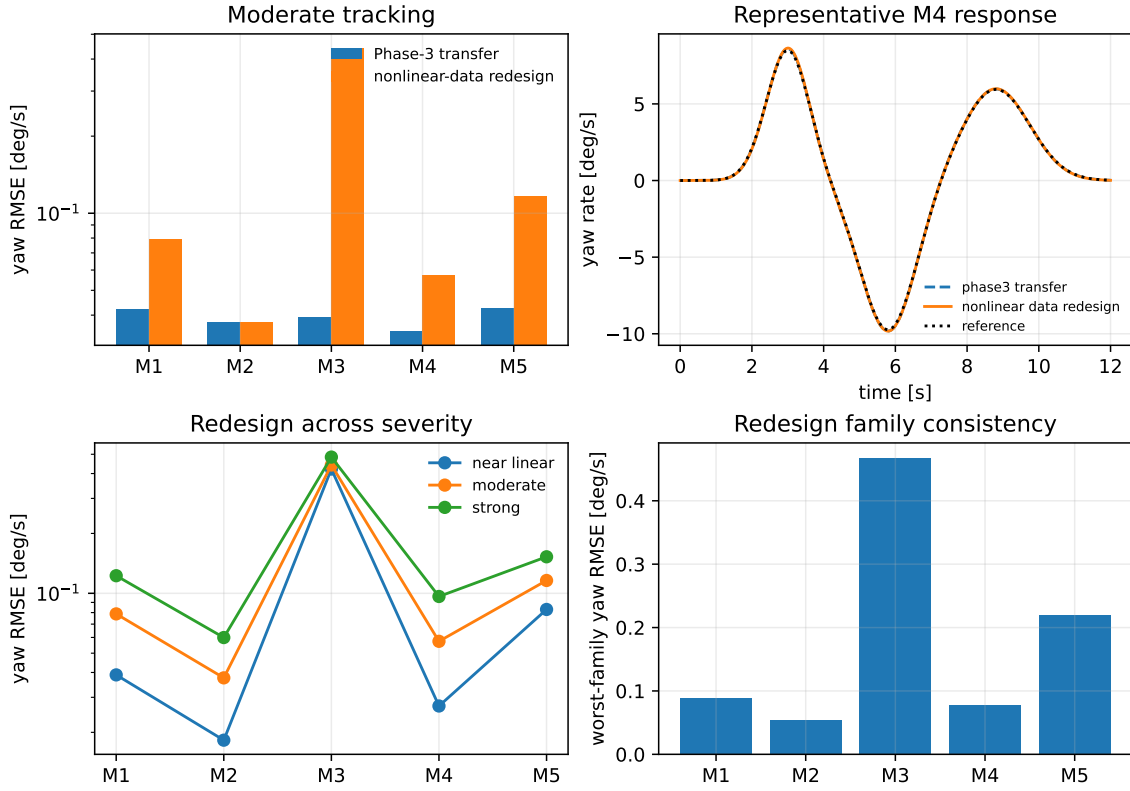


Figure 13: Experiment 4D controller transfer and nonlinear-data redesign. The unchanged M4 controller transfers successfully. Unconstrained redesign from trajectory-fitted models is unreliable, particularly for global LPV and gridded LTI, and does not improve M4 over clustered LTI. Logarithmic axes are used where failures would otherwise obscure the feasible methods.

Experiment 4D therefore records a **mixed result and a failed primary gate**. The representation result survives in 4C, and the original M4 controller transfers robustly, but direct certainty-equivalent redesign from unconstrained nonlinear-data regression is not justified. Before advancing the controller claim, a corrective Experiment 4D-R should add control-oriented regularization or robust/stability-constrained synthesis and repeat the same frozen evaluation.

11.5 Experiment 4D-R: stability-screened physics regularization

Objective. Test whether a minimal, predeclared control-oriented correction can stabilize nonlinear-data redesign while preserving useful adaptation.

Method and selection rule. For every architecture, blend the nonlinear-data model from 4C with its validated Phase-3 physics model,

$$[A_\lambda(v) \ B_\lambda(v)] = (1 - \lambda)[A_{\text{data}}(v) \ B_{\text{data}}(v)] + \lambda[A_{\text{phys}}(v) \ B_{\text{phys}}(v)]. \quad (28)$$

The candidate set is $\lambda \in \{0, 0.1, 0.25, 0.5, 0.75, 1\}$. One common λ is used for all methods. The selected value is the smallest candidate for which every M1–M5 controller has worst frozen small-signal spectral radius at most 0.99 over all training clients and 161 speeds in $[12, 28] \text{ ms}^{-1}$. No held-out tracking result enters selection. This is an oracle diagnostic: the final federated method would require uncertainty sets or another deployable robustness screen in place of known training plants.

Decision gate. M4 must use $\lambda < 1$, remain feasible for every moderate test client, and improve moderate tracking RMSE over both regularized M2 and M3 by at least 5%.

Results, interpretation, and decision. At $\lambda = 0$, M3, M4, and M5 have worst radii 5.888, 1.032, and 19.869. At $\lambda = 0.25$, M4 and M5 pass, but M3 remains unstable with radius 1.484. The common selection rule therefore chooses $\lambda = 0.5$. The resulting worst radii are 0.957, 0.962, 0.967, 0.963, and 0.961 for M1–M5. The regularization thus repairs the frozen-grid stability failure for every method.

Held-out moderate results are given in table 15. All methods are amplitude-feasible, and the M5 steering-rate peak falls from more than 800 to 24.5 deg s^{-1} .

Table 15: Experiment 4D-R results with the commonly selected physics weight $\lambda = 0.5$. Tracking values are yaw-rate RMSE in deg s^{-1} .

Method	Mean tracking	Worst family	Worst spectral radius
M1 global LTI	0.04538	0.05035	0.957
M2 clustered LTI	0.03562	0.04146	0.962
M3 global LPV	0.04612	0.05353	0.967
M4 clustered LPV	0.04045	0.04398	0.963
M5 gridded LTI	0.04535	0.04836	0.961

M4 improves on M3 by 12.3%, so family specialization remains beneficial within the LPV comparison. However, M4 is 13.5% worse than M2. The correction therefore stabilizes redesign but does not restore the full scheduling plus specialization advantage. The selection path and held-out behavior are shown in figure 14.

Experiment 4D-R therefore **fails the full performance gate**, despite passing the stability and feasibility gates. This does not support unconstrained matrix regression followed by simple post-hoc blending as the main controller-redesign method. The project now has a clean choice: retain the successful physics-model controller transfer as the Phase-4 robustness claim, or add

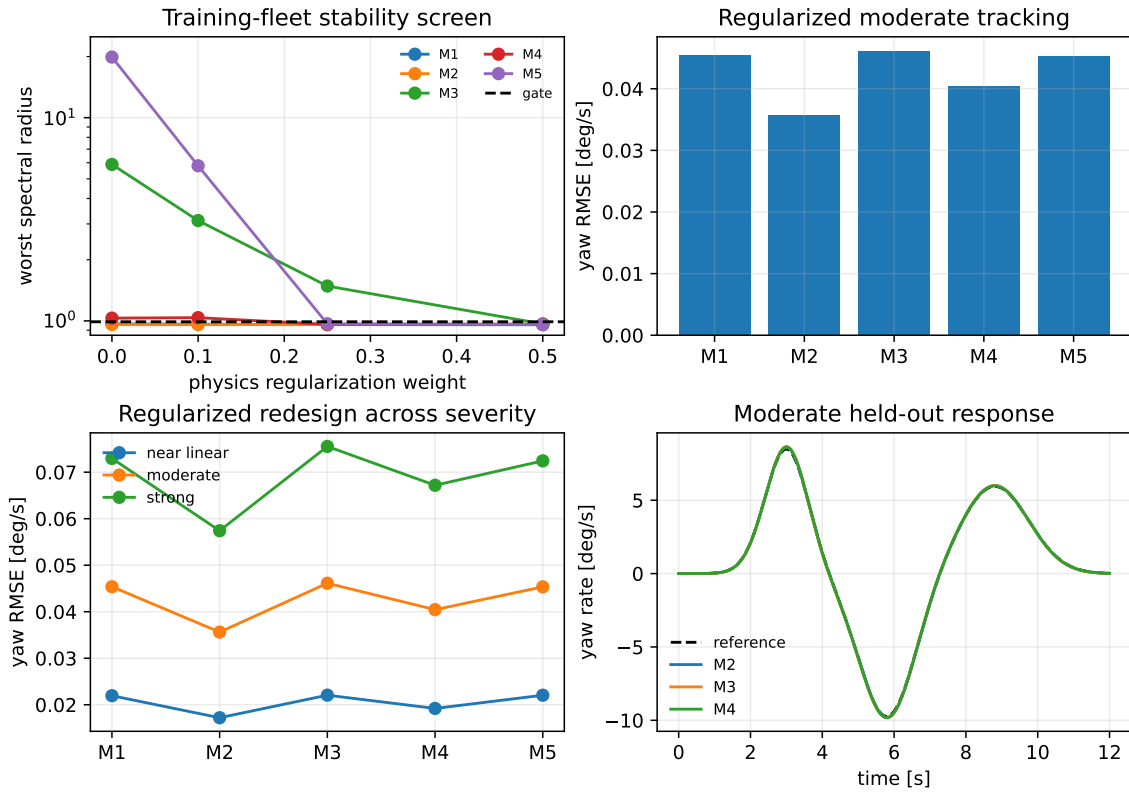


Figure 14: Experiment 4D-R stability-screened physics regularization. A common weight of 0.5 makes all redesigned controllers stable and feasible. M4 again improves over global LPV, but clustered LTI remains best.

a separate structure-preserving identification study before claiming data-driven redesign. The subsequent Experiments 4E–4F pursue that second path (sections 11.6 and 11.7).

11.6 Experiment 4E: control-relevant identification diagnosis

Question and controlled comparisons. Does redesign fail because of numerical solution error, nonlinear training amplitudes, or limitations already present in pooled linear-plant identification? The same seed-1 fleet, collection controller, two training profiles, and held-out moderate S-curve are retained. Compare original nonlinear-data regression, column-scaled SVD solving the same ridge objective, SVD using only near-linear nonlinear trajectories, and SVD using linear-plant trajectories at all three amplitudes. The last case uses exact frozen-speed ZOH propagation for training. All redesigned controllers are tested on the nonlinear plant.

For regression matrix X , define $S_{jj} = \|X_{:,j}\|_2$. The numerical check is

$$\min_W \left\| \begin{bmatrix} XS^{-1} \\ \sqrt{\epsilon}S^{-1} \end{bmatrix} W - \begin{bmatrix} Y \\ 0 \end{bmatrix} \right\|_F^2, \quad \hat{\Theta}^\top = S^{-1}W, \quad \epsilon = 10^{-10}. \quad (29)$$

Scaling changes the solver, not the regularization objective. M1, M2, and M5 retain their original solver. A small absolute ridge coefficient is not necessarily negligible relative to weak regressor directions.

Results. Table 16 shows that SVD reproduces the original M4 failure. Pooled raw condition numbers are approximately $1.46\text{--}1.80 \times 10^6$, falling to $5.86\text{--}7.76 \times 10^3$ after column scaling. All matrices have full column rank, but weak directions remain. The solver change alters M4 tracking by less than $10^{-9} \text{ deg s}^{-1}$. Numerical roundoff therefore does not explain the observed failure.

Table 16: Experiment 4E: M4 diagnosis. Tracking is moderate-test yaw RMSE in deg s^{-1} . Relative B error is averaged over clients and 161 speeds against each client’s true small-signal matrix.

Identification data and solver	Tracking	Worst ρ	Relative B error
Nonlinear, all amplitudes, original	0.05737	1.0321	0.4633
Nonlinear, all amplitudes, SVD	0.05737	1.0321	0.4633
Nonlinear, near-linear only, SVD	0.06255	0.9770	0.4715
Linear plant, all amplitudes, SVD	0.06147	0.9772	0.4679

The original worst M4 instability occurs at 12 m s^{-1} for handling_07. Near-linear training restores frozen M4 stability, but does not improve tracking. Linear-plant training also leaves substantial input-matrix error and poor tracking. Global LPV remains unstable in both ablations. Nonlinear tire mismatch is therefore not the sole explanation. Relative matrix errors use the fixed state coordinates and are not coordinate-invariant metrics.

Figure 15 relates input-channel error, gain, and stability for a representative heavy client. Complete client and speed results accompany the report as CSV files. Inspection confirms the same $r - r^*$ integrator convention, radian states and inputs, and prefilter equilibrium equation in synthesis and simulation. Existing controller tests pass, providing no evidence of a sign or unit error without claiming exhaustive software verification.

Interpretation and decision. Training uses two fixed speed–maneuver pairings, not their full cross product. Pooled clients have different dynamics even within a family, and the reciprocal basis only approximates exact discrete-time dynamics. Weak excitation, pooling mismatch, basis approximation, and ridge bias can interact. Their individual causal contributions have not all been isolated. The demonstrated finding is that a solver change or reduced tire excitation alone

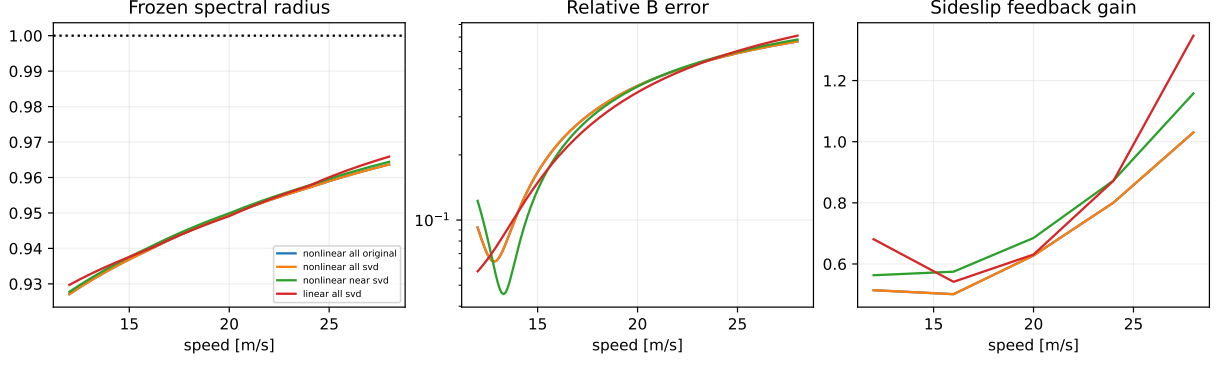


Figure 15: Experiment 4E for heavy_00. Original and SVD all-amplitude curves overlap. This representative client is not the fleet’s worst-case client.

does not recover the desired performance. Experiment 4F reduces the degrees of freedom through physical relationships while keeping controller weights fixed.

A further linear-data check fits nominal_00, heavy_00, and handling_00 separately using their own trajectories. Mean relative B errors are $2.29\text{--}4.20 \times 10^{-5}$ without ridge and $2.68\text{--}3.83 \times 10^{-4}$ with the original ridge. These are far below the pooled M4 error of 0.468. This strongly implicates heterogeneous pooling under the existing closed-loop data distribution, rather than an inability of the basis to represent each individual client. It does not isolate pooling from all interactions with excitation. In closed loop, regression coefficients can compensate along correlated state–input directions, so the pooled best predictor need not equal an average of the individually identified physical models.

11.7 Experiment 4F: structured local-dynamics identification

Method and information boundary. Fit near-linear nonlinear-plant trajectories using three positive ratios

$$p = (a_f, a_r, h) = (C_f/m, C_r/m, m/I_z). \quad (30)$$

Only the common geometry $l_f = 1.2$ m, $l_r = 1.6$ m is known to the estimator. True client mass, inertia, and stiffnesses are not supplied. The continuous matrices satisfy

$$A_c(v; p) = \begin{bmatrix} -(a_f + a_r)/v & (a_r l_r - a_f l_f)/v^2 - 1 \\ h(a_r l_r - a_f l_f) & -h(a_f l_f^2 + a_r l_r^2)/v \end{bmatrix}, \quad B_c(v; p) = \begin{bmatrix} a_f/v \\ h a_f l_f \end{bmatrix}. \quad (31)$$

This avoids attempting to recover individually unidentifiable absolute physical quantities. The estimator minimizes one-step RK4 residuals at 0.01 s, scaled by 0.05° and $0.5^\circ/\text{s}$ for sideslip and yaw rate. Logarithmic parameters enforce positivity, with bounds $a_f, a_r \in [1, 300]$ and $h \in [0.05, 3]$ in corresponding SI units. Two common starting points, $(40, 40, 0.5)$ and $(80, 60, 1)$, are used for each group. Training loss selects the solution, with no held-out tracking-based selection. These are effective group fits, not claimed exact family-center recovery.

Architecture comparison and changed estimation procedure. One structured fit is obtained globally and one per oracle family. M3 and M4 evaluate them continuously in speed. M1 and M2 freeze the corresponding fit at the preselected 20 m s^{-1} . M5 uses five frozen family fits with nearest-anchor switching. All architectures receive the same structural prior. This differs from 4D’s independently optimized constant and gridded regressions: 4F isolates deployment architecture after common identification. It does not establish dominance over every LTI fitting procedure. Earlier 4D and 4E baselines remain part of the record.

Synthesis uses exact frozen-speed ZOH matrices, unchanged Q, R , and the existing gain grid. Tests verify the reduced matrices against the validated physics and RK4 against ZOH. Scaled

local residual-Jacobian condition numbers are 1.33–1.87. Both starts agree within 9×10^{-8} relative difference. These support local numerical identifiability, not a global proof.

Results and gate. Table 17 gives the moderate results. M4 reduces mean tracking error by 8.7% relative to structured M2 and 9.1% relative to structured M3. It approaches the physics-transfer value 0.03464 without blending with oracle matrices. Near-linear and strong M4 tracking errors are 0.01644 and $0.05678 \text{ deg s}^{-1}$. All architectures are amplitude-feasible at all three severities.

Table 17: Experiment 4F: structured identification and redesign. Tracking and worst-family values are yaw RMSE in deg s^{-1} .

Architecture	Mean tracking	Worst family	Worst frozen ρ
M1	0.04065	0.04340	0.9619
M2	0.03775	0.03831	0.9646
M3	0.03792	0.04453	0.9643
M4	0.03446	0.03636	0.9663
M5	0.04228	0.04270	0.9663

The initial recovery gate passes: M4 is feasible, passes the frozen-grid audit, and improves tracking by over 5% against both structured alternatives. This is one fleet with unseen trajectories on the same clients. Frozen stability is not a time-varying LPV certificate. Next, validate independent fleet seeds and additional speed–maneuver combinations, including the original fitted LTI baselines. Only then introduce local identification, federation, and learned clustering. Structured local-dynamics identification is a promising repair, not evidence of universal M4 superiority. The planned independent-fleet validation is executed in section 11.8; the single-fleet interpretation above remains specific to Experiment 4F.

11.8 Experiment 4G: independent-fleet and trajectory validation

Purpose and frozen protocol. Experiment 4F recovered the expected architecture advantage on one fleet, but used the same clients for identification and testing. Experiment 4G tests whether that recovery transfers to independently sampled vehicles and additional trajectories. Training seeds are 11–20, paired with test seeds 111–120. Each fleet contains ten clients in each of the three unchanged physical families. Thus 300 training and 300 separate test vehicles are used. Test clients share the family distribution, not the realized physical parameters, with training clients. Repeated textual client IDs are scoped by fleet seed.

All estimator bounds, starts, residual scaling, geometry assumptions, controller weights, and speed grids remain those of 4F. The physics controller used for data collection is constructed from the training fleet only. Structured fitting uses near-linear training records only. No test trajectories or test-plant matrices enter identification, tuning, or candidate selection. True test plants are accessed only for simulation and the post-design frozen audit. Family labels remain oracle information. The JSON protocol is frozen before the runs and its SHA-256 digest is stored with the conclusions.

Comparators and additional trajectories. The five structured architectures retain the definitions in section 11.7. Two additional clustered LTI baselines fit constant matrices directly: LTI-all uses all three training amplitudes, reproducing 4D’s estimation approach, while LTI-near uses precisely the same near-linear records as structured M4. A training-physics M4 transfer controller supplies a reference. This separates the structured-LTI baseline choice from a claim against the original directly fitted LTI controllers.

Each frozen controller is evaluated on three 12 s scenarios:

1. the previous held-out S-curve with $28 \rightarrow 16 \rightarrow 24$ m/s;
2. a shifted double lane change with $16 \rightarrow 27 \rightarrow 19$ m/s, using opposite Gaussian yaw pulses of amplitude $8.5^\circ/\text{s}$, centered at 3.8 and 7.4 s with widths 0.85 and 0.65 s;
3. an unseen constant speed of 22 m/s and an enveloped sinusoidal yaw reference, in degrees/s,

$$r^*(t) = 7.5 \sin^2(\pi t/12) \sin(2\pi 0.18t + 0.35).$$

Every reference is scaled by 0.5, 1, and 1.5. The labels near-linear, moderate, and strong denote these inherited amplitude factors. The additional maneuvers have not been recalibrated to equal tire-force utilization. Speed remains in the identification support. These tests concern interpolation and new maneuver combinations, not extrapolation to new physical families or speeds.

Statistical unit and decision rule. Moderate-severity tracking is primary. For each train/test seed pair, average equally over the three scenarios and the balanced test fleet. Paired differences are baseline RMSE minus M4 RMSE. Confidence intervals resample the ten seed pairs 10,000 times with bootstrap seed 4107, preserving all within-fleet dependencies. They are pointwise 95% bootstrap intervals, not simultaneous significance statements. Individual clients and scenarios are not treated as independent replicates. Per-scenario contrasts and near-linear/strong outcomes are secondary.

The recovery gate requires at least 5% reduction in the ratio of mean primary RMSEs and a positive lower confidence bound against structured M2, structured M3, LTI-all, and LTI-near. M4 must also be amplitude-feasible in every run and have spectral radius below one on every held-out frozen audit point. The audit uses 161 speeds in $[12, 28]$ m/s and is not a time-varying LPV certificate. No failed client or seed is excluded. A vectorized simulator accelerates fleet evaluation and is tested against the original scalar RK4 simulator and metrics.

Results and interpretation. All ten seed pairs completed, yielding 21,600 closed-loop evaluations across eight methods, three scenarios, three amplitude factors, and 300 held-out clients. Table 18 reports the primary fleet averages and variation across seed pairs. Structured M4 nearly matches the training-physics controller, while outperforming both structured alternatives and the original directly fitted LTI baselines. Its average worst-family error is also smaller than those of the learned comparators.

Table 18: Experiment 4G: moderate tracking, averaged equally over three scenarios. Mean, seed standard deviation, and mean worst-family RMSE are in degrees/s. Worst-family values take the maximum family mean within each seed pair before averaging across pairs.

Method	Mean $\pm$ seed SD	Worst family	Worst frozen ρ
Structured M1	0.03723 ± 0.00067	0.04487	0.9625
Structured M2	0.03341 ± 0.00066	0.03430	0.9653
Structured M3	0.03222 ± 0.00063	0.03766	0.9648
Structured M4	0.02757 ± 0.00041	0.02842	0.9668
Structured M5	0.03342 ± 0.00057	0.03455	0.9668
Direct LTI-all	0.03528 ± 0.00242	0.05430	0.9639
Direct LTI-near	0.03314 ± 0.00224	0.05044	0.9625
Physics M4 reference	0.02752 ± 0.00043	0.02830	0.9668

Table 19 gives the paired evidence. All four primary contrasts favor M4 in all ten seed pairs, with strictly positive interval lower bounds and reductions exceeding 5%. Thus the frozen

recovery gate passes. The direct-LTI comparisons are especially relevant: the improvement is not confined to the frozen structured LTI baseline introduced in 4F.

Table 19: Experiment 4G paired moderate tracking contrasts. Δ is baseline minus M4 RMSE in degrees/s. Intervals bootstrap the ten complete seed pairs.

Baseline	M4 reduction	Mean Δ	95% interval for Δ
Structured M2	17.5%	0.00584	[0.00565, 0.00602]
Structured M3	14.4%	0.00465	[0.00433, 0.00497]
Direct LTI-all	21.9%	0.00771	[0.00637, 0.00901]
Direct LTI-near	16.8%	0.00558	[0.00433, 0.00677]

Figure 16 shows that the aggregate result is supported by each scenario. Against structured M2, M4 improves by 9.1% on the original S-curve, 31.1% on the shifted lane change, and 9.8% at constant 22 m/s. Against direct LTI-near, the reductions are 11.3%, 28.4%, and 8.5%. The last comparison favors M4 in nine of ten individual pairs, so scenario-wise dominance is not universal. Constant-speed performance reflects selecting an appropriate scheduled gain away from the LTI design speed, not time variation within that particular run. M5 chooses its lower nearest anchor at the 22 m/s tie, which explains its equality with structured M2 in this scenario.

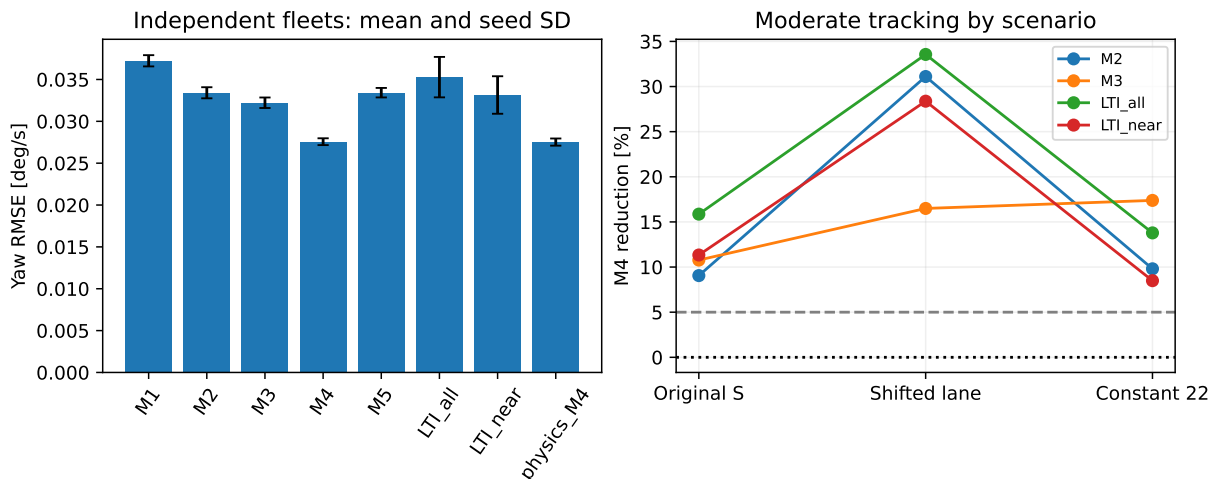


Figure 16: Experiment 4G. Left: moderate tracking averaged over scenarios, with seed standard deviations. Right: relative M4 benefits by scenario. Dashed gray marks the 5% aggregate practical threshold, not a separate scenario gate.

All methods remain amplitude-feasible in every evaluated trajectory. Structured M4 has worst held-out frozen spectral radius 0.9668. Its near-linear and strong mean RMSEs are 0.01321 ± 0.00020 and 0.04532 ± 0.00068 degrees/s. The physics reference is approximately 0.17% better on the aggregate moderate metric. Consequently, the result supports recovery of useful physics-controller performance from data, not superiority to known physics. Absolute differences are small and must be considered alongside their relative percentages.

Scope and next decision. The independent-fleet recovery is supported within the original family distribution, known geometry, noise-free state measurements, two training profile pairings, and three deterministic test scenarios. It does not establish robustness to new family centers, sensor noise, limited speed coverage, learned family assignments, or arbitrary scheduling rates. No formal time-varying stability certificate is claimed. The structured identification route now has stronger oracle support than 4F alone. The subsequent complexity and regime studies can

use this validated route, while finite local information and federated recovery remain distinct questions. No federated learning claim follows from this centralized oracle experiment.

12 Phase 5: model and controller complexity

12.1 Experiment 5A: LTI-prototype sweep

Objective and refinement of the original plan. After the independent-fleet validation in section 11.8, determine how many frozen gains approximate M4’s three scheduled families. The original generic cluster-count sweep is refined to $n \in \{1, 2, 3, 5, 8, 10\}$ speed anchors per known family, hence $3n$ frozen controllers. This isolates speed discretization while keeping structural specialization and identification fixed. It is not a learned-clustering study.

Frozen setup and selection boundary. Reuse the ten 4G training/test seed pairs and their stored structured parameter fits. Both M4 and LTI grids use these same fits and unchanged Q, R . For $n > 1$, candidate layouts are equally spaced endpoint-inclusive anchors over $[12, 28]$ m/s, equal-cell centers, and training-speed quantiles at probabilities $(j + 1/2)/n$, $j = 0, \dots, n - 1$. For $n = 1$, compare 16, 20, and 24 m/s. Each grid is shared across the three families. For every count, select the amplitude-feasible candidate with smallest mean yaw RMSE on the two original moderate training maneuvers and the training fleet. This is a finite training-based search, not an optimal placement claim. No test metric enters anchor selection.

Selected LTI grids use nearest-anchor switching, with the lower anchor at ties. M4 retains interpolation over its five fixed anchors. Test the same three scenarios and three amplitude factors as 4G on independent test clients. Report tracking, steering RMS, peak steering rate, switch counts, amplitude feasibility, and frozen stability. Neither method receives hysteresis, smoothing, steering-rate limiting, or feedback-weight retuning.

What is counted. All methods originate from three fits with three parameter ratios each. Thus this study cannot claim fewer independently learned models for M4. M4 deploys 15 gain vectors, each with three feedback coefficients and one prefilter, plus five shared speed anchors: 65 scalars. A frozen grid deploys $3n$ vectors and n shared anchors: $13n$ scalars. Counts exclude common software, units, labels, and metadata. Retaining explicit frozen A, B matrices adds $18n$ matrix scalars. Alternatively, these matrices can be reconstructed from the same nine identified ratios. Family count, gain-vector count, and storage are therefore distinct quantities.

Matching rule. Moderate tracking averaged over all three scenarios is primary. A grid matches M4 if the upper endpoint of a paired 95% percentile-bootstrap interval for $J_{\text{LTI}} - 1.05J_{\text{M4}}$ is at most zero, all trajectories are amplitude-feasible, and the worst test frozen radius is below one. Resample ten complete seed pairs 10,000 times using seed 5101. The predeclared 5% tolerance does not imply exact equivalence. Steering-rate peaks are reported as a practical cost without adding a new acceptance threshold. The minimum matching count is descriptive over tested candidates, not a universal lower bound or a deployment choice validated on a further untouched dataset.

Results, interpretation, and decision. The sweep completed 18,900 held-out closed-loop evaluations. Training selection chooses the same layout in all ten seed pairs: 24 m/s for $n = 1$, training quantiles for $n = 2, 3, 5$, and equal-cell centers for $n = 8, 10$. Table 20 reports moderate tracking and practical deployment costs. All selected grids and M4 are amplitude-feasible at all three severities and pass the held-out frozen audit, with worst radii below 0.967.

The first tested matching grid has ten anchors per family, or 30 frozen gain vectors. Its paired 95% interval for the tolerance-adjusted tracking gap is $[-0.000663, -0.000543]$ degrees/s, entirely

Table 20: Experiment 5A: moderate results. RMSE is in degrees/s, and peak rate is the maximum steering rate in degrees/s across held-out clients and scenarios. Scalars include feedback gains, prefilters, and shared speed anchors.

Design	Gain vectors	Scalars	Mean RMSE	Gap to M4	Peak rate
M4 interpolation	15	65	0.02757	—	2.29
$n = 1$	3	13	0.03181	15.4%	2.27
$n = 2$	6	26	0.03310	20.1%	34.37
$n = 3$	9	39	0.03271	18.7%	30.89
$n = 5$	15	65	0.03393	23.1%	39.72
$n = 8$	24	104	0.02925	6.1%	39.45
$n = 10$	30	130	0.02834	2.8%	14.33

below zero. Eight anchors per family do not match: their interval is $[0.000273, 0.000329]$ degrees/s. This supports a twofold gain-storage benefit for the implemented interpolated controller, not a tenfold storage benefit inferred from counting three families against 30 frozen gains. Both methods use three structural families and the same nine fitted parameter ratios.

Figure 17 shows that performance does not improve monotonically with anchor count. The grids are not nested, the selected layouts differ, and switches occur at different phases of a maneuver. Training-optimal selection within a finite candidate list does not imply test-optimal placement. In particular, the selected five-anchor grid is not the endpoint-inclusive M5 used in 4G. This prevents attributing the observed curve to model count alone.

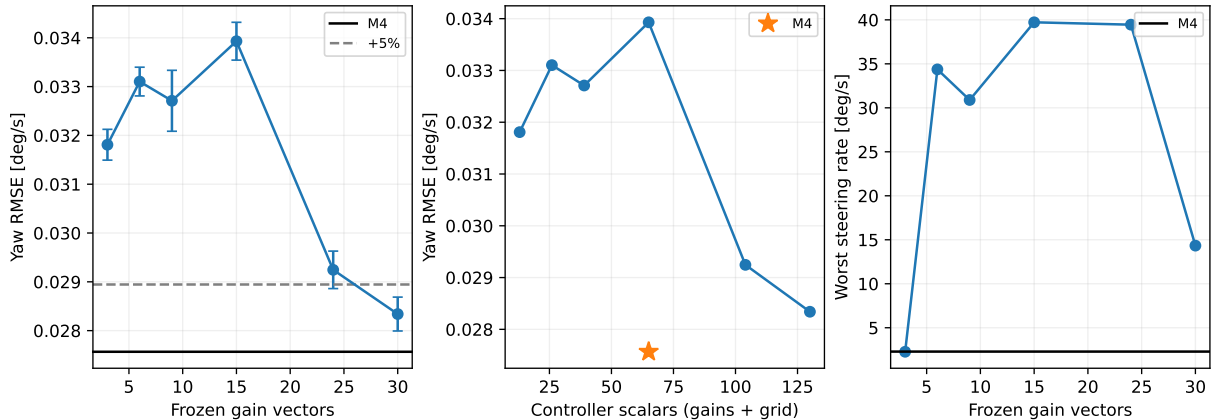


Figure 17: Experiment 5A: tracking versus frozen gain count and controller storage, followed by peak steering rate. Error bars are seed standard deviations. The horizontal tolerance line represents 5% above M4’s mean tracking error.

The ten-anchor grid’s steering RMS is 0.80576 degrees, nearly equal to M4’s 0.80578 degrees, while peak steering rate rises from 2.29 to 14.33 degrees/s. It makes up to 12 switches per varying-speed maneuver. At strong excitation, the corresponding peak rates are 3.48 and 21.58 degrees/s. Comparable RMS effort therefore does not imply comparable input smoothness.

Matching is aggregate, not universal across scenarios. With ten anchors per family, gaps are 5.03% on the original S-curve, 3.16% on the shifted lane change, and -1.11% at constant 22 m/s. The original S-curve narrowly fails the scenario-specific 5% interval criterion. A single 24 m/s gain already slightly outperforms M4 on the constant-speed case. These findings reinforce that the value of interpolation depends on the task and operating variation.

Decision and limitations. Within this frozen comparison, many more than three frozen gains are needed to meet the aggregate tracking target, and M4 achieves smoother steering with less controller storage than the first matching grid. The scheduling argument is supported for the tested nearest-switching implementation. It is not a lower bound over all LTI designs, optimized partitions, hysteretic switching, or bumpless transfer schemes, and it proves no formal time-varying stability claim. Phase 6 should now test where the advantage persists as structural heterogeneity and speed range vary, before drawing a broader model-complexity conclusion.

13 Phase 6: two-dimensional regime map

13.1 Experiment 6A: scheduling–heterogeneity sweep

Objective. Determine when each architecture is warranted rather than claiming that M4 is universally best.

Setup. Scale structural separation using

$$\vartheta_c(\gamma_h) = \vartheta_0 + \gamma_h(\vartheta_c - \vartheta_0), \quad \gamma_h \in \{0, 0.5, 1, 1.5\}, \quad (32)$$

and use speed envelopes $[18, 22]$, $[15, 25]$, and $[10, 30] \text{ m s}^{-1}$. Compare M1–M4 in every cell using model error, closed-loop performance, worst-family result, and complexity.

Predeclared expectation. Global LTI is sufficient for low heterogeneity and narrow scheduling variation. Global LPV dominates for broad scheduling variation but low heterogeneity. Clustered LTI dominates for high heterogeneity over a narrow speed range. Clustered LPV becomes necessary when both dimensions are large.

Planned evidence and decision gate. The main figure is a two-dimensional map of the best architecture and the relative advantage of M4. The map must show coherent transitions rather than one method winning due to numerical noise. Unexpected regions are investigated, not smoothed away.

Implemented protocol. The structured identification repair from Experiments 4F–4G is refitted in every cell. Training seeds 21–30 and test seeds 121–130 provide ten independent fleet pairs, each with 30 training and 30 test clients. The same random draws are reused across cells to make contrasts paired. Mass/inertia scatter remains $\pm 2.5\%$ and stiffness scatter remains $\pm 5\%$, multiplicatively around the scaled centers. Thus $\gamma_h = 0$ removes systematic family separation while retaining individual variation. Labels then partition draws from the same distribution. Family fits may differ through finite-sample variation, but have no population-level structural advantage.

For an envelope $[v_-, v_+]$, all previous speed profiles are transformed by

$$v_{\text{new}}(t) = v_- + \frac{v_{\text{old}}(t) - 12}{16}(v_+ - v_-). \quad (33)$$

The two identification profiles collectively cover each envelope. They retain their lane-change and sinusoidal references at severity 0.5. A training-fleet physics controller collects these data. Its family parameters are the average physical parameter ratios of the training clients. The estimator receives only the resulting trajectories, known geometry, and oracle family labels. The positive three-ratio fit, two starts, residual scaling, sampling time, and common LQI weights remain fixed. No test result selects a model or gain.

M1 and M2 freeze the global and family structured fits at 20 m s^{-1} . M3 and M4 interpolate gains from five equally spaced anchors spanning the current envelope. Thus the broad-speed

study does not extrapolate the previous 12–28 identification support. As in 4F, these LTI baselines are frozen structured models, not independently optimized constant regressions or the training-selected single-anchor baseline of 5A.

All methods are evaluated on the three held-out reference shapes from 4G at severities 0.5, 1, and 1.5. The former constant-22 scenario becomes constant 20.5, 21.25, and 22.5 in the three envelopes. The references and duration remain unchanged, so envelope width also changes speed rate and lateral acceleration demand. This is an operating-envelope experiment, not an isolated test of speed rate. The historical frozen-speed nonlinear plant convention remains unchanged, including omission of the longitudinal-acceleration term in the sideslip coordinate transformation. Peak speed rates across the chosen profiles are 0.79 , 1.96 , and 3.93 ms^{-2} , respectively. The broad envelope therefore exceeds the earlier illustrative 2 ms^{-2} target. Its interpretation is limited to this prescribed-speed benchmark. A time-stretched trajectory check with consistent longitudinal dynamics remains necessary for a stronger physical claim. The subsequent paired check in section 13.2 addresses this coordinate and timing question while preserving the original 6A results.

Metrics and descriptive selection rule. Primary yaw-rate RMSE averages the three moderate scenarios equally within each seed. We also retain each client/scenario/severity result, worst-family RMSE, steering RMS, peak steering rate, and amplitude feasibility. The latter uses the existing 12-degree steering and μg lateral-acceleration limits, without an actuator rate limit or clipping. A 161-point audit spans each full envelope. Frozen spectral radii below one are a diagnostic, not a certificate for varying-speed nonlinear stability.

Model errors use common moderate test trajectories collected by the training-fleet physics controller. Each fitted model predicts one step using the same RK4 convention. Separate sideslip and yaw-rate RMSE prevent a change in closed-loop state distribution from favoring a model in this comparison.

For a descriptive regime classification, let b be the empirical method with lowest mean moderate error. A method m is within tolerance when the upper paired bootstrap bound on $J_m - 1.05J_b$ is nonpositive, all its tested severities are feasible, and its worst frozen spectral radius is below one. Bootstrap resampling uses ten seed pairs, 10,000 draws, and seed 6101. We select the first eligible method in the order M1, M2, M3, M4, corresponding to 1, 3, 5, and 15 stored gain vectors (each with a prefilter). Identified parameter counts are 3, 9, 3, and 9, respectively, so this order is explicitly a controller-storage preference. Selection against an empirical best and multiple cell comparisons are descriptive, without a simultaneous confidence guarantee.

Results, interpretation, and decision. All 120 seed/cell combinations completed, giving 129,600 closed-loop client evaluations. All satisfy the amplitude-feasibility checks at all three severities. The worst audited frozen spectral radius is 0.97071. Table 21 reports the simplest eligible method, while figure 18 separates the two sources of benefit. The map broadly follows the predeclared expectation without requiring M4 everywhere.

Table 21: Experiment 6A: smallest controller bank within the descriptive 5% tolerance of the empirical best, including uncertainty and feasibility checks. The criterion concerns moderate tracking, with feasibility checked at every severity.

Family separation	18–22	15–25	10–30
0	M1	M3	M3
0.5	M2	M3	M4
1	M2	M4	M4
1.5	M2	M4	M4

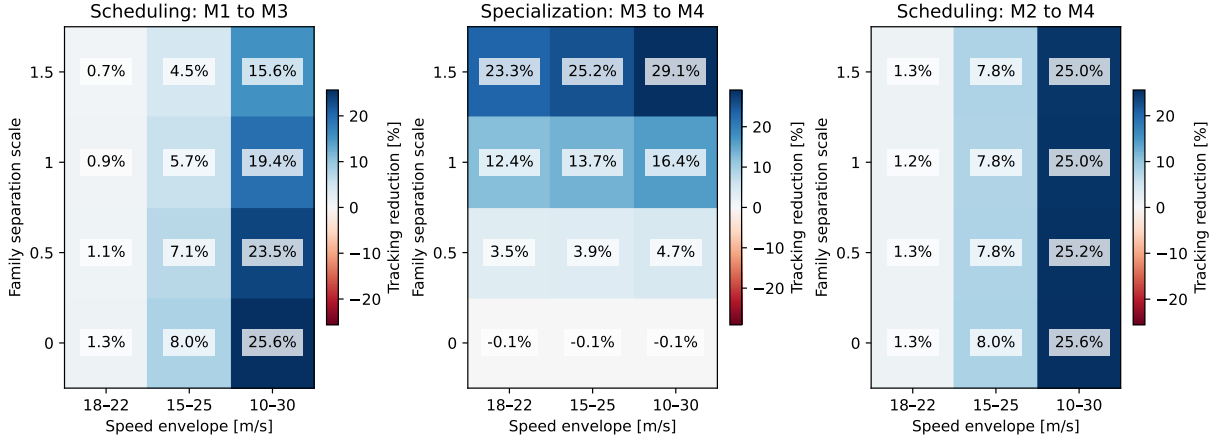


Figure 18: Experiment 6A: moderate tracking-error reductions, averaged over ten independent fleet pairs. Positive values favor the second method in each panel. The center panel isolates family specialization within LPV, and the outer panels show scheduling benefits for global and family models.

At zero separation, M4 is 0.07–0.08% worse than M3: arbitrary family labels provide no useful benefit. Global scheduling improves tracking by 1.29%, 8.03%, and 25.65% as the envelope widens. Consequently global LTI remains sufficient in the narrow cell and global LPV is selected in the wider cells. This is the intended negative control for structural specialization.

With nominal separation, M4 improves on M3 by 12.44%, 13.71%, and 16.37% across the three envelopes. Its improvement over M2 is only 1.24% in the narrow cell, increasing to 7.79% and 25.03%. The paired confidence intervals for these six mean differences exclude zero. At 1.5-times separation the specialization benefit reaches 23.29–29.10%, while the scheduling benefit within family models remains approximately 1.26–24.99%. Structural specialization and scheduling therefore address distinguishable effects, although their benefits need not be statistically independent or additive.

The boundary at half separation deserves care. In the broad envelope, M4’s mean reduction relative to M3 is 4.73%. Expressed relative to the better M4, M3’s excess is approximately 4.97%, and its confidence bound exceeds the 5% tolerance. Thus M4 is selected in that cell. This is a tolerance-boundary classification, not evidence of a sharp physical transition.

Table 22 gives the absolute results for nominal separation and the broad envelope. M4’s worst-family RMSE is 26.4% below M3’s and 24.2% below M2’s. Steering RMS differs by less than 0.05% across the methods. The tracking gain is therefore not purchased through a large increase in RMS steering, although peak steering rate is slightly higher for M4.

Table 22: Experiment 6A: nominal separation, 10–30 m/s, moderate references. Tracking is mean $\pm$ seed standard deviation in degrees/s. Worst-family values average the maximum family error within each seed. Peak rate is the maximum over the tested moderate runs, in degrees/s.

Method	Tracking	Worst family	Steering RMS [deg]	Peak rate
M1	0.04031 $\pm$ 0.00097	0.04867	0.80728	2.312
M2	0.03623 $\pm$ 0.00082	0.03728	0.80724	2.321
M3	0.03248 $\pm$ 0.00105	0.03840	0.80694	2.324
M4	0.02716 $\pm$ 0.00062	0.02826	0.80694	2.332

The common-data prediction diagnostic is consistent with an LPV advantage: in this cell the one-step yaw-rate RMSE is 0.04013, 0.04545, 0.02786, and 0.01173 degrees/s for M1–M4, respectively. M4 improves on M3 by 57.9%. However, M2’s one-step yaw error exceeds M1’s even

though its closed-loop tracking is better. This retained counterexample reinforces the lesson of 4D: model-prediction ranking alone does not determine controller quality. The separate sideslip metric and all remaining cells are retained in the prediction summary CSV.

Decision and limitations. The oracle regime study supports investing in learning persistent LPV families: both mechanisms matter when both variations are present, and simpler methods are sufficient in the appropriate limits. This remains evidence for the specified structured model class, controller weights, trajectory distribution, and prescribed-speed plant. It does not establish federation, recoverability of unknown labels, formal nonlinear stability, or superiority over optimized LTI freeze points and alternative controller designs. The next methodological step is limited local scheduling coverage and imperfect local identification, followed by testing whether aggregation and learned grouping recover the oracle benefit. Before strengthening the vehicle-level claim, the speed-rate and longitudinal-coordinate limitation above should receive a targeted check.

Reproducibility. The entry point is `experiment_6a_regime_map.py`, with its frozen JSON protocol under `code/config/`. Compressed per-seed CSVs retain client metrics, fit diagnostics, common-data prediction errors, and stability audits. The seed summaries, paired comparisons, regime classifications, and figure can be rebuilt with `--summarize-only`. The implementation and existing suite pass 30 tests, including checks that nominal separation reproduces the original fleet, collapsed centers preserve only scatter, and synthesis spans the intended speed support.

13.2 Experiment 6B: acceleration and coordinate-consistency check

Question and scope. Experiment 6A supports the scheduling–specialization decomposition but leaves an explicit prescribed-speed plant approximation. Does its principal conclusion survive a consistent lateral-momentum model, and does it persist when the speed profiles are slowed below the illustrative acceleration target? This is a paired transfer check with the 6A fitted models and controller gains frozen. It does not refit on corrected-plant data or introduce acceleration scheduling into the controller.

Coordinate correction. For the small-angle sideslip proxy used throughout this benchmark, $\beta = v_y/v_x$, the lateral-momentum and yaw equations give

$$\dot{v}_y = \frac{F_{yf} + F_{yr}}{m} - v_x r, \quad (34)$$

$$\dot{r} = \frac{l_f F_{yf} - l_r F_{yr}}{I_z}, \quad (35)$$

$$\dot{\beta} = \frac{F_{yf} + F_{yr}}{mv_x} - r - \frac{\dot{v}_x}{v_x} \beta. \quad (36)$$

The final term follows by differentiating v_y/v_x . It vanishes at constant speed, but is absent from the previous prescribed-speed implementation. We retain the same small-angle slip expressions and tanh force laws, written in lateral-velocity coordinates as

$$\alpha_f = \delta - \frac{v_y + l_f r}{v_x}, \quad \alpha_r = \frac{-v_y + l_r r}{v_x}. \quad (37)$$

The controller still receives (β, r) at each sample. This coordinate check does not replace the proxy by the exact angle $\arctan(v_y/v_x)$.

Three plant implementations. The historical implementation, denoted `legacy`, retains the 6A frozen-speed RK4 step. A diagnostic, `beta_noacc`, varies speed between RK4 stages while omitting the final term in equation (36). The coordinate-consistent implementation, `vy`, integrates (v_y, r) and converts v_y to β at sampling instants. Speed is linearly interpolated between prescribed samples during integration. Steering remains held over each sample, and the integral state and sampled controller are unchanged. Comparing the first two implementations isolates within-step speed variation. Comparing the latter two isolates the coordinate term, up to integration error.

All three retain static axle loads, pure lateral tanh tires, and externally prescribed longitudinal speed. There is no longitudinal-force allocation, combined-slip friction constraint, or acceleration-induced load transfer. Thus `vy` is coordinate-consistent under the existing bicycle assumptions, not a full longitudinal-lateral vehicle model.

Frozen protocol and contrasts. The four selected cells use $\gamma_h \in \{0, 1\}$ and speed envelopes 18–22 and 10–30 m/s. We load the committed 6A parameter fits for seeds 21–30 and reconstruct exactly the same M1–M4 gains. Test fleets remain the paired held-out realizations 121–130. These are repeated conditions on the same clients, not additional independent fleet evidence. Per-seed fit-file hashes record the model provenance.

Each of the three held-out scenarios and three severities is run with its original 12-second timing and with both speed and yaw reference stretched to 24 seconds. The sample time remains 0.01 seconds. This retains reference amplitudes and speed support while halving the rate of both signals. Broad-range peak speed acceleration decreases from 3.93 to 1.96 m/s². Stretching the yaw reference also makes lateral tracking less demanding, so we compare architectures and plants within each timing condition rather than attributing cross-duration differences solely to acceleration.

Primary metrics and amplitude-feasibility limits remain those of 6A. Paired bootstrap intervals resample ten seeds, using 10,000 draws and seed 6201. The predeclared robustness gate requires M4 to improve moderate mean tracking by at least 5% over both M2 and M3 in the nominal-separation broad-envelope `vy` condition at both durations, with positive lower confidence bounds and amplitude feasibility at all tested severities. The four-corner smallest-controller classification is a secondary descriptive result using the previous 5% upper-bound rule. The 6A frozen-speed stability audit applies to the unchanged gains and constant-speed linearizations, but does not certify the accelerating trajectories.

Validation and reproducibility. Constant-speed closed-loop trajectories and metrics agree with the previous simulator to numerical precision. The corrected derivative is checked against the coordinate transformation, and RK4 is checked against an independently implemented lateral-momentum ODE using adaptive integration and step halving. The complete suite passes 33 tests. The full legacy replay is also compared against the committed 6A raw metrics before generating conclusions. The entry point is `experiment_6b_acceleration_validation.py`, with the frozen protocol under `code/config/`. Compressed per-seed client and profile CSVs support paired summaries and a `--summarize-only` rebuild.

Results and decision. All 259,200 client evaluations complete and pass amplitude feasibility. The maximum absolute discrepancy in the retained legacy-replay metrics is 7.95×10^{-13} , confirming reproduction of the corresponding 6A runs. The unchanged frozen audit has worst spectral radius 0.96888. The primary robustness gate passes at both durations, with all four paired comparisons favoring M4 in all ten fleet pairs and all lower confidence bounds positive.

Table 23 gives the nominal-separation broad-envelope results. At the original timing, corrected-plant M4 improves on M3 by 16.46% and on M2 by 25.02%. At doubled duration, the reductions are 35.27% and 44.95%, respectively. The 95% paired intervals for the absolute mean differences

$J_{M3} - J_{M4}$ and $J_{M2} - J_{M4}$ are $[0.00491, 0.00580]$ and $[0.00893, 0.00922]$ degrees/s at 12 seconds, and $[0.00378, 0.00447]$ and $[0.00603, 0.00634]$ degrees/s at 24 seconds.

Table 23: Experiment 6B: mean moderate yaw-rate tracking RMSE in degrees/s for nominal separation and 10–30 m/s. Models and gains are identical across plant implementations and timings. The corrected plant integrates lateral momentum.

Method	12 seconds		24 seconds	
	Legacy	Corrected	Legacy	Corrected
M1	0.040306	0.040336	0.016590	0.016597
M2	0.036234	0.036246	0.013748	0.013748
M3	0.032481	0.032533	0.011676	0.011692
M4	0.027164	0.027177	0.007561	0.007568

The slower references reduce absolute error for all four methods. Although M4’s relative advantages increase, its absolute advantages over M2 and M3 decrease. This is not evidence that reducing acceleration itself increases the benefit of LPV. Both the lateral reference rate and its interaction with speed change in this timing experiment.

Figure 19 shows that the three plant implementations nearly overlap at the scale of the architecture differences. Across the four cells, two durations, and four methods, the largest absolute relative change in moderate mean tracking from **legacy** to **vy** is 0.1594%. Changing within-step speed treatment alone produces at most 0.01394% change. For broad-range nominal-family M4, the complete plant correction increases mean error by 0.0484% at 12 seconds and 0.0938% at 24 seconds. These changes are small but statistically resolvable in the paired simulations. They are not zero, and are much smaller than the architecture contrasts in this study.

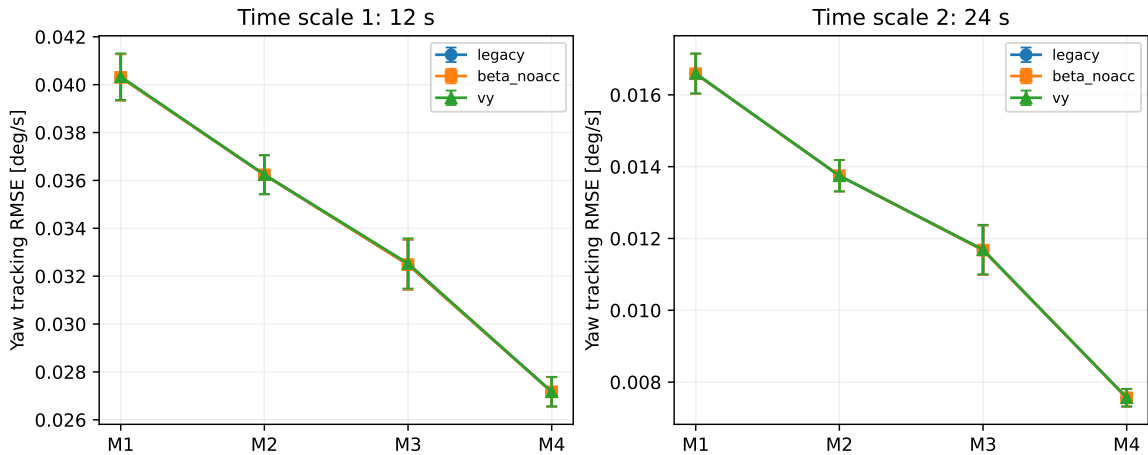


Figure 19: Experiment 6B: nominal-separation broad-range tracking across plant implementations. Error bars are standard deviations across ten seed pairs. **beta_noacc** varies speed within RK4 but omits the coordinate term, whereas **vy** integrates the consistent momentum equations. Their near overlap shows that the controller-architecture differences dominate these plant corrections under both tested timings.

The four-corner descriptive classifications also remain unchanged at both durations and for all three implementations. Global LTI is sufficient with zero separation and narrow range, global LPV with zero separation and broad range, family LTI with nominal separation and narrow range, and family LPV with nominal separation and broad range. These are the corresponding four corners of table 21. The intermediate-separation and intermediate-envelope boundaries of

6A were not rerun here.

One reason a modest effect is plausible is the size of the added term relative to linear-tire sideslip damping. Their coefficient ratio is

$$\frac{|\dot{v}_x|/v_x}{(C_f + C_r)/(mv_x)} = \frac{m|\dot{v}_x|}{C_f + C_r}. \quad (38)$$

At the nominal center, $(C_f + C_r)/m \simeq 106.7 \text{ m s}^{-2}$, so even the original peak speed acceleration gives approximately 3.7% of that linear damping coefficient. This is an explanatory scale comparison, not an error bound for tanh tires or the closed loop. The small observed tracking effect is established by the paired simulations.

The result supports continuing toward limited-data identification and federation: the tested M4 advantage is not an artifact of the omitted coordinate term or dependent on the faster timing. The remaining limitations are the prescribed longitudinal speed, pure lateral tire model, fixed controllers, and absence of a time-varying stability certificate. The result does not yet validate identification on corrected-plant data or the 5A controller-grid sweep under the new plant.

Implication for subsequent identification. The coordinate term is known kinematics shared by all clients, not a new structural family. Identification on accelerating trajectories should either use lateral-velocity coordinates or include the known term $-(\dot{v}_x/v_x)\beta$ in its predictor. The persistent unknown parameter ratios can then retain their original interpretation. This experiment tests transfer of existing speed-only controllers and does not claim that the previous identification objective is already corrected. Estimating acceleration from noisy production signals remains a separate practical question.

14 Phase 7: federated extension after the oracle gate

Only a positive oracle result justifies removing its assumptions. The first federated hypothesis is deliberately narrow:

Compatible clients with complementary speed coverage can collaboratively estimate an LPV family more accurately over the complete operating envelope than local fitting on average, while family-specific aggregation limits negative transfer.

Imperfections are introduced incrementally:

1. finite local data;
2. measurement and process noise;
3. partial and non-identical client speed coverage;
4. local LPV estimation and uncertainty quantification;
5. global aggregation and local-only baselines;
6. oracle-family federated aggregation;
7. learned family assignment;
8. uncertainty-aware clustered aggregation;
9. partial participation and communication limits.

Changing payload during a drive is excluded from the first federated study. It would make the structural state time varying and introduce change detection, concept drift, and online reassignment before the stationary problem is understood. Robust LPV synthesis, additional scheduling variables, and higher-fidelity vehicle models are likewise evidence-driven later extensions.

Experiment 7A (section 14.1) first compares local identification with centralized global and oracle-family pooling under finite data. A useful pooling benefit is the gate for implementing federated estimation with known families. Because 7A fails that gate, 7B is revised to test family-informed personalization (section 14.2). Federated estimation and learned family assignment remain deferred until a useful sharing mechanism is established. Experiment 7C (section 14.3) then tests the stronger complementary-coverage premise: each client observes only one speed, while its oracle family jointly spans the complete envelope. An equal-transition full-envelope LocalExtra oracle separates the value of shared information from the value of additional local experiments.

14.1 Experiment 7A: finite local data and the value of pooling

Question. The oracle results establish that persistent LPV families can be useful. Experiment 7A asks whether the available local evidence is insufficient enough to justify collaboration. A three-ratio physical model can potentially be identified from a narrow speed interval. Restricted coverage alone is therefore not assumed to create a federated-learning benefit. This experiment implements centralized comparisons that a subsequent federated method would need to match.

Corrected output-error estimator. Let $p = (p_1, p_2, p_3) = (C_f/m, C_r/m, m/I_z)$ and $z = (v_y, r)^\top$. The linear-tire identification model uses

$$\dot{z} = A_z(v; p)z + B_z(p)\delta, \quad \hat{y} = \text{diag}(1/v, 1)z, \quad (39)$$

where

$$A_z(v; p) = \begin{bmatrix} -(p_1 + p_2)/v & (l_r p_2 - l_f p_1)/v - v \\ p_3(l_r p_2 - l_f p_1)/v & -p_3(l_f^2 p_1 + l_r^2 p_2)/v \end{bmatrix}, \quad (40)$$

$$B_z(p) = \begin{bmatrix} p_1 \\ p_3 l_f p_1 \end{bmatrix}. \quad (41)$$

Geometry is known and unchanged. Integrating lateral velocity incorporates the coordinate correction of section 13.2. RK4 uses linearly interpolated speed within each sample, and measured steering is held constant.

For a set of independent records $\mathcal{D}$, we solve

$$\hat{p} = \arg \min_{p > 0} \sum_{d \in \mathcal{D}} \sum_{k=1}^{N_d} \|W(\hat{y}_{d,k}(p) - y_{d,k})\|_2^2, \quad W = \text{diag}(1/s_\beta, 1/s_r), \quad (42)$$

with $s_\beta = 0.05$ degrees and $s_r = 0.1$ degrees/s, converted to radians in the implementation. These scales remain fixed in the clean-data comparisons. Every simulated record starts from the known zero state. The predictor propagates from that state throughout the record and does not reset to noisy measured states. Thus measurement noise in the output is not inserted as an exact state regressor, as it would be in a naive one-step least-squares fit.

The positive log-parameter optimization retains the 4F bounds $[1, 1, 0.05] \leq p \leq [300, 300, 3]$ and two initial guesses $(40, 40, 0.5)$ and $(80, 60, 1)$. The lowest training objective selects the solution. All three stopping tolerances are 10^{-9} , with 150 function evaluations per start. Complex-step derivatives and a banded solve of the linear state recurrence accelerate computation without changing the objective. Fit status, active bounds, differences between starts, and the column-scaled Jacobian condition number are retained. The latter is a sensitivity diagnostic, not a calibrated confidence interval under nonlinear model mismatch.

Factorial data protocol. Ten new fleet realizations, seeds 31–40, each contain ten clients in each of the three nominal-separation families. Every client contributes three separate constant-speed episodes. This avoids confounding a short recording budget with an unrealistically rapid traversal of the full speed envelope. Broad coverage uses speeds (10, 20, 30) m/s. Restricted coverage assigns each client one of the three anchor sets (10, 14, 18), (16, 20, 24), or (22, 26, 30) m/s. Their interval hulls overlap, although the observations are at discrete anchors. Each family receives the same category counts, with a seed-dependent random assignment to client indices. Scheduling coverage is therefore not a proxy for the family label.

Episode durations are 2 and 8 seconds, giving total local data budgets of 6 and 24 seconds. Short records are prefixes of the corresponding long records. The three yaw references have the fixed form

$$r_j^*(t) = 3\frac{\pi}{180}(1 - e^{-t/0.3})\sin(2\pi 0.35t + 0.7j), \quad j = 0, 1, 2. \quad (43)$$

A single nominal global scheduled controller collects all records on the corrected tanh-tire plant. Its parameters are the nominal center ratios, and its gains use the existing common LQI weights. It does not use true family labels or individual parameters to adapt collection. Identification labels are supplied only for the oracle-family pooling comparison.

Noise-free outputs are compared with independent Gaussian noise of standard deviation 0.05 degrees on the sideslip proxy and 0.1 degrees/s on yaw rate. These are declared synthetic measurement assumptions, not calibrated sensor specifications. The same standardized noise draws are reused across coverage conditions and record prefixes. Noise is added to the identification records, not to the data-collection feedback. Steering and speed are exact, process noise is absent, and the initial state is known. Availability of a sideslip measurement or sufficiently reliable estimate remains an idealization.

Three comparisons and data accounting. **Local** fits one model per client using only that client’s records. **Global** fits one model to all 30 clients’ available records. **Family** fits three models using the oracle labels, ten clients per fit. All use the same estimator, bounds, and available dataset for a given cell. Pooling intentionally provides additional data: in the short condition, the local, family, and global fits use 6, 60, and 180 seconds, respectively. The long-data local fit supplies a reference for how additional individual evidence changes the comparison. The experiment does not claim a pooling benefit at equal total sample count per fitted model.

All models generate speed-scheduled LQI controllers on the common five-point 10–30 m/s grid with the previous fixed weights. Local fitting deploys 30 model/controller families, oracle-family pooling three, and global pooling one. No controller weight, model choice, or data condition is selected on test error. This is batch nonlinear least squares, with neither RLS nor federated messages nor learned clustering.

Held-out evaluation and decision rule. Each physical client is evaluated on the three held-out 24-second moderate trajectories from 6B, covering varying speed and an unseen constant speed. Training and test trajectories are separate. The evaluated clients are the same physical systems that supplied the local data, which is necessary to assess their personalized fits. The ten fleet realizations, not individual clients or scenarios, are the independent statistical units.

We report yaw tracking RMSE, worst-family tracking, steering RMS and peak rate, amplitude feasibility, and the 161-speed frozen audit. Common-input free-run prediction is evaluated separately on test trajectories collected by the nominal controller, so every candidate predicts the same steering sequence. Separate sideslip and yaw errors are retained, alongside relative error in the three true parameter ratios for diagnostic use only. The nonlinear plant is not exactly representable by the fitted linear-tire model.

The predeclared primary cell is short, noisy, restricted coverage. The collaboration gate requires oracle-family pooling to reduce mean tracking error relative to Local by at least 5%, with a positive lower paired bootstrap bound, family-controller amplitude feasibility, and frozen

radius below one. Bootstrap intervals use ten fleet seeds, 10,000 draws, and seed 7101. Global pooling, other cells, worst-family errors, and prediction errors explain the result but do not replace a failed primary gate.

Results, interpretation, and next decision. All 2,720 fits converge from both starts, with no bound hits; the largest column-scaled Jacobian condition number is 10.49 and the largest relative multistart parameter difference is 1.14×10^{-7} . All 21,600 closed-loop evaluations satisfy amplitude feasibility. The maximum frozen spectral radius is 0.96931; this remains a frozen audit, not a varying-speed stability proof.

Table 24: Experiment 7A primary cell: six seconds per client, noisy outputs, restricted coverage. Tracking is mean $\pm$ sample standard deviation across ten fleets. Prediction uses common held-out inputs. Parameter error is the root mean square of the three componentwise relative errors.

Method	Families	Tracking [deg/s]	Yaw prediction [deg/s]	Parameter error
Local	30	0.006681 ± 0.000041	0.04710	0.82%
Family	3	0.007457 ± 0.000105	0.14067	2.52%
Global	1	0.011713 ± 0.000590	0.35905	7.66%

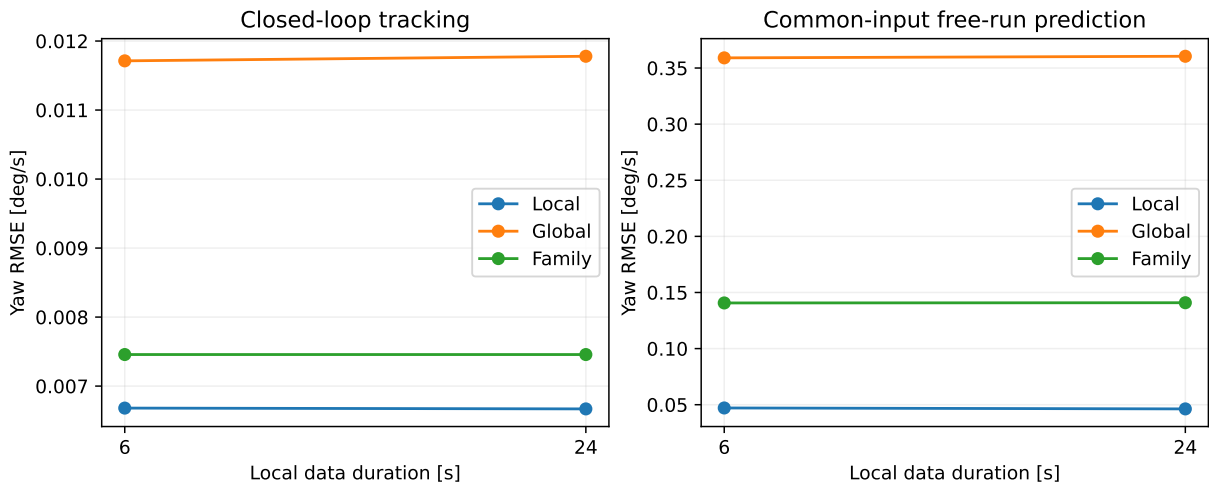


Figure 20: Experiment 7A: noisy restricted-coverage results as local recording duration increases. Closed-loop tracking and common-input prediction are different evaluations; their absolute errors should not be equated.

As shown in table 24 and figure 20, family pooling increases primary tracking error by 11.62% relative to Local. The paired difference $J_{\text{Local}} - J_{\text{Family}}$ is -0.000776 deg/s, with 95% paired bootstrap interval $[-0.000836, -0.000714]$ deg/s. None of the ten fleet pairs favors Family. The predeclared collaboration gate therefore fails. Family pooling does improve tracking by 36.33% relative to Global, but this secondary comparison cannot replace the failed Local comparison.

Local also has lower mean tracking error in all eight data conditions. For short restricted records, removing measurement noise changes Local tracking from 0.006681 to 0.006674 deg/s and its parameter error from 0.82% to 0.21%. Extending noisy restricted records from six to 24 seconds lowers Local parameter error to 0.41%, but barely changes tracking (0.006669 deg/s). Family tracking remains approximately 0.007457 deg/s. Thus local estimation variance is already small in the tested regime. The physical three-ratio structure permits generalization beyond a client’s measured speed anchors; restricted coverage does not by itself create a substantial information gap. The persistence of the pooling penalty with clean and longer records supports within-family

approximation bias as the dominant explanation here. Nonlinear model mismatch remains present, so this is evidence from controlled contrasts, not an exact bias–variance decomposition.

The primary worst-family tracking errors are 0.006823, 0.007817, and 0.013517 deg/s for Local, Family, and Global, respectively. Steering RMS is approximately 0.80961 degrees for all three, with peak rates approximately 1.163 degrees/s. The ordering is therefore not explained by a large steering-effort difference. Family deploys three scheduled families instead of Local’s 30, at an 11.62% relative tracking penalty (0.000776 deg/s absolute). This is a measurable complexity–performance trade-off, not evidence that collaboration improves individual prediction or tracking accuracy.

Consequence for the next experiment. Do not proceed to federated optimization with a claim that its accuracy benefit has already been established. A useful next diagnostic is a predeclared personalization comparison: local fitting, fixed oracle-family pooling, and family-informed local regularization, with regularization chosen using training validation only. This would test whether sharing can reduce uncertainty while retaining client-specific differences. Any harder data regime must be motivated independently and specified before testing, rather than increasing noise until pooling wins. Learned clustering and decentralized optimization remain later questions. The negative 7A gate is retained regardless of subsequent outcomes.

Reproducibility. The frozen protocol is `code/config/experiment_7a.json`; the runner is `code/experiments/experiment_7a_local_identification.py`. Per-seed compressed tables retain client metrics, fitted parameters, prediction errors, and data manifests. Summary tables, paired comparisons, and the gate decision are stored under `results/tables/experiment_7a*`. The `--summarize-only` option regenerates aggregate evidence without refitting. Predictor integration, record resets, exact-parameter recovery, and client-specific controller dispatch are covered by regression checks; the complete suite contains 36 passing tests.

14.2 Experiment 7B: family-informed personalization

Reason for the revised sequence. Experiment 7A fails the collaboration gate because accurate local fits outperform fixed family pooling. Consequently, the originally envisaged next step of federated estimation is deferred. Experiment 7B instead tests the personalization diagnostic proposed in section 14.1: can a family prior help local estimation without forcing different clients to share the same parameters? The failed 7A result is retained. This experiment remains centralized and uses oracle family labels.

Unchanged data regime and new fleets. The protocol uses ten new realizations, seeds 41–50, with 30 clients each. It retains the primary 7A regime: three two-second constant-speed episodes, restricted speed anchors, sideslip-proxy noise of 0.05 degrees, and yaw-rate noise of 0.1 degrees/s. Family separation, client scatter, nominal collection controller, references, known initial state, and exact speed and steering are unchanged. Neither noise nor scarcity is increased after observing 7A. The same three held-out 24-second scenarios and corrected tanh-tire plant are used for evaluation. These fleets are independent of 7A; within each fleet, test trajectories belong to the same physical clients as identification.

Family prior without validation leakage. For client i , first fit $\bar{p}_{c,-i}$ using only the other nine members of its true family. No record from client i contributes to this prior. This gives a fixed donor prior for all of the client’s validation folds. Using the 7A output predictor and scales, the personalized estimate is

$$\hat{p}_i(\lambda; \mathcal{D}) = \arg \min_{p \in \mathcal{P}} \frac{1}{2N_{\mathcal{D}}} \sum_{d,k} \|W(\hat{y}_{d,k}(p) - y_{d,k})\|_2^2 + \lambda \left\| \frac{\log p - \log \bar{p}_{c,-i}}{0.05} \right\|_2^2, \quad (44)$$

where $N_{\mathcal{D}}$ counts transitions, so $2N_{\mathcal{D}}$ counts scalar output residuals. The logarithm acts componentwise and $\mathcal{P}$ is the same positive bounded parameter domain as in 7A. The fixed 0.05 log scale sets the penalty units; it is not an estimated prior standard deviation. Normalizing the data term makes the meaning of λ consistent between four-second fold fits and the six-second final fit. The implementation multiplies the whole objective by the number of scalar residuals and appends the resulting prior residuals to the least-squares problem. At $\lambda = 0$, it exactly preserves the unregularized fitting objective and implementation. Both solver starts and tolerances are unchanged.

Selection using identification records only. The candidate strengths are fixed before execution:

$$\Lambda = \{0, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}. \quad (45)$$

For each candidate, fit on two complete episodes and predict the third from its known zero initial state. Repeat for all three held-out episodes and select the smallest average standardized output error. Ties select the lower strength. This validates free-run prediction, not closed-loop tracking. The selected strength is then used to refit on all three local episodes. If zero is selected, the final Local fit is reused exactly. No additional validation recordings are supplied and no test trajectory, true parameter, or test tracking metric enters selection. Since validation episodes contain different speed anchors, the selection also probes prediction beyond the two fitting anchors within the client’s assigned interval.

Comparisons and resource accounting. **Local** fits all six seconds from each client independently. **Family** fits one shared model to all ten clients’ six-second records in each true family. **Personalized** uses the donor prior and selected local estimate in equation (44). Family pooling remains an exact comparison to the deployed family architecture in 7A, whereas the prior excludes the recipient specifically to protect validation integrity. All methods access the same recorded fleet dataset. Each donor prior uses 54 seconds, each Family fit 60 seconds, and each Local final fit six seconds. Each validation candidate uses four local seconds for fitting and two for scoring. Personalization increases fitting effort and deploys 30 model and controller families, just as Local does; it does not inherit Family’s three-family deployment advantage. The leave-one-client-out priors are a diagnostic construction, not an efficient federated implementation.

Evaluation and decision rule. The common LQI weights, gain grid, held-out prediction inputs, amplitude checks, and 161-speed frozen audit are retained. The primary gate requires Personalized to improve fleet-average tracking relative to Local by at least 5%, with a positive lower 95% paired bootstrap bound, amplitude feasibility, and frozen spectral radius below one. The ten fleet seeds are the independent units; bootstrap uses 10,000 draws and seed 7201. Comparisons against Family, worst-family tracking, parameter error, common-input prediction, and selected strength counts are secondary diagnostics. Lower cross-validation error alone does not constitute held-out improvement, since it is the selection objective.

Results and interpretation. All 6,284 fits, including the 5,400 validation fits, converge from both starts without bound hits. Final fits are reused for the 46 clients selecting zero regularization. The maximum multistart relative parameter difference is 4.44×10^{-7} . All 2,700 closed-loop evaluations satisfy amplitude limits and the largest frozen spectral radius is 0.96933. These checks do not constitute a formal varying-speed stability guarantee.

Table 25 and figure 21 shows that personalization nearly recovers Local performance while improving tracking over Family by 12.79%. Against Local, however, tracking error increases by 0.0573%. The paired difference $J_{\text{Local}} - J_{\text{Personalized}}$ is -3.82×10^{-6} deg/s, with 95% bootstrap interval $[-7.37 \times 10^{-6}, -1.65 \times 10^{-7}]$ deg/s. Three of ten fleet pairs favor Personalized. The primary 5% improvement gate therefore fails. The absolute penalty is tiny, so the practical

Table 25: Experiment 7B on ten new fleets: six seconds of noisy restricted local data. Tracking reports mean $\pm$ standard deviation across fleet seeds. Parameter error uses the same component-wise relative-error convention as 7A.

Method	Tracking [deg/s]	Yaw prediction [deg/s]	Parameter error
Local	0.006669 ± 0.000016	0.04667	0.787%
Family	0.007651 ± 0.000368	0.15837	2.627%
Personalized	0.006672 ± 0.000017	0.04637	0.842%

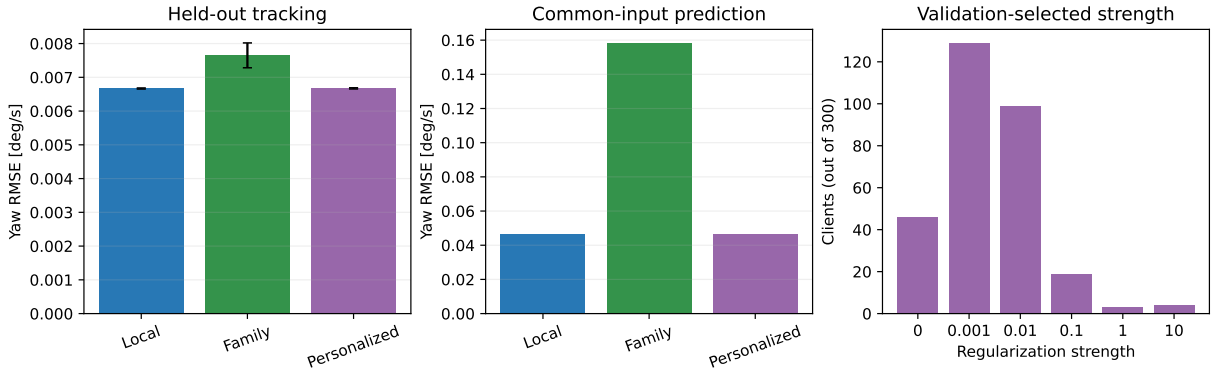


Figure 21: Experiment 7B: held-out tracking, common-input yaw prediction, and the distribution of validation-selected regularization strengths. Tracking error bars show standard deviation across ten fleets. Prediction and tracking use different inputs and should not be compared in absolute magnitude.

conclusion is near-parity, with no demonstrated tracking improvement, rather than a substantial personalization failure.

The selected-strength counts for $\lambda = 0, 0.001, 0.01, 0.1, 1, 10$ are 46, 129, 99, 19, 3, and 4, respectively. Thus 274 of 300 clients select at most 0.01. This supports weak shrinkage in this setting; it does not show that those strengths are optimal outside the tested grid or data regime. The four upper-end selections are retained without expanding the grid after seeing test outcomes. Common-input yaw prediction improves by only 0.64% relative to Local, with paired difference interval $[-0.000187, 0.000808]$ deg/s spanning zero. Relative parameter error increases from 0.787% to 0.842%. Neither secondary result establishes a reliable estimation advantage. Worst-family tracking is likewise nearly unchanged, 0.006805 versus 0.006808 deg/s. Steering RMS is approximately 0.80852 degrees for both, with peak rates about 1.1693 degrees/s.

What this establishes and what remains open. Forcing all family members to use one estimate causes a measurable penalty, and allowing local parameters largely removes it. Nevertheless, the strong three-ratio structure and available state observations already allow accurate local identification. A donor prior has little estimation uncertainty to remove here, while it can still introduce client mismatch. This interpretation is consistent with 7A’s clean-data and data-length contrasts, although 7B does not provide a formal bias–variance decomposition. Prediction-based validation also need not minimize downstream tracking error.

Experiments 7A and 7B therefore do not yet justify a claim of federated accuracy improvement. Fixed family sharing has a deployment-complexity trade-off; personalization retains 30 controllers and recovers local accuracy. Those are distinct findings. The next decision should be whether the intended application actually has a local information limitation absent here, such as unavailable sideslip measurements or insufficient excitation. Such an assumption must be motivated and specified before a new experiment. Further tuning of this regularization grid or increasing noise

solely to obtain a positive result is not the default next step. Federated optimization and learned clustering remain deferred.

Reproducibility. The protocol is `code/config/experiment_7b.json`. The runner is:

```
code/experiments/experiment_7b_personalization.py
```

Set `PYTHONPATH=code/src`. The `--summarize-only` option rebuilds the aggregates and figure. Compressed per-seed outputs retain every fold’s candidate parameters, validation loss and solver diagnostics, selected strengths, final fits, donor identities, identification-data hashes, and held-out metrics. The conclusions file records protocol and implementation hashes. The 38-test suite includes exact preservation at zero regularization, approach to the prior under a dominant penalty, explicit episode exclusion in validation, and rejection of an incorrect prior on clean identifiable data.

14.3 Experiment 7C: complementary local operating coverage

Question and frozen gate. Experiments 7A and 7B show that six seconds over three nearby speeds already suffice for accurate local structured identification. Experiment 7C therefore tests the more restrictive premise needed by the federated argument: can compatible clients compensate for operating regions that a recipient never observes? The primary gate requires oracle Family pooling to reduce mean held-out tracking error relative to Local by at least 5%, with a positive lower 95% paired-seed bootstrap bound, amplitude feasibility, and maximum frozen spectral radius below one. The gate and protocol were fixed before the ten new fleets (seeds 51–60) were evaluated.

Complementary data and controls. Each of the ten clients in an oracle family is assigned one constant collection speed from $\{10, 12, 14, 16, 18, 22, 24, 26, 28, 30\}$ m/s and records three independent one-second steering episodes only at that speed. Thus a Local fit receives 300 transitions from a single speed, whereas a Family fit receives 3,000 transitions whose union covers the full envelope. Global pools all three families. As an information-budget control, LocalExtra gives every recipient 3,000 transitions collected from that same vehicle over all ten speeds. It is an oracle for the value of additional local calibration, not a deployable federated method. All fits use the unchanged bounded three-ratio output-error estimator, noise levels, controller synthesis, three 24-second held-out scenarios, amplitude checks, and 161-speed frozen stability audit. Test trajectories are generated only after identification. No test metric enters model fitting or selection.

Results. All 640 fits converge from both starts without bound hits; the largest relative multistart difference is 2.46×10^{-7} . All 3,600 closed-loop evaluations are amplitude-feasible and the maximum frozen spectral radius is 0.96938.

Table 26: Experiment 7C on ten new fleets. Tracking reports mean $\pm$ sample standard deviation across fleet seeds. Parameter error is componentwise relative error. LocalExtra has the same transition count as Family but uses recipient-local full-envelope data.

Method	Tracking [deg/s]	Yaw prediction [deg/s]	Parameter error
Local	0.006703 ± 0.000052	0.04931	1.499%
Global	0.011755 ± 0.000465	0.36889	7.383%
Family	0.007615 ± 0.000127	0.15873	2.617%
LocalExtra	0.006683 ± 0.000051	0.04792	0.478%

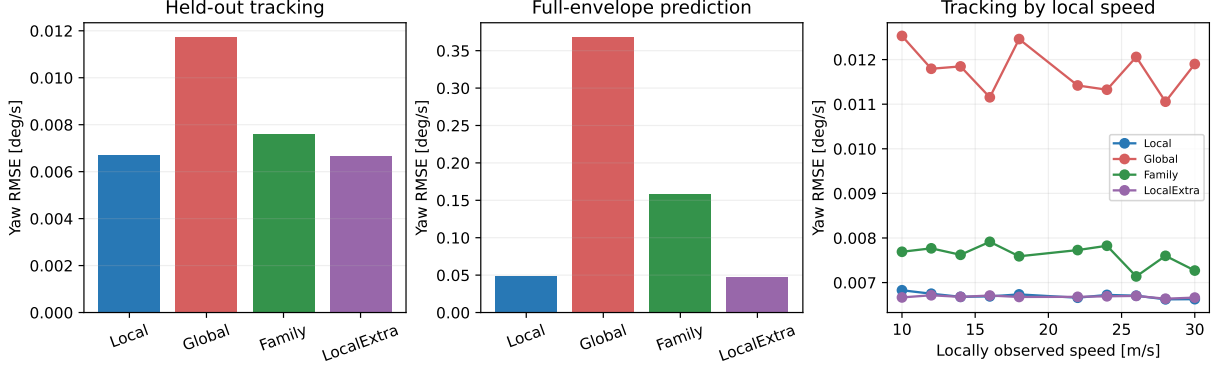


Figure 22: Experiment 7C held-out tracking, full-envelope common-input prediction, and tracking grouped by the client’s single observed speed. Tracking error bars in the first panel show standard deviation across ten fleets.

Table 26 and figure 22 shows that Family pooling improves tracking by 35.22% relative to Global, with all ten paired fleets favorable. It nevertheless increases tracking error by 13.61% relative to Local. The paired difference $J_{\text{Local}} - J_{\text{Family}}$ is -9.12×10^{-4} deg/s, with 95% bootstrap interval $[-9.76 \times 10^{-4}, -8.49 \times 10^{-4}]$ deg/s; none of the ten pairs favors Family. The primary collaboration gate therefore fails. Family also has 221.91% higher yaw-prediction error and 74.56% higher relative parameter error than Local. LocalExtra slightly improves tracking over Local and reduces parameter error to 0.478%, confirming that additional informative local data remain useful even though their effect on closed-loop tracking is small.

Interpretation and decision. Complementary speed coverage is not the missing condition in this benchmark. The known three-ratio bicycle structure lets a single-speed, three-second local record identify a controller-relevant model accurately enough to extrapolate over the full envelope. Pooling supplies speeds but also averages persistent within-family parameter differences; here that bias is larger than the variance removed by collaboration. The result is especially informative because it is not caused by instability, infeasibility, an unlucky seed, or unequal data in the Family–LocalExtra comparison.

Together, 7A–7C reject the current federated-accuracy route based only on finite data, nearby-speed restriction, personalization, or complementary speed coverage. An actual federated optimizer would approximate a centralized Family solution whose accuracy gate has now failed repeatedly, so implementing it cannot repair the statistical premise. The project should retain the oracle scheduling– heterogeneity contribution and the negative sharing boundary. A further FL study is justified only by a qualitatively different and application-grounded information constraint, such as missing state labels, excitation directions that make the local ratios unidentifiable, or a communication/deployment goal that does not claim improved accuracy.

Reproducibility. The frozen protocol is `code/config/experiment_7c.json`. The runner is:

```
code/experiments/experiment_7c_complementary_coverage.py
```

Set `PYTHONPATH` to `code/src:code/experiments`. The `--summarize-only` option rebuilds all aggregate tables and the figure. Compressed per-seed files retain client metrics, fit diagnostics, prediction errors, local-speed assignments, and transition-budget accounting. The conclusions file records hashes of the protocol, runner, and estimator.

15 Path to paper consolidation

During development, this report favors traceability over page limits. Failed experiments, revised assumptions, and negative findings remain documented. Experiment 8A performs the formal paper-readiness audit without requiring a positive federated result (section 17).

The consolidated paper should center on one precise claim and retain only the decisive evidence. The evidence now supports an LPV architecture-selection and structured-identification paper, not an uncertainty-aware federated method. The development report remains the complete experimental record and supplies supplementary details.

16 Current execution status

Phases 0–3 are complete on the linear speed-dependent bicycle plant. The plant and client population passed validation (section 7); the reciprocal-speed LPV basis passed pointwise and trajectory validation (section 8); and the oracle model and controller comparisons support complementary benefits from scheduling and family specialization (sections 9 and 10). The nonlinear tire plant passed its equation, force-limit, and integration audit in Experiment 4A. Experiment 4B then established and froze three feasible, quantitatively separated maneuver-severity levels (section 11.2). Experiment 4C shows that the clustered LPV model retains its one-step, free-run, and worst-family advantages on nonlinear-plant data and an unseen speed–maneuver combination (section 11.3). Experiment 4D finds that the unchanged clustered LPV controller transfers successfully, but direct redesign from unconstrained nonlinear-data regressions fails the primary stability and performance gate (section 11.4). Experiment 4D-R shows that a common physics regularizer restores stability and feasibility, but does not make redesigned M4 outperform clustered LTI (section 11.5). Experiment 4E shows that solver roundoff and tire nonlinearity alone do not explain the failure. Experiment 4F restores M4’s initial tracking advantage through structured parameter-ratio identification (sections 11.6 and 11.7). Experiment 4G validates that repair on ten independent train/test fleet pairs and three held-out trajectories, including the original fitted LTI baselines. Its predeclared recovery gate passes (section 11.8). This establishes oracle support within the tested distribution, not a federated result. Experiment 5A finds that the first tested frozen grid matching M4 within the aggregate 5% tolerance uses 30 gain vectors versus M4’s 15, with higher peak steering rates (section 12.1). This is a result for the tested grid search and switching rule. Experiment 6A completes the oracle regime map: global LTI is sufficient at low separation and narrow speed range, global LPV at zero separation and wider ranges, family LTI at separated families and narrow range, and family LPV at nominal separation and wider ranges (section 13.1). All 129,600 tested runs satisfy amplitude feasibility. Experiment 6B checks four corner cells with frozen 6A controllers, corrected lateral-momentum integration, and slower references. Its robustness gate passes, and the corresponding regime choices persist (section 13.2). The largest moderate aggregate tracking change from the plant correction is 0.16%, much smaller than the architecture differences. Prescribed longitudinal speed and pure lateral tire forces remain limitations. Experiment 7A completes the finite-data prerequisite comparison (section 14.1). Oracle-family pooling beats global pooling but increases primary tracking error by 11.62% relative to accurate local fits. Its collaboration gate fails. This establishes a complexity–performance trade-off, not a federated accuracy benefit. Experiment 7B tests validation-selected family-informed personalization on ten new fleets (section 14.2). It improves tracking over fixed Family pooling by 12.79%, but is 0.0573% worse than Local and fails its 5% improvement gate. Most clients select weak regularization. A realistic local information limitation must be established before pursuing a federated accuracy claim; the federated extension (section 14) remains planned. Experiment 7C then imposes single-speed local coverage with complementary full-envelope oracle-family coverage (section 14.3). Family remains 13.61% worse than Local in tracking, despite beating Global by 35.22%; the gate again fails. The

structured model extrapolates accurately from very sparse local coverage, while within-family pooling bias dominates its variance benefit. This closes the current federated-accuracy route unless a qualitatively different, application-grounded information limitation is introduced.

17 Experiment 8A: paper-readiness and contribution audit

Purpose. Experiment 8A is a scope and evidence audit rather than another performance benchmark. It asks which claims survive the complete sequence through 7C, what the paper can be about without overstating federation, and which checks remain necessary before freezing a manuscript. The audit treats failed gates as evidence and does not redefine success criteria after seeing results. Its frozen record is `code/config/experiment_8a.json`; the machine-readable claim map is `results/tables/experiment_8a_claim_audit.csv`.

Decision. The project is *conditionally ready as an LPV identification and control paper*, but it is not ready as a federated-algorithm paper. The proposed central claim is:

Measurable operating variation and persistent fleet heterogeneity are distinct design axes: scheduling addresses the former, whereas model specialization addresses the latter. Their combination is needed only when both effects are large enough relative to the required control accuracy.

This is an empirical design rule within the tested benchmark, not a universal theorem. The paper should use “fleet” or “multi-system” rather than “federated” in its title unless a later experiment supplies an actual distributed method and a separately justified benefit.

Table 27: Experiment 8A claim audit. “Boundary” denotes a supported negative or limiting result rather than a positive method contribution.

ID	Candidate claim	Status	Evidence
C1	Scheduling and specialization address different variation axes	Primary, supported	2, 3, 4G, 6A, 6B
C2	Structured family LPV identification enables redesign on nonlinear-plant data	Supported	4F, 4G, 6B
C3	Scheduled M4 uses fewer stored gains than the matching tested LTI grid	Supported with scope	5A
C4	Family sharing beats Global but need not beat accurate Local identification	Boundary, supported	7A–7C
C5	A federated LPV algorithm improves accuracy or control	Unsupported	None

Defensible contributions. The first contribution is the two-axis regime map. Experiment 6A varies speed envelope and family separation independently and recovers the four expected architectures: Global LTI, Global LPV, Family LTI, and Family LPV. Experiment 6B preserves the corner decisions under corrected varying-speed dynamics and slower maneuvers. The second contribution is methodological: unconstrained matrix regression on nonlinear-plant records can yield unsuitable control redesigns, whereas the structured positive parameter-ratio estimator restores the intended family LPV advantage and transfers across independent fleets and trajectories in 4F–4G. The third is a bounded implementation result: in the tested search, M4 stores 15 gain vectors while the first gridded family-LTI alternative meeting the 5% tracking tolerance stores 30. The fourth is a boundary result: Experiments 7A–7C consistently show that oracle-family sharing reduces error relative to Global pooling but cannot outperform a highly data-efficient Local structured fit.

Claims explicitly excluded. The present evidence does not support federated optimization, learned clustering, privacy preservation, communication efficiency, online adaptation, formal closed-loop stability, a universal model-selection law, or general superiority of LPV control. Frozen-radius audits and amplitude feasibility are empirical checks, not stability proofs. The complexity comparison is conditional on the tested gain grids and switching rule. Results use synthetic vehicle families, prescribed longitudinal speed, pure lateral tire forces, known scheduling, and measured sideslip proxy and yaw rate.

Minimum work before manuscript freeze. Only three items are classified as required. First, independently regenerate the headline numbers and figures from saved configurations and verify that every paper value traces to an artifact. Second, add one targeted higher-fidelity robustness check that introduces a genuinely absent coupling, preferably longitudinal dynamics with combined-slip tire forces; this should test the regime-map conclusion rather than launch a new method search. Third, sharpen the baseline and literature comparison against established local-LPV and multi-model identification/control approaches. Real vehicle data, learned family assignment, formal robust synthesis, and communication studies would strengthen the work but are not prerequisites for the scoped simulation paper.

Candidate manuscript blueprint. The recommended title is *Scheduling or Specialization? Structured LPV Identification and Control under Operating and Fleet Heterogeneity*. After the introduction, the paper should define the two variation axes and four candidate architectures, present the nonlinear benchmark and structured estimator, report the regime map and robustness result, then give the redesign and complexity evidence. A compact final subsection should report 7A–7C as the boundary of collaboration: model sharing is valuable relative to a global model and for model-count reduction, but does not improve accuracy over Local here. The full diagnostic path, including failed 4D and 4D-R designs, belongs in supplementary material rather than the main narrative.

Decision on an FL extension. The current manuscript should not be delayed by implementing distributed optimization of the failed Family objective. Any FL extension starts with a new oracle mechanism and an application-grounded benefit. It enters this paper only if that gate adds a result that materially changes the contribution; otherwise it should become a separate study. This separation protects both stories: the LPV paper remains evidence-complete, while a future FL method is judged against Local rather than only against an intentionally weak Global baseline.

18 Experiment 8B: federated-equivalence and secure-aggregation gate

Purpose and scope. Experiment 8B implements the first actual federated estimator in this project. Its purpose is deliberately limited: establish that the structured output-error objective can be optimized without transmitting trajectories and that an aggregate-only message path preserves the centralized solution before privacy noise is introduced. It does not revisit the failed Family-versus-Local accuracy gate and does not claim that secure aggregation is differential privacy.

Distributed objective. For a candidate log-parameter vector $q = \log p$, client i evaluates its local standardized output-error residual $e_i(q)$ and complex-step Jacobian $J_i(q)$. It returns only

$$m_i(q) = \begin{bmatrix} \frac{1}{2}e_i(q)^\top e_i(q) & J_i(q)^\top e_i(q) \end{bmatrix}^\top \in \mathbb{R}^4. \quad (46)$$

The server sums these messages to obtain the pooled objective and gradient and updates the three log parameters with bounded L-BFGS-B. The server broadcasts three candidate parameters per evaluation. Two fixed initializations are used, matching the centralized estimator’s multistart principle. Global aggregation uses all 30 clients; oracle-Family aggregation runs independently for each group of ten. The data protocol otherwise repeats 7C on fresh fleets, seeds 61–70.

Aggregate-only emulator and threat boundary. The Secure path adds deterministic pairwise random masks to the four-number client messages; masks cancel only in the sum. This numerically emulates what the server-side optimizer receives under secure aggregation. It is not a production cryptographic protocol, provides no implementation-level security claim, and does not handle client dropout or collusion. Floating-point mask cancellation can change the optimizer path at roundoff level. Consequently, the absolute Secure–Federated parameter tolerance is 10^{-6} ; an earlier 10^{-10} audit threshold was corrected because it incorrectly treated floating-point emulation as exact fixed-point cryptographic arithmetic. This correction changes only the equivalence audit, not any fitted model or performance result.

Gate and results. The gate requires convergence, centralized–federated relative parameter error below 2×10^{-4} , tracking difference below 10^{-5} deg/s, Secure–Federated absolute parameter difference below 10^{-6} , amplitude feasibility, and maximum frozen spectral radius below one. All conditions pass. All 120 fits converge and all 5,400 evaluations are feasible. The maximum centralized–federated relative parameter discrepancy is 7.65×10^{-8} ; the maximum tracking discrepancy is 6.12×10^{-10} deg/s. The maximum Secure–Federated parameter difference is 4.42×10^{-7} , and the largest frozen spectral radius is 0.96876.

Table 28: Experiment 8B tracking results across ten fresh fleets. Values are mean $\pm$ sample standard deviation across fleet seeds.

Scope	Optimization path	Tracking [deg/s]
Global	Centralized	$0.0119607082 \pm 0.0005063540$
Global	Federated	$0.0119607082 \pm 0.0005063538$
Global	Secure emulator	$0.0119607082 \pm 0.0005063538$
Family	Centralized	$0.0074997377 \pm 0.0001569473$
Family	Federated	$0.0074997377 \pm 0.0001569474$
Family	Secure emulator	$0.0074997377 \pm 0.0001569474$

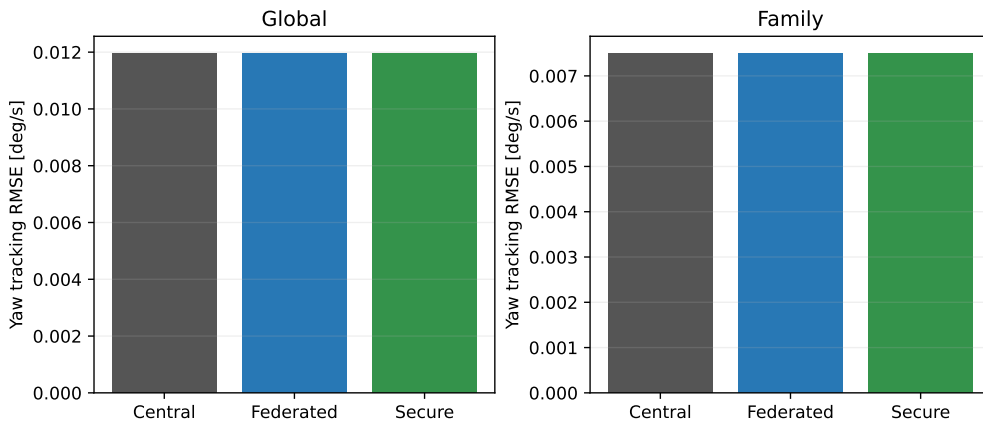


Figure 23: Centralized, federated, and aggregate-only tracking results. The bars overlap because discrepancies are many orders of magnitude smaller than the performance scale.

Table 28 and figure 23 confirms that the distributed implementation changes neither the Global nor Family conclusion. Across all federated and Secure fits, clients transmit 173,000 floating-point values and receive 129,750 values. A typical oracle-Family fit requires roughly 35–37 objective evaluations: each participating client uploads four floats and receives three per evaluation. These counts exclude identifiers, protocol metadata, cryptographic keys, mask setup, retransmission, and secure-aggregation overhead.

Decision. The federated-equivalence gate passes. The project can now truthfully claim an implemented distributed structured LPV estimator that keeps raw trajectories on clients and reproduces centralized pooled optimization. The aggregate-only emulator shows utility compatibility with secure summation, not formal privacy. Experiment 8C may therefore introduce a specified client-level differential-privacy mechanism and measure the resulting privacy–estimation–control frontier. Family remains worse than Local as established in 7A–7C; 8B does not alter that boundary result.

Reproducibility. The protocol is `code/config/experiment_8b.json`; the runner is `code/experiments/experiment_8b_federated_equivalence.py`. The reusable distributed optimizer is `code/src/federated_lpv/federated.py`. Compressed per-seed outputs retain every fitted parameter, optimizer diagnostic, client metric, and communication count. Aggregate equivalence tables and provenance hashes are stored with the other experiment results.

19 Experiment 8C: client-level privacy–control frontier

Question and threat model. Experiment 8C asks how formal client-level privacy affects structured LPV estimation and downstream control. The protected unit is one vehicle’s complete local identification dataset. Replacement adjacency allows one participating vehicle to be replaced by any other admissible vehicle. The adversary observes only the released family aggregate; secure summation is assumed from 8B. The mechanism does not hide the existence or size of a family, protect against malicious clients, or include cryptographic overhead. Unlike secure aggregation alone, differential privacy also limits information in the released aggregate.

One-shot bounded release. Every client first fits its local three-ratio model. Let $q_i = \log p_i$. Public estimator bounds define a center c and half-range vector s , giving $z_i = (q_i - c) \odot s \in [-1, 1]^3$. The contribution is clipped to $\|z_i\|_2 \leq C = \sqrt{3}$. For a family of $n = 10$ clients, the replacement sensitivity of the mean is

$$\Delta_2 = \frac{2C}{n}. \quad (47)$$

The released normalized mean is

$$\tilde{z} = \frac{1}{n} \sum_{i=1}^n z_i + \text{mathcal{N}}(0, \sigma^2 I_3), \quad (48)$$

where σ is calibrated with the analytic Gaussian mechanism for the target (ϵ, δ) guarantee [2]. Mapping $\tilde{z}$ back into the public parameter box is post-processing. This one-shot release avoids privacy composition across the iterative queries used in 8B, at the cost of aggregating local estimates rather than the exact pooled output-error objective.

Protocol. The privacy grid is $\epsilon \in \{0.5, 1, 2, 4, 8, \infty\}$ at $\delta = 10^{-5}$. Here $\epsilon = \infty$ is the non-private mean of the same bounded client contributions. Ten independent privacy draws are evaluated on each of ten fresh fleets, seeds 71–80, using the 7C single-speed local records. Every released oracle-Family model is converted to the unchanged scheduled LQI controller and tested on three

nonlinear held-out scenarios. The descriptive utility gate requires no more than 10% mean tracking increase relative to the non-private mean, full amplitude feasibility, and frozen spectral radius below one. The relative threshold was fixed before execution.

Table 29: Experiment 8C privacy-control frontier. Standard deviation and 95th percentile use the 100 fleet-privacy-draw units at each finite ϵ ; $\epsilon = \infty$ has ten fleet units.

ϵ	Tracking mean	Tracking SD	Feasible fraction	Max. ρ
0.5	1.38393	0.80278	0.89	5.3762
1	0.96284	0.75024	0.94	3.8755
2	0.45490	0.41700	0.99	3.4397
4	0.14038	0.09717	1.00	0.9896
8	0.06286	0.03668	1.00	0.9806
∞	0.007432	0.000147	1.00	0.9691

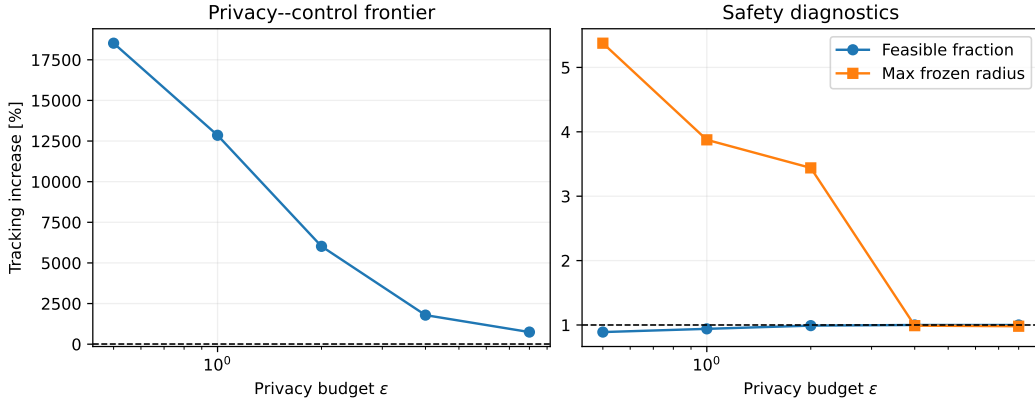


Figure 24: Tracking degradation and safety diagnostics versus the client-level privacy budget. Lower ϵ gives stronger privacy. The dashed line in the left panel is the predeclared 10% utility threshold; the dashed line in the right panel marks frozen spectral radius one.

Results and interpretation. All 300 local fits converge. The study evaluates 1,530 family releases and 45,900 client-scenario closed-loop cases. As table 29 and figure 24 shows, no finite privacy level satisfies the 10% gate. At $\epsilon = 8$, mean tracking error is 0.06286 deg/s, 746% above the unusually small non-private error of 0.007432 deg/s, although every run remains amplitude-feasible and the largest frozen radius is 0.9806. At $\epsilon = 4$, feasibility also remains complete, but mean tracking reaches 0.14038 deg/s. For $\epsilon \leq 2$, at least one privacy draw has frozen radius above one; feasibility rates are 0.99, 0.94, and 0.89 for $\epsilon = 2, 1, 0.5$, respectively. The low- ϵ results must therefore not be summarized by mean tracking alone.

This failure is structural rather than an optimizer artifact. Replacement-level sensitivity is $2\sqrt{3}/10$, so a group of ten provides limited privacy amplification. Public parameter bounds are broad, and isotropic normalized noise spends privacy budget on directions with unequal control relevance. Clipping the released value to the parameter box prevents invalid parameters but cannot recover the lost vehicle-specific information. Conversely, the large percentage increases partly reflect an exceptionally accurate non-private baseline; the absolute errors at $\epsilon = 4$ and 8 may still be operationally small, but no application-derived acceptance limit was available and none is introduced after seeing the outcomes.

Decision. Naive one-shot isotropic client-level DP is not an acceptable final method for ten-client families under the predeclared relative utility requirement. The negative result nevertheless

gives the FL extension a precise technical target. The next experiment should test mechanisms that improve privacy utility without weakening the definition: larger secure-aggregation cohorts, tighter defensible public bounds, and control-sensitivity-aware anisotropic noise or projection. Any such mechanism must retain the same (ϵ, δ) adjacency statement and be compared on fresh privacy draws. Record-level privacy or a larger ϵ must not be substituted silently.

Reproducibility. The protocol is `code/config/experiment_8c.json`; the runner is `code/experiments/experiment_8c_private_frontier.py`; analytic calibration and transformations are in `code/src/federated_lpv/privacy.py`. Per-seed compressed files retain local fits, every privacy draw, Gaussian scale, noise norm, released parameter, and client-level closed-loop metrics.

20 Experiment 8D: privacy-utility recovery

Question and fixed privacy statement. Experiment 8D tests whether the failed 8C utility gate can be recovered without weakening its privacy definition. The protected unit remains one vehicle’s complete local dataset, adjacency remains replacement of one client, and $\delta = 10^{-5}$. The same analytic Gaussian calibration and one-shot family mean release are used. Consequently, the only sensitivity reductions are a smaller public log-parameter box and larger compatible secure-aggregation cohorts; neither is a change from client-level to record-level privacy.

Protocol. The broad box from 8C is compared with the public benchmark-design box $C_f/m \in [40, 70]$, $C_r/m \in [35, 75]$, and $m/I_z \in [0.45, 0.8]$. The latter is fixed from the declared admissible fleet design and is not fitted to a participating client’s records. Clipping remains part of the mechanism, so the privacy guarantee also holds for an out-of-box contribution. No local coordinate was clipped in this run. Tight-box cohorts contain 10, 30, or 100 clients per oracle family, with complementary speeds balanced over 10–30 m/s. Five independent training fleets (seeds 81–85) are evaluated on separately generated ten-client-per-family test fleets. Each finite $\epsilon \in \{2, 4, 8\}$ uses ten privacy draws. The unchanged gate requires at most 10% mean tracking degradation relative to that mechanism’s non-private mean, full amplitude feasibility, and worst frozen spectral radius below one.

Table 30: Experiment 8D privacy recovery. Tracking is mean yaw-rate RMSE in deg/s. Increase is relative to the matching non-private cohort mean.

Bounds/cohort	ϵ	Tracking	Increase	Feasible	Max. ρ
Broad/10	2	0.37199	4632.12%	1.00	1.0457
Broad/10	4	0.14915	1797.37%	1.00	0.9913
Broad/10	8	0.06578	736.80%	1.00	0.9764
Tight/10	2	0.02162	174.99%	1.00	0.9732
Tight/10	4	0.01564	98.95%	1.00	0.9714
Tight/10	8	0.01100	39.91%	1.00	0.9703
Tight/30	2	0.01088	39.82%	1.00	0.9707
Tight/30	4	0.00908	16.60%	1.00	0.9698
Tight/30	8	0.00827	6.22%	1.00	0.9695
Tight/100	2	0.00825	6.05%	1.00	0.9692
Tight/100	4	0.00783	0.62%	1.00	0.9690
Tight/100	8	0.00779	0.09%	1.00	0.9688

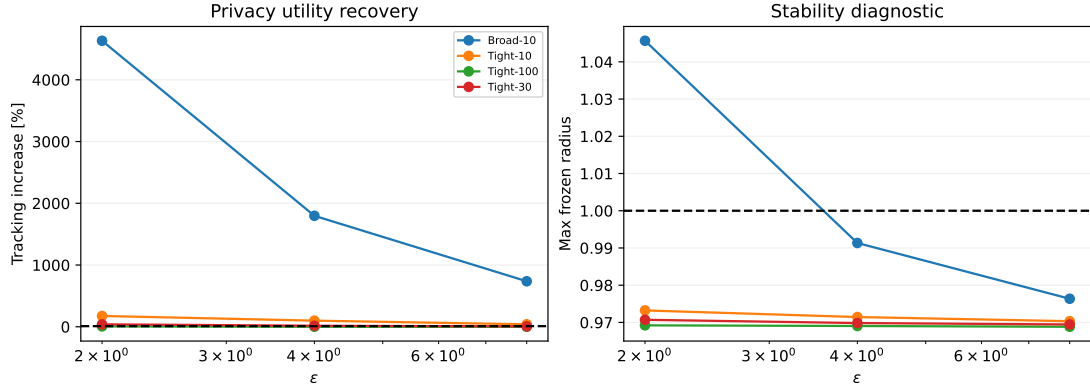


Figure 25: Recovery from public bounds and cohort scaling. The left dashed line marks the fixed 10% tracking gate; the right dashed line marks frozen radius one.

Results. All 1,500 local fits converge. The run contains 1,860 family releases and 55,800 client-scenario evaluations. Tight/100 passes at every tested privacy budget: degradation is 6.05% at $\epsilon = 2$, 0.62% at $\epsilon = 4$, and 0.09% at $\epsilon = 8$. Tight/30 first passes at $\epsilon = 8$ with 6.22% degradation. Tight bounds alone greatly improve the ten-client result, but its best finite cell still degrades tracking by 39.91%. All cells are amplitude-feasible, although Broad/10 at $\epsilon = 2$ fails the frozen-radius criterion. Thus the recovery gate passes because sensitivity scales as $1/n$ and the defensible box reduces the physical magnitude represented by a fixed normalized perturbation.

Interpretation and limitations. The result supplies a concrete FL privacy benefit: compatible vehicles can share a useful client-level-DP family model while retaining their trajectories locally, provided the aggregation cohort is sufficiently large. It is not evidence that ten-client fleets are adequate, that family labels can be exposed safely, or that production secure aggregation has been implemented. Cohort sizes 30 and 100 are simulated availability assumptions with full participation; dropout, malicious clients, privacy accounting across repeated releases, and learned private clustering remain open. The relative gate is also stringent because non-private tracking is about 0.0078 deg/s; application-specific safety limits should be established independently rather than inferred from this sweep.

Decision and reproducibility. Experiment 8D passes its recovery gate. The strongest tested passing point is $\epsilon = 2$ with 100 clients per family. This justifies proceeding to a federated system study that treats cohort formation, partial participation, and repeated releases explicitly. The frozen protocol is `code/config/experiment_8d.json`; the runner is `code/experiments/experiment_8d_privacy_recovery.py`; and aggregate results are retained in `results/tables/experiment_8d_summary.csv` with compressed per-seed fits, releases, and client metrics available from a full run.

21 Experiment 8E: control-aware private aggregation

Question. Experiment 8E tests whether client-level privacy noise can be redistributed toward directions that matter less for closed-loop control, while retaining the replacement-adjacent (ϵ, δ) guarantee from 8C–8D. The comparison does not change the public parameter box, cohort, privacy budget, estimator, or controller. It changes only the public geometry in which client contributions are clipped and Gaussian noise is calibrated.

Mechanism and privacy argument. Let $z_i \in [-1, 1]^3$ be client i 's bounded normalized log-parameter vector and let $W_g = \text{diag}(w_g)$ be a fixed positive diagonal matrix for public compatibility class g . Define

$$y_i = \Pi_{C_g}(W_g z_i), \quad C_g = \|w_g\|_2, \quad (49)$$

where Π_{C_g} is Euclidean projection onto the radius- C_g ball. This ball contains the transformed public box, since $\|W_g z\|_2 \leq \|w_g\|_2$ for every $z \in [-1, 1]^3$. The release is

$$\tilde{z}_g = W_g^{-1} \left(\frac{1}{n} \sum_{i=1}^n y_i + \xi \right), \quad \xi \sim \mathcal{N}(0, \sigma_g^2 I). \quad (50)$$

Under replacement adjacency, the mean in transformed coordinates has ℓ_2 sensitivity $2C_g/n$. Calibrating σ_g with the same analytic Gaussian mechanism as 8C therefore gives (ϵ, δ) client-level DP. Multiplication by W_g^{-1} , conversion to physical parameters, parameter-box projection, and controller synthesis are deterministic post-processing and do not weaken the guarantee. Isotropic aggregation is the special case $W_g = I$.

The family-specific weights are public and frozen before private evaluation. They are the normalized fourth roots of central finite-difference curvatures of mean nonlinear yaw-tracking RMSE at the public family-center plants, using the three declared design maneuvers and a step of 0.1 in normalized log coordinates. For example, the nominal weights are (1.636, 1.820, 0.336). Relative to isotropic noise, this reduces normalized standard deviations in the two tire/mass ratios to approximately 0.87 and 0.78, while increasing that of m/I_z by a factor of 4.25. This allocation is justified only if the public curvature correctly reflects downstream control relevance; Experiment 8E tests that premise.

Frozen protocol and gate. Ten fresh training fleets, seeds 86–95, contain 100 clients per family and use the complementary single-speed recording protocol. Separate nonlinear test fleets contain ten clients per family. Cohorts of 10, 30, and 100 are evaluated at $\epsilon \in \{2, 4, 8, \infty\}$ and $\delta = 10^{-5}$ over three held-out maneuvers. Every finite cell uses ten privacy draws. Isotropic and control-aware runs use common standard-normal draws for a paired comparison. Fleet means, not individual clients or privacy draws, are the independent statistical units. The gate, fixed before execution, requires at least one finite cell to achieve all of: at least 15% tracking improvement over isotropic DP, at most 10% degradation from the common non-private mean, complete amplitude feasibility, and worst frozen radius below one.

Table 31: Experiment 8E tracking results. “Cost” is degradation from the common non-private mean; “gain” is improvement over isotropic DP.

n	ϵ	Isotropic	Control-aware	Cost	Gain
10	2	0.02169	0.01949	154.22%	10.16%
10	4	0.01523	0.01349	76.03%	11.39%
10	8	0.01094	0.01005	31.16%	8.10%
30	2	0.01165	0.01063	39.52%	8.73%
30	4	0.00886	0.00851	11.74%	3.97%
30	8	0.00810	0.00796	4.48%	1.73%
100	2	0.00812	0.00797	4.72%	1.86%
100	4	0.00775	0.00771	1.30%	0.55%
100	8	0.00767	0.00765	0.52%	0.19%

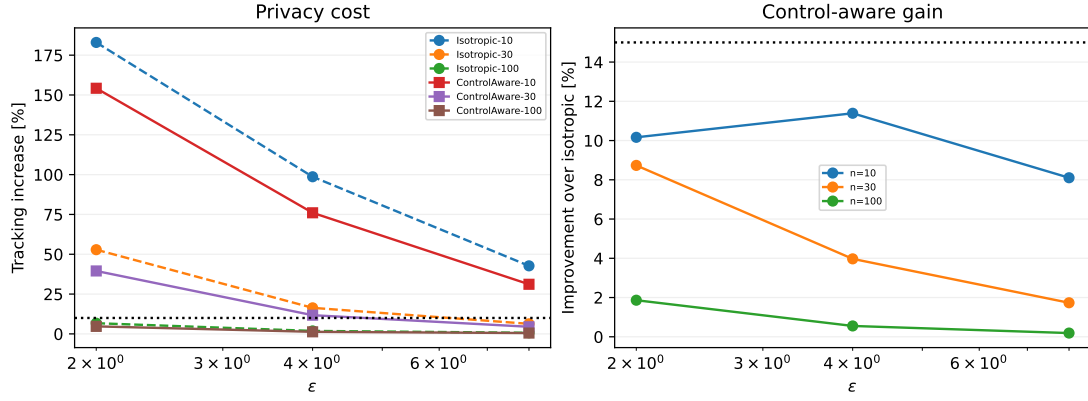


Figure 26: Privacy cost and improvement from public control-aware noise shaping. The dashed thresholds are the frozen 10% utility-cost and 15% improvement requirements.

Results. All 3,000 local fits converge. The study retains 5,580 family releases and 167,400 client–scenario evaluations. No parameter coordinate or transformed contribution is clipped, and every closed-loop evaluation is amplitude-feasible. The control-aware method improves mean tracking in all nine finite cells and in all ten independent fleets within every cell. A one-sided paired Wilcoxon test on the ten fleet means gives $p = 0.000977$ for every cell. The largest mean gain, 11.39%, occurs for $n = 10$, $\epsilon = 4$; it does not reach the predeclared 15% requirement. At the strategically important $n = 30$, $\epsilon = 4$ cell, privacy cost falls from 16.36% to 11.74%, narrowly missing the 10% utility threshold. The $n = 30$, $\epsilon = 8$ and all $n = 100$ cells satisfy the utility and diagnostic conditions, but their improvements over isotropic DP are below 15%. Consequently, the joint control-aware gate fails.

Interpretation and decision. Public curvature shaping has a real and highly consistent directional effect, but its practical magnitude is insufficient under the frozen criterion. The benefit diminishes with cohort size because both mechanisms approach the same non-private baseline. The result also shows why post-hoc threshold changes would be misleading: $n = 30$, $\epsilon = 4$ is close to useful, yet it remains outside the declared region. Experiment 8E must therefore be reported as a negative gate, not as the final proposed-method validation.

The next step should not tune the diagonal weights on these private outcomes. A defensible successor may derive a full public covariance from a stated quadratic control-loss approximation and test it on new fleets, or retain the simpler isotropic mechanism and focus the paper on the empirically established cohort/privacy operating map. Partial participation and bound misspecification remain necessary deployment tests under either choice.

Reproducibility. The frozen weights and protocol are in `code/config/experiment_8e.json`; the shaped mechanism is implemented in `code/src/federated_lpv/privacy.py`; and the runner is `code/experiments/experiment_8e_control_aware_privacy.py`. Aggregate, paired, and seed–draw summaries retain the complete reported evidence.

22 Experiment 8F: deployment robustness of private federated LPV estimation

Question. Experiment 8F tests deployment effects omitted from Experiments 8C–8E: partial participation, imperfect public clipping bounds, bounded outlying contributions, and reduced tire–road friction. It replaces Experiment 8E’s arbitrary 15% improvement criterion with an

operational quantity: the minimum cohort that keeps tracking degradation from its matching non-private aggregate at or below 10%.

Frozen protocol and gate. Ten fresh training fleets (seeds 96–105) contain 100 clients per family, with separate nonlinear ten-client test fleets. Cohorts $n \in \{10, 20, 30, 50, 100\}$ are selected by balanced, random, or deliberately low-speed-biased policies. Isotropic and public control-aware mechanisms retain replacement adjacency, $\delta = 10^{-5}$, and $\epsilon \in \{2, 4, 8, \infty\}$; common random draws enable paired comparisons. A cell is viable when tracking cost is at most 10%, every nonlinear run is amplitude-feasible, and the worst frozen radius is below one.

Stage B fixes three $\epsilon = 4$ points: $n = 10$ (nonviable), $n = 30$ (boundary), and $n = 100$ (viable). It repeats them with the correct box, a box narrowed by 10%, a box shifted upward by 5%, and 5% or 10% bounded outlier contributions, at friction coefficients $\mu = 0.9$ and 0.7. The deployment gate requires the viable point to pass all stresses. Outliers are adversarial aggregation inputs projected by the mechanism, not new physical vehicle models.

Table 32: Minimum viable cohort. All participation policies give the same minima.

ϵ	Isotropic	Control-aware	Cohort reduction
2	100	100	0%
4	50	50	0%
8	30	20	33.3%

Table 33: Tracking-cost ranges across both friction levels and all five stresses at $\epsilon = 4$.

Point	Isotropic	Control-aware	All viable
$n = 10$ nonviable	82.05–104.54%	63.28–81.74%	no
$n = 30$ boundary	14.86–19.99%	10.67–14.33%	no
$n = 100$ viable	0.18–2.02%	−0.04–1.51%	yes

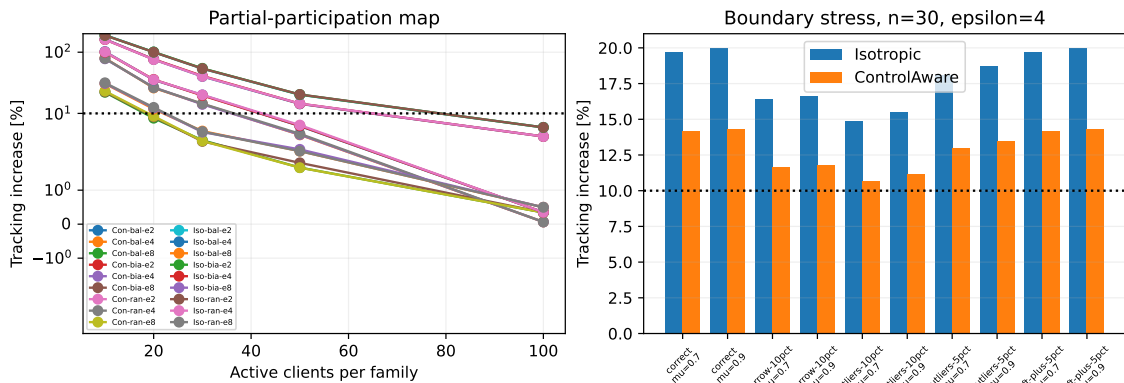


Figure 27: Minimum viable cohort and deployment-stress results under the fixed 10% tracking-cost boundary.

Results. All 3,000 local fits converge. The experiment evaluates 25,200 private releases and 756,000 nonlinear client–scenario combinations. All closed-loop runs are amplitude-feasible and the largest frozen radius is below one. Minimum cohort is insensitive to the tested participation policies. For control-aware privacy it is 100, 50, and 20 at $\epsilon = 2, 4, 8$; isotropic privacy requires 100, 50, and 30. Hence shaping reduces the required cohort by one third at $\epsilon = 8$.

At $n = 100, \epsilon = 4$, both mechanisms pass every Stage B stress. Their largest tracking costs are 2.02% (isotropic) and 1.51% (control-aware), including at $\mu = 0.7$. Neither mechanism makes the $n = 30$ or $n = 10$ point viable under all stresses, preserving the boundary of the supported region. In total 16,224 released coordinates are projected to a public box, so clipping is active.

Interpretation and limitations. Experiment 8F passes its deployment gate. The supported claim is an empirical cohort–privacy operating region that remains valid under the stated bounded perturbations. Control-aware shaping reduces cohort size in one regime; it is not universally superior. The participation finding is benchmark-specific, the outlier test is not Byzantine security, and privacy covers one release rather than repeated-round composition.

Reproducibility. The design is in `code/config/experiment_8f.json`; the runner is `code/experiments/experiment_8f_deployment_robustness.py`. Friction is propagated by `code/experiments/experiment_6b_acceleration_validation.py`. Aggregate and seed-level evidence is retained under `results/tables`.

23 Experiment 8G: blind end-to-end confirmation

Question and frozen predictions. Experiment 8G prevents the operating map in Experiment 8F from serving as both discovery and confirmation. On ten new fleet seeds (106–115), it tests two points chosen before execution. The primary prediction is that control-aware DP is viable and isotropic DP is not at $n = 20, \epsilon = 8$. The secondary prediction is that both are viable at $n = 50, \epsilon = 4$. Viability retains the 10% tracking-cost limit, full nonlinear amplitude feasibility, and frozen radius below one. The primary gate additionally requires control-aware improvement in all ten fleet seeds.

Each test fleet is separate from its training fleet. The comparison includes a three-second Local fit for every test vehicle, the non-private compatible-family aggregate, isotropic client-level DP, and control-aware client-level DP. Ten privacy draws use common random numbers. Local superiority is neither required nor claimed; its inclusion makes the collaboration tradeoff explicit.

Table 34: Blind Experiment 8G results. Tracking cost is relative to the matching non-private family aggregate.

Method	$n = 20, \epsilon = 8$	$n = 50, \epsilon = 4$	Feasible
Local	−11.90%	−11.32%	yes
Family non-private	0.00%	0.00%	yes
Isotropic DP	14.40%	5.08%	yes
Control-aware DP	10.43%	3.48%	yes

Results. All 1,800 local fits converge. The run contains 1,200 private family releases and 39,600 nonlinear client–scenario evaluations. All methods are amplitude-feasible and have worst frozen radius below one. At the primary point, control-aware DP improves on isotropic DP in every fleet seed, with a 3.47% mean gain and one-sided paired Wilcoxon $p = 0.000977$. Nevertheless, its 10.43% cost narrowly exceeds the frozen limit; isotropic DP costs 14.40%. The primary confirmation gate therefore fails. At the secondary point, both mechanisms confirm as viable: control-aware DP costs 3.48% and isotropic DP 5.08%.

Local identification has the best absolute tracking and lower mean parameter error. Its tracking is 11.90% and 11.32% better than the respective non-private family aggregates. This repeats the earlier boundary finding that federation does not improve a highly efficient Local

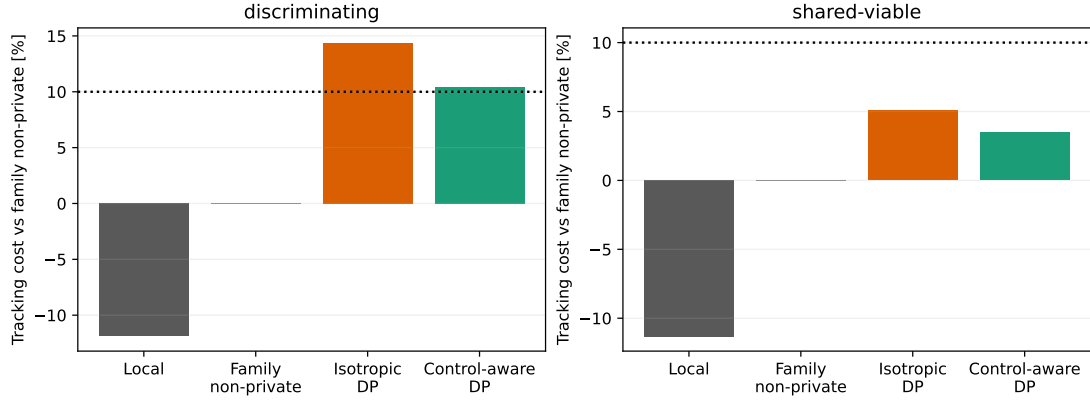


Figure 28: Blind comparison at the two frozen deployment points. The dotted line is the unchanged 10% tracking-cost boundary.

estimator in this benchmark; the supported benefits are privacy-preserving shared calibration and model consolidation.

Decision. Experiment 8G is a negative primary gate and a positive secondary confirmation. The manuscript must not claim that 20 clients suffice at $\epsilon = 8$ under a strict 10% criterion. Its headline confirmed point is instead $n = 50, \epsilon = 4$, while the 20-client outcome defines a close but unsupported boundary. The all-seed directional improvement supports control-aware shaping, but does not override the failed absolute-utility criterion. No post-hoc threshold or weight adjustment is warranted.

Reproducibility. The frozen protocol is in `code/config/experiment_8g.json`; the runner is `code/experiments/experiment_8g_blind_confirmation.py`. Aggregate, seed-draw, paired, and conclusion files are stored under `results/tables`.

24 Experiment 9A: learned compatibility groups

Question. Experiment 9A asks whether useful sharing groups and their number can be learned from compact local LPV estimates rather than supplied as oracle family labels. It is deliberately non-private: privacy-aware cluster sizing and protection of cluster discovery are deferred until useful grouping itself passes a hard gate. Training vehicles determine the partition and cluster models; separate unseen vehicles are assigned by their local three-ratio estimate. Simulator family labels enter only the oracle baseline and post-hoc adjusted Rand index (ARI).

Methods and frozen selection. Ten fresh fleets (seeds 116–125) contain 60 training and ten unseen test vehicles per simulated family. All clients receive three one-second, single-speed records. The methods are Local (30 test-vehicle models), Global (one model), Oracle family (three models), ordinary K -means with fixed $K = 3$, and two learned- K variants. Parameter clustering operates on publicly bounded normalized log-ratios. Control clustering multiplies all clients by one public, label-independent diagonal metric derived at the nominal public plant; it does not use family-specific weights.

Learned K maximizes training silhouette over $K \in \{2, \dots, 8\}$, subject to at least ten training clients per cluster. If no candidate exceeds 0.25, the method selects $K = 1$. Unseen assignments use the nearest training centroid. The hard gate requires learned- K control clustering to track no worse than the oracle-family model, remain amplitude-feasible with frozen radius below one, and deploy fewer models than Local.

Table 35: Experiment 9A unseen-client results.

Method	Tracking	Models	Parameter error	Worst client
Local	0.006692	30.0	0.0230	0.007397
Global	0.011700	1.0	0.1299	0.018969
Oracle family	0.007556	3.0	0.0469	0.010510
Fixed $K = 3$	0.007554	3.0	0.0469	0.010500
Learned- K parameter	0.007554	3.0	0.0469	0.010500
Learned- K control	0.007757	2.9	0.0497	0.011256

Table 36: Learned partition diagnostics averaged over ten fleets.

Method	K range	Train ARI	Test ARI	Silhouette
Fixed $K = 3$	3–3	0.987	1.000	0.537
Learned- K parameter	3–3	0.987	1.000	0.537
Learned- K control	2–3	0.915	0.951	0.572

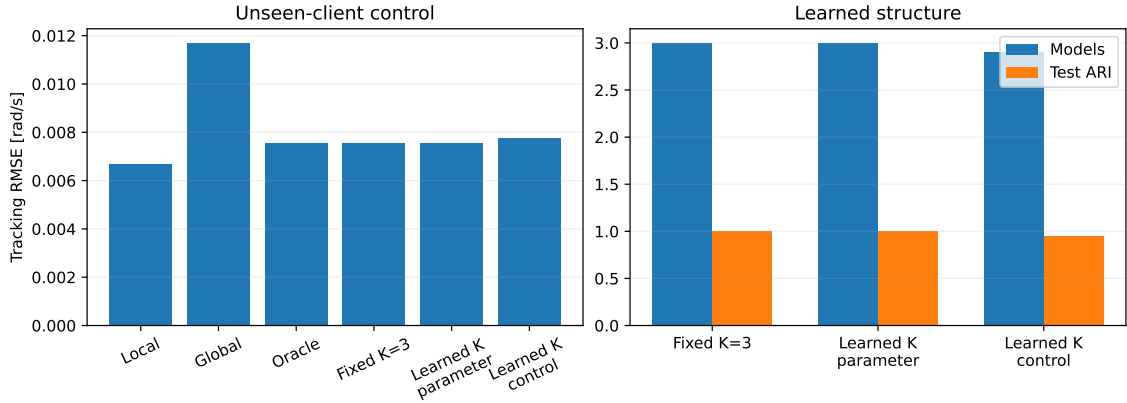


Figure 29: Unseen-client tracking and learned grouping structure in Experiment 9A. ARI is diagnostic only and is not used for model selection.

Results. All 2,100 local fits converge and all 5,400 nonlinear evaluations are amplitude-feasible with frozen radius below one. Learned- K parameter clustering selects three groups in every fleet. Its training ARI is 0.987, while nearest-centroid assignment reaches test ARI 1.0 on all unseen vehicles. Its tracking RMSE, 0.007554 rad/s, is effectively equal to and slightly below the oracle-family value of 0.007556 rad/s. It therefore removes the label assumption for this discrete-family benchmark while retaining three deployed controllers instead of 30 Local controllers. Local remains 11.4% better in tracking, and Global is 54.8% worse than oracle family.

The learned- K control variant fails the hard gate. It chooses $K = 2$ on seed 124, merges incompatible groups (test ARI 0.506), and increases that fleet’s tracking error by 25.5% relative to oracle. Across all fleets it is 2.66% worse than oracle. Its higher silhouette shows why geometric compactness alone is insufficient: the public curvature used to allocate DP noise suppresses a parameter direction that remains important for deciding which clients should share a controller.

Decision and limitations. Experiment 9A supports label-free parameter-space compatibility discovery, but rejects direct reuse of the control-aware privacy geometry for clustering. The positive result is presently limited by the simulator’s clearly discrete three- family construction. Before it becomes a strong paper contribution, Experiment 9B must test shorter/noisier identification, unbalanced populations, and continuous heterogeneity. Moreover, the server currently observes individual local estimates; this is trusted-server clustering, not end-to-end private group discovery. Privacy-aware clustering should proceed only if 9B confirms stable useful partitions beyond this favorable discrete setting.

Reproducibility. The frozen protocol is in `code/config/experiment_9a.json`; implementation is in `code/experiments/experiment_9a_learned_groups.py`. Seed summaries, candidate scores, selected-cluster diagnostics, and conclusions are retained under `results/tables`.

25 Experiment 9B: robustness of learned compatibility groups

Question and frozen gate. Experiment 9B tests whether the successful label-free parameter clustering of Experiment 9A survives four departures from its favorable benchmark: very short local records, increased measurement noise, strongly unbalanced family sizes, and a population with continuous rather than discrete heterogeneity. Privacy and the rejected control-weighted metric remain excluded so that this experiment isolates the reliability of group discovery. Ten fresh training and test fleet pairs (seeds 126–135) are used in every regime.

For the discrete regimes, Learned- K must remain within 5% of Oracle family, reach unseen-client ARI of at least 0.9, remain feasible and frozen-stable, and use fewer models than Local. For the continuous regime, where no family labels exist, the reference partition is selected from the unobserved true parameter ratios by the same rule. Learned- K must remain within 5% of that oracle- information partition and improve on Global by at least 20%. These gates were fixed before running the experiment.

Protocol. The baseline uses 180 balanced training clients, 30 unseen test clients, three one-second single-speed records per client, and output-noise standard deviations of 0.05° and 0.1° . Short-data reduces each record to 0.25 s; high-noise increases the two noise levels to 0.2° and 0.4° . Unbalanced populations use 100, 60, and 20 training clients and 15, 10, and 5 test clients in the three latent families. The continuous population interpolates physical vehicle parameters between handling and heavy endpoints, adds small independent scatter, and supplies no class labels. The selection rule remains silhouette maximization over $K = 2, \dots, 8$, with minimum silhouette 0.25 and at least ten training clients per cluster. All final control tests use the nonlinear tire plant and three held-out maneuvers.

Table 37: Experiment 9B learned-group robustness. Tracking cost is relative to Oracle family for discrete regimes and the true-ratio partition for the continuous regime.

Regime	Learned tracking	K range	Reference cost	Global gain	Gate
Baseline	0.007583	3–3	+0.08%	35.66%	pass
Short data	0.012441	2–7	+63.87%	−7.46%	fail
High noise	0.009184	2–6	+21.90%	21.82%	fail
Unbalanced	0.007551	2–3	+1.49%	21.08%	pass
Continuous	0.007662	2–2	−0.42%	17.25%	fail

Table 38: Partition diagnostics for the learned estimator. ARI is defined only for the discrete-family regimes and is used only for their frozen gate.

Regime	Mean K	Test ARI	Silhouette
Baseline	3.0	0.980	0.538
Short data	3.2	0.082	0.349
High noise	2.8	0.390	0.356
Unbalanced	2.9	0.954	0.530
Continuous	2.0	—	0.513

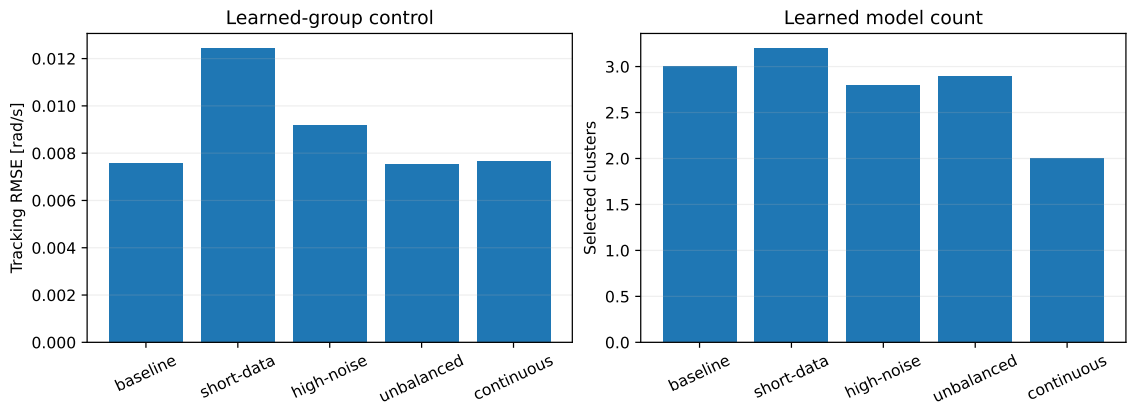


Figure 30: Tracking and selected model count under the five Experiment 9B regimes. The continuous population is represented by two learned regions, not interpreted as two physical families.

Results. All 10,500 local identification fits converge. All 21,600 nonlinear closed-loop evaluations are amplitude-feasible, and every audited frozen closed-loop radius is below one. Baseline replicates the 9A result: three groups are always selected, unseen ARI is 0.980, and tracking is within 0.08% of Oracle family. The highly unbalanced case also passes, despite selecting two groups in one fleet; its mean unseen ARI is 0.954 and its tracking cost is 1.49%.

Short-data and high-noise fail decisively. With 0.25 s records, selected K ranges from two to seven and unseen ARI falls to 0.082. Learned grouping is 63.87% worse than Oracle family and 7.46% worse than Global. It is nevertheless 6.70% better than the highly uncertain Local models, showing variance reduction without reliable compatibility discovery. Under high noise, unseen ARI is 0.390 and the tracking cost relative to Oracle family is 21.90%.

For continuous heterogeneity, both estimated-parameter and true-ratio selection choose two regions in every fleet. Learned- K is 0.42% better than the oracle- information partition and uses two rather than 30 Local controllers. Its 17.25% improvement over Global, however, misses the frozen 20% requirement. This is evidence that coarse region sharing remains useful without natural families, but not a passed robustness claim. Local tracking remains best in absolute terms in this regime.

Decision and paper implication. The omnibus robustness gate fails: ordinary point-estimate clustering is credible for adequately informative records, including strong population imbalance, but is not robust to severe local estimation uncertainty. Thus the paper must not claim generally reliable automatic family discovery. A defensible claim is narrower: label-free compatibility discovery can reproduce oracle sharing in informative regimes and reveals useful coarse regions under continuous heterogeneity. The next methodological experiment should make the partition uncertainty-aware—for example by repeated local estimates or fit-covariance-aware distances—and test recovery on the frozen short-data and high-noise cases. Private clustering should follow, rather than compound, this identified weakness.

Reproducibility. The frozen protocol is in `code/config/experiment_9b.json`; implementation is in `code/experiments/experiment_9b_group_robustness.py`. Aggregate tables, per-seed summaries, gate decisions, and provenance hashes are retained under `results/tables`.

26 Experiment 9C: uncertainty-aware compatibility discovery

Question and staged gate. Experiment 9C asks whether explicit local-identification uncertainty repairs the short-data and high-noise failures of Experiment 9B. The experiment uses the 9B seeds 126–135 as a locked development replication. Seeds 136–145 are reserved as an untouched confirmation set and are opened only if both development regimes pass. The frozen gate requires the proposed method to remain within 5% of Oracle family, strictly improve ordinary point clustering, remain amplitude- feasible and frozen-stable, and use fewer shared base models than Local. Hard ARI is retained as a diagnostic rather than a gate because the method explicitly permits uncertain, soft membership. Privacy remains excluded.

Method. For client i , the output-error Jacobian J_i is evaluated at the fitted log-ratios. Given the known output-noise scaling, the Gauss–Newton covariance is

$$P_i = (J_i^\top J_i)^{-1}, \quad (51)$$

with eigenvalue regularization only for numerical inversion. The covariance is mapped into the same normalized log-parameter coordinates used in 9A–9B. A heteroscedastic mixture then models

$$z_i \sim \sum_{k=1}^K \pi_k \mathcal{N}(\mu_k, P_i + S_k), \quad (52)$$

where P_i is client-specific measurement uncertainty and S_k captures within-group variation. Generalized EM estimates the weights, means, and full intrinsic covariances. BIC selects $K \in \{1, \dots, 8\}$ subject to at least ten hard-assigned training clients per component. For an unseen client, posterior component probabilities blend the K shared scheduled controllers. Thus the server retains K base models plus three client membership weights, rather than a separately learned model for every vehicle.

Table 39: Experiment 9C development results on the two failed 9B regimes.

Regime and method	Tracking	Mean K	Oracle cost	Point gain
Short data: Oracle family	0.007597	3.0	—	—
Short data: Point- K	0.011887	3.6	56.47%	—
Short data: uncertainty-soft	0.011810	1.0	55.45%	0.65%
High noise: Oracle family	0.007632	3.0	—	—
High noise: Point- K	0.009715	2.3	27.29%	—
High noise: uncertainty-soft	0.008038	2.0	5.31%	17.27%

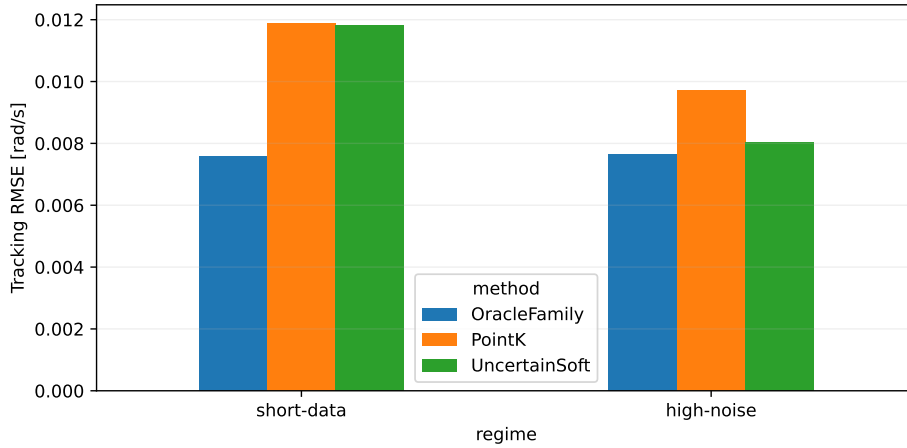


Figure 31: Development tracking for ordinary point clustering and uncertainty-aware soft grouping relative to Oracle family.

Results. All 4,200 local fits converge, and all 10,800 nonlinear closed-loop evaluations are amplitude-feasible and frozen-stable. In the high-noise regime, uncertainty-aware soft grouping reduces the point-clustering tracking error by 17.27%. Its oracle cost falls from 27.29% to 5.31%, narrowly missing the predeclared 5% threshold. BIC selects two groups in every development fleet; the mean unseen hard-label ARI improves to 0.522, compared with 0.390 for the 9B point method, but still does not recover the three simulated families.

The 0.25 s regime is qualitatively different. BIC selects $K = 1$ in every fleet, and the proposed method improves point clustering by only 0.65%. Its 55.45% oracle gap shows that uncertainty representation cannot manufacture missing client-level information. This conclusion is supported by the diagnostics: the median normalized covariance trace rises from 0.0247 under high noise to 0.141 with 0.25 s records, while the median information-matrix condition number rises from 55.5 to 248.

Decision. The development gate fails in both regimes, so the reserved confirmation seeds 136–145 are not evaluated. This is a deliberate anti-overfitting decision, not missing evidence.

Experiment 9C supports a narrower result: uncertainty-aware mixture modeling substantially repairs high-noise grouping, but does not resolve structural non-identifiability from extremely short records. For the paper, the automatic-grouping claim should remain limited to informative local records. The next useful experiment should change the information available—for example active excitation, longer or repeated calibration, or an abstain-and-request- data policy—rather than applying another clustering rule to the same uninformative 0.25 s estimates. The untouched seeds remain available for a genuinely revised method.

Reproducibility. The staged protocol is in `code/config/experiment_9c.json`; covariance and mixture routines are in `code/src/federated_lpv/uncertainty_clustering.py`; the experiment driver is `code/experiments/experiment_9c_uncertainty_groups.py`. Development summaries, cluster diagnostics, gates, and provenance hashes are stored under `results/tables`.

27 Experiment 9D: uncertainty-triggered adaptive calibration

Question and staged protocol. Experiment 9D tests the implication of the 9C failure: if 0.25 s records are structurally uninformative, can uncertainty trigger a small amount of targeted additional excitation and recover useful label-free sharing? Seeds 126–135 are used for development. The final acquisition policy and the unchanged hard gate are then evaluated once on previously untouched seeds 136–145. Confirmation is opened only after development passes.

Every client starts with three 0.25 s records at one locally assigned speed, corresponding to 0.75 s total calibration. If the normalized Gauss–Newton covariance trace exceeds 0.05, the client records a one-second maneuver at a complementary speed and refits. At most two such follow-ups are permitted, so the full fixed-budget comparator uses 2.75 s, still below the three-second 9B baseline. The final policy was selected on development after a preliminary variant with additional 0.25 s bursts showed that repeatedly restarting a very short maneuver did not provide sufficient transient information. No confirmation seed was accessed during that revision.

The proposed Adaptive-soft method applies the 9C heteroscedastic mixture and posterior controller blending to each client’s stopping-time estimate. It is compared with Short-point, which applies ordinary learned- K clustering to the initial 0.75 s data; Fixed-full-soft, which always consumes 2.75 s; Local- adaptive; Global-adaptive; and Oracle-adaptive, which uses simulated family labels only for reference. The frozen gate requires Adaptive-soft to remain within 5% of Oracle-adaptive, strictly improve Short-point, use less mean data than Fixed-full-soft, remain feasible and frozen-stable, and retain fewer shared models than Local-adaptive.

Table 40: Experiment 9D development and untouched-confirmation results.

Phase	Method	Tracking	Models	Data [s]	Parameter error
Development	Oracle-adaptive	0.007610	3.0	1.75	0.0494
Development	Short-point	0.012299	2.4	0.75	0.1791
Development	Adaptive-soft	0.007886	2.9	1.75	0.0524
Development	Fixed-full-soft	0.007594	3.0	2.75	0.0486
Confirmation	Oracle-adaptive	0.007542	3.0	1.75	0.0464
Confirmation	Short-point	0.011666	2.7	0.75	0.1759
Confirmation	Adaptive-soft	0.007758	2.8	1.75	0.0534
Confirmation	Fixed-full-soft	0.007513	3.0	2.75	0.0451

Results. The development gate passes. Adaptive-soft is 3.62% above Oracle-adaptive and 35.88% better than Short-point. It selects 2.9 shared models on average, achieves unseen hard-label ARI 0.935, and uses 1.75 s per client rather than the 2.75 s fixed full budget. Every client

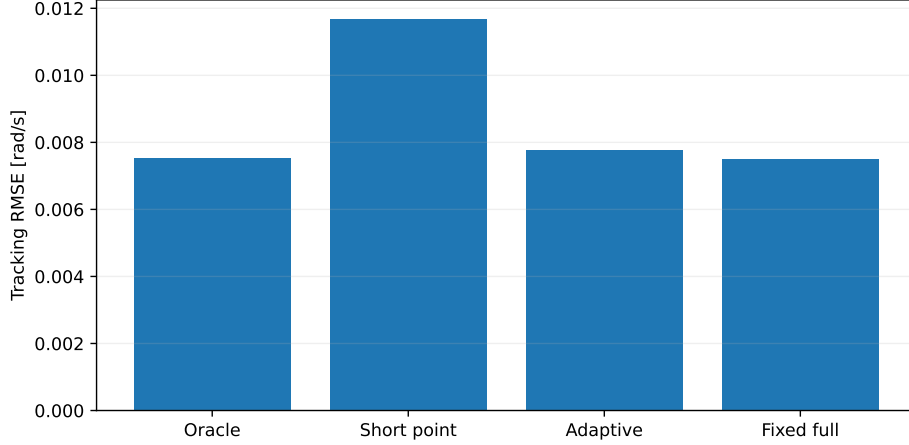


Figure 32: Untouched-confirmation tracking for the oracle, initial short-data, adaptive, and fixed full-budget methods.

stops after the first one-second follow-up, which means the covariance criterion gives a consistent stopping decision in this benchmark rather than merely reducing the fleet-average budget.

The untouched confirmation gate also passes. Adaptive-soft tracking is 0.007758 rad/s, 2.86% above Oracle-adaptive and 33.50% better than Short-point. It retains 2.8 shared models on average versus 30 Local-adaptive models, and again stops at 1.75 s for every client. Confirmation hard-label ARI is lower, 0.820, yet soft membership preserves near-oracle control. This distinction is important: exact recovery of simulator labels is neither necessary nor used by the proposed control policy. Fixed-full-soft reaches 0.007513 rad/s, but needs one additional second per client.

Across both stages, all 4,200 final client fits converge and all 10,800 nonlinear closed-loop evaluations are amplitude-feasible. The worst frozen radius remains below one. Local-adaptive still gives the lowest absolute tracking error (0.006907 rad/s in confirmation), so the result is not an accuracy advantage over a separate per-vehicle controller. Its contribution is a calibration–deployment tradeoff: label-free sharing recovers near-oracle family performance with 2.8 shared controllers and 1.75 s of local calibration.

Decision and FL interpretation. Experiment 9D resolves the short-data failure in the intended operational sense. The federated system should not demand a fixed calibration budget from every client or cluster estimates known to be uninformative. Instead, clients expose compact estimates and uncertainty, the server learns shared compatibility models, and uncertainty triggers additional local excitation only when needed. This gives the paper a stronger control-specific FL message: federation changes the data-acquisition decision as well as the estimator. Privacy is still absent from group discovery and uncertainty reporting. A subsequent experiment may combine this confirmed policy with the established private aggregation mechanism, but should not alter the 9D acquisition rule.

Reproducibility. The frozen final protocol is in `code/config/experiment_9d.json`; the implementation is in `code/experiments/experiment_9d_adaptive_calibration.py`. Development and confirmation seed summaries, budget statistics, group diagnostics, gate decisions, and provenance hashes are retained under `results/tables`.

28 Experiment 9E: causal audit of the Local advantage

Question. The preceding experiments consistently show that a separately fitted Local controller gives the lowest tracking error. Experiment 9E asks whether this is caused by genuine vehicle-specific information, an identification artifact that compensates the nonlinear plant, an inappropriate parameter metric, or favorable calibration–deployment matching. This is a diagnostic experiment and has no positive-result gate: its purpose is to determine what must be redesigned before claiming an accuracy benefit from information sharing.

Ten fresh fleets (seeds 146–155) each contain 30 test vehicles. Every vehicle receives the frozen 9D adaptive calibration protocol and stops after 1.75 s. Five controllers isolate distinct effects: Local uses the fitted client ratios; Exact-individual uses its true simulator ratios; Swap-within-family uses the Local fit of the next vehicle in the same true family; Exact-family uses the geometric mean of the true ratios within each family; and Global-exact uses the population geometric mean. Thus Exact-family is a deliberately stronger sharing reference than a learned or federated family estimate: it has neither clustering nor estimation error. An Exact-deployment reference is additionally recomputed after drift and is not a deployable method.

All fixed controllers are evaluated on the nominal plant and after two common post-calibration changes: a 10% loss of both axle stiffnesses and a load shift that raises mass by 10% and yaw inertia by 12%. The same drift is applied to every method. Besides closed-loop tracking, the audit records unweighted log-parameter error, scheduled-gain error relative to Exact-individual, and normalized held-out nonlinear prediction error.

Table 41: Experiment 9E tracking RMSE on ten fresh fleets. Parentheses report change relative to Local in the same condition.

Method	Nominal	Tire fade	Load shift
Exact-deployment	0.006702 (−3.89%)	0.006868 (−10.34%)	0.006995 (−9.48%)
Exact-individual	0.006702 (−3.89%)	0.007348 (−4.07%)	0.007425 (−3.91%)
Local	0.006974	0.007660	0.007728
Exact-family	0.007521 (+7.85%)	0.008286 (+8.18%)	0.008339 (+7.91%)
Swap-within-family	0.008569 (+22.88%)	0.009448 (+23.33%)	0.009477 (+22.64%)
Global-exact	0.011638 (+66.89%)	0.012814 (+67.28%)	0.012795 (+65.58%)

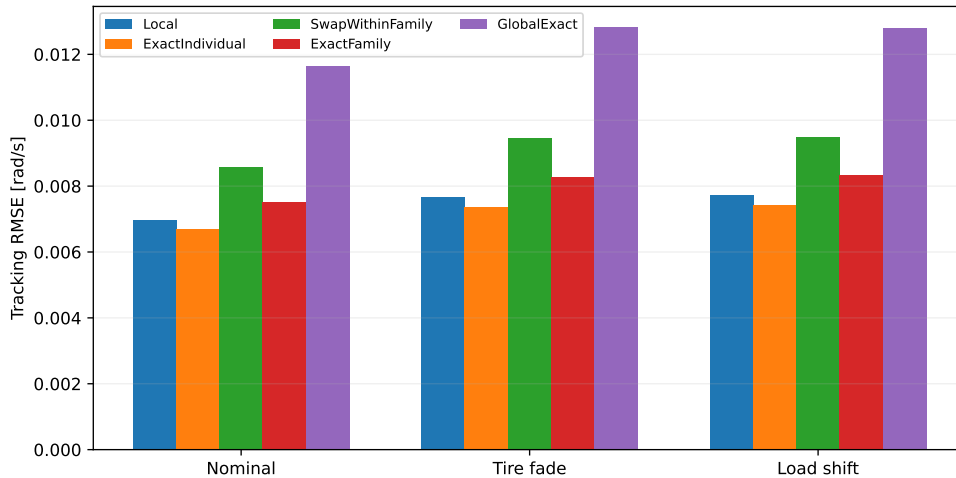


Figure 33: Local-advantage decomposition under matched and shifted deployment.

Attribution. Exact-individual improves nominal tracking by 3.89% over Local. Consequently, the Local fit is not winning by learning a beneficial but physically incorrect surrogate for the nonlinear plant; finite-data estimation still leaves a small recoverable loss. In contrast, substituting another correctly classified family member worsens Local by 22.88%. Even Exact-family, which knows the true family and its exact mean ratios, is 7.85% worse. Most of the remaining Local advantage therefore comes from real within-family specialization, not federated optimization, clustering, or privacy error.

The diagnostic measures explain why scalar parameter error was previously misleading. Across clients, Local log-parameter error has Spearman correlation $\rho = 0.029$ with tracking, whereas scheduled-gain error has $\rho = 0.216$ and held-out nonlinear prediction error has $\rho = 0.325$. For the shared comparators the control-aligned measures are much more predictive: Exact-family gain and prediction correlations are 0.757 and 0.708, and Global-exact values are 0.873 and 0.937. Parameter distance treats all three identified ratios as equally important even though their closed-loop consequences differ.

The two deployment shifts increase absolute errors but preserve the ordering and nearly the same relative gaps. Fixed Local remains about 4% behind the drift-unaware Exact-individual controller and approximately 8% ahead of Exact-family. The tested drift therefore does not provide evidence for replacing Local with a shared controller on robustness grounds. All 300 fits converge and all 16,200 nonlinear closed-loop evaluations remain feasible.

Decision. Experiment 9E rejects two tempting repairs: improving family discovery alone cannot erase the exact-family gap, and changing only the reporting metric cannot erase the closed-loop gap. It also shows why deliberately shortening or degrading Local data would move the goalposts rather than solve the scientific problem.

The defensible next model is a shared family backbone plus a low-dimensional local, control-relevant residual. Federation should learn the common scheduled dynamics or controller basis, while each vehicle retains only the few local coordinates that materially change its gain schedule. The resulting comparison should be against full Local using three separate axes: closed-loop quality, local calibration burden, and deployed degrees of freedom. Identification and aggregation should select residual directions using gain sensitivity or closed-loop validation, not isotropic log-parameter distance. A useful next experiment must therefore ask how many local residual coordinates are required to close the 7.85% exact-family gap; it should not ask whether a single shared family model can outperform a correctly specialized Local controller.

Reproducibility. The protocol is frozen in `code/config/experiment_9e.json` and implemented in `code/experiments/experiment_9e_local_advantage_audit.py`. Seed-level results, diagnostic correlations, comparisons, provenance hashes, and the figure source data are retained under `results/tables`.

29 Experiment 9F: control-aware personalized LPV manifold

Question and representation. Experiment 9E shows that exact family averaging cannot remove the Local gap, because control-relevant within-family variation is genuine. Experiment 9F asks whether that variation nevertheless lies on a low-dimensional manifold. In log-parameter coordinates $z_i = \log \theta_i$, the personalized model is

$$z_i = \mu_g + U_g a_i, \quad \theta_i = \exp(z_i), \quad (53)$$

where the group center μ_g and directions U_g are shared, while $a_i \in \mathbb{R}^r$ is retained by client i . Rank zero is the shared family model, rank one gives a scalar local head, and rank three is the unreduced three-parameter model.

Two bases are learned from 180 training clients per fresh fleet. Parameter-PCA uses Euclidean variation in z . Control-PCA instead applies PCA after whitening by

$$W_g = \frac{1}{n_K} J_g^\top J_g + \eta I, \quad J_g = \left. \frac{\partial \text{vec } K(v, \exp z)}{\partial z} \right|_{z=\mu_g}, \quad (54)$$

where scheduled LQI gains are sampled at five speeds and each gain output is scaled by its root-mean-square sensitivity. This makes the first retained direction preserve variations that most change the controller rather than those that are merely largest in parameter space.

The bases are evaluated on 30 unseen clients for each of seeds 156–165 and on the three held-out trajectories used in Experiment 9E. True family labels, true training ratios, and oracle projections of the true test ratios are used. This deliberate oracle protocol tests representation capacity only; it does not yet claim that a client can estimate a_i from finite data or that U_g has been learned by a federated protocol. The unchanged 9D Local fit is retained as the practical comparator.

Table 42: Experiment 9F unseen-client results. “Gap closed” measures recovery from Exact-family toward Exact-individual.

Method	Local dim.	Tracking	vs. Local	Gap closed
Exact-family	0	0.007522	+7.54%	0.0%
Parameter rank 1	1	0.007468	+6.77%	6.6%
Control rank 1	1	0.006952	−0.62%	69.5%
Parameter rank 2	2	0.006704	−4.15%	99.6%
Control rank 2	2	0.006720	−3.93%	97.7%
Local fit	3	0.006995	—	64.2%
Exact-individual	3	0.006701	−4.20%	100.0%

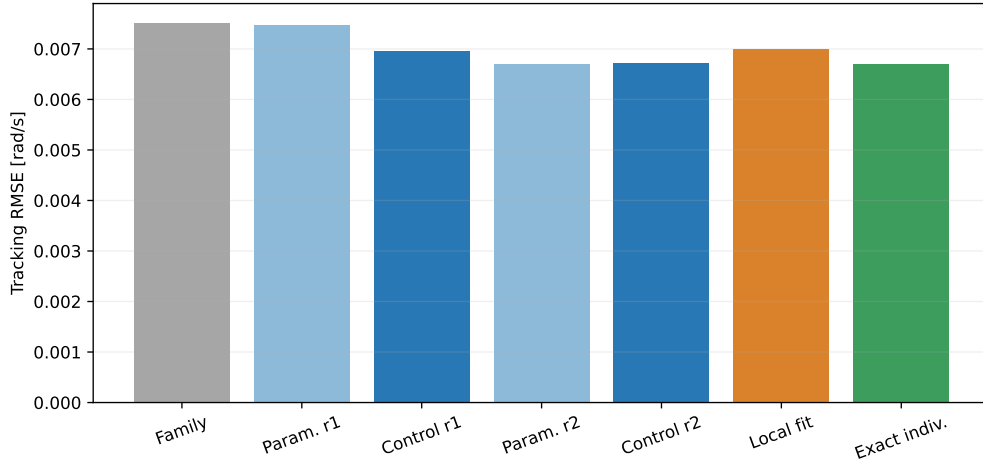


Figure 34: Tracking as local personalization rank increases. Control awareness is decisive at rank one; two coordinates are sufficient for either basis.

Results. The rank-one outcome confirms the central hypothesis. Parameter-PCA reduces parameter error but closes only 6.6% of the control-performance gap and loses to Local in all ten fleets. Control-PCA rank one closes 69.5%, improves the fresh-fleet mean over Local by 0.62%, and beats Local in six of ten fleets. It therefore provides mean parity rather than uniform

superiority, but does so with one client-specific scalar instead of a full three-parameter local model.

At rank two, both constructions recover essentially all useful individual behavior. Control rank two is only 0.28% above Exact-individual, closes 97.7% of the gap, and beats fitted Local in every fleet. Parameter rank two is 0.05% above Exact-individual and closes 99.6%. Its slight advantage over the control-aware version shows that the local quadratic gain metric is most useful for ordering the first direction, not universally optimal at every rank. Rank-three reconstruction error is 2.24×10^{-16} , and both bases exactly recover Exact-individual as required. All 300 Local fits converge and all 8100 nonlinear evaluations are feasible.

The representation changes the deployment accounting. Per family, rank one shares a three-parameter center and one three-element direction, while each vehicle stores one coordinate. Rank two shares nine scalars per family and retains two per vehicle. This is not a reduction in the number of personalized controllers actually executed: each client still obtains a distinct gain schedule. It is a reduction in client-specific model dimension and a structured way to transfer population information.

Decision and next gate. Both predeclared structural gates pass: Control rank one is no worse than Local on the fleet mean, and Control rank two lies within 1% of Exact-individual with fewer than three local coordinates. Personalized FL is therefore a reasonable direction, but Experiment 9F alone is not yet an FL result. The next experiment must learn μ_g and U_g from client messages and estimate a_i directly from the same finite local records. It must compare calibration time as well as tracking, because obtaining an oracle coefficient from the true test parameter vector would otherwise hide the original identification problem.

A defensible personalized protocol alternates local scalar or two-coordinate output-error updates with server aggregation of centers and basis directions. Unknown groups can use the already validated uncertainty-aware soft membership. The primary gate should require rank-one or rank-two personalization to retain its advantage under finite-data coefficient estimation; privacy can then be applied only to the shared updates, while a_i remains on the vehicle.

Reproducibility. The frozen protocol is in `code/config/experiment_9f.json`; its implementation is `code/experiments/experiment_9f_personalized_manifold.py`. Aggregate and seed-level results, learned-basis summaries, comparisons, and provenance hashes are stored under `results/tables`.

30 Experiment 9G: finite-data personalized federated identification

Question and protocol. Experiment 9F establishes that oracle client coordinates on a shared manifold can preserve individual control performance. Experiment 9G removes that oracle: can shared backbones be learned from client messages and can an unseen client identify only its local coordinate from finite records?

For each of ten fresh seeds (166–175), 180 training clients apply the frozen 9D adaptive acquisition policy. They transmit their three-dimensional structured estimate and normalized covariance, not raw trajectories. The server selects an unknown number of groups with the heteroscedastic mixture from Experiment 9D and learns a control-aware center and rank-one or rank-two basis from the estimates assigned to each group. This is a one-shot federated-summary protocol rather than iterative gradient FL; privacy is not applied in 9G.

Each of 30 unseen clients begins with the same three 0.25 s records and one one-second complementary record, for 1.75 s total. For every candidate group it fits only $a_i \in \mathbb{R}^r$ on the three initial records. It selects the group with the lowest error on the held-out complementary record, then refits only that group’s r coefficients on all records. No true test parameter is

used. Learned-rank methods use the unknown- K groups; Oracle-group variants replace only the training grouping by true simulator families. Local fits all three ratios from the same records, and Exact-individual remains an unattainable reference.

The rank-one gate requires Learned-rank-1 to lie within 2% of Local. The rank-two gate is stricter: Learned-rank-2 must be no worse than Local. These criteria were fixed before inspecting the fresh seeds.

Table 43: Experiment 9G finite-data tracking on ten fresh fleets.

Method	Groups	Local dim.	Tracking	vs. Local
Exact-individual	—	3	0.006679	−4.25%
Local	—	3	0.006975	—
Learned rank 1	3.0	1	0.007474	+7.16%
Learned rank 2	3.0	2	0.007018	+0.61%
Oracle-group rank 1	3	1	0.007512	+7.70%
Oracle-group rank 2	3	2	0.007018	+0.62%

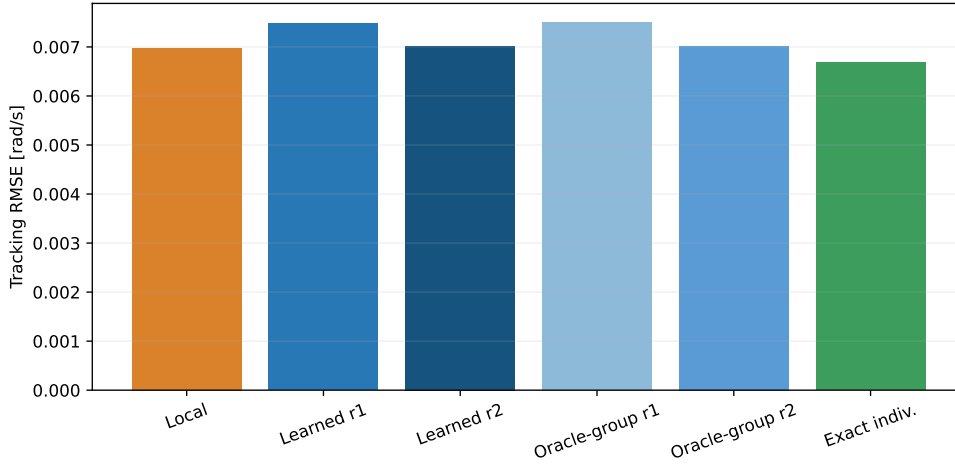


Figure 35: Finite-data personalized models compared with full Local and the exact individual reference.

Results. The server selects three groups in every fleet and reaches mean training ARI 0.951. Learned-rank-2 is only 0.61% worse than Local on average and wins four of ten paired fleets; its fleet-wise relative difference ranges from −1.25% to +2.57%. Its scheduled-gain error is 0.00577, essentially the same as Local’s 0.00578, even though its unweighted parameter error is larger (0.0745 versus 0.0699). This is practically near-parity with one fewer locally estimated coordinate, but it fails the predeclared requirement of being no worse than Local.

Rank one does not preserve its oracle promise. Learned-rank-1 is 7.16% worse than Local, exceeds the 2% tolerance, and wins only one fleet. The corresponding Oracle-group result is 7.70% worse. For rank two, Learned and Oracle-group tracking differ by less than 0.01 percentage points relative to Local. Hence the remaining loss is not caused by unknown group discovery. It arises because finite short-data estimation of the manifold coordinates and backbones does not realize the oracle projections used in 9F.

All 2100 adaptive client fits and all personalized coefficient fits converge. All 5400 nonlinear closed-loop evaluations remain feasible. Both formal gates fail, so 9G supports a compression–performance tradeoff, not a claim that personalized federation improves absolute tracking.

Interpretation and decision. The result materially changes the FL conclusion. A personalized formulation is reasonable: rank two reproduces Local gain quality and nearly reproduces its tracking while transferring a shared group backbone. However, with only three original parameters, reducing the local head from three coordinates to two is a modest compression. It is not by itself a sufficiently strong headline unless the two-coordinate estimator can also demonstrate less calibration, improved uncertainty, privacy benefits, or better adaptation for new clients.

Further algorithmic work should target the actual bottleneck exposed here rather than clustering. Candidate improvements are joint uncertainty-aware estimation of backbone and coefficients, direct control-loss validation of a_i , and adaptive acquisition targeted along the missing manifold direction. A particularly clean next test would compare how much data Local and rank two require to reach a fixed tracking or gain-error tolerance. If rank two reaches Local-quality control with materially less excitation, personalized FL gains an operational purpose. If it needs the same data, the current three-parameter benchmark is probably too small to sustain a strong personalized-FL contribution on compression alone.

Reproducibility. The frozen configuration is `code/config/experiment_9g.json`; the implementation is `code/experiments/experiment_9g_finite_personalized.py`. Aggregate and seed-level results, group diagnostics, assignments, comparisons, and provenance hashes are stored under `results/tables`.

31 Experiment 9H: calibration-efficiency frontier

Question and frozen protocol. Experiment 9G shows near-Local tracking with a two-coordinate personalized head, but does not establish a calibration advantage. Experiment 9H therefore asks whether previously learned fleet structure helps a newly arriving vehicle when its local excitation is limited. The experiment evaluates the complete calibration curve rather than weakening Local at one selected operating point.

Ten fresh fleets (seeds 176–185) contain a mature cohort of 180 training clients and 30 unseen test clients. Training clients provide 4.75 s of local data and transmit fitted parameter and covariance summaries. The server learns the unknown groups and control-aware rank-one and rank-two backbones once; these quantities are frozen before test clients appear. The server selects exactly three groups in every fleet with training ARI 1.000.

Every test method receives identical cumulative prefixes of 0.25, 0.50, 0.75, 1.25, 1.75, 2.75, and 4.75 s. The first 0.75 s consists of three independent 0.25 s records, followed by truncated prefixes of two complementary-speed records. Local fits all three ratios at every budget. Learned-backbone selects a shared group but estimates no continuous local coordinate. Learned-rank-1 and Learned-rank-2 jointly select a group by normalized output error and fit one or two client coefficients. Oracle-group-rank-2 uses the true family only to diagnose group selection.

The predeclared primary gate asks whether Learned-rank-2 at 1.25 s lies within 2% of Local at 1.75 s. In addition, the full matched-budget curves are reported to avoid making the conclusion depend on that single tolerance.

Results: sharing solves cold start, not time to full quality. The primary gate fails. The 1.75 s Local reference is 0.007103 and its 2% tolerance is 0.007245, whereas Learned-rank-2 reaches only 0.007620 at 1.25 s. Learned-rank-1, Learned-rank-2, and Local all first cross the target at the measured 1.75 s budget. The experiment therefore does not show that personalization reaches mature Local quality sooner.

At matched short budgets, however, sharing gives a large and consistent advantage. At 0.50 s the shared backbone and rank-one model improve Local by 39.95% and 31.79%, respectively. At 0.75 s, rank one improves Local by 20.92%; at 1.25 s, rank two improves it by 0.94%. At 1.75 s

Table 44: Experiment 9H mean tracking across the calibration frontier.

Data [s]	Shared backbone	Rank 1	Rank 2	Local
0.25	0.011628	0.017693	0.049853	0.494610
0.50	0.011621	0.013199	0.019004	0.019351
0.75	0.010699	0.010591	0.013263	0.013392
1.25	0.007901	0.007652	0.007620	0.007693
1.75	0.007758	0.007100	0.007099	0.007103
2.75	0.007474	0.006730	0.006701	0.006702
4.75	0.007444	0.006688	0.006681	0.006684

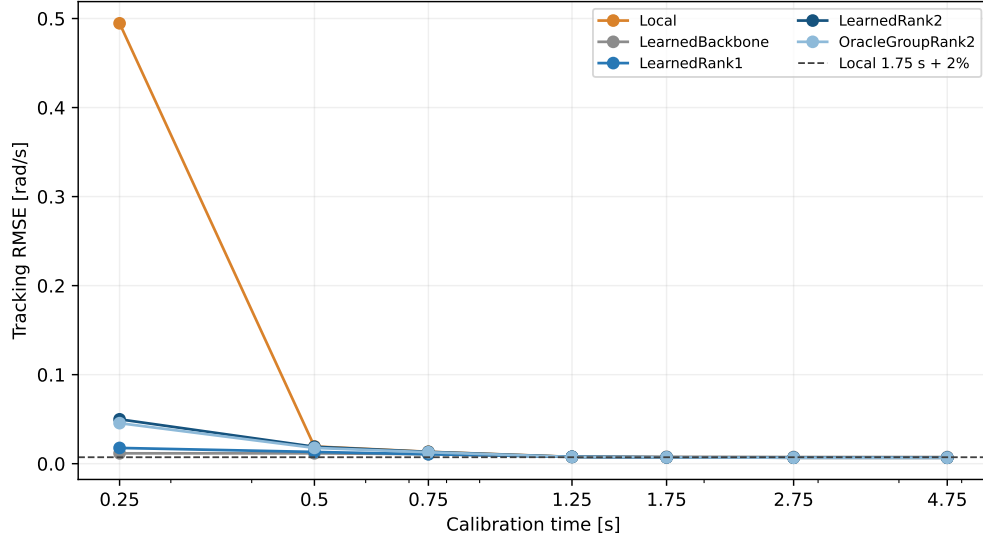


Figure 36: Matched-prefix calibration frontier. Personalization protects the information-limited client, while Local catches up once enough excitation is available.

both personalized models are within 0.05% of Local. Thus the benefit decreases continuously as the local record becomes informative, exactly as expected for transferable fleet knowledge.

The useful personalization rank is itself data dependent. With 0.25–0.50 s, estimating two coordinates amplifies uncertainty and the zero-coordinate shared backbone is best among deployable sharing methods. Rank one becomes best near 0.75 s, and rank two becomes competitive at 1.25 s. By 2.75 s, rank two and Local both essentially reach Exact-individual tracking. This reveals a more useful design than a fixed personalized head: uncertainty should activate local degrees of freedom only when the available excitation supports them.

The very large 97.65% relative improvement of the shared backbone over Local at 0.25 s should not be used as a headline percentage: the unconstrained three- parameter Local fit is nearly unidentified and has tracking 0.4946. The 0.50– 0.75 s results are more representative evidence of cold-start protection. All 3900 parameter fits and all coefficient fits converge, and all 37,800 nonlinear closed-loop evaluations remain feasible.

Decision. Experiment 9H supplies the missing reason to share information, but narrows the claim. Personalized fleet learning does not outperform a sufficiently calibrated Local model and does not reduce the observed time to the mature 1.75 s quality threshold. It instead prevents severe identification and control degradation during new-client cold start, then converges smoothly to Local performance.

The next method should therefore use uncertainty-gated personalization order: start from a shared group backbone, activate the first client coordinate when supported, and activate the second only after complementary excitation. This directly combines the uncertainty-triggered acquisition of 9D with the control-aware manifold of 9F. An end-to-end federated experiment should next verify that distributed sufficient-statistic aggregation reproduces the shared backbones under partial participation and should report the communication cost. Privacy should be added only after that non-private protocol is frozen.

Reproducibility. The configuration is frozen in `code/config/experiment_9h.json`; the implementation is `code/experiments/experiment_9h_calibration_frontier.py`. Aggregate and seed-level curves, equal-budget improvements, log-time areas, efficiency thresholds, assignments, and provenance are retained under `results/tables`.

32 Experiment 9I: fleet-informed active identification

Question and protocol. Experiment 9H shows that shared knowledge protects an information-limited new client. Experiment 9I asks whether the fleet can also tell that client which data to acquire next. Ten fresh fleets (seeds 186–195) use a mature 180-client cohort to learn three latent groups and rank-two personalized backbones. Thirty unseen clients each receive three initial 0.25 s records and exactly one 0.5 s follow-up maneuver.

The frozen candidate library contains low-, medium-, and high-speed maneuvers at 0.22 and 0.50 Hz. Selection uses predicted Fisher information based on the current estimate and nominal closed-loop input templates; realized candidate outputs and evaluation trajectories are unavailable to every deployable policy. Parameter-active policies minimize normalized parameter or coefficient covariance. Control-active policies minimize covariance weighted by scheduled-gain sensitivity. Local policies fit all three parameters; fleet policies use the learned rank-two backbone. Fixed policies use the medium-speed, low-frequency maneuver. Oracle-best-fleet tries all realized candidate records and minimizes true scheduled-gain error; it is an unattainable diagnostic, not an oracle for tracking itself.

The predeclared aggregate gate requires Fleet-control-active to improve both Fleet-fixed and Local-control-active. This gate is accompanied by a mechanistic check: a meaningful

personalized active policy should not collapse to one universal maneuver or become identical to its parameter-only ablation.

Table 45: Experiment 9I equal-budget active-identification results.

Method	Tracking	Parameter error	Gain error
Local fixed	0.006846	0.0518	0.00445
Local parameter-active	0.006843	0.0508	0.00439
Local control-active	0.006736	0.0577	0.00376
Fleet fixed	0.006857	0.0477	0.00443
Fleet parameter-active	0.006724	0.0466	0.00351
Fleet control-active	0.006724	0.0466	0.00351
Oracle gain selector	0.006735	0.0397	0.00184

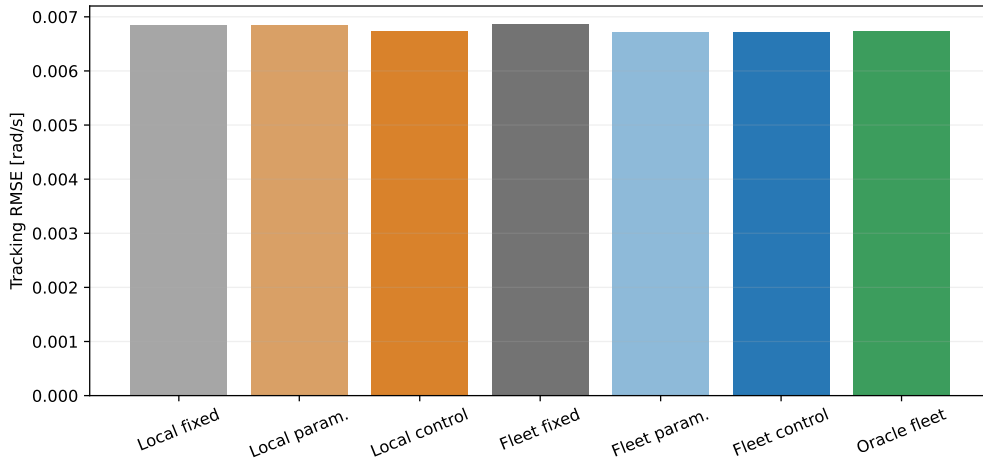


Figure 37: Tracking after one equal-duration follow-up maneuver. Active excitation helps, but the tested fleet criteria collapse to the same universal choice.

Aggregate result. Fleet-control-active improves Fleet-fixed by 1.94% and Local-fixed by 1.78%, so it passes the literal aggregate gate. Local-control-active already improves Local-fixed by 1.60%. Consequently, the incremental fleet advantage over an active Local client is only 0.18% on average. It wins that paired comparison in five of ten fleets; the fleet-wise difference ranges from 0.93% better to 1.02% worse. This is parity, not a reliable federated advantage.

The mechanism check is negative. Fleet-parameter-active and Fleet-control-active select the high-speed, high-frequency maneuver for all 300 clients and produce identical results. Local-control-active also makes that universal selection. Thus the candidate library contains one maneuver that dominates both predicted information criteria. The improvement over the deliberately fixed comparator demonstrates the value of more informative excitation, but not individualized fleet-informed experiment selection or control-aware weighting.

The oracle gain-error selector chooses all six candidates: high-fast for 77 clients, high-slow for 69, medium-fast for 47, medium-slow for 56, low-fast for 23, and low-slow for 28. This confirms that client-dependent best maneuvers exist under the oracle gain metric. However, the local linearized criteria do not predict that diversity. The oracle selector also need not minimize nonlinear tracking, which explains why its aggregate tracking is slightly worse than the universal high-fast fleet policy despite much lower gain error.

The server selects three groups with mean training ARI 0.998. All 3900 fits converge, and all

6300 nonlinear evaluations remain feasible. Clustering, optimization failure, and unsafe candidate maneuvers therefore do not explain the collapse.

Decision. Experiment 9I does not justify pivoting the paper toward active learning in its current form. The literal gate is too weak: it rewards discovering that one globally informative maneuver is better than the fixed comparator. A genuine federation-specific active result would need heterogeneous admissible maneuver sets, operating conditions, or excitation costs, together with a posterior criterion capable of predicting client-dependent value. Adding those elements now would substantially broaden the IEEE IV paper and risk appearing designed to manufacture differentiation.

The recommended paper core therefore remains the empirically supported result from 9H: shared group backbones and progressive personalization protect cold-start clients, while sufficiently informed Local models catch up. Active experiment selection may be retained as a transparent negative boundary or developed later as a separate contribution. The immediate next requirement for the current paper is an explicit distributed implementation of shared-backbone learning, including partial participation and communication accounting, followed by a frozen confirmation. Privacy can then be integrated into the shared updates while keeping personalization coefficients local.

Reproducibility. The protocol is frozen in `code/config/experiment_9i.json`; the implementation is `code/experiments/experiment_9i_active_identification.py`. Aggregate and seed-level results, selected-maneuver counts, diagnostics, and provenance hashes are stored under `results/tables`.

33 Experiment 9J: distributed personalized backbone learning

Question and protocol. Experiment 9J closes the implementation gap left by the centralized summary experiments: can the unknown-group personalized LPV backbone be learned from federated aggregate messages, and does it remain useful with partial client participation? Ten development fleets (seeds 196–205) each contain 180 mature training clients and 30 unseen test clients. Training clients retain their structured parameter estimates and uncertainty locally. In each communication round they compute mixture responsibilities and transmit additive precision, information-vector, and scatter summaries. The server learns the number of groups from candidates one through six and constructs group centers and control-aware principal directions from the aggregated moments. Personalized coordinates and short calibration records never leave a test client.

Fed100, Fed50, and Fed20 sample 100%, 50%, or 20% of clients per round for 30 rounds. They are compared with the established centralized-summary method, full three-parameter Local identification, and an exact-individual diagnostic. The progressive policy frozen after Experiment 9H uses rank one after 0.75 s and rank two after 1.25 s. Evaluation uses three unseen nonlinear closed-loop scenarios. Gate A requires a deterministic full-participation rerun to reproduce the same groups and moments to numerical precision. Gate B limits degradation relative to Central to 1% for Fed50 and 2% for Fed20; improvements do not count as failures. Gate C requires Fed20 to beat Local at both budgets.

Result. All three gates pass. Fed100 agrees with Central to below $6 \times 10^{-8}\%$ in tracking, and the independent deterministic aggregation rerun reproduces the selected group count, means, and covariances to below 10^{-12} . Fed50 remains within 0.53% of Central at both budgets. Fed20 is 2.34% better at 0.75 s and only 0.04% worse at 1.25 s. The apparent Fed20 improvement is not claimed as a benefit of dropping clients; it is consistent with sampling-induced regularization and finite-fleet variation.

Table 46: Experiment 9J nonlinear tracking RMSE. Percentages compare with Central.

Method	0.75 s		1.25 s	
	RMSE	vs. Central	RMSE	vs. Central
Local	0.013330	+23.76%	0.007620	+0.52%
Central	0.010771	0.00%	0.007580	0.00%
Fed100	0.010771	0.00%	0.007580	0.00%
Fed50	0.010828	+0.53%	0.007581	+0.01%
Fed20	0.010518	-2.34%	0.007584	+0.04%

Table 47: Experiment 9J federated protocol audit (mean per fleet). Payload counts exclude transport headers and cryptographic overhead.

Method	Groups	ARI	Coverage	Upload	Download
Fed100	3.0	1.000	1.000	1.160 MB	0.670 MB
Fed50	3.1	0.989	1.000	0.610 MB	0.346 MB
Fed20	3.1	0.892	0.999	0.258 MB	0.138 MB

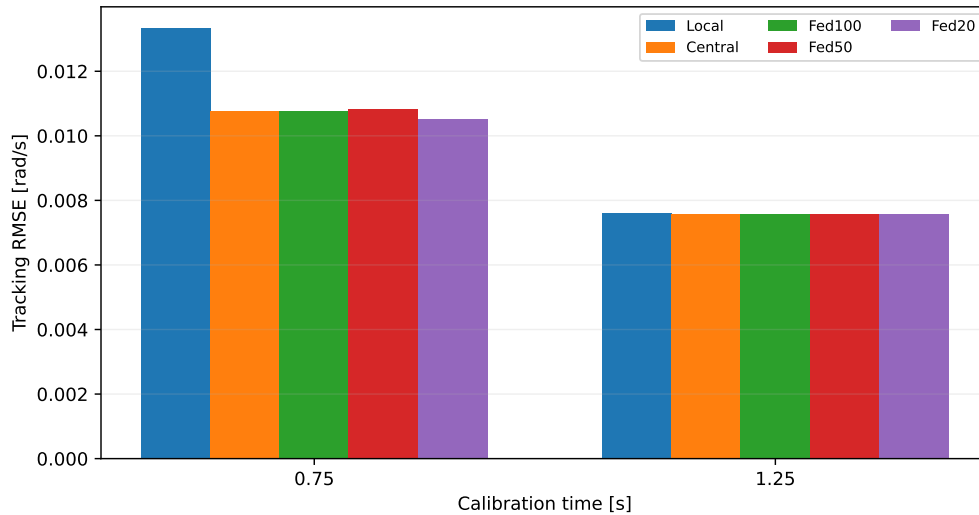


Figure 38: Distributed shared-backbone learning preserves the cold-start benefit under partial participation. At 1.25 s the Local estimator has largely caught up.

Relative to Local, Fed20 improves aggregate tracking by 21.09% at 0.75 s and 0.47% at 1.25 s. It wins all ten fleet-wise comparisons at 0.75 s and seven of ten at 1.25 s. Thus the main operational result is concentrated in cold start: sharing is valuable when the new client is information limited, while Local identification catches up as its own excitation becomes adequate. All 2400 fits converge and all 10800 nonlinear evaluations remain feasible.

FL interpretation and limitations. This experiment supplies an explicit federated mechanism rather than merely calling centralized pooling federated. The server receives only additive sufficient statistics; raw trajectories, test-client coefficients, and local calibration records remain local. The aggregation interface is compatible with secure aggregation, but this simulation does not implement cryptography or claim resistance to inference from released aggregate models. Communication figures count 32-bit numerical payloads only. Partial participation is simulated by random cohorts; stragglers, asynchronous updates, packet loss, and nonrandom availability remain outside the present scope.

The ten development fleets support the paper mechanism but are not the final confirmatory evidence. Seeds 206–215 remain untouched for a frozen confirmation of the complete 9J protocol. Privacy perturbation should subsequently be applied to these aggregate updates while keeping personalized coordinates local.

Reproducibility. The frozen development protocol is in `code/config/experiment_9j.json`. The distributed aggregation implementation is `code/src/federated_lpv/federated_manifold.py`, and the experiment driver is `code/experiments/experiment_9j_distributed_backbone.py`. Aggregate, seed-level, communication, diagnostic, and provenance records are stored under `results/tables`.

34 Experiment 9K: blind confirmation of distributed personalization

Frozen protocol. Experiment 9K evaluates the complete Experiment 9J protocol on the ten seeds 206–215 that were reserved before 9J was executed. A machine-checked freeze requires every inherited data volume, algorithmic setting, model-order candidate, communication round, participation rate, progressive rank, and nonlinear scenario to equal 9J. No development result or confirmation result is used for tuning.

Five primary gates were declared before execution: Fed100 must degrade by less than 0.1% relative to the established Central implementation; Fed50 and Fed20 must degrade by less than 1% and 2%, respectively; Fed20 must beat Local at the 0.75 s cold-start budget; and every nonlinear closed-loop evaluation must remain feasible. The 1.25 s Fed20–Local comparison was deliberately excluded from the positive gate because Experiment 9J predicted parity after Local identification acquired sufficient data.

Table 48: Experiment 9K blind-confirmation tracking RMSE. Percentages compare with the Central implementation.

Method	0.75 s		1.25 s	
	RMSE	vs. Central	RMSE	vs. Central
Local	0.012714	+18.47%	0.007557	+0.36%
Central	0.010731	0.00%	0.007530	0.00%
Fed100	0.010762	+0.28%	0.007531	+0.02%
Fed50	0.010648	-0.77%	0.007548	+0.25%
Fed20	0.010518	-1.99%	0.007562	+0.42%

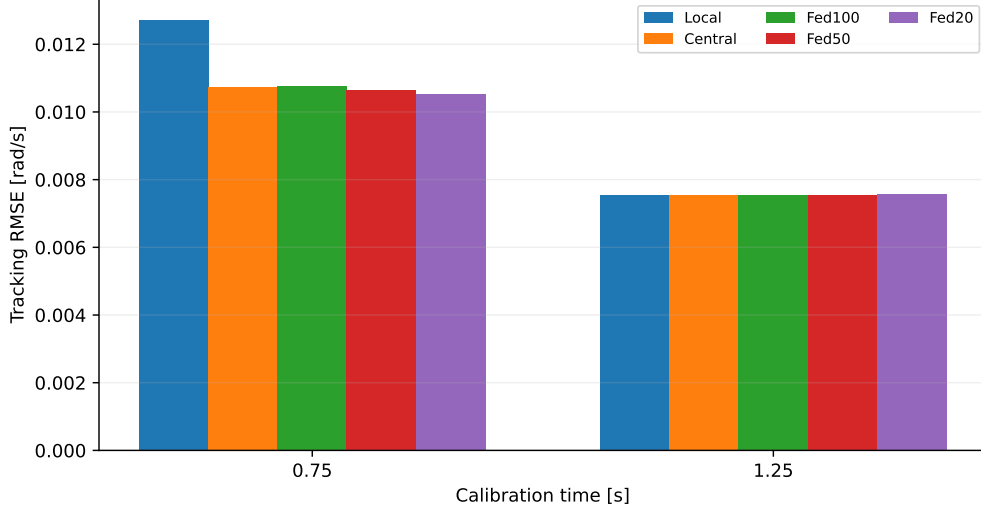


Figure 39: Blind confirmation preserves the large cold-start benefit and the later convergence toward Local performance.

Gate outcome. Four of the five primary gates pass. Fed50 and Fed20 stay below their permitted degradation, Fed20 beats Local at 0.75 s, and all 10800 nonlinear evaluations remain feasible. The strict Fed100 equivalence gate fails: its aggregate tracking is 0.281% worse than Central at 0.75 s, exceeding the predeclared 0.1% tolerance. At 1.25 s the difference is 0.020%. This threshold is not changed after observing the data.

The failed gate separates two notions that were nearly indistinguishable in 9J. A deterministic rerun of the federated aggregation reproduces its selected group count, means, and covariances below 10^{-12} and therefore passes the internal implementation check. Central and Fed100, however, use different mixture-fitting implementations and need not produce exactly the same finite-sample optimum. The confirmation shows that they are control-equivalent to well below 1%, but it does not support a 0.1% numerical-equivalence claim.

Confirmed scientific result. At 0.75 s, Fed20 improves aggregate tracking over Local by 17.27%; the mean paired fleet improvement is 16.90%, and Fed20 wins nine of ten fleet-wise comparisons. Fed50 improves the aggregate result by 16.24% and also wins nine of ten fleets. The effect is therefore smaller than the 21.09% development estimate but remains large and consistent on untouched fleets.

At 1.25 s, Fed20 is 0.06% worse than Local and wins five of ten fleets; Fed50 is 0.11% better and wins seven. This confirms the predicted transition to parity rather than continued federated superiority. The evidence supports a specific deployment claim: a mature fleet representation protects a new client during cold start, while sufficiently informative local identification catches up.

Fed100 selects three groups on every fleet with mean ARI 0.997. Fed50 and Fed20 obtain mean ARI 0.937 and 0.917; occasional under- or over-partitioning does not destroy the control result. All 2400 fits converge. Together with the 9J development results, the confirmation rules out a favorable-seed explanation for the main cold-start finding.

Decision. Experiment 9K confirms the paper’s main personalized-FL claim but narrows the implementation statement. The manuscript should claim sub-percent agreement between full-participation federated and Central learning, not exact equivalence. The failed 0.1% gate should be reported transparently. The next robustness question is nonrandom availability: random

partial participation retains almost complete client coverage and approximately balanced groups, whereas a deployed fleet may systematically omit a family or operating region.

Reproducibility. The frozen configuration and gates are in `code/config/experiment_9k.json`. The confirmation driver is `code/experiments/experiment_9k_blind_confirmation.py`; it refuses to run if inherited 9J settings or the reserved seed set differ. Aggregate, seed-level, communication, gate, and provenance records are stored under `results/tables`.

35 Experiment 9L: nonrandom participation robustness

Question and protocol. Experiments 9J–9K sample clients uniformly and reach almost every training client across 30 rounds. Experiment 9L asks whether the personalized federated LPV mechanism survives structured availability. Ten new development fleets (seeds 216–225) retain the 9J training population, progressive 0.75/1.25 s personalization policy, and three nonlinear evaluation scenarios. All methods nominally invite 20% of clients per round.

Uniform20 is the reference. Persistent20 reuses one fixed 20% subset in every round. FamilySkew20 makes the third generating family twenty times less likely to communicate. OperatingSkew20 decreases availability with the local calibration-speed category to a minimum relative weight of 0.05. Dropout50 uses the exact Uniform20 invitation trace but independently loses half of the selected messages. Generating-family labels and calibration categories create the stress traces only; neither is supplied to the estimator. Model-order selection and final backbone moments use only clients that actually communicated.

The aggregate development gate permits at most 5% cold-start degradation from Uniform20 and requires each stress regime to beat Local at 0.75 s. Fleet-wise win rate and worst-case degradation are retained as secondary reliability diagnostics.

Table 49: Experiment 9L tracking RMSE and cold-start comparisons.

Method	0.75 s	vs. Uniform	vs. Local	1.25 s	Cold wins
Local	0.013641	+17.82%	0.00%	0.007622	–
Uniform20	0.011577	0.00%	-15.13%	0.007586	10/10
Persistent20	0.011890	+2.70%	-12.83%	0.007564	9/10
FamilySkew20	0.011710	+1.14%	-14.16%	0.007599	10/10
OperatingSkew20	0.011299	-2.40%	-17.17%	0.007611	10/10
Dropout50	0.011418	-1.38%	-16.30%	0.007645	10/10

Table 50: Experiment 9L availability and learned-group audit. Coverage denotes distinct clients observed at least once.

Method	Coverage	Min. family	Min. speed	Groups	ARI
Uniform20	0.999	0.997	0.989	3.3	0.834
Persistent20	0.200	0.155	0.056	2.1	0.600
FamilySkew20	0.807	0.420	0.644	3.7	0.854
OperatingSkew20	0.933	0.907	0.478	3.5	0.886
Dropout50	0.972	0.953	0.894	2.3	0.588

Aggregate result. Every predeclared aggregate gate passes. Persistent and family-skewed participation degrade cold-start tracking from Uniform20 by only 2.70% and 1.14%; operating skew and matched dropout happen to improve the aggregate mean. These improvements are not

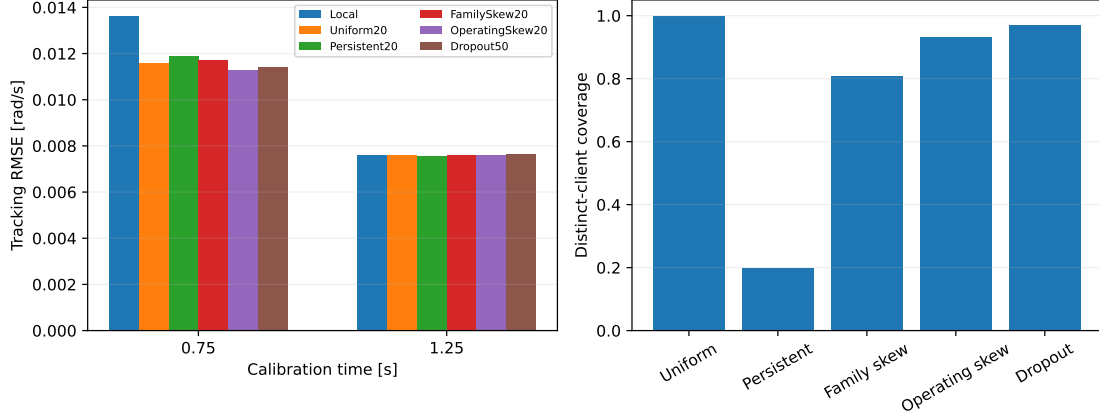


Figure 40: Aggregate tracking remains robust under all availability stresses, but the persistent subset exposes a fleet-wise failure hidden by the mean.

interpreted as benefits of biased or missing updates: stochastic cohorts alter the selected group count and finite-sample regularization. Their defensible interpretation is absence of aggregate harm. All stressed methods retain a 12.83–17.17% cold-start improvement over Local. At 1.25 s every method is again within 0.8% of Uniform20 and Local.

Dropout50 uses only 745 messages per fleet on average, versus 1296 for Uniform20, yet covers 97.2% of clients and preserves cold-start performance. Family and operating skew also retain broad distinct-client coverage. Exact recovery of the generating families is not required for useful control: Dropout50 has mean ARI 0.588 while still improving Local in all ten fleets.

Persistent-subset failure. The aggregate result does not imply worst-case reliability. Persistent20 sees only 20% of clients, with mean minimum family and speed-category coverage of 15.5% and 5.6%. On seed 221 it selects a single group and becomes 34.43% worse than Local at 0.75 s. It nevertheless wins nine of ten fleets, so its aggregate mean passes. This is a concrete failure mode: repeated participation cannot compensate for missing population support.

The other stresses win every cold-start fleet comparison; their worst fleet still improves Local by 5.47% (family skew), 8.47% (operating skew), and 9.00% (dropout). Thus temporary message loss and strong but rotating bias are substantially safer than a permanently narrow cohort.

Decision. Experiment 9L strengthens the partial-participation claim while defining its boundary. The method tolerates reduced messages, dropout, and rotating availability bias when cumulative population coverage remains broad. It is not reliable when the same small subset participates indefinitely. A deployable server should therefore monitor group/operating-region coverage and solicit underrepresented clients or fall back to Local identification when support is insufficient. This coverage-aware safeguard is a more direct next improvement than increasing the number of communication rounds.

All 2400 fits converge and all 12600 nonlinear closed-loop evaluations remain feasible. Seeds 226–235 remain reserved; 9L is a development robustness study, not an additional blind confirmation.

Reproducibility. The protocol is in `code/config/experiment_9l.json`; the driver is `code/experiments/experiment_9l_nonrandom_participation.py`. The reusable cohort-controlled aggregation path is implemented in `code/src/federated_lpv/federated_manifold.py`. Aggregate, seed-level, coverage, gate, and provenance records are stored under `results/tables`.

36 Experiment 9M: unified confirmation-fleet ablation

Purpose and protocol. Experiment 9M reconstructs the principal baselines under one matched protocol on the ten Experiment 9K fleets. It is a retrospective ablation of already executed confirmation fleets, not a second blind confirmation. A freeze guard requires the 9K clients, data volumes, identification settings, nonlinear scenarios, mixture candidates, and federated settings. No model or controller hyperparameter is retuned.

GlobalK1 shares one fleet-wide parameter vector and ignores heterogeneity. LearnedCenter discovers an unknown number of groups but performs no continuous personalization. CentralRank1/2 add one or two local coordinates to centralized learned backbones. Fed20Rank1/2 learn the corresponding backbones from 20% random participation. OracleFamilyRank2 supplies the true generating family as an unattainable diagnostic. Local fits all three parameters independently, and ExactIndividual uses the true parameters.

Statistical comparisons first average all clients and scenarios within each fleet. Ten independent fleet differences are then resampled 50000 times. This fleet-level bootstrap avoids treating correlated trajectories or clients as independent replicates. Reported intervals are percentile 95% confidence intervals for mean relative improvement.

Table 51: Experiment 9M matched nonlinear tracking ablation.

Method	Local dim.	0.75 s		1.25 s	
		Tracking	Param. err.	Tracking	Param. err.
Local	3	0.012714	0.2063	0.007557	0.1197
GlobalK1	0	0.011544	0.1285	0.011544	0.1285
LearnedCenter	0	0.010826	0.0977	0.007798	0.0549
CentralRank1	1	0.010731	0.1007	0.007433	0.0499
CentralRank2	2	0.012709	0.1464	0.007530	0.0635
Fed20Rank1	1	0.010518	0.0976	0.007462	0.0520
Fed20Rank2	2	0.012774	0.1459	0.007562	0.0632
OracleFamilyRank2	2	0.012205	0.1322	0.007496	0.0533
ExactIndividual	—	0.006664	0.0000	0.006664	0.0000

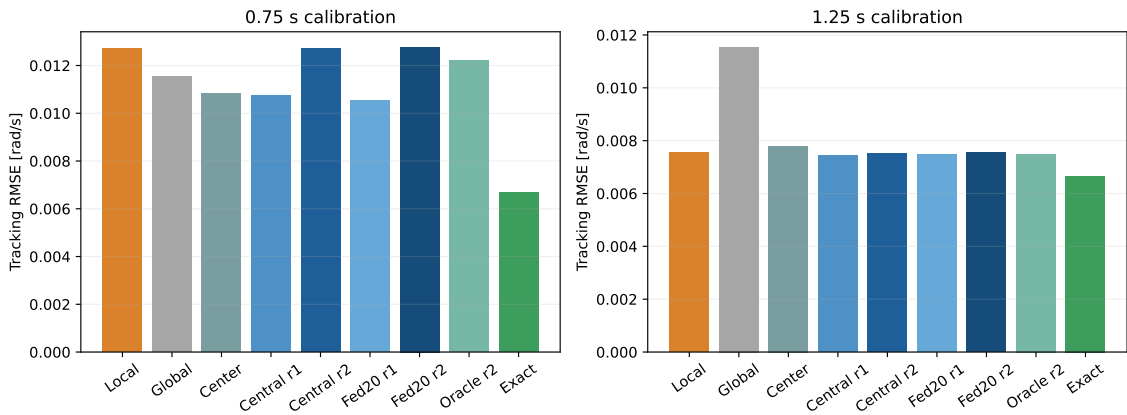


Figure 41: The matched ablation separates global sharing, discrete group structure, continuous personalization, and distributed implementation.

Cold-start decomposition. At 0.75 s, even one global fleet model improves Local by 8.43%, although it wins only six fleets. Learned group centers improve Local by 14.47%, win all ten

Table 52: Fleet-level paired bootstrap analysis. Positive values favor the first method.

Budget	Comparison	Mean	95% CI	Wins
0.75 s	Fed20Rank1 vs. Local	16.90%	[11.79, 21.10]%	9/10
0.75 s	LearnedCenter vs. Local	14.47%	[9.73, 19.48]%	10/10
0.75 s	GlobalK1 vs. Local	8.43%	[2.50, 14.57]%	6/10
0.75 s	LearnedCenter vs. GlobalK1	5.95%	[-0.44, 11.66]%	7/10
0.75 s	Fed20Rank1 vs. LearnedCenter	2.20%	[-4.56, 8.79]%	6/10
1.25 s	Fed20Rank1 vs. LearnedCenter	4.21%	[2.40, 6.36]%	10/10
1.25 s	Fed20Rank2 vs. Local	-0.06%	[-0.71, 0.58]%	5/10
1.25 s	Fed20Rank1 vs. Fed20Rank2	1.17%	[-1.27, 3.54]%	6/10

fleets, and have a confidence interval bounded away from zero. Thus the dominant cold-start mechanism is a heterogeneous fleet prior, not estimation of continuous client coordinates.

Fed20Rank1 improves Local by 16.90%, confirming the 9K headline with a fleet-bootstrap interval of 11.79–21.10%. Its incremental 2.20% improvement over LearnedCenter is uncertain: the interval spans zero and it wins six fleets. The paper must therefore not attribute the entire cold-start gain to the personalized head. Federation itself incurs no observed penalty here: Fed20Rank1 is 1.95% better than CentralRank1, with a 0.70–3.18% interval; this favorable difference is treated as finite-sample cohort variation rather than a general advantage of partial participation.

Personalization with increasing information. At 1.25 s, the group center alone is no longer competitive with Local, whereas Fed20Rank1 improves the center by 4.21%, wins all ten fleets, and has a 2.40–6.36% interval. The role of the local head therefore emerges as the client obtains enough information to move reliably within the fleet-supported subspace. Fed20Rank1 is 1.12% better than Local on average, but its interval includes zero; this is parity rather than confirmed superiority.

Rank two is unnecessary in this matched analysis. At 0.75 s it overfits and is approximately as poor as Local. At 1.25 s it is non-inferior to Local within the chosen 1% margin, but rank one has lower mean tracking and the rank-order interval contains zero. Experiment 9H showed only a small development advantage for rank two at 1.25 s; that ordering does not replicate. The robust conclusion is that a very low-dimensional head suffices, not that rank two is required.

Scientific interpretation. The unified evidence supports a staged personalized-FL mechanism. Discrete group structure supplies a useful prior immediately; one local coordinate then adapts that prior when more client excitation becomes available. This is more precise than claiming generic personalization superiority. Central learns exactly three groups with ARI 1.000; Fed20 selects 3.1 groups on average with ARI 0.917 and 99.94% cumulative coverage, yet preserves the control result.

Both retrospective statistical gates pass: the Fed20Rank1 cold-start interval excludes zero, and Fed20Rank2 at 1.25 s is non-inferior to Local under the 1% margin. Because the confirmation outcomes were already known before this ablation was specified, these intervals quantify consistency and uncertainty; they must not be presented as newly preregistered confirmatory tests. All 2400 fits converge and all 16200 nonlinear evaluations remain feasible.

Reproducibility. The matched configuration is in `code/config/experiment_9m.json`; the driver and fleet-level bootstrap are in `code/experiments/experiment_9m_unified_ablation.py`. Aggregate, seed-level, cluster, statistical, and provenance records are stored under `results/tables`.

37 Experiment 10A: nonlinear basis-order and coverage audit

Why this experiment is necessary. The earlier cold-start study identifies only three positive physical ratios. That model extrapolates unusually well from short local recordings because its speed dependence is imposed analytically. Restricting the amount of data alone therefore does not create a defensible need for cross-client information. The new experiment sequence asks a narrower question: can complementary client coverage help identify a richer LPV map that is genuinely required by nonlinear lateral dynamics?

Experiment 10A is a representation and identifiability gate before any new FL comparison. It must establish three facts. First, the frozen physics basis must remain sufficient where the plant is effectively linear. Second, nonlinear operating points must justify additional speed dependence under validation on entirely withheld speed regions. Third, a client restricted to one part of the speed envelope may be locally underdetermined while the union of complementary clients is identifiable. Failure of either of the last two conditions stops the planned Experiment 10B.

Operating-point construction. The data-generating plant is the existing nonlinear tanh-tire bicycle model; 10A does not introduce an artificial high-order plant. For speed v and road curvature κ , the steady-cornering point satisfies

$$r^* = v\kappa, \quad f_{\text{nl}}\left([\beta^*, r^*]^\top, \delta^*, v\right) = 0. \quad (55)$$

The continuous Jacobians with respect to state and steering are evaluated by centered finite differences and discretized exactly under zero-order hold at $T_s = 0.01$ s. This gives speed-indexed local models along physically meaningful turning equilibria rather than only around straight-line motion. The audit uses $\kappa \in \{0, 0.0025, 0.005\} \text{ m}^{-1}$ and $v \in \{10, 12, \dots, 30\} \text{ m s}^{-1}$. The most severe point has $a_y = v^2\kappa = 4.5 \text{ m s}^{-2}$; across all clients, the equilibrium range remains within $|\beta^*| \leq 2.37^\circ$ and $|\delta^*| \leq 2.27^\circ$.

Nested basis family and selection rule. The baseline order is the previously validated physics basis

$$\Phi_3(v) = [1, v^{-1}, v^{-2}]. \quad (56)$$

Orders $L = 4, \dots, 7$ append the nested normalized-speed terms z, z^2, z^3, z^4 , where $z = (v - 20)/10$. Thus each candidate is a Laurent polynomial that retains the known reciprocal physics while adding smooth residual freedom. To avoid the severe collinearity of raw reciprocal and polynomial columns, each candidate space is transformed by QR orthogonalization on a fixed 401-point grid over $10\text{--}30 \text{ m s}^{-1}$. This is an invertible change of coordinates: it improves numerical conditioning without changing the functions contained in a candidate space. It also avoids an arbitrary ordering of hand-placed radial-basis centers.

Five cross-validation folds withhold contiguous speed regions: $\{10, 12\}$, $\{14, 16\}$, $\{18, 20, 22\}$, $\{24, 26\}$, and $\{28, 30\}$. This is deliberately harder than random sample splitting because no neighboring sample at the same withheld speed is available. The score is relative Frobenius error of the stacked $[A_d, B_d]$ entries. It is first averaged over clients and folds within each independent fleet. The chosen order is the smallest order within one fleet-level standard error of the best mean, subject to conditioning. Errors below 10^{-4} relative, or 0.01%, are treated as practically equivalent. This floor prevents numerical differences in an already exact model from rewarding unnecessary basis terms.

The development population comprises 300 vehicles from ten independent fleet seeds. In total, 22500 blocked validation fits are evaluated. No controller tracking result or later test fleet enters the order selection.

Table 53: Experiment 10A blocked-speed matrix error. Entries are percentages; “worst” is the largest client-fold error over all development fleets.

κ [m ⁻¹]	L	Mean	Fleet SE	Worst	κ_2 worst
0	3	0.0008	4.89×10^{-6}	0.0044	27.47
0	7	0.00004	2.50×10^{-7}	0.0003	240.38
0.0025	3	0.6604	0.0011	2.3584	27.47
0.0025	6	0.0047	8.69×10^{-6}	0.0230	68.95
0.0025	7	0.00006	3.04×10^{-7}	0.0004	240.38
0.0050	3	2.6607	0.0044	9.5054	27.47
0.0050	4	0.9669	0.0016	4.1013	28.46
0.0050	5	0.2081	0.0004	0.9593	32.24
0.0050	6	0.0210	0.00004	0.1048	68.95
0.0050	7	0.0007	3.59×10^{-6}	0.0047	240.38

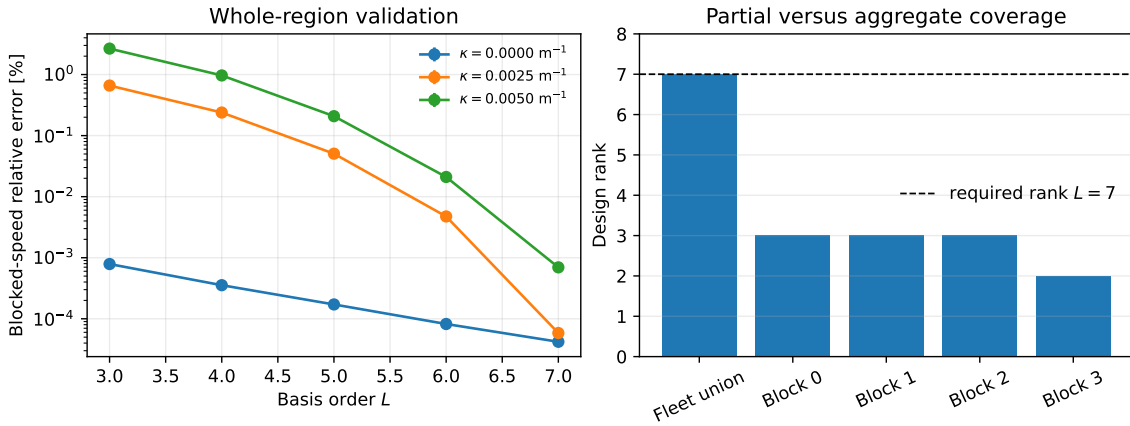


Figure 42: Blocked-speed basis audit and design-rank comparison. The nonlinear cornering cases require residual speed dependence, while no restricted local block can identify the seven selected coefficients.

Results. The straight-line negative control behaves as required. All candidate bases are already far below the practical error floor, so the rule retains $L = 3$ despite the numerically smallest value occurring at $L = 7$. Extra flexibility is not justified for the original small-signal model.

The nonlinear operating points require more structure. At $\kappa = 0.0025 \text{ m}^{-1}$, $L = 6$ is the smallest model below the 0.01% practical floor; its mean blocked-speed error is 0.0047%, compared with 0.6604% for $L = 3$. At $\kappa = 0.005 \text{ m}^{-1}$, the rule selects $L = 7$. Mean error falls monotonically from 2.6607% at $L = 3$ to 0.0007% at $L = 7$, and the worst client-fold error falls from 9.51% to 0.0047%. The strongest audited operating condition therefore determines the common order $L = 7$. The orthogonalized blocked-training condition number remains below 241, so this conclusion is not purchased by numerically singular regression.

Partial-coverage rank audit. For the selected $L = 7$ model, a client assigned to one of the four speed blocks $\{10, 12, 14\}$, $\{16, 18, 20\}$, $\{22, 24, 26\}$, or $\{28, 30\}$ has design rank three, three, three, or two, respectively. Every local block is therefore rank deficient. The union of the eleven fleet speeds has rank seven and a condition number of 3.41 in the fixed orthogonalized coordinates; the worst blocked-training condition number across validation folds is 240.38. Thus complementary coverage supplies structurally missing scheduling information rather than merely more repetitions of the same local experiment.

Decision and limitation. Experiment 10A passes its representation and identifiability gate and freezes

$$L = 7, \quad \text{span } \Phi_7(v) = \text{span}[1, v^{-1}, v^{-2}, z, z^2, z^3, z^4], \quad z = (v - 20)/10. \quad (57)$$

This conclusion is narrower than an FL benefit. The targets are exact local Jacobians, the operating points use known curvature, and rank deficiency alone does not prove improved estimation or closed-loop control. Experiment 10B must fit the frozen model from noisy complementary recordings and compare Local, pooled compatible sharing, and an equal-budget local-information oracle on unseen speeds. If sharing does not improve those outcomes, the new direction must again be rejected or reframed.

Reproducibility. The frozen protocol, implementation, aggregate tables, client-fold records, operating points, coverage diagnostics, and provenance hashes are stored in `code/config/experiment_10a.json`, `code/experiments/experiment_10a_basis_coverage_audit.py`, and `results/tables`.

38 Experiment 10B: oracle complementary-coverage gate

Purpose. Experiment 10A established that the nonlinear cornering LPV map requires $L = 7$, that a client restricted to one speed block cannot identify all seven basis directions, and that the fleet union is full rank. Those structural facts do not guarantee that sharing finite noisy data improves either identification or control. Experiment 10B therefore tests the missing causal link under the most favorable compatibility assumption: true vehicle-family labels are known and aggregation is centralized. Clustering and federated optimization are deliberately absent. If oracle compatible sharing fails here, they cannot rescue the accuracy claim.

Finite-data protocol. Ten new development fleets, seeds 321–330, contain 30 clients each. Every client is assigned to exactly one of the four speed blocks

$$\{10, 12, 14\}, \quad \{16, 18, 20\}, \quad \{22, 24, 26\}, \quad \{28, 30\} \text{ m s}^{-1}, \quad (58)$$

while each physical family jointly covers the full envelope. At each available speed, 36 independent one-step perturbation transitions are generated around the nonlinear steady-cornering equilibrium at $\kappa = 0.005 \text{ m}^{-1}$. The perturbations have standard deviations 0.25° in sideslip, 2° s^{-1} in yaw rate, and 0.3° in steering. The next state is corrupted by independent noise with standard deviations 0.01° and $0.03^\circ \text{ s}^{-1}$. A local least-squares regression first estimates $[A_d, B_d]$ at each observed speed. Consequently, 10B uses finite nonlinear transitions rather than adding noise to the exact Jacobians from 10A.

Each ordinary client supplies 72 or 108 transitions. LocalFull is an information-budget oracle: the same recipient vehicle is measured across all eleven speeds using approximately the total transition budget of its ten-client family pool, averaging 997 transitions. Test matrices and closed-loop trajectories are generated only after every model and regularization strength has been selected.

Compared estimators. All $L = 7$ models use the basis frozen in Experiment 10A.

Local fits the locally identifiable physics span $L = 3$ and extrapolates it across speed.

LocalRegularized fits $L = 7$ around the client’s $L = 3$ prediction; leave-one-local-speed-out validation selects the ridge strength.

Global pools all clients and ignores structural heterogeneity.

FamilyPool fits one $L = 7$ map from the complementary data of each oracle-known physical family.

FamilyPersonalized regularizes a client-specific $L = 7$ map toward its FamilyPool backbone, with strength selected from local validation.

LocalFull fits $L = 7$ from the recipient’s equal-budget full-envelope recordings and is not deployable under the restricted-coverage premise.

ExactLPV fits $L = 7$ to the recipient’s exact full-grid Jacobians.

For deployment outside the calibrated speed block, Local and LocalRegularized hold the controller designed at the nearest calibrated speed. This is a safe, finite baseline and avoids interpreting a failed Riccati equation from an unsupported extrapolation as an arbitrarily large performance gain. Their prediction metric still evaluates the actual extrapolated model.

Evaluation and gate. Matrix error is the relative Frobenius error of stacked $[A_d, B_d]$ entries. The control test initializes the nonlinear plant away from each steady-cornering equilibrium and applies a discrete LQR controller for 1.5 s. The reported score is RMS state error normalized by the initial perturbation. A controller is feasible only if the state remains finite and its normalized final error is below 0.1. Seen and locally unseen speeds are reported separately.

The primary gate requires FamilyPersonalized to beat Local on both unseen-speed prediction and nonlinear recovery with fleet-level paired bootstrap intervals bounded above zero, while remaining feasible on every evaluation. Ten fleet means, rather than clients or time samples, are the independent units; 20000 bootstrap resamples form the 95% intervals. No arbitrary minimum percentage improvement is imposed.

Interpretation. The primary oracle-sharing gate passes. FamilyPersonalized reduces unseen-speed matrix error from 66.12% to 1.91%, improves nonlinear recovery by 1.67%, wins every fleet on both metrics, and remains feasible throughout. This is the first experiment in the project where the benefit comes directly from complementary operating-point information: local regularization alone is ineffective because it cannot create the missing speed directions.

Table 54: Experiment 10B finite-data results. Matrix errors are percentages; recovery is normalized nonlinear closed-loop RMS.

Method	Seen speeds		Locally unseen speeds		
	Error [%]	Recovery	Error [%]	Recovery	Feasible
Local	2.027	0.17510	66.121	0.18067	100%
LocalRegularized	2.027	0.17510	66.121	0.18067	100%
Global	3.853	0.17506	4.296	0.17767	100%
FamilyPool	1.385	0.17498	1.816	0.17760	100%
FamilyPersonalized	1.499	0.17503	1.907	0.17765	100%
LocalFull	0.966	0.17501	0.952	0.17759	100%
ExactLPV	6.72×10^{-6}	0.17497	6.76×10^{-6}	0.17757	100%

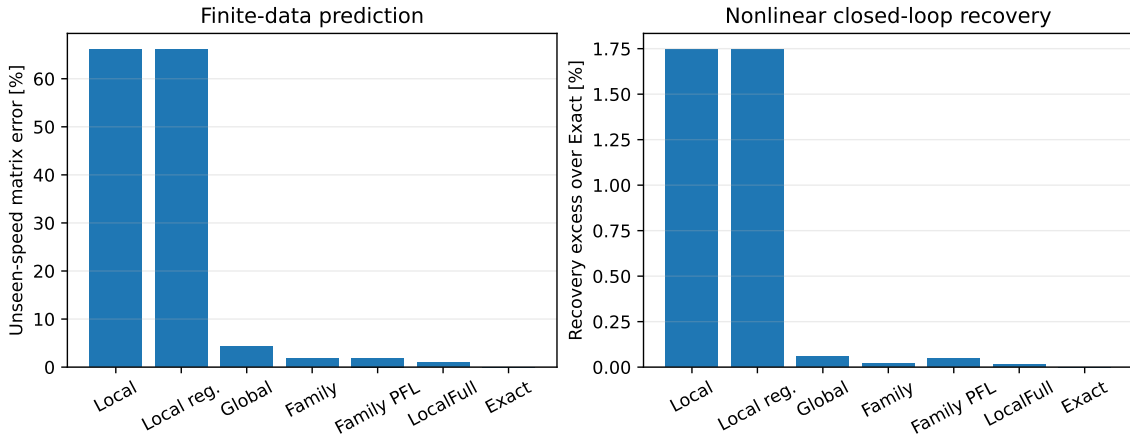


Figure 43: Unseen-speed finite-data prediction and nonlinear recovery excess over the ExactLPV controller. Sharing largely closes the model-generalization gap, whereas closed-loop differences are smaller because feedback is robust to the remaining model errors.

Table 55: Fleet-level paired improvements on locally unseen speeds. Positive values favor the first method.

Comparison	Metric	Mean	95% CI	Wins
FamilyPersonalized vs. Local	Prediction	97.11%	[96.93,97.27]%	10/10
FamilyPersonalized vs. Local	Recovery	1.67%	[1.53,1.82]%	10/10
FamilyPool vs. Local	Prediction	97.24%	[97.03,97.42]%	10/10
FamilyPool vs. Local	Recovery	1.70%	[1.55,1.85]%	10/10
FamilyPool vs. Global	Prediction	57.70%	[55.32,59.71]%	10/10
FamilyPool vs. Global	Recovery	0.04%	[-0.01,0.08]%	8/10
FamilyPersonalized vs. FamilyPool	Prediction	-5.13%	[-8.35,-2.48]%	1/10
FamilyPersonalized vs. FamilyPool	Recovery	-0.027%	[-0.047,-0.011]%	0/10
LocalFull vs. FamilyPool	Prediction	47.21%	[44.57,50.28]%	10/10
LocalFull vs. FamilyPool	Recovery	0.003%	[-0.046,0.053]%	4/10

The mechanism is nevertheless simpler than the intended personalized story. FamilyPool is 5.13% more accurate than FamilyPersonalized and slightly better in control on every fleet. At this cold-start data volume, estimating a local head adds variance rather than useful adaptation. The evidence therefore supports a shared compatible-family backbone, not yet personalized FL. A local head may become useful only after more recipient data are available, which must be tested separately rather than asserted.

Family structure matters strongly for identification: FamilyPool reduces error by 57.70% relative to Global. That gain barely changes recovery, whose confidence interval includes zero. The LQR feedback is robust enough that several imperfect shared models give nearly identical regulation. Conversely, LocalFull halves FamilyPool’s remaining matrix error but provides no detectable control improvement. Thus the main demonstrated benefit is accurate full-envelope model availability without requiring each vehicle to collect full-envelope data; the control benefit over a conservative nearest-speed Local controller is positive but modest.

Decision. Experiment 10B justifies continuing to a learned compatibility mechanism and an explicit federated implementation, but the next experiment should preserve FamilyPool as the primary shared-backbone baseline. Personalization should be treated as a data-dependent extension and must earn its place through a calibration-budget sweep. Claims that family sharing beats Global should be made for model quality, not for control performance under this recovery test.

Reproducibility. The protocol and implementation are stored in `code/config/experiment_10b.json` and `code/experiments/experiment_10b_oracle_complementary_gate.py`. Per-fleet client records, selected strengths, aggregate tables, paired bootstrap results, and provenance hashes are stored under `results/tables`.

39 Experiment 10C: backbone-to-personalization transition

Purpose. Experiment 10B showed that complementary oracle-family sharing solves the restricted-coverage identification problem, but that immediate personalization is worse than the shared family backbone. Experiment 10C asks whether this is a cold-start effect: as a recipient covers more of the speed envelope, does a client-specific head eventually become useful, and does any model improvement translate into control performance?

This experiment is a development sweep, not a confirmation run. Its purpose is to determine which estimator structure deserves a federated implementation. The result may support a staged backbone-to-personalization policy, a permanently shared family model, or a transition directly from the family model to Local.

Frozen and progressive information. Ten fresh fleets, seeds 341–350, use the 10B plant, excitation, noise, curvature, $L = 7$ basis, and nonlinear recovery test. Initial speed-block assignment is balanced independently inside each oracle family; hence every family jointly covers all four blocks. The FamilyPool backbone is fit once from the initial client blocks and then frozen.

Each recipient starts with its original two- or three-speed block, averaging 2.8 speeds and 100.8 transitions. It then adds the nearest missing speeds until covering 5, 8, and all 11 grid speeds. These stages contain 180, 288, and 396 local transitions. The added recipient records never update FamilyPool. This separates a fixed fleet prior from increasing client information.

The comparison contains four models. FamilyPool is the fixed oracle-family backbone. FamilyPersonalized fits an $L = 7$ client model around that backbone, with ridge strength selected by leave-one-local-speed-out validation. LocalRegularized uses the same local records but an

$L = 3$ locally estimated physics prior instead of fleet knowledge. ExactLPV remains the exact-Jacobian reference. Metrics cover the complete speed envelope at every stage, including the locally measured speeds.

Safety guard. Personalization must not sacrifice deployability. If a personalized controller fails the nonlinear recovery feasibility check, deployment falls back to the FamilyPool controller at that speed. An infeasible LocalRegularized controller falls back to its nearest-observed-speed $L = 3$ physics controller. Every fallback is recorded. This guard prevents unsafe adaptation from appearing as an accuracy improvement and distinguishes nominal estimator quality from the deployed safe policy.

Statistical criterion. At each stage, paired differences are formed after averaging all clients and speeds within each fleet. A crossover is declared only when the 95% bootstrap interval of the mean improvement over FamilyPool is positive. Prediction and control crossovers are tested separately; a personalized-control claim requires both. As in 10B, 20000 resamples use the ten fleets as independent units.

Table 56: Experiment 10C full-envelope results. Errors are percentages; recovery is normalized nonlinear RMS.

Speeds	Trans.	FamilyPool		Personalized		Local	
		Error	Recovery	Error	Recovery	Error	Recovery
2.8	100.8	1.631	0.17671	1.717	0.17676	43.694	0.19585
5	180	1.631	0.17671	1.656	0.17679	13.392	0.18265
8	288	1.631	0.17671	1.505	0.17676	2.565	0.17755
11	396	1.631	0.17671	1.406	0.17672	1.236	0.17675

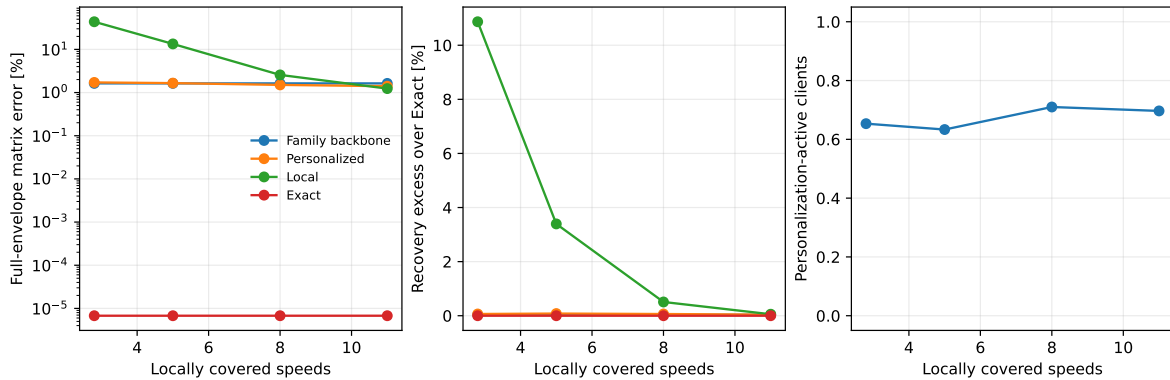


Figure 44: Full-envelope identification, recovery excess over ExactLPV, and the fraction of clients whose validation-selected personalization strength does not exceed one. Personalization improves model quality only after sufficient speed coverage and never improves recovery over the backbone.

Identification transition. At cold start, personalization is significantly worse than FamilyPool by 5.37%. Five covered speeds are still insufficient to resolve an advantage. The first reliable prediction crossover occurs at eight speeds: personalization improves full-envelope error by 7.74%, with a [5.69,10.01]% interval and wins all ten fleets. At full coverage the improvement grows to 13.66% and again wins every fleet. The shared backbone therefore becomes a useful prior for client adaptation once local data span most of the scheduling envelope.

Table 57: FamilyPersonalized relative to the fixed FamilyPool. Positive values favor personalization.

Speeds	Trans.	Metric	Mean	95% CI	Wins
2.8	100.8	Prediction	-5.37%	[-7.31,-3.57]%	0/10
2.8	100.8	Recovery	-0.030%	[-0.050,-0.014]%	1/10
5	180	Prediction	-1.61%	[-3.93,0.77]%	4/10
5	180	Recovery	-0.047%	[-0.079,-0.018]%	2/10
8	288	Prediction	7.74%	[5.69,10.01]%	10/10
8	288	Recovery	-0.030%	[-0.072,-0.004]%	1/10
11	396	Prediction	13.66%	[10.62,16.81]%	10/10
11	396	Recovery	-0.006%	[-0.020,0.008]%	5/10

The locally selected ridge strength alone does not provide a trustworthy activation rule. Between 63% and 71% of clients are classified as active at all stages, including cold start where personalization is harmful. A deployment rule must therefore incorporate coverage or aggregate information rank rather than relying only on local validation loss.

Control result. There is no control crossover. Personalization is slightly but significantly worse at two, five, and eight covered speeds. At full coverage it reaches parity with FamilyPool: the mean difference is -0.006% with a $[-0.020, 0.008]\%$ interval. Feedback suppresses the model-quality differences that are clear in open-loop matrix error. All guarded personalized evaluations are feasible; fallback is required for only 0.030% of evaluations at the five-speed stage and never at the other stages.

LocalRegularized provides a second boundary. It is unusable without its safety fallback at low coverage, but at all eleven speeds it becomes the most accurate finite-data model: 1.236% error versus 1.406% for FamilyPersonalized. It is 13.79% more accurate than personalization with a confidence interval bounded away from zero, while their recovery remains statistically equivalent. Thus the eventual endpoint is fully local identification, not indefinite personalization.

Decision. Experiment 10C supports a three-stage identification policy:

1. use the compatible shared backbone at cold start;
2. enable a personalized head only after broad local scheduling coverage, with eight of eleven speeds as the observed development crossover;
3. switch to a fully local $L = 7$ model once full-envelope local data are available.

This policy is supported for model accuracy, not for improved closed-loop recovery. The paper should therefore frame personalization as progressive model ownership and local fidelity, while the demonstrated control advantage remains the approximately 1.7% gain of complementary sharing over the conservative restricted Local controller from 10B.

The next experiment should replace oracle family labels with learned compatible groups while retaining FamilyPool as the cold-start anchor and the coverage gate for personalization. Only after that test should the aggregation be implemented federatively.

Reproducibility. The configuration and implementation are in `code/config/experiment_10c.json` and `code/experiments/experiment_10c_personalization_transition.py`. Per-fleet records, fit diagnostics, fallback rates, aggregate summaries, paired bootstrap comparisons, and provenance hashes are stored under `results/tables`.

40 Experiment 10D: learned compatible groups with unknown cardinality

Question and hypothesis. Experiment 10B established the value of complementary coverage when the physical family of every client is supplied to the estimator. That oracle partition is not a credible deployment assumption. Experiment 10D therefore asks whether clients can be grouped from their restricted local records, while also selecting the number of groups, and still retain the cold-start benefit of family sharing. The intended claim is deliberately task based: the method must discover clients that can share an LPV backbone, not reconstruct the simulator’s physical labels.

Protocol. The experiment uses ten new 30-client fleets (seeds 361–370), the nonlinear tanh-tire plant, and the order-seven basis frozen in Experiment 10A. Each client observes one contiguous block of the 10–30 m/s grid: three speeds in the first three blocks and two in the final block. At every observed speed, 36 noisy perturbation transitions are used to estimate the six entries of the local discrete-time state/input matrix. Blocks are stratified within physical families solely to keep the simulated information-coverage comparison balanced. The learned estimator receives neither those family labels nor the oracle partition.

Fitting order-seven models independently before clustering would be invalid: every local scheduling block is rank deficient by construction. We instead fit a hard mixture of LPV regressions directly to the observed speed–matrix pairs. For a candidate group count K , initialization uses K -means on each client’s mean residual relative to the global LPV fit. The algorithm alternates between fitting one pooled LPV regression per group and assigning every client to the model with the smallest variance-normalized residual on its own record. We use 12 deterministic restarts, require at least five clients per group, and consider $K \in \{1, \dots, 5\}$. The selected count minimizes

$$\text{BIC}_K = N \log(\text{SSE}_K/N) + (42K + K - 1) \log N, \quad (59)$$

where N is the number of scalar matrix observations, 42 is the coefficient count of one six-output order-seven LPV map, and SSE is computed after scaling each matrix entry by its across-client observation standard deviation.

Comparators and gates. LearnedCluster is compared with the restricted Local model, one Global LPV model, OracleFamily, the equal-transition-budget LocalFull reference, and the exact projected LPV model. Evaluation uses full-envelope relative matrix error and the guarded nonlinear recovery metric from Experiment 10B. Paired uncertainty intervals bootstrap the ten fleet-level means with 20,000 resamples. The primary gates require LearnedCluster to improve both metrics over Local with positive 95% intervals and to remain feasible in every closed-loop evaluation. Recovery of three labels is reported only as a diagnostic: a compatibility partition may legitimately merge physical families when their control-relevant maps are sufficiently similar.

Table 58: Experiment 10D full-envelope results across ten fleets. Matrix errors are percentages; recovery is the dimensionless nonlinear recovery metric.

Method	Matrix error	Recovery	Feasible fleets
Local	45.344%	0.179441	10/10
Global	4.105%	0.177013	10/10
LearnedCluster	2.691%	0.176932	10/10
OracleFamily	1.687%	0.176928	10/10
LocalFull	0.938%	0.176869	10/10
ExactLPV	$6.76 \times 10^{-6}\%$	0.176842	10/10

Table 59: Paired fleet-level improvements in Experiment 10D. Positive values favor the first method.

Method	Baseline	Metric	Mean	95% CI	Wins
LearnedCluster	Local	Matrix error	94.06%	[93.35,94.78]%	10/10
LearnedCluster	Local	Recovery	1.398%	[1.341,1.459]%	10/10
LearnedCluster	Global	Matrix error	34.27%	[24.66,42.65]%	9/10
LearnedCluster	Global	Recovery	0.046%	[0.022,0.070]%	8/10
LearnedCluster	OracleFamily	Matrix error	-62.25%	[-96.50,-38.61]%	0/10
LearnedCluster	OracleFamily	Recovery	-0.003%	[-0.024,0.022]%	4/10

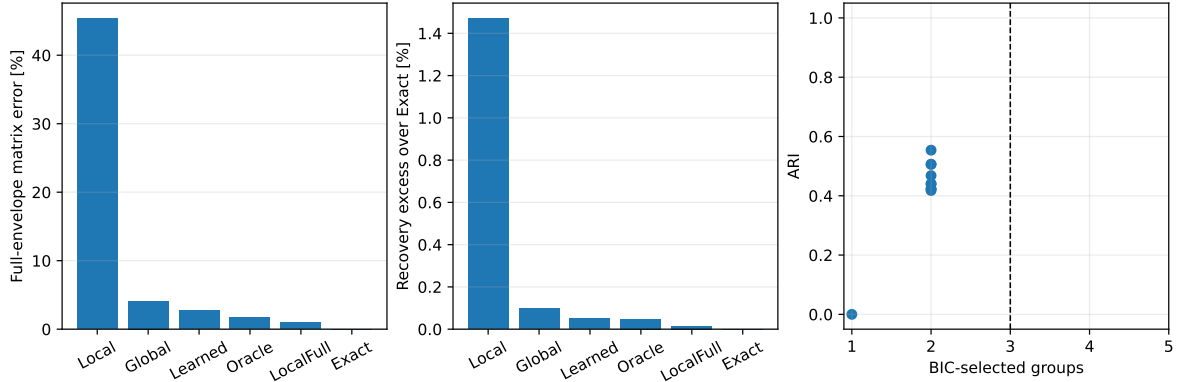


Figure 45: Experiment 10D model error, recovery excess over ExactLPV, and the BIC-selected group count versus adjusted Rand index (ARI).

Result. The primary label-free sharing gate passes. LearnedCluster reduces matrix error by 94.06% relative to Local and improves nonlinear recovery by 1.398%; both intervals are bounded away from zero and all ten fleets are feasible. It also improves matrix error by 34.27% relative to Global. The associated recovery gain over Global is only 0.046%, although its interval is positive, so this comparison should be interpreted as near-equivalent closed-loop behavior with substantially better model fidelity.

Learned grouping does not reach the oracle identification accuracy. Its matrix error is 62.25% higher than OracleFamily, although their recovery is statistically indistinguishable. Relative to Local, LearnedCluster retains 97.7% of the matrix-error reduction delivered by OracleFamily and essentially all of its recovery reduction. LocalFull remains the most accurate finite-data reference because it gives every client an equal total transition budget over the complete speed envelope; it is a data-acquisition upper reference rather than a cold-start competitor.

What the groups mean. BIC selects two groups in nine fleets and one group in the remaining fleet. It never selects the simulator’s three-family count, and mean ARI is 0.418. In the nine two-group solutions, the handling and heavy families occupy opposite groups, while nominal clients are merged with or split between them. This is not a failed family classifier. It shows that the penalized prediction task supports fewer shared backbones than the data generator has physical labels. The distinction is important: Experiment 10D learns a compact compatibility partition for LPV estimation, not vehicle taxonomy.

Decision and limitation. The experiment removes oracle labels from the positive complementary-sharing result and supports moving to a federated implementation of latent compatible backbone learning. It does not yet demonstrate federation: the current alternating fit is centralized and accesses clients’ matrix observations. Moreover, block stratification uses simulator labels to balance

the evaluation, although labels are absent from the estimator. A final federated experiment must reproduce the alternating sufficient-statistic updates, account for candidate-count selection and communication, and test non-stratified coverage or partial participation. Personalization need not be part of that central claim; Experiment 10C already showed no closed-loop benefit for it.

Reproducibility. The configuration and implementation are in `code/config/experiment_10d.json` and `code/experiments/experiment_10d_learned_compatible_groups.py`. Per-fleet client records, candidate BIC scores, assignments, aggregate summaries, paired bootstrap comparisons, and provenance hashes are stored under `results/tables`.

41 Experiment 10E: federated latent-group LPV identification

Question. Experiment 10D learned compatible groups without using family labels, but its alternating regression accessed all client observations centrally. Experiment 10E asks whether the same latent-group solution can be obtained through a genuine federated protocol and whether the complementary-sharing benefit survives when operating-point coverage is assigned without family labels. This is an equivalence and feasibility gate. It is not yet the final control-relevance experiment.

Federated sufficient-statistic method. For client i , let $\Phi_i \in \mathbb{R}^{n_i \times 7}$ contain the frozen LPV basis evaluated at its locally observed speeds and let $Y_i \in \mathbb{R}^{n_i \times 6}$ contain its estimated state/input matrix entries. The client retains (Φ_i, Y_i) and computes

$$G_i = \Phi_i^\top \Phi_i, \quad H_i = \Phi_i^\top Y_i. \quad (60)$$

For the clients currently assigned to group k , the server obtains the additive aggregates and solves

$$\hat{\Theta}_k = \left(\sum_{i:z_i=k} G_i + \lambda I \right)^{-1} \sum_{i:z_i=k} H_i. \quad (61)$$

The server broadcasts the candidate backbones. Every client evaluates their variance-normalized residuals on its own observations and returns one score per candidate group. The server reassigns clients, repeats equation (61), and evaluates all feasible $K \in \{1, \dots, 5\}$ using the BIC from Experiment 10D. Twelve deterministic initializations are used for $K > 1$, and groups must contain at least five clients. Raw perturbation trajectories and speed–matrix observations are not transmitted. This is a federated protocol emulator rather than a cryptographic secure-aggregation implementation; the sufficient statistics and assignment scores can themselves disclose information.

Three elementary properties explain the method. First, additivity of G_i and H_i makes the federated ridge solution algebraically identical to the pooled centralized solution for a fixed partition. Second, the group model is identifiable whenever $\sum_{i:z_i=k} G_i$ has full rank, even if each G_i is singular. This is precisely the complementary-coverage mechanism established in Experiment 10A. Third, model refitting minimizes the objective for fixed assignments and reassignment minimizes each client’s residual for fixed models, so every feasible alternating step is non-increasing in the fitting objective.

Protocols and gates. Ten fresh 30-client fleets (seeds 371–380) use the nonlinear tanh-tire plant, 36 noisy perturbation transitions per observed speed, and the frozen order-seven basis. Two coverage protocols are tested:

Controlled Speed blocks are stratified within simulator families as in Experiment 10D. This isolates centralized–federated equivalence.

Label-free The 30 clients are randomly permuted and the four contiguous speed blocks are balanced globally, without inspecting family labels. A physical family is no longer guaranteed to cover the full speed envelope.

The numerical gate requires identical selected partitions and maximum model and BIC discrepancies below 10^{-8} . The scientific gate requires positive fleet-bootstrap intervals for FederatedLearned relative to restricted Local in both full-envelope matrix error and nonlinear recovery, with feasible control on all fleets.

Table 60: Experiment 10E mean results across ten fresh fleets. Matrix errors are percentages. An infinite recovery mean indicates at least one infeasible fleet.

Coverage	Method	Matrix error	Recovery	Feasible
Controlled	Local	45.972%	0.179531	10/10
Controlled	Global	4.122%	0.177021	10/10
Controlled	FederatedLearned	2.446%	0.176934	10/10
Controlled	OracleFamily	1.689%	0.176924	10/10
Label-free	Local	49.562%	0.179232	10/10
Label-free	Global	4.250%	0.177107	10/10
Label-free	FederatedLearned	3.249%	0.177052	10/10
Label-free	OracleFamily	1.966%	∞	9/10

Table 61: FederatedLearned improvements in Experiment 10E. Positive values favor FederatedLearned; intervals bootstrap paired fleet means.

Coverage	Baseline	Metric	Mean	95% CI	Wins
Controlled	Local	Matrix error	94.67%	[94.32,95.02]%	10/10
Controlled	Local	Recovery	1.446%	[1.338,1.570]%	10/10
Controlled	Global	Matrix error	40.64%	[36.44,44.75]%	10/10
Controlled	Global	Recovery	0.049%	[0.037,0.063]%	10/10
Label-free	Local	Matrix error	93.41%	[92.60,94.21]%	10/10
Label-free	Local	Recovery	1.216%	[1.128,1.311]%	10/10
Label-free	Global	Matrix error	23.74%	[15.39,31.04]%	8/10
Label-free	Global	Recovery	0.031%	[0.020,0.042]%	8/10

Federated-equivalence result. The gate passes in all 20 fleet-coverage conditions. Federated and centralized latent grouping select identical partitions and group counts. The maximum coefficient difference is 3.20×10^{-14} and the maximum BIC difference is 1.87×10^{-9} . CentralLearned and FederatedLearned consequently have identical reported prediction and control metrics. The result verifies the distributed sufficient-statistic implementation rather than merely assuming that centralized clustering transfers to FL.

Label-free sharing result. Removing family-stratified coverage weakens but does not remove the benefit. FederatedLearned reduces matrix error by 93.41% and recovery by 1.216% relative to Local, with positive intervals and ten wins. Against Global it reduces matrix error by 23.74%. The recovery improvement over Global is only 0.031%, despite its positive interval, and is too small to carry a practical control-superiority claim.

The label-free protocol also exposes why compatibility should not be equated with physical family. In seed 375, no nominal-family client observes the final 28–30 m/s block. OracleFamily

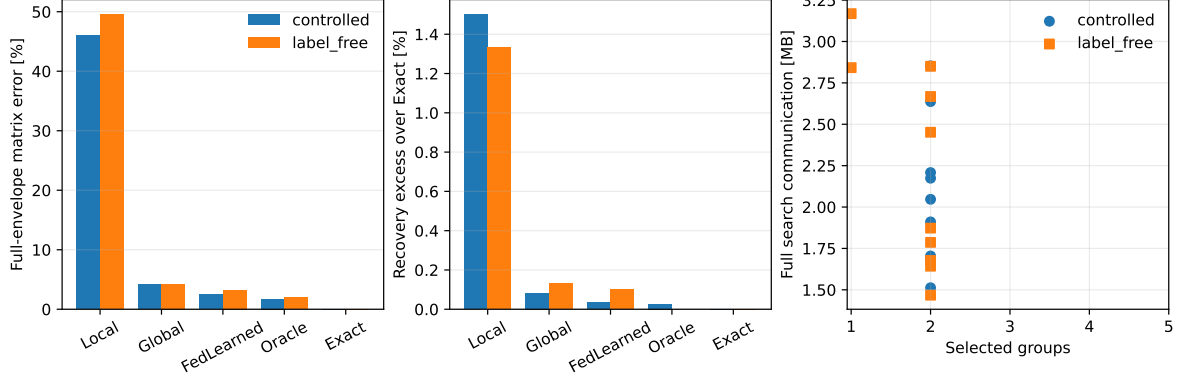


Figure 46: Experiment 10E identification, control, and full-search communication results for controlled and label-free coverage.

therefore lacks the required scheduling coverage and produces an infeasible 30 m/s controller, whereas FederatedLearned remains feasible. Learned compatibility accounts jointly for dynamics and information coverage; a correct physical label alone does not guarantee an identifiable group model.

Selected groups and communication. Controlled coverage selects two groups in all ten fleets. Label-free coverage selects two groups in eight fleets and a single group in two, with mean ARI 0.352. The full candidate search requires on average 45.0 model/assignment rounds and 2.04 MB per controlled fleet, or 48.9 rounds and 2.24 MB per label-free fleet, assuming 64-bit scalars. These totals include every restart and every candidate K , not only the final selected model. The absolute load is modest, but the exhaustive multistart search is not communication efficient and should not be advertised as such.

Decision. Experiment 10E establishes the missing genuine-FL component: locally retained data, additive client updates, iterative label-free group assignment, unknown group count, and explicit end-to-end communication accounting. It also confirms that complementary sharing remains beneficial without family-aware coverage. It does not resolve the principal control-story weakness. FederatedLearned and Global remain almost indistinguishable under the recovery test. The next experiment must therefore use speed-varying tracking, steering constraints, and worst-client metrics to determine whether the substantially better learned LPV models provide a practically meaningful control advantage.

Reproducibility. The configuration and implementation are in `code/config/experiment_10e.json` and `code/experiments/experiment_10e_federated_latent_groups.py`. Per-fleet records, assignments, candidate scores, communication counts, aggregate summaries, paired intervals, and provenance hashes are stored under `results/tables`.

42 Experiment 10F: control-relevance stress test

Motivation and claim gate. Experiment 10E reduced full-envelope matrix error by 23.74% relative to Global, but its isolated recovery metric improved by only 0.031%. Better identification therefore did not yet imply a practically distinguishable control result. Experiment 10F tests the missing link under nonlinear, speed-varying yaw-rate tracking with actuator constraints. A valid control-relevance claim requires FederatedLearned to improve over Global on fresh fleets after the maneuver and controller have been frozen without using their pairwise result.

Controller audit without method-directed tuning. Twelve LQI configurations vary sideslip, yaw-rate, and integral weights and the steering penalty. On development fleets 381–385, every configuration is evaluated using only each client’s ExactLPV model. Neither Global nor FederatedLearned is evaluated during selection. A candidate must retain feasibility on every development trajectory, have less than 5% mean actuator saturation, and satisfy the dense-grid small-signal stability test. Among passing candidates, the controller with minimum mean ExactLPV tracking RMSE is selected. The frozen choice is

$$Q = \text{diag}(25, 50, 1000), \quad R = 0.15. \quad (62)$$

All twelve candidates pass the constraints. The selected controller obtains 0.0954 deg/s mean tracking RMSE across the two development maneuvers, compared with 0.1818 deg/s for the most conservative candidate.

An initial maneuver-intensity pilot on seeds 391–400 is excluded from all reported confirmation results. That pilot produced only about 2 deg/s peak steering rate, no saturation, and approximately 0.01 deg/s tracking error. It therefore did not constitute a meaningful stress test. The final protocol was redefined using physically interpretable lateral-acceleration requests, and a new untouched confirmation set was reserved.

Final protocol. Confirmation uses ten fresh fleets (seeds 411–420), globally randomized label-free coverage, and the complete federated latent-group estimator from Experiment 10E. The reference curvature is generated as

$$\kappa_{\text{ref}}(t) = \frac{a_{y,\text{ref}}(t)}{v(t)^2}, \quad r_{\text{ref}}(t) = v(t)\kappa_{\text{ref}}(t), \quad (63)$$

which prevents speed variation from accidentally changing maneuver severity. The moderate multisine requests up to approximately 4 m/s². The hard three-pulse maneuver requests up to approximately 7 m/s², close to the nonlinear tire envelope. Both use smooth speed profiles spanning 11–29 m/s or 12–29 m/s.

At each sample, the controller interpolates the scheduled LQI gain and prefilter from the eleven design speeds. Steering is limited to 12 deg and 120 deg/s. Conditional integration prevents windup. Feasibility requires finite trajectories and peak sideslip below 6 deg. We report mean and worst-client yaw-rate tracking RMSE, peak tracking error, sideslip, steering, saturation, and the maximum dense-grid spectral radius. Local retains the conservative nearest-observed-speed rule outside its local block.

Table 62: Experiment 10F confirmation results. Tracking values are yaw-rate RMSE in deg/s, averaged across ten fresh fleets.

Maneuver	Method	Mean	Worst client	Feasible	ρ_{max}
Moderate	Local	0.05396	0.08870	10/10	0.9761
Moderate	Global	0.04290	0.05637	10/10	0.9756
Moderate	FederatedLearned	0.04108	0.05303	10/10	0.9759
Moderate	ExactLPV	0.04002	0.04072	10/10	0.9759
Hard	Local	0.16981	0.21845	10/10	0.9761
Hard	Global	0.15644	0.18317	10/10	0.9756
Hard	FederatedLearned	0.15457	0.17714	10/10	0.9759
Hard	ExactLPV	0.15074	0.16332	10/10	0.9759

Table 63: Paired FederatedLearned improvements over Global. Positive values favor FederatedLearned. Absolute changes are in deg/s.

Maneuver	Metric	Relative	95% CI	Absolute
Moderate	Mean tracking	4.26%	[1.95,6.55]%	0.00182
Moderate	Worst client	5.90%	[1.03,11.20]%	0.00335
Hard	Mean tracking	1.21%	[0.47,2.01]%	0.00187
Hard	Worst client	3.31%	[1.16,5.77]%	0.00603

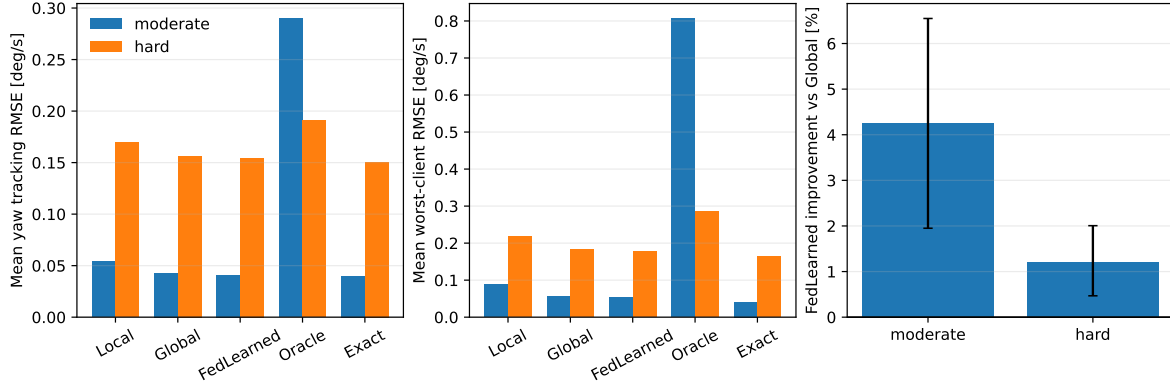


Figure 47: Experiment 10F mean and worst-client tracking under the frozen controller, with paired improvement of FederatedLearned over Global.

Result. The predeclared control-relevance gates pass. FederatedLearned improves mean tracking relative to Global by 4.26% in the moderate maneuver and 1.21% in the hard maneuver. The paired 95% intervals are positive. Worst-client tracking improves by 5.90% and 3.31%, respectively, also with positive intervals. FederatedLearned improves over restricted Local by 23.86% in the moderate maneuver and 8.96% in the hard maneuver. Every federated trajectory is feasible, and the maximum scheduled small-signal spectral radius is 0.9759.

BIC selects two groups in six confirmation fleets and one group in four. In the single-group fleets, FederatedLearned and Global are identical. Accordingly, the six positive fleet-level differences against Global occur exactly where the selector retains two groups. Across all fleets, learned grouping closes 63.3% of the mean Global-to-Exact tracking gap in the moderate maneuver and 32.9% in the hard maneuver. For worst-client tracking it closes 21.4% and 30.4%, respectively.

Critical interpretation. Experiment 10F establishes that the improved LPV models can propagate into statistically supported closed-loop benefits under a frozen, independently selected controller. It therefore resolves the binary control-relevance gate. The practical magnitude relative to Global remains small: the mean absolute improvements are only 0.00182 and 0.00187 deg/s. The paper may claim a repeatable control effect and a meaningful fraction of the remaining Global-to-Exact gap, but should not claim a large practical tracking gain.

The physical-family oracle is not uniformly reliable under label-free coverage. In one confirmation fleet its pooled observations do not identify a stable full-envelope scheduled controller, producing a maximum spectral radius above three. This reinforces the Experiment 10E conclusion that compatibility must account for information coverage as well as physical family membership.

Decision. The evidence chain now supports a genuine FL-and-control paper: local speed blocks are individually insufficient, federated latent grouping reconstructs full-envelope LPV models

without labels, and the model improvement produces a confirmed closed-loop benefit over both Local and Global. Before manuscript consolidation, the remaining high-value robustness test is partial participation or missing-block participation under the frozen 10F controller. This should be treated as a robustness extension, not as another attempt to increase the reported control gap.

Reproducibility. The configuration and implementation are in `code/config/experiment_10f.json` and `code/experiments/experiment_10f_control_relevance.py`. The frozen controller selection, discarded-pilot disclosure, development audit, confirmation records, aggregate summaries, paired intervals, and provenance hashes are stored under `results/tables`.

43 Experiment 10G: vehicle-class control-relevance gate

Purpose and precondition. Experiment 10F established a statistically repeatable but practically modest control benefit of FederatedLearned over Global. Experiment 10G tests a proposed route to a stronger effect: introduce physically interpretable vehicle classes and steering-actuator dynamics. This experiment is an oracle prerequisite gate, not an FL comparison. Learned clustering and finite-data estimation are intentionally excluded. A federated extension is justified only if known vehicle classes first provide a meaningful control advantage over a single pooled model.

Extended nonlinear plant. The tanh-tire bicycle plant is augmented by the applied steering angle δ_a ,

$$x = [\beta, r, \delta_a]^\top, \quad \dot{\delta}_a = \frac{\delta_{\text{cmd}} - \delta_a}{\tau_\delta}. \quad (64)$$

The compact, SUV, and sport centers in Table 64 vary mass, yaw inertia, geometry, axle cornering stiffness, and actuator time constant. Each confirmation fleet contains ten clients per class. Independent bounded scatter is $\pm 5\%$ in mass, $\pm 7\%$ in inertia, $\pm 2\%$ in geometry, $\pm 8\%$ in stiffness, and $\pm 15\%$ in τ_δ .

Table 64: Experiment 10G vehicle-class centers. Stiffnesses are in kN/rad.

Class	m kg	I_z kg m ²	l_f m	l_r m	C_f	C_r	τ_δ s
Compact	1300	1900	1.10	1.50	70	75	0.10
SUV	2100	3900	1.35	1.70	90	100	0.18
Sport	1550	2350	1.25	1.45	105	115	0.07

Oracle models and controller protocol. At each of eleven speeds from 10 to 30 m/s, the exact discrete Jacobian of the three-state plant is computed. Global averages the matrices of all 30 clients, OracleClass averages within the known class, and ClientExact retains each client’s matrices. All three use the same scheduled four-state LQI structure, including yaw-error integration. Six weight candidates are evaluated on five development fleets using ClientExact only. The selected candidate is

$$Q = \text{diag}(100, 100, 10, 2000), \quad R = 0.1. \quad (65)$$

It minimizes development mean tracking among candidates with all trajectories feasible and less than 5% mean saturation. Architecture comparison then uses ten untouched fleets, seeds 441–450, and the frozen moderate and hard 10F maneuvers. The proposed gate requires at least

5% Global degradation relative to ClientExact, at least 40% recovery of that gap by OracleClass, a positive paired confidence interval, and complete class-controller feasibility.

Table 65: Experiment 10G confirmation yaw-rate RMSE in deg/s, averaged over ten fleets.

Maneuver	Method	Mean client	Worst client	Mean saturation
Moderate	Global	0.10281	0.14903	0.000
Moderate	OracleClass	0.10471	0.14996	0.000
Moderate	ClientExact	0.10294	0.13688	0.000
Hard	Global	0.36339	0.65306	0.0169
Hard	OracleClass	0.38063	0.67296	0.0197
Hard	ClientExact	0.42267	2.05197	0.0246

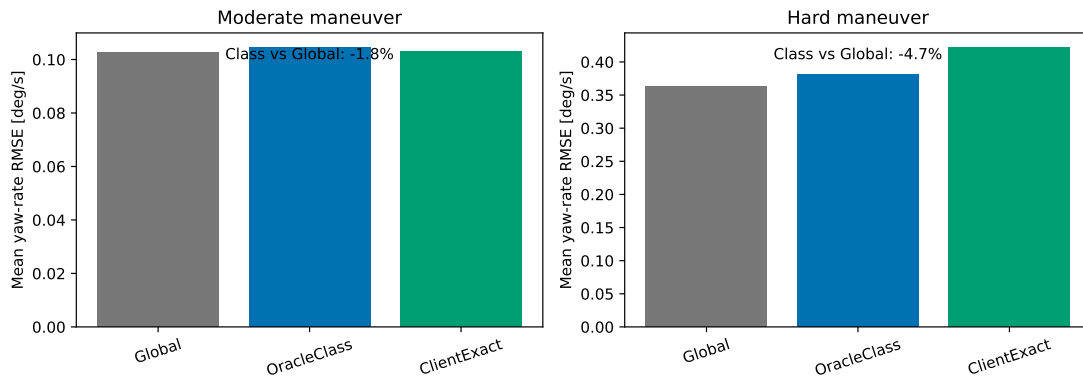


Figure 48: Experiment 10G oracle architecture comparison. Known vehicle classes do not improve mean nonlinear closed-loop tracking over the pooled controller.

Result. The gate fails. OracleClass is 1.85% worse than Global in moderate mean tracking and 4.74% worse in hard mean tracking. The paired 95% intervals are $[-2.15, -1.56]\%$ and $[-5.09, -4.39]\%$, respectively, where positive would favor OracleClass. OracleClass loses to Global on mean tracking in every confirmation fleet. All trajectories remain within the sideslip feasibility limit, so the result is not caused by numerical divergence.

ClientExact also does not establish the assumed oracle advantage. In the hard maneuver, three SUV clients enter sustained command saturation and increase the fleet-level worst-client result. Their exact nominal LQI controllers are more aggressive near the nonlinear tire and actuator constraints, whereas matrix averaging produces a more conservative Global controller. A disclosed post-hoc diagnostic replaces the selected controller by the adjacent, more conservative development-approved candidate $Q = \text{diag}(50, 75, 5, 1500)$, $R = 0.15$. It removes the extreme SUV outliers, but Global still has lower mean error than both OracleClass and ClientExact in both maneuvers. Thus the conclusion is not rescued by a small change in LQI weights.

Interpretation and decision. Adding vehicle classes and a first-order steering actuator does not by itself create a stronger knowledge-sharing argument. The present LQI design is robust to the structured plant variation, while nominal specialization can reduce robustness under nonlinear constraints. Increasing parameter ranges or tuning against the opened confirmation fleets would manufacture a gap rather than establish one. We therefore stop this branch before adding FL.

The next defensible route is to change the control question rather than merely the fleet: use a constraint-aware or robust synthesis objective, or quantify the amount of local excitation needed to certify a controller over unseen operating regions. Such a study must first show an oracle advantage on new development data and reserve another untouched confirmation set.

Reproducibility. The configuration and implementation are in `code/config/experiment_10g.json` and `code/experiments/experiment_10g_vehicle_class_gate.py`. Controller selection, client-level confirmation data, summaries, comparisons, gate decisions, the figure, and provenance hashes are stored under `results`.

44 Experiment 10H: higher-order LPV sharing gate

Redesign objective. Experiment 10G showed that broader vehicle classes alone do not strengthen the FL argument: a pooled LQI controller remained robust, while exact nominal specialization could be less robust near nonlinear constraints. Experiment 10H therefore changes the information problem without changing the control paradigm. It retains speed-scheduled LQI but introduces dynamic tire-force and steering-actuator states that require richer speed–frequency excitation. The claim under test is that fleet sharing can compensate for realistically incomplete local calibration coverage.

Higher-order LPV plant. The state is

$$x = [\beta, r, f_{yf}, f_{yr}, \delta_a]^\top, \quad f_{yf} = F_{yf}/m, \quad f_{yr} = F_{yr}/m. \quad (66)$$

Normalizing tire forces by mass keeps the state numerically well scaled. The lateral motion is

$$\dot{\beta} = \frac{f_{yf} + f_{yr}}{v_x} - r, \quad \dot{r} = \frac{m(l_f f_{yf} - l_r f_{yr})}{I_z}. \quad (67)$$

Front and rear force dynamics use relaxation lengths σ_f, σ_r ,

$$\dot{f}_{yf} = \frac{v_x}{\sigma_f} \left[\frac{C_f}{m} \left(\delta_a - \beta - \frac{l_f r}{v_x} \right) - f_{yf} \right], \quad (68)$$

$$\dot{f}_{yr} = \frac{v_x}{\sigma_r} \left[\frac{C_r}{m} \left(-\beta + \frac{l_r r}{v_x} \right) - f_{yr} \right], \quad (69)$$

and $\dot{\delta}_a = (\delta_{\text{cmd}} - \delta_a)/\tau_\delta$. Compact, SUV, and sport centers retain the 10G mass, inertia, geometry, stiffness, and actuator values. Their front/rear relaxation-length centers are (0.42, 0.50), (0.82, 0.95), and (0.25, 0.30) m, respectively, with bounded within-class scatter.

The primary simulation plant is the exact time-varying LPV system, evaluated by interpolating its exact discrete matrices over 10–30 m/s. This intentionally removes tire saturation as a confounder. Nonlinear robustness is deferred until the sharing mechanism has passed an oracle gate.

Complementary information audit. Each client observes one of four contiguous speed blocks and one of three steering-frequency bands. The matrix-valued LPV regression uses five speed basis functions and the six-dimensional regressor $[x^\top, u]^\top$, hence it requires rank 30. Even with noise-free full-state data, local designs have rank 12–18 and are structurally deficient. Pooling ten clients within a class produces rank 30 in every fleet. After column normalization, aggregate design conditions have median 1.28×10^5 and maximum 2.58×10^5 . Thus sharing adds missing directions rather than merely averaging repeated data.

Controller and protocol. The LQI state augments the five plant states with integrated yaw-rate error. Four weight candidates are selected on development seeds 461–465 using only ClientExact models. The frozen choice is

$$Q = \text{diag}(50, 75, 0.05, 0.05, 5, 1500), \quad R = 0.2. \quad (70)$$

All candidates are stable and feasible. Evaluation uses a speed-varying broadband reference containing 0.35, 0.9, and 1.35 Hz components and a three-pulse transient.

An initial oracle-class pilot on seeds 471–480 found that Global was already better than OracleClass. Those seeds are excluded from the final sharing comparison and retained in the repository. The revised endpoint compares Global with LocalRestricted, which uses each client’s exact noise-free matrices at its observed speeds and nearest-speed extrapolation elsewhere. This is an optimistic local baseline. The revised protocol was frozen before running untouched seeds 481–490.

Table 66: Experiment 10H confirmation yaw-rate RMSE in deg/s, averaged over ten fresh fleets.

Maneuver	Method	Mean client	Worst client
Broadband	LocalRestricted	0.16820	0.25026
Broadband	Global	0.15235	0.21319
Broadband	OracleClass	0.15575	0.22165
Broadband	ClientExact	0.15504	0.21141
Transient	LocalRestricted	0.13484	0.19952
Transient	Global	0.13140	0.17205
Transient	OracleClass	0.13823	0.19779
Transient	ClientExact	0.13719	0.18804

Table 67: Global improvement over LocalRestricted. Positive values favor sharing.

Maneuver	Endpoint	Improvement	95% CI	Fleet wins
Broadband	Mean client	9.42%	[9.13,9.71]%	10/10
Broadband	Worst client	14.71%	[12.17,17.21]%	10/10
Transient	Mean client	2.55%	[2.43,2.67]%	10/10
Transient	Worst client	13.65%	[11.09,16.23]%	10/10

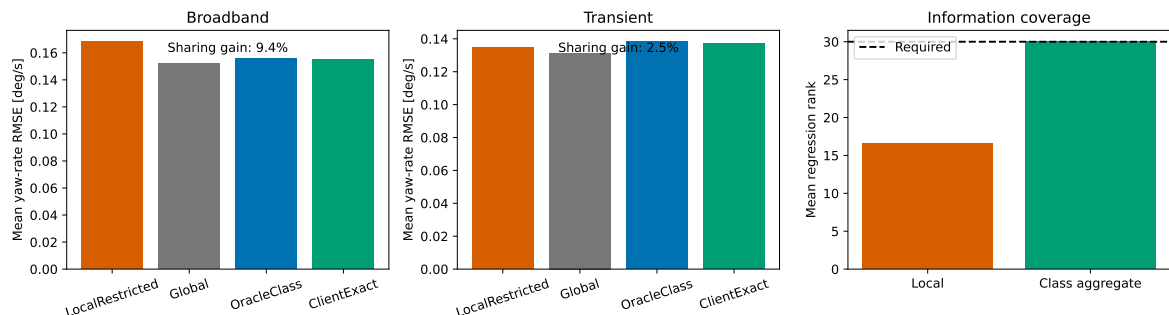


Figure 49: Experiment 10H matched-LPV control and information-coverage audit.

Result and interpretation. Complementary sharing produces a genuine closed-loop benefit. Global beats the optimistic LocalRestricted baseline in every fleet and endpoint, with positive

paired intervals. The effect is strongest in the fleet tail: worst-client tracking improves by 14.71% and 13.65%. Mean broadband tracking improves by 9.42%. Mean transient tracking improves by only 2.55%, so the deliberately strict requirement of at least 5% on every endpoint fails. Experiment 10H therefore supports a strong calibration-coverage and worst-client claim, not a uniformly large nominal-mean claim.

Global is also within the predefined 5% envelope of ClientExact on every endpoint and sometimes has lower yaw-only RMSE because pooling regularizes the fixed LQI design. Oracle-Class is worse than Global, confirming that physical class specialization is not needed for this control objective. The emerging FL story is consequently shared full-envelope LPV backbone identification, not clustered controller deployment.

Decision. The higher-order model resolves the core information-sharing problem without leaving LPV/LQI control. The next experiment should estimate this model from finite noisy local trajectories and compare LocalRestricted, centralized pooled, federated pooled, LocalFull, and ClientExact under a calibration-time budget. The primary endpoint should be worst-client tracking and the amount of local excitation needed to reach a target performance. Learned clustering should be reintroduced only if finite-data diagnostics show that Global pooling becomes biased.

Reproducibility. Configuration, implementation, and tests are in `code/config/experiment_10h.json`. The implementation is in `code/experiments/experiment_10h_higher_order_lpv_gate.py`; tests are in `code/tests/test_experiment_10h.py`. Development selection, the disclosed pilot, final client-level records, coverage diagnostics, paired comparisons, gate decisions, and provenance hashes are stored under `results`.

45 Experiment 10H-A: tire-force observer feasibility

Question. Experiment 10H assumes access to $x = [\beta, r, f_{yf}, f_{yr}, \delta_a]^\top$, where $f_y = F_y/m$. Physical tire forces and sideslip are not ordinary production measurements. Experiment 10H-A asks whether the latent states can be reconstructed from yaw rate, lateral acceleration, applied steering, and longitudinal speed, and whether the estimates are adequate for the frozen 10H LQI controller. This is an observer feasibility gate, not a federated identification result.

Measurement model and mass interpretation. The available output is

$$y = \begin{bmatrix} r & a_y & \delta_a \end{bmatrix}^\top = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} x + v. \quad (71)$$

Thus f_{yf} and f_{yr} are estimated directly as lateral-acceleration contributions. Their state definition does not require an externally supplied mass. Mass remains embedded in the unknown LPV dynamics.

For reference, an algebraic reconstruction follows from $a_y = f_{yf} + f_{yr}$ and $(I_z/m)\dot{r} = l_f f_{yf} - l_r f_{yr}$,

$$f_{yf} = \frac{l_r a_y + (I_z/m)\dot{r}}{l_f + l_r}, \quad f_{yr} = \frac{l_f a_y - (I_z/m)\dot{r}}{l_f + l_r}. \quad (72)$$

The implemented algebraic baseline is optimistic: it receives the true client geometry and I_z/m and uses a Savitzky–Golay yaw-acceleration estimate.

Observers and protocol. ExactKF uses each client’s exact scheduled matrices. GlobalKF uses one matrix schedule pooled across the fleet and therefore includes plant-model mismatch. Both measure r , a_y , and δ_a . The simulated standard deviations are 0.03 deg/s, 0.03 m/s², and 0.02 deg, with trajectory-constant bias standard deviations of 0.05 deg/s, 0.04 m/s², and 0.03 deg. Five process-noise scales are selected on development seeds 501–505 by a normalized GlobalKF sideslip- plus-force score. The frozen scale is 0.01. Confirmation uses seeds 511–520.

Every scheduled exact system has observability rank five. The maximum column-scaled observability condition over all clients and speeds is 196.9, so the problem is observable and not numerically singular under the assumed measurements.

Table 68: Experiment 10H-A confirmation estimation accuracy. Force errors are normalized by the RMS magnitude of the corresponding true contribution.

Method	β RMSE [deg]	Front NRMSE	Rear NRMSE
Algebraic	–	6.95%	6.47%
GlobalKF	0.0483	7.96%	5.32%
ExactKF	0.0091	3.72%	3.12%

The predefined estimation gates pass: GlobalKF sideslip RMSE is below 0.3 deg, and both axle-force NRMSE values are below 25%. The class breakdown identifies the model-mismatch tail. GlobalKF sideslip RMSE is 0.0208 deg for compact vehicles, 0.0416 deg for sport vehicles, and 0.0825 deg for SUVs; front-force NRMSE rises to 13.21% for SUVs.

Output-feedback control test. The frozen 10H Global LQI controller is evaluated with FullState, ExactKF, and GlobalKF feedback. No controller retuning is allowed. The integrator uses measured yaw rate in the observer cases. Table 69 reports the paired loss of GlobalKF relative to FullState.

Table 69: GlobalKF output-feedback degradation relative to full-state LQI.

Maneuver	Endpoint	Degradation	95% CI
Broadband	Mean client	13.08%	[12.37,13.94]%
Broadband	Worst client	37.34%	[34.67,40.38]%
Transient	Mean client	20.55%	[19.75,21.39]%
Transient	Worst client	44.76%	[40.16,50.09]%

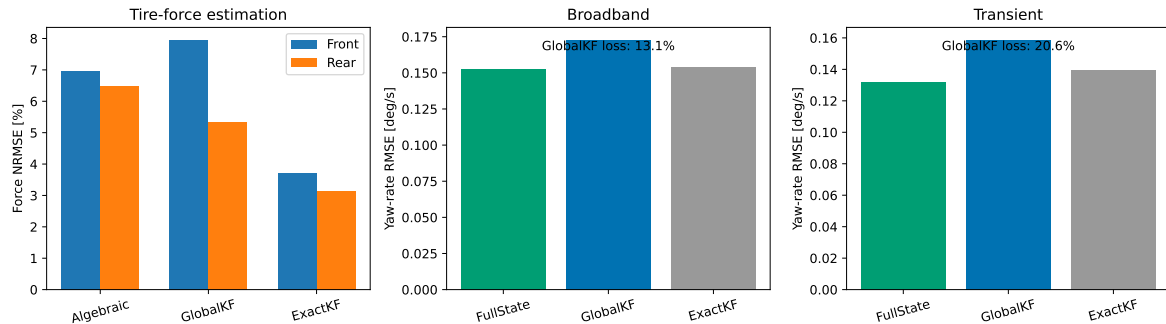


Figure 50: Experiment 10H-A estimation accuracy and output-feedback LQI performance.

All output-feedback trajectories remain finite, but the control gates fail. Even ExactKF introduces a smaller control loss, showing that the full-state LQI gain amplifies estimation noise

and bias. GlobalKF adds structured model mismatch, concentrated in SUV clients, and produces the larger fleet-tail degradation.

Conclusion and decision. Normalized axle-force estimation is feasible from the assumed production-like signals; unknown absolute mass is not a blocker. However, good state RMSE does not guarantee acceptable closed-loop performance with the existing aggressive full-state LQI. We must not use GlobalKF pseudo-states in the next federated identification experiment without resolving this interface.

The clean next diagnostic is a class-compatible or uncertainty-aware scheduled observer evaluated with the same frozen LQI, together with an ablation of sensor bias and observer-model mismatch. If compatible observer sharing closes the SUV tail without controller retuning, it provides a natural CFL role. If not, the five-state full-state formulation should be replaced by output-error LPV identification rather than increasingly complex observer tuning.

Reproducibility. Configuration and implementation are in `code/config/experiment_10ha.json` and `code/experiments/experiment_10ha_tire_force_observer.py`. Development selection, client-level estimation and control records, observability results, paired intervals, gate decisions, and provenance hashes are stored under `results`.

46 Experiment 10H-B: observer-mechanism audit

Question and preregistered logic. Experiment 10H-A established that normalized tire forces and sideslip are observable, but a fleet-global observer caused a large closed-loop tail loss. Experiment 10H-B separates three possible causes: sensor bias, mismatch within a compatible vehicle class, and mismatch caused by pooling incompatible classes. The key question is whether class-compatible observer sharing removes enough of the GlobalKF loss to justify a clustered-federated observer stage.

The fleet, ten confirmation seeds (511–520), H03 Global LQI controller, measurement covariance, observer process covariance, maneuvers, and all controller parameters are frozen from Experiments 10H and 10H-A. No tuning is performed in 10H-B. The predefined gates require the oracle-class observer to recover at least 25% of the GlobalKF-to-ExactKF tracking gap, remain within 10% of ExactKF on average, outperform GlobalKF at all four mean/worst and broadband/transient endpoints, and keep every trajectory finite.

Factorial observer comparison. The same measurements $y = [r, a_y, \delta_a]^\top$ are supplied to three scheduled Kalman observers:

- ExactKF uses the exact client matrices and isolates the observer and measurement interface;
- OracleClassKF uses matrices averaged only within the known compact, sport, or SUV class and isolates within-class mismatch;
- GlobalKF averages all classes and adds cross-class mismatch.

Each observer is tested with measurement noise only and with the same noise plus trajectory-constant sensor biases. Common random numbers are used for every paired observer comparison. FullState is retained as a reference, but the ExactKF–FullState contrast must be interpreted as the net effect of filtered output feedback, not as a pure additive noise effect: filtering also changes the effective behavior of the aggressive full-state controller.

Table 70: Experiment 10H-B mean yaw-rate tracking RMSE [deg/s].

Maneuver	Measurement condition	ExactKF	OracleClassKF	GlobalKF
Broadband	Noise only	0.14535	0.14646	0.16642
Broadband	Noise plus bias	0.15335	0.15441	0.17402
Transient	Noise only	0.13121	0.13291	0.14880
Transient	Noise plus bias	0.13931	0.14094	0.15647

Mechanism attribution. Table 71 reports paired percentage changes. Bias is material but is not the main explanation for GlobalKF’s poor tail. Within a true class, residual model mismatch costs only 0.69–2.42%. Moving from the class observer to one fleet-global observer adds 11.02–33.79%. Thus the damaging mismatch is predominantly between classes, not between clients within a class.

Table 71: Incremental observer mechanisms. Positive values denote worse tracking than the preceding baseline.

Maneuver	Endpoint	Sensor bias	Within class	Cross class
Broadband	Mean	5.50%	0.69%	12.70%
Broadband	Worst	6.32%	2.42%	33.79%
Transient	Mean	6.18%	1.16%	11.02%
Transient	Worst	11.29%	2.13%	28.97%

Table 72: OracleClassKF recovery of the GlobalKF-to-ExactKF gap. Intervals are paired seed-bootstrap 95% intervals.

Maneuver	Endpoint	Gap recovery	95% CI
Broadband	Mean	94.87%	[94.36,95.40]%
Broadband	Worst	93.83%	[90.74,96.77]%
Transient	Mean	90.52%	[89.19,92.01]%
Transient	Worst	94.34%	[90.48,98.37]%

Every output-feedback trajectory is finite and all predefined gates pass.

Interpretation and decision. This audit reverses the broad conclusion one might draw from 10H-A. Shared observers are not intrinsically unsuitable. A single observer pooled across incompatible vehicle classes is unsuitable; a class-compatible shared observer is nearly as effective as a client-exact observer. This identifies a specific and control-relevant role for clustered FL: learn compatible groups and aggregate their higher-order LPV observer models without centralizing client trajectories. Sensor-bias estimation remains a separate engineering improvement and should not be credited to FL.

The result is still oracle evidence because the class labels are known. The next experiment should replace those labels by learned assignments using only federated model updates or privacy-preserving update summaries, freeze the number of groups from data, and test whether the learned clustered observer retains the oracle gap recovery on untouched fleets. A failure there would limit the claim to known-class sharing; a success would connect complementary coverage, clustered FL, state estimation, and LQI control in one chain.

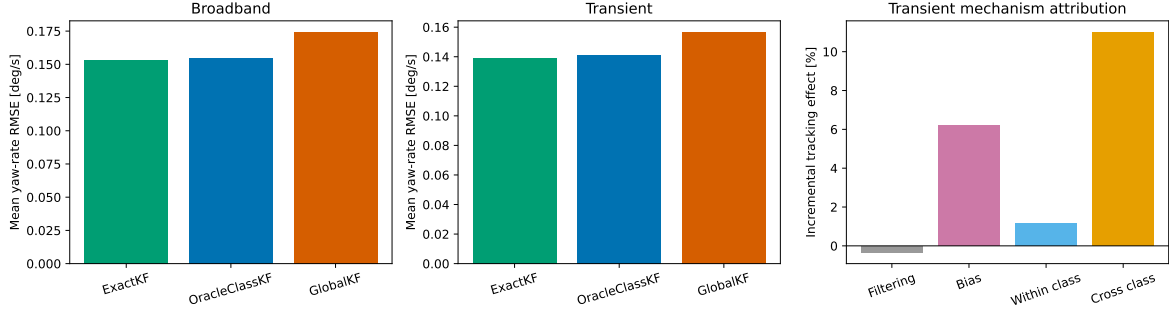


Figure 51: Biased output-feedback performance and incremental mechanism attribution in Experiment 10H-B.

Reproducibility. Configuration and implementation are in `code/config/experiment_10hb.json` and `code/experiments/experiment_10hb_observer_mechanism.py`. Client-level records, seed summaries, paired mechanism effects, gap-recovery intervals, gate decisions, and provenance hashes are stored under `results`.

47 Experiment 10H-C: learned observer groups with unknown cardinality

Question. Experiment 10H-B showed that an oracle-class observer recovers most of the control loss caused by one fleet-global observer. Experiment 10H-C removes the known-class assumption. It asks whether clients can be grouped from partial observer-model updates alone, whether the number of groups can be inferred rather than prescribed, and whether the resulting observers preserve the oracle control benefit on independent fleets.

Federated update abstraction. Each client observes only its frozen Experiment 10H speed block. It constructs discrete LPV matrix updates for $[A(\rho) \ B(\rho)] \in \mathbb{R}^{5 \times 6}$ at those speeds and transmits the update summaries rather than trajectories. Independent Gaussian perturbations with standard deviation 0.5% of the fleetwise variation of each matrix entry represent finite local update uncertainty. The server never receives physical class labels. Those labels are retained only for the OracleClassKF comparator and retrospective adjusted Rand index (ARI).

This is deliberately an update-level grouping gate. The updates are generated from the simulated client matrices with controlled perturbations, not by an end-to-end output-error or EM observer fit from raw r , a_y , and steering measurements. Consequently, 10H-C establishes the clustering and control mechanism but does not yet establish the complete local estimation pipeline.

Unknown-cardinality mixture. Each group is represented by the same five-function LPV basis used in 10H. For a candidate number of groups $K \in \{1, \dots, 5\}$, the algorithm alternates between ridge fitting group schedules from the assigned partial updates and assigning every client to the schedule with minimum standardized residual on its locally observed speeds. Sixteen deterministic restarts are used, groups must contain at least five clients, and BIC selects K . Development seeds 521–525 freeze this rule; confirmation uses untouched seeds 531–540.

The controller, H03 weights, observer process covariance, measurement noise and trajectory-constant bias, maneuvers, and common-random-number protocol are frozen from 10H-A/B. The complete H03 Global LQI schedule is designed from the same fleet-global reference construction as in 10H-B. Therefore, ExactKF, OracleClassKF, LearnedClusterKF, and GlobalKF differ only in their observer model and scheduled observer gains.

Group discovery. BIC selects three groups in four of five development fleets and nine of ten confirmation fleets. The remaining fleet in each phase is split into four groups. Confirmation mean ARI is 0.986 and the minimum is 0.859. Thus the method usually recovers the three physical classes, while occasionally over-partitioning one class rather than merging incompatible classes.

Table 73: Experiment 10H-C confirmation yaw-rate tracking RMSE [deg/s] with sensor noise and trajectory-constant bias.

Maneuver	Observer	Mean client	Worst client
Broadband	ExactKF	0.15434	0.21571
Broadband	OracleClassKF	0.15547	0.22105
Broadband	LearnedClusterKF	0.15539	0.22105
Broadband	GlobalKF	0.17134	0.28257
Transient	ExactKF	0.13953	0.18882
Transient	OracleClassKF	0.14249	0.19612
Transient	LearnedClusterKF	0.14248	0.19612
Transient	GlobalKF	0.15892	0.25434

LearnedClusterKF improves over GlobalKF by 9.31% [9.00,9.65]% in broadband mean tracking and 10.34% [9.89,10.74]% in transient mean tracking. Worst-client improvements are 21.74% [19.97,23.53]% and 22.92% [20.74,24.81]%, respectively. All intervals are paired seed-bootstrap 95% intervals.

Table 74: LearnedClusterKF retention of the GlobalKF-to-ExactKF control gap.

Maneuver	Endpoint	Gap retained	95% CI
Broadband	Mean	93.89%	[91.26,96.27]%
Broadband	Worst	92.48%	[89.83,94.48]%
Transient	Mean	85.19%	[81.19,88.95]%
Transient	Worst	89.75%	[86.33,93.41]%

At the displayed precision, LearnedClusterKF matches OracleClassKF on both worst-client endpoints and is slightly better on both mean endpoints. This does not imply superiority to oracle grouping: the occasional four-group split gives the learned method additional specialization. The appropriate claim is oracle-level control performance with a data-selected model count.

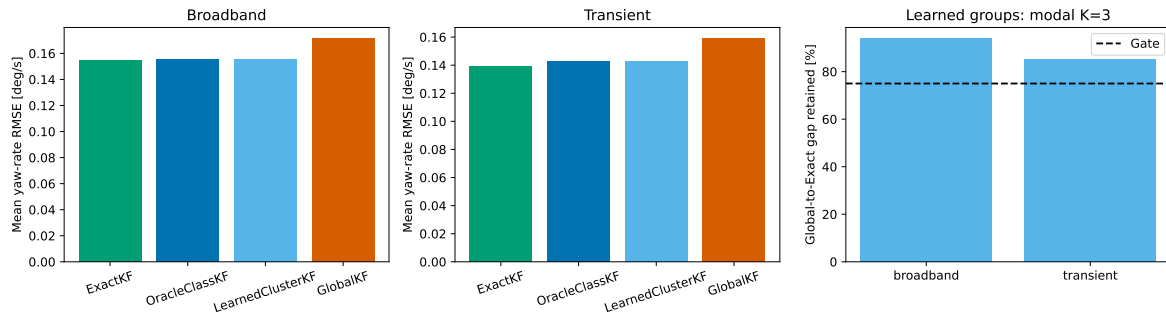


Figure 52: Confirmation output-feedback performance and retained observer gap in Experiment 10H-C.

Gate decision and scientific meaning. All predefined gates pass: modal $K = 3$, positive gap-recovery intervals, at least 75% retained gap at every endpoint, no more than 5% tracking excess over OracleClassKF, and finite trajectories throughout. Unlike the earlier LPV experiments where collaboration was mainly a model-count compromise, 10H-C exhibits a direct control benefit from label-free compatible sharing. Its mechanism is clear: sharing completes each client’s operating-point coverage, while clustering prevents the damaging cross-class observer bias identified in 10H-B.

The remaining step before claiming an end-to-end CFL observer is to replace the perturbed matrix-update oracle by locally learned observer updates from the available production-like outputs. That experiment must preserve the unknown- K discovery and closed-loop benefit under local estimation error. A privacy or secure-aggregation study should follow only after this end-to-end gate, because privacy noise would otherwise be applied to an optimistic update interface.

Reproducibility. Configuration and implementation are in `code/config/experiment_10hc.json` and `code/experiments/experiment_10hc_learned_observer_groups.py`. Client-level control records, candidate BIC scores, selected cardinalities, ARI diagnostics, paired comparisons, gap-retention intervals, gate decisions, and provenance hashes are stored under `results`.

48 Experiment 10H-D: observer groups from measured outputs

Question and gate. Experiment 10H-C learned useful unknown-cardinality observer groups from controlled noisy matrix summaries, but those summaries were derived from the known simulation matrices. Experiment 10H-D replaces that optimistic interface by local identification from production-like measurements. Its development gate asks whether measured-output updates recover the physical grouping and improve all mean and worst-client control endpoints over a measured-output GlobalKF by at least 5%. Confirmation seeds 551–560 remain sealed unless the development gate passes.

Local measured-output pipeline. Each client records yaw rate r , lateral acceleration a_y , applied steering δ_a , longitudinal speed, and steering command over its single 10H speed block and frequency band. The latent states β , f_{yf} , and f_{yr} are never supplied to learning. One fixed nominal physics model, obtained by averaging the three public class specifications without client labels, defines a scheduled Kalman filter for every client. Offline centering removes the estimated constant sensor offset from each zero-mean calibration trajectory.

The nominal filter produces local pseudo-states. At each observed speed, the client fits

$$\hat{x}_{k+1} \approx \hat{A}_i(\rho)\hat{x}_k + \hat{B}_i(\rho)u_k \quad (73)$$

with a prior penalty toward the same nominal matrices. Only the fitted partial matrix summaries are transmitted. The unknown- K BIC mixture from 10H-C is then applied unchanged. True states, matrices, and class labels are used only for retrospective diagnostics and ExactKF/OracleClassKF comparisons.

Development result. The gate fails consistently on seeds 541–545. BIC selects $K = \{2, 2, 2, 1, 2\}$, giving modal $K = 2$ rather than three and zero three-group selections. Mean ARI is 0.365, with one fleet at zero. The locally fitted matrices have 52.21% mean relative error. The very small pseudo-state one-step RMSE of 1.50×10^{-3} is therefore misleading: the nominal filter creates trajectories that are predictable in its own distorted latent coordinates but do not recover the true physical state transition. Measured outputs alone do not fix that latent-coordinate ambiguity under this simple one-pass pseudo-state regression.

Table 75: Experiment 10H-D development tracking RMSE [deg/s].

Maneuver	Observer	Mean client	Worst client
Broadband	ExactKF	0.15441	0.21499
Broadband	OracleClassKF	0.15642	0.23922
Broadband	LearnedClusterKF	0.15590	0.24422
Broadband	GlobalKF	0.15706	0.25139
Broadband	NominalKF	0.16199	0.26107
Transient	ExactKF	0.14024	0.19042
Transient	OracleClassKF	0.14158	0.20843
Transient	LearnedClusterKF	0.14287	0.21577
Transient	GlobalKF	0.14385	0.22114
Transient	NominalKF	0.14574	0.22467

LearnedClusterKF improves over GlobalKF by only 0.73% in broadband mean and 0.68% in transient mean tracking. Worst-client improvements are 2.82% and 2.45%. Their paired intervals are positive, so the effect is reproducible, but every endpoint misses the predefined 5% relevance threshold. Moreover, LearnedClusterKF is 2.14% worse than OracleClassKF on the broadband tail and 3.54% worse on the transient tail. All trajectories remain finite.

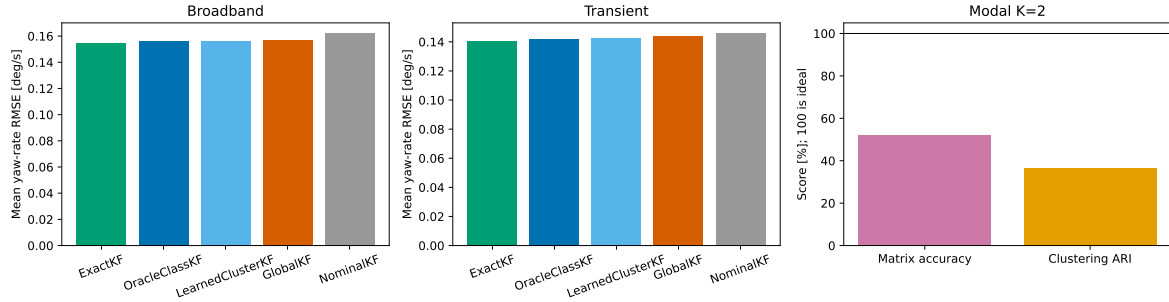


Figure 53: Development-only measured-output performance, local matrix accuracy, and learned grouping quality in Experiment 10H-D.

Interpretation and decision. The positive 10H-C result does not survive the simplest end-to-end local identification mechanism. This is an informative boundary: observability of the five-state model does not imply that a mismatched nominal filter followed by pseudo-state regression identifies physically comparable client updates. Low pseudo-state prediction error can coexist with poor matrix accuracy and poor clustering because all clients inherit the nominal observer’s coordinate and regularization bias.

The reserved confirmation fleets are not opened. Further progress requires a joint state-parameter method that respects the fixed physical coordinates, such as structured EM/prediction-error identification with the known measurement map, or direct clustering of output innovations with a subsequent groupwise model update. Simply reducing the ridge penalty or increasing the number of mixture restarts would tune symptoms rather than resolve the identification ambiguity. Privacy experiments are premature until this gate is recovered.

Reproducibility. Configuration and implementation are in `code/config/experiment_10hd.json` and `code/experiments/experiment_10hd_measured_output_observer_groups.py`. Development client records, candidate BIC scores, local-fit diagnostics, paired comparisons, failed gate decisions, reserved confirmation seeds, and provenance hashes are stored under `results`.

49 Experiment 10H-E: leakage-safe innovation separability

Question and leakage constraint. Experiment 10H-D showed that matrices regressed from a single nominal observer’s pseudo-states are not physically comparable. Experiment 10H-E asks the prerequisite question for a mixture-of-state-space-models approach: do the locally measured innovation sequences already contain enough information to distinguish compatible vehicle groups before state or matrix learning?

The grouping pipeline is deliberately leakage-safe. It never receives a client’s true mass, inertia, tire stiffness, geometry, relaxation length, actuator constant, latent state, LPV matrix, or physical class label. Every client uses the same fixed nominal observer specified independently in the configuration as an engineering design prior. Its values are not the true parameters of any client and are not adapted from fleet labels. True class labels are opened only after unsupervised fitting for retrospective ARI, balanced accuracy, and class-recall diagnostics.

Innovation signatures. For every locally observed speed, the client calculates

$$\nu_{i,k} = y_{i,k} - C\hat{x}_{i,k|k-1}, \quad y = [r, a_y, \delta_a]^\top, \quad (74)$$

using only measured outputs, steering command, and speed. The trajectory is centered and whitened by the fixed sensor-noise covariance. The communicated signature contains aggregate covariance, lag-one autocorrelation, normalized innovation-energy statistics, innovation–input cross moments, and three-band spectral energies. Raw measurements, pseudo-states, and trajectories remain local.

Because every client covers only one speed block, known block effects are removed using blockwise fleet means and standard deviations. PCA retains at most eight components. Diagonal-covariance Gaussian mixtures with 24 deterministic restarts test $K \in \{1, \dots, 5\}$; BIC selects the cardinality subject to at least five clients per group. Development uses seeds 561–570. Confirmation seeds 571–580 remain reserved.

Result. BIC selects $K = 1$ in all ten development fleets. The resulting ARI is zero and balanced accuracy is 1/3. The entropy gate passes trivially because a one-group solution has no membership ambiguity; it must not be interpreted as confident physical classification.

A retrospective fixed- $K = 3$ diagnostic separates model-selection failure from feature-separability failure. It obtains mean ARI 0.455 and mean balanced accuracy 0.747. Results are inconsistent: two fleets reach ARI above 0.80 and balanced accuracy 0.933, whereas several fleets completely lose one physical class. The minimum class recall over all development fleets is zero. BIC for $K = 3$ is 24–54 points worse than for $K = 1$, so the evidence for three groups is not merely just below a selection threshold.

Table 76: Experiment 10H-E development innovation-separability diagnostics.

Diagnostic	Unknown- K result	Fixed- $K = 3$ diagnostic
Modal selected groups	1	3 fixed
Mean ARI	0.000	0.455
Mean balanced accuracy	0.333	0.747
Minimum class recall	0.000	0.000
Fleets with ARI > 0.8	0/10	2/10

Interpretation and decision. The nominal observer’s innovation process contains some class information, but the aggregate signature is not a reliable stand-alone initializer for a three-component EM algorithm. The failure is not caused by access to hidden parameters: none

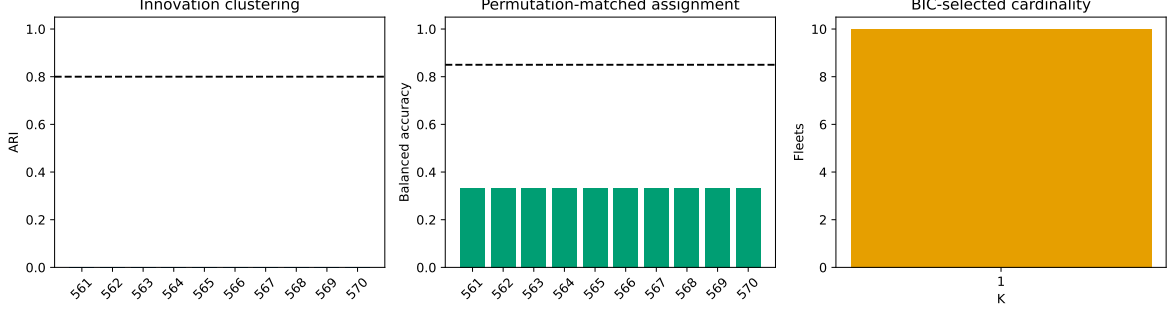


Figure 54: Development ARI, permutation-matched balanced accuracy, and BIC-selected group count for leakage-safe innovation clustering.

enters the grouping pipeline. Instead, restricted excitation, sensor bias, speed-block variation, and the common observer compress different physical mismatches into partially overlapping innovation statistics.

The development gate fails and confirmation is not opened. This result does not rule out joint state–parameter EM. It rules out using one-shot clustering of hand-designed aggregate innovations as its sole initialization. A stronger next test should compare complete trajectory likelihoods under several deliberately diverse candidate physical models, or initialize a joint structured EM with multiple restarts and allow likelihood-based soft memberships to evolve together with the group models. Those candidate models must be fixed design hypotheses or learned from outputs; they must not be constructed from the clients’ hidden physical parameters.

Reproducibility. Configuration and implementation are in `code/config/experiment_10he.json` and `code/experiments/experiment_10he_innovation_separability.py`. Client assignments, candidate BIC values, fixed- K retrospective diagnostics, failed gate decisions, reserved confirmation seeds, leakage declaration, and provenance hashes are stored under `results`.

50 Experiment 10H-F: output identifiability and physical-score audit

Question and parameterization. Experiment 10H-E found that generic innovation summaries do not reliably separate the vehicle classes. Experiment 10H-F asks two more fundamental questions: are physically meaningful LPV coefficients identifiable from the complete measured-output prediction error, and do likelihood scores at a common nominal model provide a better label-free grouping signal?

The estimator does not receive mass, inertia, tire stiffness, geometry, relaxation lengths, actuator constants, latent states, client matrices, or class labels. Instead, it operates in nine effective continuous-time LPV coordinates:

$$\theta = [a_f^r, a_r^r, q_f^\beta, q_f^r, q_r^\beta, q_r^r, s_f, s_r, s_\delta]^\top, \quad (75)$$

where the first two coefficients map normalized axle forces to yaw acceleration, the next four govern front/rear force response to sideslip and yaw rate, s_f and s_r are relaxation decay rates, and s_δ is the actuator rate. These combinations generate the structured $A(v; \theta)$ and $B(v; \theta)$ matrices directly. They avoid claiming separate identification of quantities such as C_f and m when only their combinations affect the model.

Measured-output information. Each calibration segment begins with 100 zero-command samples to estimate the trajectory-constant sensor offset, followed by the client’s frozen 10H

frequency-band excitation. A KF at one fixed nominal design produces whitened innovations. Central finite differences with a 0.02 log-parameter step give the innovation Jacobian J_i . The local observed-information approximation and physical score are

$$F_i = J_i^\top J_i, \quad s_i = J_i^\top \bar{v}_i. \quad (76)$$

All quantities come from r , a_y , applied steering, steering command, and speed. True classes are opened only afterward to aggregate compatible-client information and evaluate clustering.

Across 300 development clients, 296 local information matrices have numerical rank nine and four have rank eight at a relative 10^{-6} tolerance. Thus the expected strict local-rank-deficiency mechanism does not occur. Local identification is nevertheless weakly conditioned: the median condition number is 5.95×10^4 and the maximum is 9.64×10^5 . Every compatible-class aggregate is full rank, with condition numbers between 1.16×10^4 and 2.15×10^4 . Sharing therefore improves numerical conditioning and expected estimation variance rather than restoring literal rank.

Physical-score grouping. Client scores are whitened by the fleet information matrix, residualized for known speed-block effects, standardized, and clustered with the same unknown- K diagonal-GMM/BIC protocol as 10H-E. BIC selects one group in two fleets, two groups in six fleets, and three groups in two fleets. Modal $K = 2$ and the three-group rate is 20%.

The retrospective fixed- $K = 3$ diagnostic obtains mean ARI 0.209, mean balanced accuracy 0.593, and minimum class recall zero. This is worse than the generic 10H-E innovation signature. A first-order score evaluated at one common nominal point therefore does not represent the nonlinear distance between a client trajectory and the eventual group models.

Table 77: Experiment 10H-F development diagnostics.

Diagnostic	Result
Local full-rank fraction	98.67%
Median local information condition	5.95×10^4
Maximum local information condition	9.64×10^5
Compatible aggregates full rank	30/30
Maximum compatible condition	2.15×10^4
Modal BIC group count	2
Fixed- $K = 3$ mean ARI	0.209
Fixed- $K = 3$ balanced accuracy	0.593

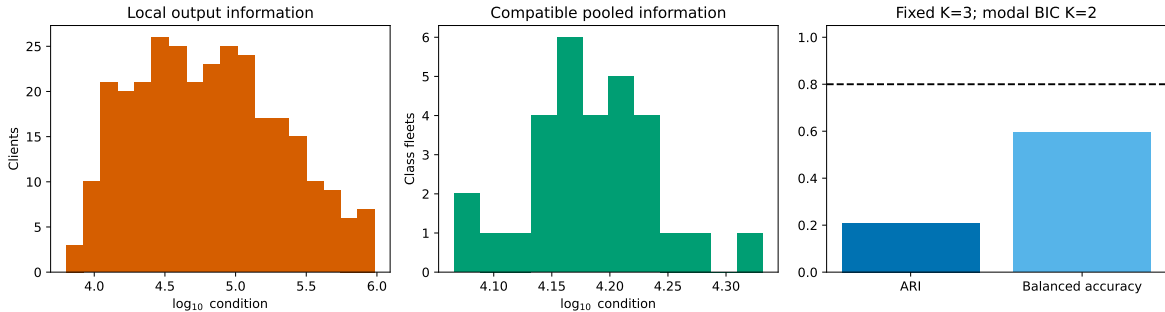


Figure 55: Local and compatible-pooled information conditioning together with fixed- K physical-score clustering performance.

Gate decision and interpretation. The development gate fails. Its two parts have different implications. The output-identifiability audit is encouraging: the selected effective coordinates are visible, compatible aggregation is full rank, and pooling improves conditioning. The clustering audit is negative: neither generic innovation summaries nor a single nominal-point score reliably initializes the physical groups.

Consequently, the next method should not hard-cluster first-order scores before model learning. The defensible next gate is a structured global prediction-error fit from measured outputs, with held-out likelihood and parameter-profile diagnostics. If that optimizer reliably learns one physical model without hidden-parameter initialization, it can be extended to a known- K multi-start mixture in which assignments and models evolve jointly. Candidate mixture models should be generated by physically bounded split-and-perturb restarts around the learned global model, not by true class centers or client parameters. Confirmation seeds 591–600 remain sealed.

Reproducibility. Configuration and implementation are in `code/config/experiment_10hf.json` and `code/experiments/experiment_10hf_output_identifiability.py`. Client information diagnostics, compatible and fleet aggregates, candidate BIC scores, fixed- K clustering metrics, gate decisions, leakage declaration, reserved confirmation seeds, and provenance hashes are stored under `results`.

51 Experiment 10H-G: global measured-output prediction-error gate

Purpose. Experiments 10H-E and 10H-F showed that neither generic innovation summaries nor first-order scores at a fixed nominal model reliably reveal the latent vehicle groups. Before attempting a mixture or federated EM method, Experiment 10H-G asks a narrower prerequisite question: can one structured global LPV model be learned reliably from measured input–output trajectories without true states, physical parameters, matrices, or class labels?

The nine effective coordinates and structured $A(v; \theta), B(v; \theta)$ model from 10H-F are retained. A steady-state KF produces normalized innovations from measured yaw rate, lateral acceleration, applied steering, steering command, and speed. The objective is the mean squared normalized innovation. The fit uses 20 clients per fleet; every third client is held out by list position, giving ten label-blind held-out clients. Each development fleet is optimized from three physically bounded initializations. The best training solution is evaluated on the untouched clients. Conditional profiles perturb each fitted log-coordinate by $\{-0.16, -0.08, 0, 0.08, 0.16\}$ while holding the others fixed.

The bounds are fixed multipliers, 0.35 and 3.0, of one public nominal design; they do not use the simulated client truths. Class labels and hidden physical parameters are opened only after fitting for retrospective interpretation. Seeds 611–620 remain sealed.

Prediction result. The learned model improves held-out normalized prediction error in every development fleet. The improvement ranges from 9.50% to 21.86%, with a mean of 15.76%. Thus measured outputs contain useful information beyond the nominal observer, and the structured global-fit direction is not rejected on predictive grounds.

Optimization and identifiability result. The stronger optimizer gate fails. Every selected run reaches the declared 25-iteration limit, so the optimizer-success rate is zero. Across the three starts, the worst objective spread is 4.70% and the worst coordinate coefficient of variation is 0.330, both above the declared tolerances of 0.5% and 0.08. Three of five selected fits place one of nine coordinates on a bound. The minimum conditional edge increase is -0.057% , meaning that at least one selected point is not even a coordinate-wise minimum at the audited profile radius.

Retrospectively, the fitted vector has mean log-RMSE 0.641 relative to the arithmetic mean of the training clients’ effective truths. That diagnostic is not used for fitting or gate decisions.

Table 78: Experiment 10H-G development diagnostics.

Diagnostic	Result
Mean held-out prediction improvement	15.76%
Minimum held-out prediction improvement	9.50%
Optimizer success rate	0/5
Maximum restart objective spread	4.70%
Maximum restart coordinate CV	0.330
Minimum conditional-profile edge increase	−0.057%
Maximum bound-active fraction	1/9
Development gate	Fail

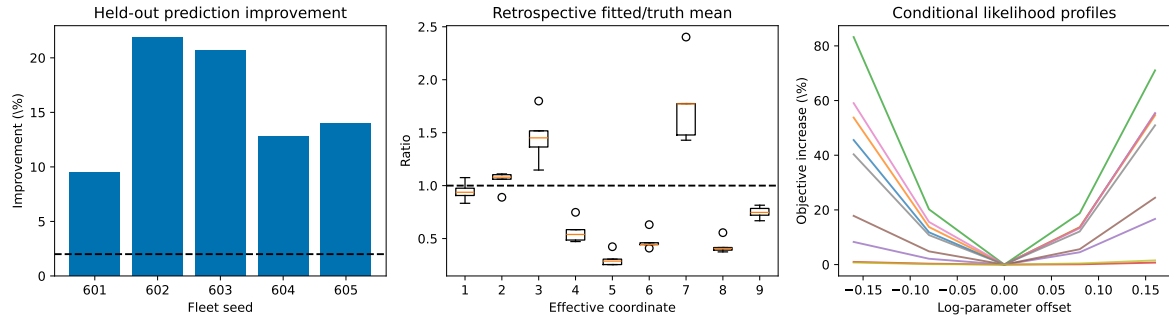


Figure 56: Held-out prediction improvement, retrospective fitted-to-fleet-mean ratios, and conditional objective profiles for the global structured fit.

Decision. The implementation and scoring convention in this experiment are superseded by the covariance and full-likelihood repair in Section 52. The archived results are retained for provenance; their objective percentages are not likelihood improvements.

The positive held-out result and failed optimizer gate must not be conflated. They establish that raw-output fitting is informative, but not that the nine-coordinate solution is unique or sufficiently reproducible to seed a mixture. Starting multi-model EM now would mix physical grouping with initialization artifacts and weak directions.

The next experiment should therefore repair the $K = 1$ estimator before any clustering: introduce exact or automatic-differentiation gradients, scale the coordinates using the 10H-F information matrix, and compare a staged fit (actuator/relaxation dynamics followed by force/yaw coefficients) with the joint fit. A successful repair must converge before its iteration budget, collapse restart dispersion, retain held-out improvement, avoid active bounds, and show positive local profile curvature. Only then is a known- K joint membership/model experiment warranted.

Reproducibility. The preregistered configuration and implementation are in `code/config/experiment_10hg.json` and `code/experiments/experiment_10hg_global_prediction_error.py`. Per-fleet summaries, fitted coordinates, restart outcomes, profile values, gate decisions, leakage declaration, and provenance hashes are stored under `results`.

52 Experiment 10H-H: repair and validation of the single-model estimator

Purpose and separation. Experiment 10H-G improved a normalized innovation objective, but its selected fits exhausted their iteration budget. Source inspection also found an extra covariance prediction after the prior Riccati solution and omission of the covariance log-determinant. Experiment 10H-H repairs these issues before any mixture or joint membership/model estimation. It reuses the five opened development fleets 601–605, their measured records, the label-blind 20/10 training/held-out client split, and the three deterministic initializations. This is a paired development repair, not a new blind confirmation. Seeds 611–620 remain sealed.

The estimator receives only yaw rate, lateral acceleration, applied steering, steering command, and speed. True states, client parameters, class labels, and exact client matrices remain behind the simulation boundary. A fixed public nominal design supplies initial coordinates and bounds. The inherited process-noise design and measurement covariance are frozen. Local trajectory constant bias is estimated from the same quiet segment as in 10H-G.

Corrected covariance and objective. At each recorded speed, let $A(\theta), B(\theta)$ denote the structured five-state model and let C be its fixed measurement matrix. The filtering Riccati equation is

$$P = A(P - PC^\top(CPC^\top + R)^{-1}CP)A^\top + Q. \quad (77)$$

Its solution is the *prior* covariance. Consequently,

$$S = CPC^\top + R, \quad K = PC^\top S^{-1}, \quad (78)$$

without the additional $APA^\top + Q$ prediction used in 10H-G. Every segment starts with zero posterior mean and stationary covariance. With $\nu_t = y_t - C(A\hat{x}_{t-1} + Bu_t)$, the repaired objective is

$$\mathcal{L}(\theta) = \frac{1}{3N_s} \sum_{t=1}^{N_s} \left[\nu_t^\top S^{-1} \nu_t + \log \det S - \log \det R \right]. \quad (79)$$

The sensor-noise determinant is a parameter-independent reference. It makes the reported objective positive and fixes the scale of its percentage changes, but does not affect its minimizer. This is a stationary Gaussian *working likelihood*: one shared model is fitted to heterogeneous vehicles, Q is design tuning rather than identified process noise, and quiet-segment bias-estimation uncertainty is not modeled explicitly. It is not a correctly specified stochastic model of the entire fleet. The fixed process-noise scale 0.01 was originally selected in 10H-A using latent-state estimation errors. No 10H-H client truths are used to retune it, but this inherited tuning means the complete covariance-selection pipeline has not yet been demonstrated using measured outputs alone. The present experiment validates the measured-array estimator interface with that design setting held fixed. Subsequent covariance selection must use measured-output validation if an entirely output-only training pipeline is claimed.

An independent predictive endpoint uses fixed sensor weights,

$$\mathcal{E}_R(\theta) = \frac{1}{3N_s} \sum_t \nu_t^\top R^{-1} \nu_t. \quad (80)$$

Improvement in this endpoint cannot be obtained merely by increasing S . Both endpoints concern one-step measured-output prediction, not closed-loop tracking or latent-state estimation.

Derivatives and optimization. Exact zero-order-hold matrix derivatives are evaluated with matrix-exponential Fréchet derivatives. Put $M = I - KC$, $U = MP$, and $F = AM$. For log-coordinate θ_j , implicit Riccati differentiation gives

$$P_j - FP_jF^\top = A_jUA_j^\top + AU A_j^\top. \quad (81)$$

This discrete Lyapunov equation supplies S_j and K_j . Forward sensitivities propagate through the innovation and posterior-mean recursion. The likelihood gradient includes all these derivatives and $\log \det S$. Training-only diagonal information scaling preserves the original public box bounds, from 0.35 to 3.0 times the nominal coordinates.

The frozen comparison includes a direct joint fit and a staged fit: actuator rate first, relaxation rates second, and force/yaw coefficients third, followed by full joint optimization. Both have a 300-iteration joint budget; each restricted stage has an additional 80-iteration budget. This larger budget is part of the repair, so numerical improvement is not attributed to gradients alone. The full physical gate retains the declared restart, stationarity, held-out, interior-bound, Hessian, and conditional-profile requirements.

Numerical and predictive results. Analytic gradients agree with central differences on every real development fleet, with maximum norm-relative error 6.39×10^{-8} . Independent tests also verify matrix derivatives, the Riccati fixed point, likelihood agreement with a separate filter recursion, and constraint-gradient signs.

Every final optimizer call returns a success flag, but one joint-fit restart on seed 605 terminates by small function reduction with projected gradient 0.551. Such a flag does not establish convergence. Its resulting coordinate dispersion causes the joint numerical gate to fail. Staging repairs this: all 15 staged restarts pass the projected-gradient check, their worst coordinate CV is 1.71×10^{-6} , and their maximum objective spread is $1.79 \times 10^{-10}\%$. Joint polishing takes 40–107 iterations, below its 300-iteration budget. No likelihood evaluation is invalid.

Table 79: 10H-H results on five opened development fleets, with three starts per method and fleet. Prediction improvement is relative to the corrected nominal model.

Diagnostic	Joint	Staged
Stationary final restarts	14/15	15/15
Mean held-out $\mathcal{L}$ improvement	18.57%	18.57%
Minimum held-out $\mathcal{L}$ improvement	12.24%	12.24%
Mean held-out $\mathcal{E}_R$ improvement	18.02%	18.02%
Minimum held-out $\mathcal{E}_R$ improvement	11.90%	11.90%
Maximum restart parameter CV	0.1365	1.71×10^{-6}
Maximum projected gradient	0.551	1.25×10^{-5}
Fleets with an active bound	5/5	5/5
Numerical repair gate	Fail	Pass
Full development gate	Fail	Fail

The selected joint and staged solutions have effectively identical objectives on every fleet. Mean fixed-weight prediction improvement is 18.02%, with fleet gains from 11.90% to 25.38%. Re-scoring the archived 10H-G vectors with the corrected filter and objective gives 16.02% mean likelihood and 15.71% fixed-weight prediction improvement. These are valid paired comparisons under the repaired scoring rule; the original 10H-G percentage should not be compared directly with the new likelihood percentage.

Boundary interpretation: a post-hoc diagnostic. The rear sideslip-to-force coefficient reaches its lower bound in all five fleets. The corresponding front coefficient also reaches its upper bound in three fleets. The full Hessian has a negative eigenvalue and clipped conditional-profile edges can have zero increase, so the predeclared interior-solution gate fails. These observations must not be misread as proof that the staged optimizer has missed a constrained minimum.

A separate post-hoc audit leaves fitted vectors and all gates unchanged. Every active-bound gradient has the appropriate Karush–Kuhn–Tucker sign. The Hessian restricted to free

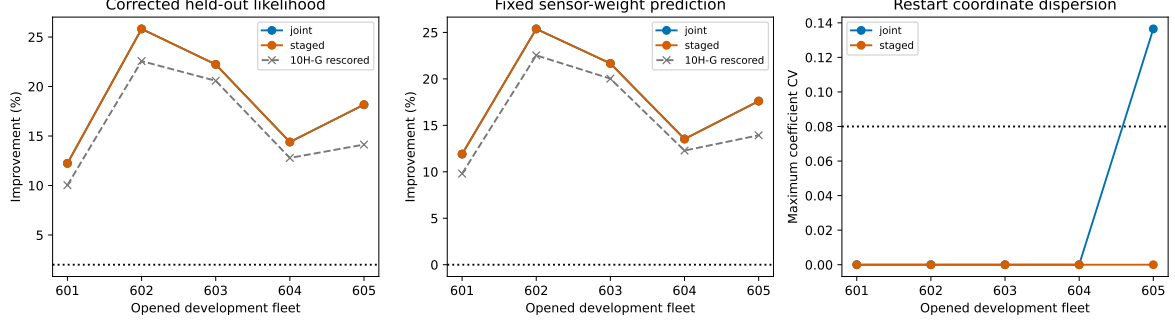


Figure 57: Corrected held-out likelihood and fixed-weight prediction improvement, plus restart dispersion. Archived 10H-G vectors are re-scored without refitting.

coordinates is positive definite in every selected fit, with minimum eigenvalue 0.0201. Thus the selected fits satisfy the audited local constrained-minimum conditions. A zero outward profile after clipping reflects the boundary, not an experimentally observed flat direction.

Physical-model limitation. The same audit reveals that the nine effective coordinates do not all vary independently in a physical bicycle model. Their definitions imply

$$\frac{\varphi_{\text{yaw},f}\varphi_{\text{front},\beta}}{\varphi_{\text{front},\text{yaw}}} = \frac{\varphi_{\text{yaw},r}\varphi_{\text{rear},\beta}}{\varphi_{\text{rear},\text{yaw}}} = \frac{m}{I_z}. \quad (82)$$

Here $\varphi = \exp(\theta)$ denotes the positive effective coefficients. The identity can be checked from fitted coefficients alone: actual mass, inertia, geometry, and simulator truths are unnecessary. The two fitted inertial-ratio expressions disagree by 29.52–79.42%, normalized by their mean. Consequently, an independently fitted nine-coordinate LPV model is structurally anchored but need not correspond to one physical bicycle. The development audit does not establish that this inconsistency causes the bound activity; heterogeneous global pooling and the working covariance may also contribute. These explanations require a controlled comparison.

Decision and next gate. The numerical repair succeeds with staged initialization: gradients are validated, final fits are stationary and repeatable, and independently weighted held-out prediction improves. The full physical-model gate remains closed. The next experiment should enforce the exact common-inertial-ratio identity through an eight-coordinate representation, using only measured signals and public nominal bounds, and separate the effects of this physical coupling from heterogeneous global pooling. It should retain the current prediction and stationarity audits. Bounds should not be widened merely to make the interior gate pass. Joint membership/model learning remains deferred until this physical-model prerequisite is resolved. The controlled eight-coordinate comparison and measured-output covariance selection are subsequently tested in section 53.

The covariance convention also occurs in the historical 10H-A observer helper. Its older observer/control results therefore need rechecking with the corrected gain before those numerical results support the final manuscript. This experiment repairs the identification interface and does not re-evaluate those closed-loop studies.

Reproducibility. The protocol was fixed before fitting. The fit implementation and protocol are archived in commit 112f35b. Configuration, estimator, runner, and numerical tests are stored in `code/config/experiment_10hh.json`, `code/src/federated_lpv/innovation_likelihood.py`, `code/experiments/experiment_10hh_single_model_repair.py`, and `code/tests/test_experiment_10hh.py`. Per-fleet and aggregate CSVs retain restarts, phase outcomes, parameters, Hessian eigenvalues, conditional profiles, gradient comparisons, data-split

metadata, and measured-array hashes. The post-hoc boundary/coupling audit has separate outputs and is explicitly excluded from initial method selection and gate definitions.

53 Experiment 10H-I: physically coupled fitting and output-only covariance selection

Purpose and controlled separation. The numerical repair in Experiment 10H-H produced repeatable staged fits, but its nine independently fitted coefficients violated an exact bicycle identity. Experiment 10H-I enforces that identity and distinguishes two questions. Part A changes only the physical parameterization, retaining the inherited process covariance to isolate model structure. Part B chooses the process-noise scale using measured training outputs alone. Both parts use opened development fleets 601–605, the same 200-sample records and 20/10 training/held-out client split as 10H-H. Part A verifies exact measured-array SHA-256 agreement with the archived 10H-H records. This is development evidence, not blind confirmation. Seeds 591–600 and 611–620 remain sealed.

Eight coordinates and exact coupling. The positive physical coordinates are

$$w = (g, \ell_f, \ell_r, b_f, b_r, d_f, d_r, a_\delta), \quad g = \frac{m}{I_z}, \quad b_j = \frac{C_j}{m\sigma_j}, \quad d_j = \frac{1}{\sigma_j}, \quad a_\delta = \frac{1}{\tau_\delta}. \quad (83)$$

Their physical definitions specify the model; true client values are not inputs to estimation. The existing nine coefficients become

$$\varphi(w) = (g\ell_f, g\ell_r, b_f, \ell_f b_f, b_r, \ell_r b_r, d_f, d_r, a_\delta). \quad (84)$$

Equation (82) therefore holds identically. The relation reduces the physically admissible coefficient space from nine to eight dimensions. It does not identify absolute mass, inertia, or cornering stiffnesses: common scaling of m, I_z, C_f, C_r leaves the measured-output model unchanged. Positive coordinates and this coupling ensure realizability in the adopted bicycle structure, not that every feasible geometry represents a passenger vehicle with plausible dimensions.

Let $\eta = \log(w/w_{\text{nom}})$, where the nominal design is public. Products in (84) imply a constant full-column-rank map

$$z = z_{\text{nom}} + T\eta, \quad \nabla_\eta \mathcal{L} = T^\top \nabla_z \mathcal{L}, \quad \mathcal{I}_8 = T^\top \mathcal{I}_9 T. \quad (85)$$

The nine coefficient bounds are preserved exactly as

$$z_{\text{lower}} - z_{\text{nom}} \leq T\eta \leq z_{\text{upper}} - z_{\text{nom}}, \quad (86)$$

with the existing 0.35–3.0 nominal multipliers. Independent boxes on the eight new quantities would change the comparison and are not introduced. Starts use the nominal and two deterministic perturbations in eight log coordinates. A perturbation is shrunk along its nominal-centered ray until all nine inequalities are satisfied. Coefficient-wise clipping is avoided because it would break the physical identity.

Likelihood, optimization, and diagnostics. The repaired covariance, exact discretization, complete analytic derivatives, and working likelihood in (79) are unchanged. Training-only diagonal information scaling is calculated in the eight coordinates. SLSQP handles the linear inequalities with analytic gradients. Part A compares joint and staged fitting. Stages fit the actuator, relaxation rates, and the five mechanical quantities, followed by joint refinement. Restricted stages have 80 iterations and final joint optimization has 300, with function tolerance 10^{-12} and three starts.

An independent KKT audit fits nonnegative multipliers to the outward normals of active coefficient constraints. It reports stationarity, feasibility, and complementarity in physical log coordinates. A box projected gradient is inappropriate for (86). Curvature is reported both in the full eight-dimensional space and on the tangent space of the active constraint face. Positive face curvature supports a local constrained interpretation, not a claim of global uniqueness.

Profiles fix one physical log coordinate at offsets $-0.16, -0.08, 0, 0.08, 0.16$ and reoptimize the seven nuisance coordinates. A feasibility linear program precedes a nearest-feasible projection. Infeasible targets are recorded rather than clipped to the optimum. Profile KKT and feasibility checks include the fixed-coordinate equality. These are nuisance-reoptimized profiles, stronger than the conditional slices in 10H-H. The predeclared numerical/prediction gate retains gradient, restart, feasibility, stationarity, likelihood, and fixed- R prediction checks. The separate physical-readiness gate also requires informative two-sided profiles, positive active-face curvature, and no active coefficient bounds. Boundary solutions do not automatically imply optimizer failure, and no bounds or thresholds are changed after inspecting results.

Measured-output covariance selection. Part A freezes the historical Q scale 0.01, originally selected using latent observer errors, solely for its controlled structure comparison. Part B does not read that selection artifact. The measurement covariance R and base Q diagonal are held as public sensor/design assumptions. The scalar Q grid is $10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10$, declared before fleet fitting.

The 20 training clients are divided into two folds by alternating their sorted array positions. Every client’s speed records stay together. For each scale and fold, a staged model is fitted with three deterministic starts on the other ten clients. The training likelihood chooses the best restart. The mean inner validation $\mathcal{E}_R$ chooses the scale. The model is then refitted with three starts on all 20 training clients before the ten outer clients are scored. Neither class labels nor any hidden physical quantity or latent-state error enters these decisions. Inner candidate stationarity and an interior grid selection are audited separately. Selection targets one-step prediction and does not identify the true stochastic process covariance.

Homogeneous reference checks. A known, physically consistent public synthetic fixture uses the training input/speed design and the same measurement budget, with all records generated by one planted model. The learner receives only measured arrays and public nominal initialization. Planted coordinates are used only retrospectively. The noiseless reference uses $Q = 0$ and fixed sensor weights. Three staged fits recover all eight planted quantities, with maximum relative error 1.14×10^{-8} . Adding sensor noise and quiet-segment bias-estimation uncertainty, and fitting with the frozen working Q , gives maximum relative error 0.99%. Both references have no active coefficient bounds, positive eight-dimensional curvature, and reproducible stationary fits. The noisy reference’s maximum restart CV is 9.17×10^{-7} . These finite fixtures check implementation and local recovery; they do not prove fleet-wide identifiability.

Controlled structure result: Part A. The exact coupling holds to roundoff on every fit, with maximum absolute log residual 8.89×10^{-16} . Analytic physical-coordinate gradients agree with central differences, with maximum norm-relative error 8.34×10^{-8} across both parts. All 15 joint and 15 staged restarts pass the final KKT and feasibility checks. The largest Part A KKT residual is 1.18×10^{-5} and the largest parameter CV is 1.36×10^{-6} . Joint and staged selected objectives agree to numerical precision. Both methods pass the numerical/prediction gate.

The coupling repair retains prediction performance; the mean gain over nominal increases by only 0.21 percentage points. It does not resolve the bound activity. The rear sideslip-to-force coefficient remains at its lower bound in all five fleets, and the front counterpart reaches its upper bound on seeds 602 and 605. Active-face curvature remains positive, with minimum eigenvalue 0.0255. All feasible nuisance profiles are stationary and feasible, but direct force-gain profiles

Table 80: Controlled Part A comparison, with the same measured records, sensor weights, and inherited Q scale 0.01. Percentages are means over five opened fleets relative to the same corrected nominal model.

Diagnostic	10H-H staged	10H-I joint	10H-I staged
Coordinates	9	8	8
Mean held-out $\mathcal{L}$ improvement	18.57%	18.86%	18.86%
Mean held-out $\mathcal{E}_R$ improvement	18.02%	18.23%	18.23%
Minimum held-out $\mathcal{E}_R$ improvement	11.90%	12.98%	12.98%
Stationary final restarts	15/15	15/15	15/15
Physical coupling enforced	No	Yes	Yes
Fleets with active coefficient bounds	5/5	5/5	5/5
Full physical-readiness gate	Fail	Fail	Fail

at active bounds are one-sided. The weakest feasible profile edge on seed 604 increases loss by only 0.0158%, below the declared 0.02% requirement. No outward edge is replaced by a clipped zero displacement. The numerical audits support a converged local constrained fit. Physical calibration remains unresolved. The eight-coordinate aggregate information is full rank in every selected Part A fit, with conditions from 7.21×10^3 to 4.13×10^4 . This local working-model calculation is not a global identifiability proof.

Output-only covariance result: Part B. All five training-only selections choose scale 1, an interior grid point and 100 times the inherited scale. All 15 final refit restarts are stationary, with maximum KKT residual 6.56×10^{-7} and maximum parameter CV 1.72×10^{-6} . Mean absolute held-out $\mathcal{E}_R$ falls from 5.010 in Part A to 1.996 in Part B. The mean paired decrease is 59.91%, ranging from 56.60% to 64.20%. This combines covariance adaptation and refitting; it must not be attributed to the physical coupling, which is already present in both parts.

Table 81: Part B prediction and covariance selection. The nominal and fitted models in each row use the same selected Q scale. These nominal references differ from Part A, so percentage gains across parts are not interchangeable.

Fleet	Selected scale	Fitted $\mathcal{E}_R$	$\mathcal{E}_R$ gain	$\mathcal{L}$ gain
601	1	1.907	5.25%	1.23%
602	1	1.982	12.63%	2.81%
603	1	1.981	9.49%	1.97%
604	1	2.083	7.31%	1.68%
605	1	2.027	7.05%	1.72%
Mean	1	1.996	8.35%	1.88%

The nominal model itself improves substantially with the changed observer covariance: mean $\mathcal{E}_R$ changes from 6.145 to 2.179 before parameter refitting. The refitted Part B model adds a mean 8.35% improvement over this same-covariance nominal reference, with positive gains on every fleet. However, its held-out likelihood improvement is below the predeclared 2% minimum on four fleets, so the combined numerical/prediction gate fails despite reliable numerical convergence. The separate covariance pipeline gate also fails: 179/180 inner restarts satisfy the KKT threshold, but seed 603, fold zero, smallest scale 10^{-4} has one residual 2.15×10^{-4} against the 10^{-4} limit. It was not a selected scale. This exception is retained and no threshold is relaxed.

Every final Part B model still has an active coefficient bound. The rear sideslip-to-force gain is at its lower bound throughout; on seed 605 the rear yaw-to-force gain also reaches its lower

bound. Active-face curvature is positive, with minimum eigenvalue 0.00693, but profile edges are weaker than the declared requirement. The full physical-readiness gate remains closed.

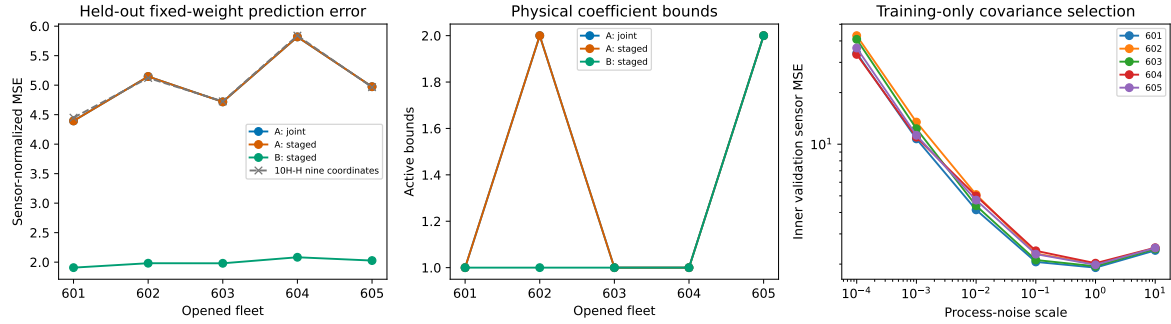


Figure 58: Absolute held-out prediction error, active coefficient bounds, and training-only covariance scores. Part A uses the inherited scale; Part B uses its inner-selected scale. The left panel uses absolute errors to avoid comparing percentage gains against different nominal covariances.

Post-hoc innovation calibration. A separate measured-output audit leaves models, selections, and gates unchanged. Correctly specified Gaussian innovations would have unit mean normalized innovation squared, identity whitened covariance, and no serial correlation. The fitted Part A models instead have mean normalized NIS 4.013 and maximum absolute lag-one correlation 0.879. Part B reduces these to 0.445 and 0.346. Its whitened covariance eigenvalues are all below one, with ranges 0.173–0.183 for the smallest and 0.684–0.728 for the largest. The prediction-selected covariance therefore produces conservative innovation variances and leaves appreciable serial structure. Better one-step prediction does not validate a calibrated likelihood for assigning vehicle groups. Heterogeneous pooling, the fixed covariance shape, and quiet-bias uncertainty remain possible contributors. These diagnostics do not isolate their causes.

Decision. The eight-coordinate physical relation and the estimator implementation are validated on the declared development checks. Measured-output prediction remains useful and final fitting is reproducible. The homogeneous references recover interior parameters, while heterogeneous pooled fits remain bounded. This argues against missing physical coupling being the sole explanation for the earlier boundary solutions. It does not prove that heterogeneity alone causes them.

The complete physical/covariance gate fails. Joint group/model learning is still deferred under that gate. The next focused work must resolve the inner stationarity exception and distinguish predictive covariance tuning from likelihood calibration, while diagnosing the pooled-model boundary/profile limitations. Parameter bounds should remain unchanged unless a new public physical argument justifies a separately declared protocol. These results do not establish latent-state accuracy, closed-loop improvement, or blind confirmation performance.

Reproducibility. The protocol, learner, runner, and tests were archived before fleet fitting in commit 0381587. The implementation is in `code/src/federated_lpv/physical_coupled_fit.py`, `code/experiments/experiment_10hi_physically_coupled_fit.py`, and `code/config/experiment_10hi.json`. Numerical checks cover the exact mapping, likelihood gradients, information transformation, coupled KKT normals, feasible nuisance profiles, and client-blocked splits. Per-fleet CSVs retain parameters, starts, phases, profiles, curvature, measured-array hashes, inner scores, and covariance selection. No confirmation execution option exists. One initial Part A job on seed 602 stopped when a redundant projection of the zero-offset profile

center rejected bound roundoff. Commit 9f08abc reuses the already audited center and resumes that job. The fit, nonzero profile targets, configuration, and gates are unchanged. An execution manifest retains the original and corrected runner revisions and their source hashes; completed jobs were preserved. The innovation audit is explicitly post-hoc. Validation passes 124 tests and lint, and the full report compiles.

54 Experiment 10H-J: bound sensitivity and single-model validation

Purpose and status of the bounds. Experiments 10H-H/I produced repeatable constrained optima but left active bounds, shallow profiles and miscalibrated innovations. The inherited coefficient multipliers 0.35 and 3 were introduced in 10H-G without a documented engineering-range derivation. Their preservation made the earlier comparisons controlled; it did not make them calibrated physical limits. Physical coupling and positive coordinates establish realizability in the adopted bicycle structure, not plausibility for the intended vehicle envelope.

For a uniform coefficient box with nominal multipliers a, b , axle distances and normalized stiffnesses have marginal relative ranges $[a/b, b/a]$; m/I_z has range $[a^2/b, b^2/a]$. Relaxation lengths and the actuator time constant have relative ranges $[1/b, 1/a]$. Thus the original box permits a nominal 2.8m wheelbase to range from 0.327 to 24m even under exact coupling. These marginal intervals are not independent physical prior boxes. The provenance and required engineering evidence are recorded in `docs/experiment_10hj_bound_provenance.md`. Public model definitions support the quantities but do not justify fleet-wide numerical ranges [8].

Predeclared diagnostic. Before execution, commit aadfc7f fixes three nested coefficient boxes, $[0.35, 3]$, $[0.2, 5]$, and $[0.1, 10]$, in two separate fixed- Q panels at scales 0.01 and 1. All boxes retain eight positive coordinates and exact coupling. The same opened fleets 601–605, measured records, 200-sample segments and training/held-out client positions are used. Array hashes must match 10H-I before fitting. The 0.01 panel is a historical controlled setting; it is not a newly validated covariance-selection rule. Neither a bound nor a covariance setting is selected in this study. Three deterministic staged starts use 80 iterations per restricted stage and 500 for joint refinement; only training working likelihood selects a restart. Confirmation seeds 591–600 and 611–620 remain sealed.

The numerical audit checks independent KKT residuals, feasibility, gradients, coupling, restart consistency, full/active-face curvature, and monotonic best training likelihood under nested feasible sets. Every widest-box model has eight nuisance-reoptimized profiles at offsets $0, \pm 0.08, \pm 0.16$. A separate profile fixes $b_r/b_{r,\text{nom}}$ at 0.1, 0.15, 0.2, 0.25, 0.35, 0.5, 1 and reoptimizes the seven other coordinates. Infeasible targets are recorded. Working-objective profiles are not calibrated confidence intervals for this heterogeneous model.

Causal output validation. All predeclared models are scored after training selection. Forecast horizons are 1, 5, 20 and 50 samples, or 0.01, 0.05, 0.2 and 0.5 s. With a 20-sample observation burn-in, each horizon uses identical forecast origins. The filtered state uses outputs strictly before its origin; subsequent propagation uses known commands with no intervening output corrections. Zero-initialized full-record simulation is a separate diagnostic. This separates measurement-assisted one-step prediction from longer-horizon dynamic validation [7].

Innovation covariance, normalized NIS, means, and correlations at lags 1–20 are also reported. Positive lags pair the present innovation with past innovations or past steering commands, without crossing record boundaries. Pooled and within-record-centered correlations expose sensitivity to constant bias. They are descriptive; no independent-sample significance threshold is applied to heterogeneous, temporally dependent records. Residual–input correlation complements residual whiteness in model validation [9].

Interpretation. The wider boxes are diagnostic domains, not proposed engineering priors. No new zero-active-bound acceptance gate is introduced, and historical gate results remain unchanged. An interior solution alone does not establish physical calibration; an active bound alone does not invalidate a constrained optimum. Persistent boundary motion or weak profiles indicates unresolved sensitivity, but cannot isolate fleet heterogeneity from covariance or bias mismatch. No latent-state accuracy or closed-loop-control gain is claimed.

Numerical result. All 90 model restarts satisfy the declared independent KKT and feasibility checks, with maximum stationarity residual 1.17×10^{-5} , maximum parameter CV 2.83×10^{-6} and coupling error 1.78×10^{-15} . No fitting evaluation is invalid. Every nested best training objective is nonincreasing. All feasible local and extended rear-gain profiles also satisfy stationarity and feasibility checks. The original-box fits reproduce the 10H-I staged objectives and held-out errors. Table 82 reports the complete predeclared comparison.

Table 82: Mean full-record held-out one-step fixed- R error and fleets with active coefficient bounds. All five fleets are included in each row.

Q scale	Coefficient multipliers	Mean E_R	Boundary fleets
0.01	[0.35, 3]	5.010	5/5
0.01	[0.2, 5]	4.729	5/5
0.01	[0.1, 10]	4.126	5/5
1	[0.35, 3]	1.996	5/5
1	[0.2, 5]	1.955	3/5
1	[0.1, 10]	1.945	1/5

At scale 0.01, every rear gain follows the changing floor exactly: $b_r/b_{r,\text{nom}} = 0.35, 0.2, 0.1$. The widest-box gain is 9.596 s^{-2} throughout. Rear axle distances move to 2.558–2.913 m and normalized rear stiffnesses change with the bound. The mean paired one-step error reduction is 17.55%, with range 14.46–19.77%, but an interior physical calibration has not emerged. Outward profiles remain infeasible at the new rear-gain floor. Correct KKT signs and positive face curvature establish valid constrained optima, not that the floor has a physical justification.

At scale 1, four widest-box models are interior; seed 604 still hits the rear-gain floor. Seeds 601 and 603 already settle in the middle box, and further widening preserves their optima. The widest rear gains range from 9.596 to 21.401 s^{-2} , rear axle distances from 2.174 to 2.909 m and rear relaxation lengths from 1.754 to 3.020 m. These are fitted working-model quantities, not comparisons with true client parameters. Mean paired one-step error reduction versus the original box is 2.54% (1.18–3.97%). Weak profiles remain: the smallest feasible local profile-edge increase is 0.00194% of working likelihood. For seed 605 the negative force-gain offset -0.16 is infeasible even though the fit is interior; its negative profile edge is -0.08 . Thus interiority does not imply adequate distance from the limit or useful calibrated uncertainty.

Longer-horizon result. Table 83 uses the identical 131 origins per record at all forecast horizons. Its one-step values differ from full-record scores because burn-in and common-origin truncation change the scoring window. The full simulation uses all 200 samples and no output corrections.

At scale 0.01, widening improves the 50-step error in all five fleets, with mean paired reduction 7.99%, but the widest model is worse than the nominal model in four fleets at that horizon and in four under simulation. At scale 1, the widest model improves one-step and five-step prediction, while 50-step error worsens in every fleet versus the original box. Mean paired 50-step degradation is 29.82% (7.57–58.88%); four fleets also perform worse than nominal at that horizon. Full simulation deteriorates in all five fleets versus both the original box and the nominal model. These are output dynamics checks, not latent-state or closed-loop-control tests.

Table 83: Mean fixed- R output error for causal forecasts and full-record simulation. Nominal simulation is independent of Q ; filtered initialization makes nominal finite-horizon forecasts depend on Q .

Q scale	Model/box	1 step	20 steps	50 steps	Simulation
0.01	Original	4.909	98.069	178.614	167.904
0.01	Widest	3.962	98.657	164.766	156.155
1	Original	1.968	88.571	170.535	170.155
1	Wider	1.921	92.264	192.734	203.233
1	Widest	1.907	98.289	222.477	236.497
1	Nominal	2.171	93.114	160.011	150.066

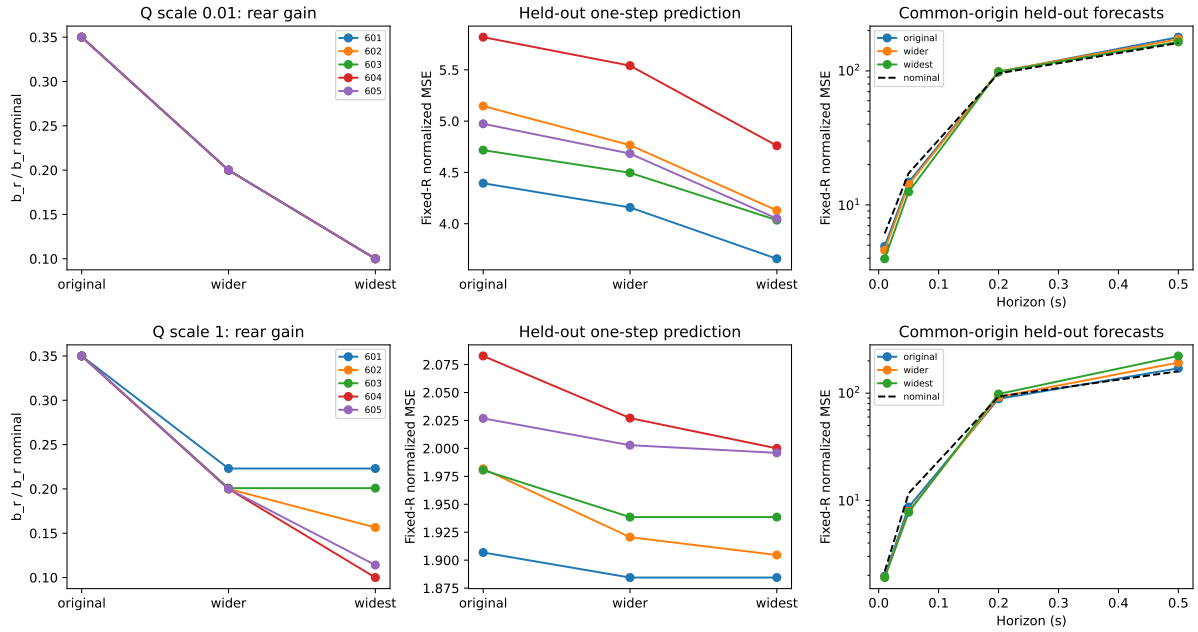


Figure 59: Rear-gain boundary motion, one-step errors and common-origin forecast errors. The curves expose covariance-dependent boundary behavior and the difference between short-horizon prediction and dynamic validation.

Residual calibration and scientific decision. For widest-box models, mean normalized NIS is 3.285 at scale 0.01 and 0.437 at scale 1, rather than the correctly specified Gaussian reference of one. Maximum within-record innovation autocorrelation over lags 1–20 is 0.812 and 0.347, respectively; maximum correlation with past commands is 0.322 and 0.173. Scale-1 whitened covariance eigenvalues remain below one, with minimum eigenvalues 0.174–0.184 and maximum eigenvalues 0.694–0.731. Removing record means does not eliminate the serial correlation. Every fitted model is stable at the measured scheduling speeds (maximum discrete plant spectral radius 0.994), so forecast degradation is not explained by an unstable fitted pole at these speeds. This finite-speed check is not a continuous-speed stability certificate.

The diagnostic answers the bound question: the original box was restrictive for some optima, but expanding it is insufficient to establish a calibrated physical model. Numerical repair is successful. The scientific bottleneck is now prediction-focus, pooled-model mismatch and uncertainty calibration. No widest box is adopted as an engineering prior and no historical gate is reclassified. Actual physical intervals require independent evidence for a defined vehicle envelope, using the transformed geometry, normalized stiffness, relaxation and actuator quantities.

The next bounded study should compare label-blind $K = 1$ and $K = 2$ jointly learned models under common covariance/bias assumptions, with client-blocked training selection and predeclared multi-step and simulation validation. A one-step working-likelihood gain alone must not establish improved dynamics or calibrated grouping probabilities. This comparison should test whether multiple compatible models resolve the observed mismatch; it should not require the heterogeneous $K = 1$ model to become physically perfect first.

Reproducibility. The configuration and numerical source were committed before execution. A subsequent reporting-only correction fixes access to the forecast `mode` column in the figure. All ten execution manifests retain commit `aadfc7f`, archived source fingerprints and output hashes. Summary reconstruction verifies that all fitting code and other execution sources are unchanged before accepting this reporting update. No fit, profile, bound, covariance, data split or numerical criterion is changed. Independent tests cover forecast timing, absence of future-output leakage, known-model recovery, one-step scoring equivalence and record boundaries. The full suite passes 130 tests; lint passes for the new Python files.

55 Experiment 10H-K: jointly learned plants and client memberships

Question and predeclared design. Experiment 10H-J shows that releasing heuristic bounds does not validate a single fleet plant: short prediction improves while longer propagation can deteriorate. Experiment 10H-K tests whether two learned plants improve future measured outputs under the same physical structure, covariance and bias assumptions. Protocol commit `080a6aa` precedes fitting; implementation commit `d5e1b11` is recorded in all ten execution manifests. The same opened development fleets 601–605 contain 20 training and ten held-out clients each. Confirmation 591–600 and 611–620 remains sealed.

Both component counts retain the eight positive coordinates, exact bicycle coupling, coefficient box $[0.1, 10]$, fixed public R , public Q shape, quiet-start bias correction and stationary-covariance/zero-state initialization from 10H-I/J. The two fixed Q scales, 0.01 and 1, are reported separately; neither is selected here. The box remains a diagnostic domain, not a validated engineering envelope. The audited 10H-J widest-box $K = 1$ solutions and their three-restart evidence are reused. Their output hashes, measured-array hashes and training objectives are independently verified before $K = 2$ fitting.

Joint plant–membership objective. For client i and candidate plant g , let

$$D_{ig} = \sum_{j \in \mathcal{R}_i} \sum_t \left[e_{jgt}^\top S_{jg}^{-1} e_{jgt} + \log \det S_{jg} - \log \det R \right],$$

where $\mathcal{R}_i$ contains every measured speed record associated with that client. The models recompute $A_g(v)$, $B_g(v)$, stationary covariance, Kalman gain, innovations and their total derivatives at each proposal. With N record-time samples, the minimized objective is

$$L_K = -\frac{2}{3N} \sum_i \log \left(\sum_{g=1}^K \pi_g \exp[-D_{ig}/2] \right).$$

Stable log-sum-exp gives the objective and current working membership weights. For $K = 1$ this is exactly the previous objective; duplicating its plant in both components supplies an available nested reference. $K = 2$ has sixteen plant coordinates and one mixing weight, with $\pi_1 \in [0.01, 0.99]$ to keep log weights finite. The limit does not force any hard client occupancy. These weights are not calibrated class probabilities: covariance mismatch and serial correlation established in 10H-I/J remain relevant.

Three deterministic random directions produce symmetric log-coordinate perturbations of amplitude 0.3 around the learned global plant. Nearest feasible projections retain exact coupling. No nominal-score clustering or true class information initializes the models. Analytic-gradient SLSQP moves both plants through actuator, relaxation and mechanics stages (80 iterations each), then all seventeen variables jointly (800 iterations, tolerance 10^{-12}). Memberships evolve at every objective evaluation. The selected restart minimizes training likelihood only.

Causal unseen-client validation. Only the first 50 samples, or 0.5 s, of a held-out client’s first listed speed record determine its membership. That membership is frozen across all its speed records. The primary deployment selects its maximum-weight component; a secondary diagnostic averages separately propagated outputs with the same fixed weights. No matrices or observer gains are interpolated, and no full held-out score chooses a component.

All models, including the nominal reference, use common forecast origins 50–150 inclusive: 101 origins per 200-sample record, at horizons 1, 5, 20 and 50 samples. Filtered states use outputs strictly before each origin; subsequent propagation uses known commands without intervening output corrections. Separate zero-start simulations propagate from the record start but score only samples 50–199, after membership calibration. Consequently the $K = 1$ errors here differ from the earlier full-record/20-sample-burn-in tables despite identical fitted coefficients. Prediction and simulation validate different aspects of an application model [7].

Numerical and provenance result. All 30 independent $K = 2$ restarts report success and satisfy KKT and feasibility limits: maximum KKT residual 2.86×10^{-6} , zero constraint violation, zero invalid fitting evaluations and coupling error 1.78×10^{-15} . Every selected two-component training objective is below its global reference; neither a collapsed reference nor a mixing-weight bound is selected. Selected active-face Hessians have positive minimum eigenvalues, at least 0.00164. Training hard occupancy is at least seven clients per component at $Q = 0.01$ and six at $Q = 1$.

One of ten predeclared finite-difference gradient checks narrowly misses the 10^{-5} relative-error limit: seed 601, $Q = 0.01$, has error 1.14×10^{-5} at step 10^{-5} . At the unchanged initial point, steps 5×10^{-6} and 2.5×10^{-6} give 2.84×10^{-6} and 7.81×10^{-7} , respectively. The approximately quadratic refinement supports a finite-difference truncation explanation. The original missed check remains recorded; no fit, criterion or protocol is changed.

An independent audit verifies all execution sources against committed revision `d5e1b11`, all checkpoint hashes and unchanged historical gate artifacts. The original single-model evaluator

reproduces prefix memberships to 4.13×10^{-14} and independently computed component forecasts agree to 2.10×10^{-12} . Six focused tests cover client aggregation, mixture gradients and permutation invariance, first-record prefix isolation, forecast equivalence, forecast causality and coupled feasibility. The full suite has 136 passing tests; all four new Python files pass scoped lint.

Fleet-average future-output result. Table 84 and Figure 60 show substantial improvement beyond one-step filtering. The primary $K = 2$ deployment improves every reported horizon and suffix simulation relative to $K = 1$ in all five fleets, in both covariance panels. Mean paired training-objective improvement is 35.14% at $Q = 0.01$ and only 1.31% at $Q = 1$; the latter still produces large longer-horizon output gains. Working training likelihood and future fixed- R dynamics error measure different quantities.

Table 84: Held-out output errors with frozen prefix-MAP memberships. Errors are means across five fleets. Gains average the five paired fleet reductions, not the ratio of the displayed means. All horizons share forecast origins.

Q scale	Diagnostic	$K = 1$ E_R	$K = 2$ E_R	Paired gain (%)
0.01	0.01 s forecast	4.089	2.833	30.40
0.01	0.05 s forecast	12.664	7.089	43.51
0.01	0.2 s forecast	100.066	38.195	61.52
0.01	0.5 s forecast	165.009	49.796	70.07
0.01	Zero-start suffix	164.418	46.389	71.98
1	0.01 s forecast	1.917	1.778	7.26
1	0.05 s forecast	7.673	4.534	40.81
1	0.2 s forecast	100.262	30.868	69.93
1	0.5 s forecast	228.686	61.572	73.31
1	Zero-start suffix	263.852	63.323	75.90

At 0.5 s, paired fleet gains range 63.13–74.75% for $Q = 0.01$ and 67.15–76.23% for $Q = 1$. Two-model forecasts and suffix simulations also improve the public nominal reference in every fleet. The frozen-weight ensemble gives similar results at $Q = 0.01$, where prefix weights are almost hard. At $Q = 1$ it gives 74.40% mean paired 0.5 s gain, versus 73.31% for MAP; both predeclared rules are reported without selecting one afterward.

Individual clients do not all benefit. Fleet-average improvement is not a universal personalization guarantee. At 0.5 s, 39/50 clients improve at $Q = 0.01$ and 43/50 at $Q = 1$. Median client improvement is 73.64% and 70.76%, respectively. The worst client error nevertheless increases by factors 14.06 and 3.71 relative to $K = 1$. At $Q = 1$, for example, seed 602 client position 5 increases from $E_R = 22.56$ to 83.68. These positions are measured-record associations, not vehicle class labels. Figure 61 exposes the unfavorable tail. At $Q = 0.01$, the arithmetic mean of client percentage gains is actually -16.47% despite the 70.07% mean paired fleet-error reduction: percentage ratios place large weight on clients with small baseline errors. Both aggregations must remain visible when deciding whether to deploy grouping.

Model and partition repeatability remains unresolved. Numerical stationarity does not establish a unique two-model solution. At $Q = 0.01$, fleets 603–605 reproduce models and partitions across all three starts; fleets 601 and 602 have minimum restart ARI 0.622 and 0.043 and objective spreads 1.59% and 9.67%. ARI here compares learned partitions with each other after fitting; it never compares them with true labels. At $Q = 1$, all five fleets have distinct stationary alternatives, despite objective spreads only 0.17–0.27%. Minimum restart ARI ranges -0.011 –0.799, and maximum aligned parameter CV ranges 0.387–1.051. High working

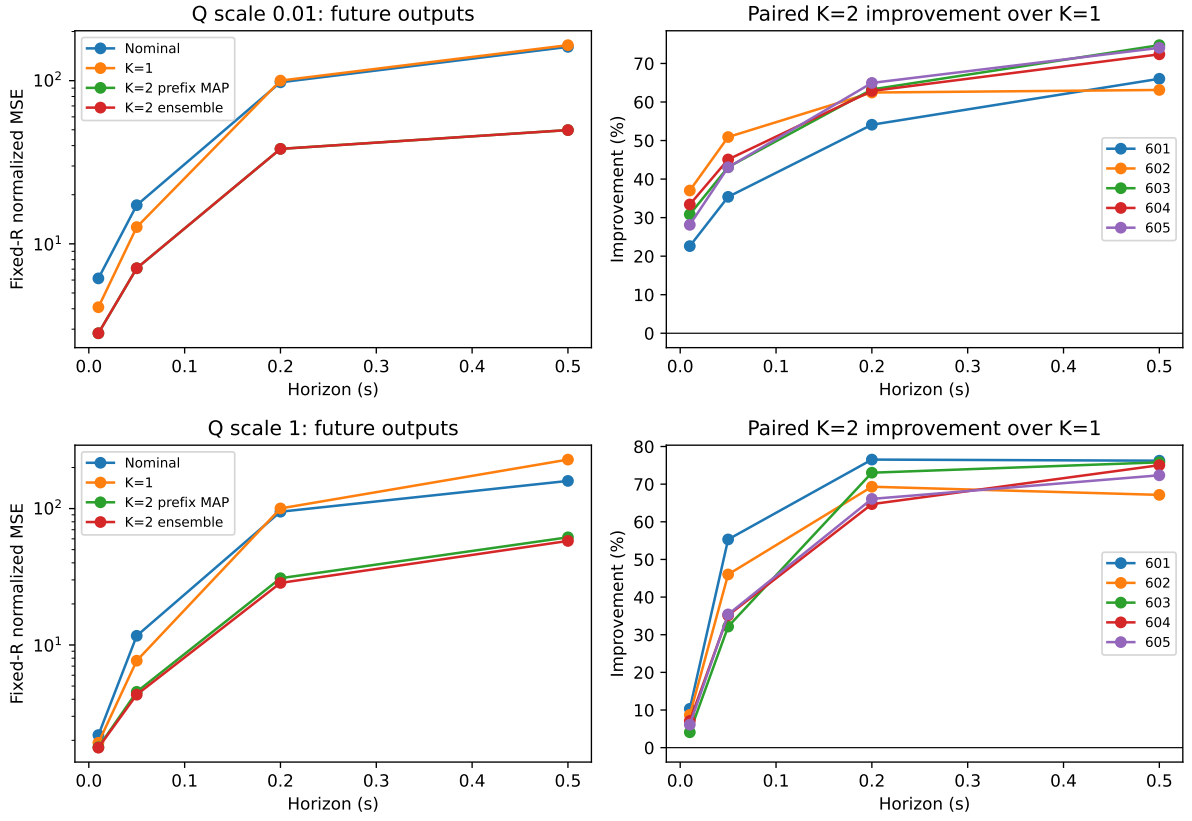


Figure 60: Future-output errors and paired fleet gains under the predeclared calibration/forecast split. Every horizon is assessed without intervening output corrections. Both fixed covariance panels and both deployment rules are retained.

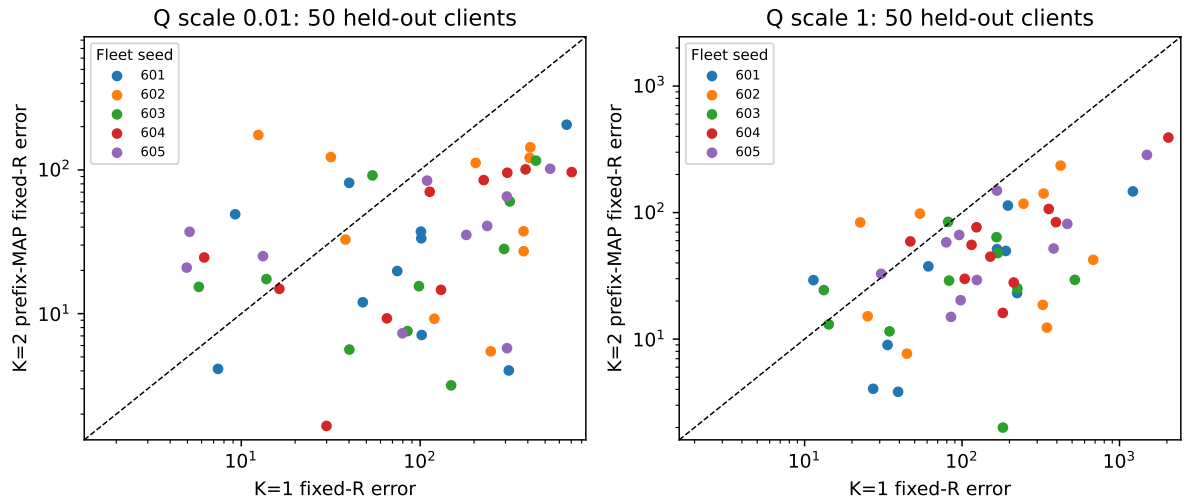


Figure 61: Per-client 0.5s fixed- R errors. Points below the identity line benefit from the two-model prefix-MAP deployment; points above it deteriorate. All 50 held-out clients per panel are shown without class labels.

membership confidence therefore cannot certify a reproducible physical grouping. No true family-count or class-recovery claim is made.

All five selected $Q = 0.01$ mixtures retain one coefficient bound; at $Q = 1$ four selected mixtures are interior and one reaches a rear-decay upper bound. Exact coupling and interiority do not validate the broad engineering domain. No physical confidence interval or calibrated client parameter is established.

Residual calibration remains unresolved. On the held-out suffixes, $Q = 0.01$ changes mean normalized NIS from 3.373 to 2.393 and the mean fleet maximum within-record autocorrelation from 0.783 to 0.719. At $Q = 1$, NIS changes from 0.433 to 0.407, while maximum autocorrelation increases from 0.339 to 0.412. The corresponding mean minimum/maximum whitened covariance eigenvalues are 0.171/0.681, below one. Mean maximum past-input correlations fall from 0.315 to 0.249 and from 0.154 to 0.133, respectively, but remain present. These are descriptive record-aware diagnostics, not independent-sample significance tests [9]. Two plants improve measured-output dynamics without repairing the working noise model.

Consequence for the KF/controller learning story. This is positive development evidence for learning multiple compatible plant models directly from measured outputs. The improvement persists when outputs no longer correct the propagated state, supporting the plant-model part of the proposed iterative learning loop. It is stronger evidence than nominal score clustering or one-step error alone. The same learned $A_g(v), B_g(v)$ can supply plant dynamics to a KF and a controller, with their different noise assumptions and control augmentations.

The result does not establish latent-state accuracy, calibrated memberships, an acceptable worst-client outcome, a federated communication benefit or closed-loop performance. Before an observer/controller deployment comparison, the next development work should stabilize the mixture solutions, investigate the covariance/objective mismatch and protect clients for whom specialization degrades future prediction. A subsequent controlled closed-loop experiment can separate observer-model and controller-model updates. Frozen-speed plant poles and stationary KFs are not LPV/adaptive stability certificates. Historical oracle/covariance gates remain unchanged, and confirmation remains unopened.

Reproducible artifacts. The protocol is `docs/experiment_10hk_protocol.md`; the execution entry point is `experiment_10hk_joint_client_mixture.py` and the independent reporting audit is `experiment_10hk_result_audit.py`. Results include all restart parameters and phases, initial splits, client memberships, per-client/aggregate forecasts, residuals, refined gradient checks and immutable source/output manifests under prefix `results/tables/experiment_10hk`. No labels, true client quantities, latent trajectories or exact client matrices enter learning or this audit.

56 Experiment 10H-L: validation-selected fits and global fallback

Question and fixed design. Experiment 10H-K establishes an average future-output benefit from two jointly learned plants, with unfavorable client outcomes and competing stationary solutions. Experiment 10H-L tests whether client-blocked validation can select a reproducible fitting strategy and whether causal calibration can support group choice with a global fallback. Protocol commit `6cc2ad0` precedes execution; every fitting manifest records implementation `d744941`. Fleets 601–605 and their outer clients remain opened development evidence. Confirmation seeds 591–600 and 611–620 remain sealed.

The measured arrays, 20 training/ten outer clients per fleet, eight-coordinate coupling, coefficient multiplier box $[0.1, 10]$, public R , fixed Q shape and scales $0.01/1$, bias correction and numerical limits are unchanged. Neither bounds nor covariance panels are selected here. Two inner folds block all speed records of each training client together: sorted client rank modulo

two assigns ten fitting and ten validation clients to each fold. Each inner global plant uses three starts; each two-plant fit uses six. All contexts share six deterministic split directions around their own learned global plant. On the full 20-client training set, the archived global fit and first three mixture starts are audited and reused; only three additional mixture starts are fitted. A direction is eligible only when both inner fits and its full fit pass success, KKT and feasibility checks.

Causal decision and selection rules. Each validation or outer client supplies the first 100 samples (1s) of its first listed record. Each of the global and two group plants scores fixed- R forecasts at horizons 5, 20 and 50 samples, using common prefix origins 20–50 inclusive. Every forecast filters strictly earlier measurements and then propagates commands without intervening output corrections. The mean three-horizon error is the candidate’s calibration score. The group with the smaller score is deployed only if its score is strictly below $(1 - \delta)$ times the global score; otherwise the client retains the global plant. The predeclared margins are $\delta \in \{0, 0.1, 0.25\}$. One choice is frozen across all that client’s subsequent speed records.

For every restart direction and margin, inner validation assesses future outputs at horizons 5, 20 and 50 from origins 100–150 inclusive. Its selection score averages $\log(E_{ig,h}/E_{i,\text{global},h})$ equally across clients, horizons and folds. Only this inner score selects direction and margin; ties within 10^{-12} prefer a larger margin, then a lower restart index. The full fit transfers the selected initialization direction, which does not identify an invariant optimizer basin across training subsets.

Outer comparisons include the global plant, six-start training-likelihood winner with prefix likelihood MAP, validation-selected fit with the same MAP, that selected fit with forecast group choice, and the primary forecast choice with fallback. The archived three-start winner and public nominal plant are also retained. All seven methods have the same 100-sample calibration budget, forecast origins 100–150 and horizons 1, 5, 20 and 50. Zero-start simulations propagate from the record start but score samples 100–199. These windows differ from 10H-K’s 50-sample calibration and are compared only within 10H-L.

Numerical execution and independent audit. All 250 fit records pass success and stationarity/feasibility checks: 210 new fits and 40 archived fits. Maximum KKT residual is 1.95×10^{-5} , maximum constraint violation 6.30×10^{-13} , and maximum physical coupling residual 1.78×10^{-15} . There are no invalid fitting evaluations. All six strategies are numerically eligible in every fleet/panel. Selection chooses a 25% margin in eight jobs and zero in two; no 10% margin is selected. Numerical convergence therefore does not explain the adverse validation transfer described below.

The independent reporting audit verifies ten job manifests, 30 context caches, committed source hashes, all measured arrays and whole-client fold isolation. Aggregate tables reproduce the verified individual jobs exactly. Saved inner scores reproduce selection without outer information. The earlier independent single-model evaluator reproduces first-record calibration scores to 9.38×10^{-12} , frozen-choice future errors to 2.04×10^{-12} and zero-start suffix errors to 2.12×10^{-12} . Historical 10H-I/J/K gate artifacts remain byte-identical. Six additional tests cover prefix isolation, causal propagation, forecast equivalence, strict fallback decisions, inner selection/eligibility and the sealed-seed configuration. The full suite passes 142 tests, and all four new Python files pass scoped lint.

Mixed future-output result. Table 85 separates fitting-strategy choice from client assignment. At $Q = 0.01$, the primary rule reduces mean 0.5s error from 49.081 for six-start likelihood MAP to 37.357; its mean paired fleet gain is 22.24%, with improvement in every fleet. The primary gain relative to the global plant is 76.64%. At $Q = 1$, mean error instead rises from 56.430 to 87.738; mean paired gain is -60.19% , with deterioration in every fleet relative to likelihood

MAP. The primary still improves the global plant in every fleet, with mean paired gain 59.60%. Positive global-relative gain alone would conceal the failed improvement over the available mixture baseline.

Table 85: Outer 0.5 s fixed- R output errors, averaged across five fleets. The final column averages paired fleet reductions relative to the six-start likelihood-MAP rule. Calibration and future windows are identical for all rows.

Q scale	Deployment	Mean E_R	Gain vs likelihood MAP (%)
0.01	Global	161.187	—
0.01	Archived three-start MAP	47.320	2.47
0.01	Six-start likelihood MAP	49.081	0.00
0.01	Validation fit + MAP	46.606	6.82
0.01	Validation fit + forecast group	46.580	6.59
0.01	Validation fit + fallback	37.357	22.24
1	Global	221.235	—
1	Archived three-start MAP	58.452	−3.87
1	Six-start likelihood MAP	56.430	0.00
1	Validation fit + MAP	87.279	−58.88
1	Validation fit + forecast group	104.405	−91.70
1	Validation fit + fallback	87.738	−60.19

The horizon/simulation results agree with this split. At $Q = 0.01$, primary one-step and 0.2 s errors fall from 3.100/42.256 to 2.883/31.143, and suffix simulation from 47.019 to 35.971. At $Q = 1$, the corresponding errors rise from 1.809/32.900 to 1.825/42.970 and simulation from 66.296 to 110.070. Both fixed covariance panels and every predeclared ablation remain visible in the saved tables and Figure 62.

Fallback reduces global-relative losses but supplies no guarantee. At $Q = 0.01$, likelihood MAP improves 37/50 clients and degrades 13/50 relative to the global plant. The primary improves 36, leaves 13 unchanged through global fallback, and degrades one. Its worst global-relative error ratio falls from 6.98 to 1.73. At $Q = 1$, likelihood MAP improves 42 and degrades eight; the primary improves 31, leaves 16 unchanged, and degrades three. The worst ratio falls only from 2.19 to 2.13.

Absolute error tails give a further qualification. At $Q = 0.01$, the likelihood/primary 90th-percentile client errors are 113.67/108.00 and maxima 275.79/190.48. At $Q = 1$, maxima fall from 453.67 to 402.03, while the 90th percentile rises from 123.21 to 272.34 and median from 36.36 to 46.72. Against likelihood MAP, the worst primary client error ratio is 1.80 at $Q = 0.01$ and 67.16 at $Q = 1$. A better extreme maximum or fewer losses against the global reference does not establish better personalization across the error distribution. Prefix margins remain empirical thresholds, with no future per-client guarantee.

Restart transfer is the exposed failure mechanism. For seed 602 at $Q = 1$, validation selects direction 2 with margin 25%. Its selected-fit MAP error on outer clients is 205.413, versus 52.254 for the likelihood winner (direction 5). Forecast group choice increases error to 291.378; fallback reduces it to 177.412. The bad result is already present with unchanged MAP assignment, then worsens with forecast group choice. Thus assignment mismatch alone cannot explain it.

Directions 2 and 5 converge to essentially the same inner objective values in both folds, yet reach distinct full-fit objectives 2.015082 and 2.002854. Their averaged inner selection scores differ by less than 2×10^{-6} . A tiny ranking of initialization indices therefore transfers to a

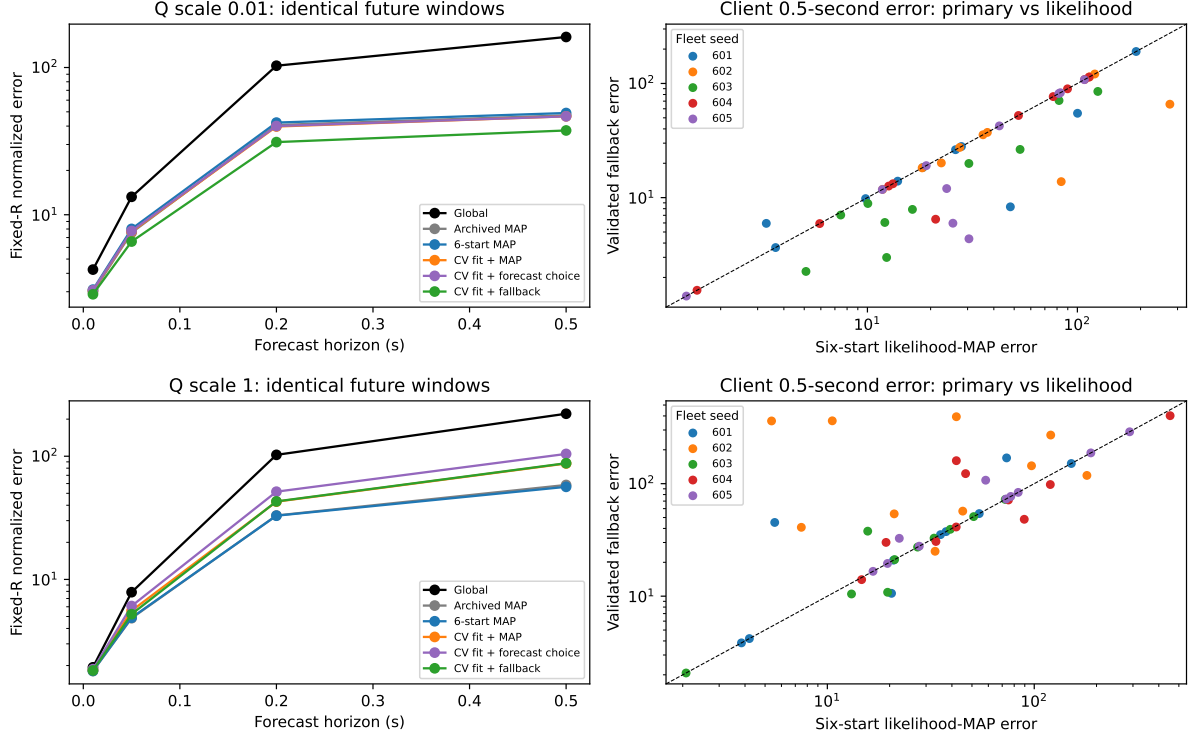


Figure 62: Left: mean future errors on common post-calibration origins. Right: individual-client 0.5s errors of the primary rule against six-start likelihood MAP. Points above the identity line deteriorate. Colors identify fleet seeds, without vehicle class labels.

materially different final model. This supports a basin-transfer explanation; it does not establish that two-fold validation can reliably rank those full models. Descriptive fold winners agree on both direction and margin in only one of the ten jobs. No outer score changes the original selection.

Across six full starts, minimum partition ARI to the selected fit ranges 0.211–1 at $Q = 0.01$ and -0.011 –0.214 at $Q = 1$. Alignment minimizes distance between the two eight-coordinate log vectors; the maximum aligned physical-coordinate CV ranges 7.92×10^{-7} –0.693 and 0.406–1.806, respectively. Seed 605 at $Q = 0.01$ retains identical hard partitions but parameter CV 0.342: partition repeatability does not imply model repeatability. Across all six descriptive prefix-MAP predictions, the median client maximum/minimum error ratio is 1.73 at $Q = 0.01$ and 3.64 at $Q = 1$; maxima are 62.33 and 115.07. These outer audits describe variability and never rank candidates for deployment.

Remaining physical/noise limitations and next development step. All five selected $Q = 0.01$ mixtures retain coefficient bounds (one or two active constraints); one selected $Q = 1$ mixture has an active bound. The broad box remains heuristic. Primary suffix NIS means are 2.444 and 0.414, and mean fleet maximum within-record autocorrelations are 0.708 and 0.399. Whitened covariance eigenvalue means are 1.249/3.931 and 0.174/0.688. The working likelihood remains miscalibrated despite valid numerical fits.

The next useful selection experiment should fit a library of complete models on fitting clients, freeze those models, and select among the actual fitted candidates on separate validation clients without refitting after selection. It should retain causal assignment/fallback and a separate outer test set. This isolates model selection from initialization-to-refit basin transfer. Only after this behavior is reproducible should new covariance/objective changes or observer/controller comparisons be introduced. The present result supports measured-output learning of multiple

predictive LPV plants, but leaves reliable model selection, engineering bounds, latent-state accuracy, closed-loop performance and federated communication benefit to be validated.

Reproducible artifacts. The protocol is `docs/experiment_10hl_protocol.md`; fitting uses `experiment_10hl_validated_model_selection.py` and independent reporting uses `experiment_10hl_result_audit.py`. Prefix `results/tables/experiment_10hl` stores all fold/full fits, phases, initializations, measured-array and source/output hashes, inner scores, calibration choices, every outer ablation, restart prediction variability, aligned parameter spread and residual diagnostics. No true client physical quantity, class label, latent trajectory or exact client matrix enters learning, selection or the independent audit.

57 10H-M reconstruction: actual frozen donor libraries

The original 10H-M-P implementation and learned coefficients were not recovered. This section records a fresh reconstruction from the unchanged, audited 10H-L fold fits; it does not reproduce or replace the historical reported numbers. For each development fleet and fitting panel, two ten-donor libraries were selected on separate donor-validation clients. The actual fitted plants were frozen without a full-donor refit or matrix averaging. Each held-out recipient chose its plant using only its first-record 100-sample output prefix.

All 20 libraries passed coefficient-bound, physical-coupling, provenance, disjoint-client and causal-assignment audits. Mean recipient percentage gains at horizon 50 relative to the same library’s donor-global plant were 49.91% for $Q_{\text{fit}} = 0.01$ and 46.94% for $Q_{\text{fit}} = 1$. These are averages of recipient gains after averaging the two deployment errors per recipient, not percentage changes in pooled fleet error. The mean of each fleet’s worst recipient gain was negative in both panels (−18.82% and −22.03%). Thus mean forecast improvement still does not establish uniformly beneficial transfer. Confirmation fleets remain sealed.

58 10H-N reconstruction: multihorizon refinement

Starting from the 20 reconstructed M libraries, each donor-global plant and two components was refined with an equal-weight sensor-normalized measured-output loss at horizons 1, 5, 20 and 50. Component training memberships were held fixed at their M values, an explicit reconstruction choice. Derivatives included both the KF recursion and future open-loop forecast. Bounds, covariances and the eight-coordinate physical coupling were unchanged.

Of 60 proposed refinements, 59 passed success, normalized KKT, feasibility and training-loss checks; the remaining plant retained its original coefficients. Validation selected the actual refined bundle for 16 of 20 libraries. At horizon 50, mean paired recipient gains relative to M were 6.49% for $Q_{\text{fit}} = 0.01$ and 26.99% for $Q_{\text{fit}} = 1$. Short-horizon gains were mixed: at horizon 1 they were −4.04% and −3.55%, and at horizon 5 they were 3.26% and −11.16%. Individual long-horizon harms remain substantial. These fresh development results support a feedback diagnostic, not a uniform transfer claim. The libraries and prefix-only recipient assignments are frozen before closed-loop evaluation; no controller costs or bounds are retuned.

59 10H-P reconstruction: core output feedback

The four core methods were rebuilt from audited reconstructed N libraries: public-nominal KF/LQI, learned KF only, learned LQI only and learned Both. The 1,600 matched linear-LPV rollouts use the historical H03 LQI costs, existing 12-second maneuvers, clipped commands, causal measured-yaw integration and the correct DARE-prior KF gain. Noise draws are freshly declared; these are not the lost historical P trajectories. Scalar-Q calibration and local/global extensions were deferred to focus the bottleneck diagnostic.

Method	$Q_{\text{fit}} = 0.01$		$Q_{\text{fit}} = 1$	
	Broadband	Transient	Broadband	Transient
Learned KF	2.14	−9.20	−6.26	−12.19
Learned LQI	−176.26	−370.17	−19.92	−23.41
Learned Both	3.02	0.04	−1.57	−2.42

Entries are mean paired fleet tracking gains in percent after averaging the two deployments per recipient. Both improves only three of five primary-panel broadband fleets and two of five transient fleets. Prediction gains and better beta reconstruction still do not imply a useful controller/observer combination.

All 1,600 rollouts were finite. Independent saved-step and metric errors were below 2.0×10^{-15} . Strict independent full-path replay passed for 1,545 records; 55 were sensitive to floating-point ordering under clipped feedback, with maximum tracking-RMSE change 0.11949 degrees/s. The tolerance remains 10^{-8} ; these failures are retained separately from saved-step integrity. Finite trajectories and frozen-speed radius checks do not certify scheduled stability. The next frozen diagnostic compares controller models with true states, partial truth corrections and common-input observer replay.

60 10H-Q reconstruction: the feedback bottleneck

The frozen diagnostic tests all combinations of controller and state source using the audited reconstructed P parent at 6afb002. Controllers are public nominal, learned N and simulator oracle; supplied states are nominal KF, learned KF or true plant state. Three additional learned-Both cases replace only beta, both axle forces, or beta plus both forces in the controller input. These interventions never overwrite the observer’s internal state. The measured yaw integrator, noise draws, clipping, costs, speed schedules and assignments remain fixed. Q performs no fitting or membership selection.

The 4,800 rollouts cover five development fleets, ten recipients, two separate donor deployments, two maneuvers, two fitting panels and twelve cases. A further 2,400 common-input replays compare nominal, learned and oracle KFs on the same nominal and Both records. The original M-P artifacts were unavailable: this is a fresh reconstruction from intact L inputs, not reproduction of lost historical results. Confirmation seeds 591–600 and 611–620 remain sealed.

60.1 Tracking and interventions

Controller / supplied state	$Q_{\text{fit}} = 0.01$		$Q_{\text{fit}} = 1$	
	Broadband	Transient	Broadband	Transient
Nominal / nominal KF	0.00	0.00	0.00	0.00
Nominal / learned KF	2.14	−9.20	−6.26	−12.19
Nominal / true state	4.55	6.18	2.78	4.66
Learned / nominal KF	−176.26	−370.17	−19.92	−23.41
Learned / learned KF (Both)	3.02	0.04	−1.57	−2.42
Learned / true state	−159.75	−306.38	−22.85	−25.27
Oracle / nominal KF	1.17	3.42	0.20	2.14
Oracle / learned KF	5.90	−5.46	−4.11	−10.43
Oracle / true state	5.89	7.75	4.15	6.26
Both + true beta	3.00	−0.59	−1.80	−2.70
Both + true forces	−158.04	−304.12	−16.93	−18.74
Both + true beta and forces	−158.21	−306.18	−23.44	−26.43

Entries are mean paired fleet tracking gains in percent against nominal output feedback. First average the two deployment RMSEs per recipient, then ten recipients per fleet; percentage gains

are paired within each of five fleets. Positive entries mean lower tracking RMSE. The two panels change both learned plants and KF design settings and are not a causal comparison of fitting Q .

The prespecified full-state comparison is decisive in the primary panel: learned versus nominal control gives -172.36% broadband and -333.60% transient mean fleet gains. Only one of five fleets improves; recipient counts are 19/50 and 12/50. Supplying true states does not rescue learned control. Correcting Both's forces instead gives -160.93% and -294.92% against Both, with only 14/50 and 13/50 recipients improving. Beta correction has little effect. In the comparison panel, learned full-state control improves four of five fleets, but large harm in the fifth produces negative fleet-average gains of -26.39% and -31.34% against nominal full-state control. The tails matter.

Oracle full-state control improves on nominal full-state control by only 1.42% broadband and 1.68% transient in the primary panel. All five fleet means improve, but only 25/50 and 26/50 recipients do. Its gains against nominal output feedback are 5.89% and 7.75% . Thus headroom is modest under these fixed tasks and LQI costs; the oracle is not an optimal tracking bound.

60.2 Common-input reconstruction and compensation

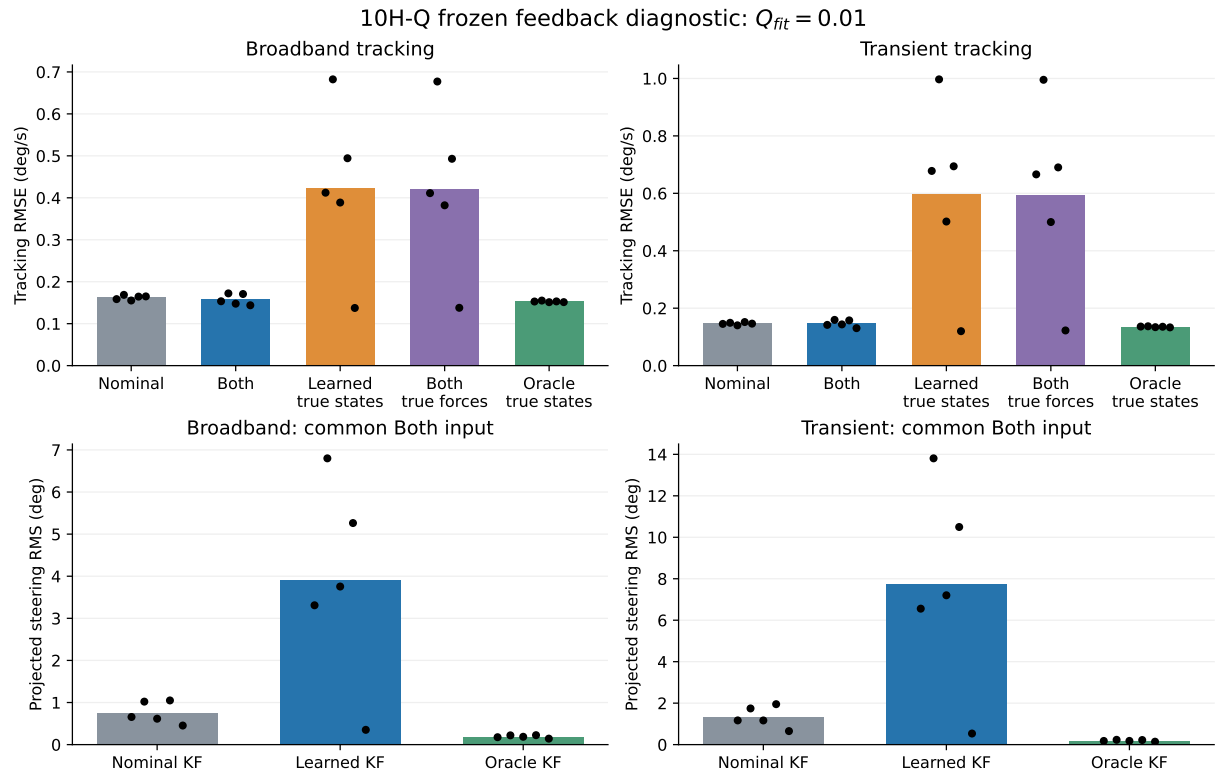


Figure 63: Primary-panel tracking and common-input projected steering error. Bars are means of five fleet summaries; dots show individual fleets. Lower panels hold Both's recorded inputs and controller gains fixed. Their projection is an instantaneous command-error diagnostic, not a decomposition of tracking loss across different closed-loop trajectories.

On the same Both records, learned KF beta RMSE improves from 0.03672 to 0.02581 degrees broadband and from 0.06199 to 0.04558 degrees transient. However, front/rear force-per-mass RMSEs increase from 0.02325/0.01174 to 0.05325/0.05438 m/s^2 broadband and from 0.03366/0.01392 to 0.09006/0.08881 m/s^2 transient. These are mean recipient RMS values after deployment averaging, not a pooled trajectory RMS.

With the same learned state gain, total projected steering RMS is 0.759/3.896/ 0.190 degrees for nominal/learned/oracle KF broadband and 1.336/7.719/0.190 degrees transient. The learned

KF’s signed joint-force contributions are 3.795 and 7.639 degrees. The same ordering holds on nominal records and in the comparison panel. All five component second moments and their cross terms are retained: summing component RMS values would discard cancellation.

The combination of poor learned full-state control, harmful force corrections and larger common-input force projections is consistent with compensation between controller-model and observer errors. It does not establish an additive causal tracking decomposition or prove that bounds alone caused the failure. Better beta reconstruction is insufficient evidence of a useful complete physical state estimate for the learned LQI.

60.3 Validation, limits and consequence

All 4,800 rollouts are finite, with no design failures or beta/steering limit violations. Saturation is retained in the per-case tables. All 1,600 overlapping P records agree exactly in every array. Independent saved-step and metric errors are below 2.0×10^{-15} . An independent observer replay and projection audit covers 7,200 diagnostic rows, with maximum metric error 2.85×10^{-14} . A separate verification checks 620 provenance references, the two-stage paired aggregation, and 400 cross-panel nominal/oracle full-state trajectory pairs, which agree exactly despite differing unused KF internals.

Strict independent full-path replay at tolerance 10^{-8} passes for 4,684 records; 116 remain sensitive to floating-point ordering, with maximum tracking-RMSE change 0.11949 degrees/s. The principal primary-panel learned Both, learned full-state and force-correction paths are not sensitive. Nominal full-state paths include sensitive cases. Conservative bounds using each record’s observed alternate-path RMSE change preserve the signs of the three primary prespecified contrasts above. These bounds are neither confidence intervals nor a general robustness guarantee. Tolerances were not relaxed.

Frozen-speed checks do not certify scheduled stability. Primary learned full-state control has maximum frozen radius 0.9964 despite poor tracking; comparison-panel learned full-state control has five of 100 deployed pairs with radius at least one and maximum 4.426. Finite clipped rollouts therefore do not establish a usable controller family.

The inherited coefficient bounds are heuristic, even with exact physical coupling. Post hoc inspection of deployed primary models gives front lengths down to 0.223 m and rear lengths up to 3.713 m. These are inferred model coordinates, not measurements of client geometry. Output prediction and the coupling identity do not ensure plausible control-design models. The next priority is independently justified physical-coordinate constraints and controller-relevant validation before adding observer complexity. This diagnostic does not demonstrate the intended iterative federated improvement of KF and controller matrices.

Execution, restoration and audit commands are in `docs/experiment_10hq.execution.md`; result tables, second moments, recipient tails and provenance are committed. The numerical environment is Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0 and pandas 2.2.3 with one BLAS thread. The repository suite passes 170 tests, and Q sources pass scoped lint.

References

- [1] Pierre Apkarian, Pascal Gahinet, and Greg Becker. Self-scheduled H_∞ control of linear parameter-varying systems: A design example. *Automatica*, 31(9):1251–1261, 1995.
- [2] Borja Balle and Yu-Xiang Wang. Improving the gaussian mechanism for differential privacy: Analytical calibration and optimal denoising. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 394–403, 2018.

- [3] Matteo Corno, Giulio Panzani, Federico Roselli, Marco Giorelli, Davide Azzolini, and Sergio Matteo Savaresi. An LPV approach to autonomous vehicle path tracking in the presence of steering actuation nonlinearities. *IEEE Transactions on Control Systems Technology*, 29(4):1766–1774, 2021.
- [4] Avishek Ghosh, Jichan Chung, Dong Yin, and Kannan Ramchandran. An efficient framework for clustered federated learning. In *Advances in Neural Information Processing Systems*, volume 33, pages 19586–19597, 2020.
- [5] Alex Gimondi, Matteo Corno, and Sergio Matteo Savaresi. Linear parameter varying path tracking control for over-actuated electric vehicles. *Frontiers in Control Engineering*, 2:750190, 2021.
- [6] Christian Hoffmann and Herbert Werner. A survey of linear parameter-varying control applications validated by experiments or high-fidelity simulations. *IEEE Transactions on Control Systems Technology*, 23(2):416–433, 2015.
- [7] MathWorks. Simulate and predict identified model output. <https://www.mathworks.com/help/ident/ug/definition-simulation-and-prediction.html>, 2026. Documentation accessed 2026-10-06.
- [8] MathWorks. Vehicle body 3dof. <https://www.mathworks.com/help/vdynblks/ref/vehiclebody3dof.html>, 2026. Documentation accessed 2026-10-06.
- [9] MathWorks. What is residual analysis? <https://www.mathworks.com/help/ident/ug/what-is-residual-analysis.html>, 2026. Documentation accessed 2026-10-06.
- [10] H. Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, pages 1273–1282, 2017.
- [11] Hans B. Pacejka. *Tire and Vehicle Dynamics*. Butterworth-Heinemann, 3 edition, 2012.
- [12] Felix Sattler, Klaus-Robert Müller, and Wojciech Samek. Clustered federated learning: Model-agnostic distributed multitask optimization under privacy constraints. *IEEE Transactions on Neural Networks and Learning Systems*, 32(8):3710–3722, 2021.
- [13] Virginia Smith, Chao-Kai Chiang, Maziar Sanjabi, and Ameet S. Talwalkar. Federated multi-task learning. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [14] Roland Tóth. *Modeling and Identification of Linear Parameter-Varying Systems*. Springer, 2010.
- [15] Han Wang, Leonardo F. Toso, and James Anderson. FedSysID: A federated approach to sample-efficient system identification. In *Proceedings of the 5th Annual Learning for Dynamics and Control Conference*, volume 211 of *Proceedings of Machine Learning Research*, pages 1308–1320, 2023.